Fundamental Fractal-Geometric Field Theory (FFGFT) or T0 Theory – Time-Mass Duality

Part 7: The Lattice in Hilbert Space, Spectral Theory and Galois Structure

Johann Pascher

2026

Contents

A Introduction to Volume 7 . . . 3

I Part I — Four onto Three, the Closure Scale and the Lattice in Hilbert Space 5

B Doc. 311 — Four onto Three — How does a 4D lattice arrive at three spatial dimensions? . 7

C Doc. 312 — The Closure Scale: Λ as a Derived Quantity of the Compact Geometry . . . 21

D Appendix A to Doc. 312 — Altermagnetism as a Laboratory Analogy . . . 41

E Doc. 313 — No Beginning: The Thermal History on the Time Cycle . . . 44

F Doc. 314 — The Lattice in Hilbert Space: Geometry, Spectrum and Fibre . . . 60

G Doc. 315 — The Form of K_{frak}: Additive or Multiplicative? . . . 79

H Doc. 316 — FFGFT and the Riemann Zeta Function . . . 87

II Part II — Leptons, Spectral Theory and Gauge Structure 113

I Doc. 317 — Topological Origin of Lepton Generations: KSAU and FFGFT . . . 115

J Doc. 318 — Scope of Mass Derivations in FFGFT: Rest Masses, Hadron Dynamics, and Open Bridges . . . 122

K Doc. 319 — The Proton as a Vibrating Torus: Geometric Foundations of Mass Emergence in FFGFT . . . 130

L Doc. 320 — Detailed Field-Geometric Spectral Theory . . . 136

M Doc. 321 — Algebraic Derivation of the $SU(3)_c$ Gauge Structure . . . 144

N Doc. 322 — Spectral Theory and Hilbert-Space Embedding of the FFGFT Tori . . . 152

O Doc. 323 — Derivation of the Weinberg Angle at M_Z from FFGFT Geometry . . . 159

III Part III — Hyperbits, the Hawking Mechanism, Coupling and the Continuum 163

P Doc. 324 — $G(6, \mathbb{Z}_3\mathbb{C})$ Vacuum Structure and FFGFT . . . 165

Q Doc. 325 — The FFGFT Hawking Mechanism . . . 173

R Doc. 326 — Black Holes from Hyperbits and from FFGFT . . . 180

S Doc. 327 — Self-Adjointness of the Fundamental Fractal Operator $\hat{F}$ . . . 191

T Doc. 328 — Coupling Regimes, Resonance, and Synchronisation . . . 196

U Doc. 329 — Is the Stability Threshold n_{thresh} Universal? . . . 202

V Doc. 330 — Three Operations from T^4 to Observable Space . . . 211

W Doc. 332 — Two Roads from the Discrete to the Continuum . . . 219

X Doc. 333 — K_{frak}, Unrolling and the $\tilde{T} \cdot m$ Duality . . . 227

Y Doc. 334 — Superposition Without Time 233
Z Doc. 335 — Mass-Bound Time Scales and Interferometry 238

IV Part IV — The Galois Bridge: GF(9), GF(27) and the Zeta Function of the Torus 242

aa Doc. 336 — The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$ 244
ab Doc. 337 — Time Emergence: FFGFT and GALG Compared 255
ac Doc. 338 — Mass Calculations and Galois Comparison 262
ad Doc. 339 — Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$ 268
ae Doc. 340 — Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure 276
af Doc. 341 — GF(27) in GALG and FFGFT: Why G(3) Contains No GF(27) but G(6) Does 284
ag Doc. 342 — Factorisation Classes in G(6) and the Harmonics of the Primes 297
ah Doc. 343 — The Zeta Function of the T^4/Z_3 Torus: Orbifold Projection and Dirichlet L-Functions 304

Chapter A

Introduction to Volume 7

Seventh Volume of the Document Collection

Volume 7 continues the extension of the collection begun in Volume 6 and comprises Documents 311 to 343 from August and early September 2026. In this period the focus of the corpus shifts from the projection into observable space to the algebraic structure of the compact geometry itself.

Content and Focus

The first part begins with the question of how a four-dimensional lattice arrives at three spatial dimensions (Doc. 311), and leads via the closure scale, in which Λ appears as a derived quantity of the compact geometry, and the thermal history on the time cycle to the lattice in Hilbert space (Doc. 314). There the lattice is read as the index set of the Fourier modes; geometry becomes spectrum and fibre. The investigation of the form of K_{frak} and the comparison with the Riemann zeta function complete this part.

The second part brings together the lepton generations in a topological reading, the scope of the mass derivations, the spectral theory of the FFGFT tori with its Hilbert-space embedding, the algebraic derivation of the $SU(3)_c$ gauge structure from $\mathbb{Z}_3$ triality, and the derivation of the Weinberg angle at M_Z with the complete renormalisation-group running. The third part contains the comparison with the hyperbit vacuum structure $G(6, \mathbb{Z}_3\mathbb{C})$, the FFGFT Hawking mechanism, the self-adjointness of the fundamental operator $\hat{F}$, the coupling regimes of resonance and synchronisation, and the two roads from the discrete to the continuum.

The fourth part contains the Galois bridge. Starting from the finite field $GF(9)$ (Doc. 336) it is shown that the Frobenius automorphism and the FFGFT sector pairing are structurally identical; via $GF(27)^*$ there follow the comparison of the mass ratios with Galois group orders, the Frobenius separation of the massive and massless sectors, the neutrino mass hierarchy, the factorisation classes in $G(6)$ with the harmonics of the primes, and the zeta function of the $T^4/\mathbb{Z}_3$ torus.

Organisation

The 33 documents are divided into four parts: four onto three, the closure scale and the lattice in Hilbert space (Docs. 311–316); leptons, spectral theory and gauge structure (Docs. 317–323); hyperbits, the Hawking mechanism, coupling and the continuum (Docs. 324–335); the Galois bridge (Docs. 336–343).

Reading the Documents

Documents 311 to 314 were originally typeset as multi-chapter treatises; in the book their chapters appear as sections of the respective document. Status markers and the corrections register apply as in the preceding volumes; in each case the most recent version of a result in the corpus is authoritative.

Status

The documents in this volume reflect their respective versions in the corpus as of October 2026.

Part I

Part I — Four onto Three, the Closure Scale and the Lattice in Hilbert Space

Chapter B

Doc. 311 — Four onto Three — How does a 4D lattice arrive at three spatial dimensions?

What this is about

A lattice in $\mathbb{R}^4$ has four directions on an equal footing. Measurement takes place in three. How does one get from the four to the three?

There are exactly three routes, and their costs differ greatly. This note walks through them, condenses them at the end into two conditions, and applies those conditions to three cases: the author's own construction, generic projection constructions, and string theory.

Status: working note. Not part of the A-Series. The chapters on the routes and on the decomposition finding are computed (`ffgft_311_vier_auf_drei.py`); the chapter "Application" poses a question and makes no assertion.

B.1 The three routes

Route 1 — Compactify

The fourth direction becomes a circle of radius R. Kaluza 1921, Klein 1926.

What it buys. On a circle only integer windings are admitted, and from that a mode tower arises with $m_n \propto n/R$. The mode index *is* a scale. Nothing has to be attached.

What it costs. R is a free modulus. Whoever does not say what fixes R has gained a scale and acquired a parameter — that is, nothing. This is the price string theory pays with its landscape.

The modulus disappears only when R is bound to another quantity. In FFGFT this happens through $\tilde{T} \cdot m = 1$: once the mass is fixed, the cycle is fixed, and nothing is left to tune.

Route 2 — Project

The 4D lattice is projected onto a 3D section. The standard route for quasicrystals, cut-and-project.

What it costs. The projection has a window parameter, and the lattice invariants do not survive it. For D4 onto the first three coordinates:

$$24 \text{ shortest vectors} \longrightarrow 18 \text{ distinct images.}$$

Six pairs coincide. What arrives after projection is no longer a lattice with kissing number 24, but an aperiodic point pattern with different invariants.

Route 3 — The fourth is not space

The fourth coordinate carries a different quantity: time, mass, charge, an internal index. Then only three directions are space, and there is nothing to reduce.

What it costs. One must say *what* it carries, and the statement must bear weight. This is no formality: a fourth direction called “mass” without anything binding it to a mass is merely a label.

B.2 Routes 1 and 3: each incomplete on its own

The two look alike because both take the fourth direction out of space. But they do different things, and the difference is precisely the point at which a construction holds or does not.

Route 1 makes the direction compact and leaves open what it carries. Pure Kaluza-Klein. A circle of radius R enforces integer windings, from that a mode tower $m_n \propto n/R$ arises, and the mode index is a scale. But R itself is free. One has gained a scale and bought a parameter along with it.

Route 3 says what the direction carries and leaves open whether it is compact. It is called “mass” or “charge” or “internal index”. Without compactness this is a label: nothing enforces integer values, without integrality there is no mode tower, and without a mode tower no scale.

Each is incomplete on its own, in opposite ways:

Route 1: mechanism without meaning
Route 3: meaning without mechanism

What closes the gap

Both at once, and the binding between them.

	compact	bound	result
Kaluza-Klein	yes	no	scale present, but free modulus
mere label	no	in name only	no mode tower, no scale
FFGFT	**yes**	**yes, via $\tilde{T} \cdot m = 1$**	**scale without modulus**

The reciprocity closes both gaps in one step. The mode tower acquires a meaning because the direction is the mass direction. And the free radius disappears because R is no longer choosable once m is fixed — the cycle *is* the mass, not a container for it.

The sharper question

The real distinction is therefore not “which of the three routes”, but:

> Is the fourth direction compact **and** bound to something — or only one of the two?

Whoever only compactifies has a modulus to stabilise. Whoever only names has a word. Both together and bound to each other is the only case in which a scale arises without a parameter following behind.

This is also the touchstone for other people’s constructions, and it can be put in a single question: *what fixes the radius?* If no answer comes, the scale has merely been displaced.

B.3 Route 2 in detail: what projecting presupposes and costs

Route 2 differs from the other two not in degree but in kind.

Projection needs a standpoint outside

Compactifying is a property of the object: the circle is there or it is not, regardless of who is looking. Interpreting the fourth direction as the mass direction is a statement about the structure.

Projecting is a map **from outside**. It needs a cut direction and a window, and both are chosen by the observer. They say nothing about the lattice; they say something about the access to it.

This is a different type of free parameter from the radius in Route 1. A radius is a feature of the object and can be bound to another quantity of the object — exactly what $\tilde{T} \cdot m = 1$ achieves. A cut direction cannot be bound. It can only be fixed, and the fixing remains a stipulation.

What happens to the 24 vectors

Under projection onto the first three coordinates:

24 vectors	$\longrightarrow$	18 images
12 FCC vectors ($x_4 = 0$)	$\longrightarrow$	12 images, length $\sqrt{2}$, untouched
12 winding vectors	$\longrightarrow$	6 images, length 1, merged in pairs

For $(1, 0, 0, +1)$ and $(1, 0, 0, -1)$ have the same image $(1, 0, 0)$. The difference between two opposite windings disappears.

And the image contains two different lengths, 1 and $\sqrt{2}$. The image is no longer a lattice with a uniform kissing number; packing density in the usual sense loses its meaning there.

This is the real loss. Under compactification the twelve winding neighbours remain distinguishable — $+1$ and -1 are different windings and carry different quantum numbers. Under projection they are made indistinguishable. What leads out of space is erased rather than reinterpreted.

Where the irrational numbers come from

For a rational cut direction, say a coordinate axis, finitely many image lengths remain — here two. The shortening of the winding vectors by $1/\sqrt{2}$ is irrational, but it repeats.

In genuine cut-and-project the cut direction is chosen irrational, and then the projection lengths are generically irrational. For the direction $(1, \varphi, 0)$:

$$\begin{aligned}
(1, 0, 0) &\longrightarrow 0.525731 \\
(0, 1, 0) &\longrightarrow 0.850651 \\
(1, 1, 0) &\longrightarrow 1.376382 \\
(1, -1, 0) &\longrightarrow -0.324920
\end{aligned}$$

Every length a new number, and the image is **dense**. Only the window makes it discrete again — the second free parameter.

This is exactly how quasicrystals arise. And exactly how φ enters there:

> The stretching factors are irrational, yes. But they are properties of the **chosen direction**, not of the lattice. Choose differently and they are different numbers.

The consequence for other constructions

Whoever brings φ in through a cut-and-project construction has put it in. It does not come from the lattice but from the choice of projection direction. That is admissible, provided it is said — and it is something fundamentally different from a number that follows from the structure.

The touchstone from the previous chapter thus acquires a second question beside *what fixes the radius?*:

Who chose the cut direction, and from what?

If no answer comes, all numbers originating from the projection are properties of the observer.

B.4 The reversal: where φ is found, the direction is no longer free

If φ appears in cut-and-project because the cut direction was chosen accordingly — what holds when φ is not chosen but *found*?

Where φ stands in the corpus

A120 derives $\theta = 2/9$ from the redistribution of the lepton modes under a five-fold rotation. The three weights:

$p_0 = 2/9$	$= 0.2222222$	**rational**	— no φ in it
$p_2 = (5 - 3\varphi)/9$	$= 0.0162109$	contains φ	
$p_1 = (2 + 3\varphi)/9$	$= 0.7615669$	contains φ	
sum	$= 1.0000000$		

Remarkably, the share passing into the trivial mode is **rational**. φ sits solely in the split between the two non-trivial modes. All three carry the denominator $9 = 3^2$, the same Z_3 structure in which $2/9$ sits as an orbit address.

φ is not a choice here but an eigenvalue

In cut-and-project φ appears because the observer laid the direction that way. Here φ is the eigenvalue of the structure itself:

$$\varphi^2 = \varphi + 1.$$

A_5 is the icosahedral group. Its five-fold axes are the twelve vertices with coordinates $(0, \pm 1, \pm\varphi)$ and cyclic permutations. **The directions are completely fixed by** φ — no angle remains choosable, only which of the six axes, and A120 shows that this is a discrete two-fold choice ($2/9$ or $4/9$), not a free parameter.

The uniqueness has been tested there as well: 200 random rotation axes at 72° give values between 0.036 and 0.452, not a single one hitting $2/9$. For a three-fold axis the result is $2/5$, at 144° it is $4/9$.

And the window as well

An icosahedron has no shape parameter. Its circumradius follows from the edge:

$$|\text{vertex}| = \sqrt{1 + \varphi^2} = 1.9021130, \qquad \frac{\text{circumradius}}{\text{edge}} = 0.9510565 = \frac{\sqrt{\varphi\sqrt{5}}}{2}.$$

All size ratios are determined by φ. What remains is one scale factor — and that is exactly the single quantity A085 carries as the anchor.

The juxtaposition

	cut direction	window shape	window size
cut-and-project, generic	two free angles	free	free
icosahedral	6 axes, discrete	the icosahedron, shape-fixed	one scale only

A continuum of freedoms on one side, a discrete selection plus one scale on the other.

What this means

Where φ is chosen, it has been put in.
Where φ is found, it fixes direction and window.

This is not wordplay but the difference between a parameter and a structure. A construction in which φ is set as the projection direction has a free angle. One in which φ occurs as the eigenvalue of a group has a discrete selection.

And it reverses the touchstone. Instead of only asking *who chose the direction*, one can ask:

Is there a structure whose eigenvalues enforce the direction?

If an answer comes, the direction is no longer a stipulation.

B.5 Two conditions instead of three routes

If φ enforces direction and window, Route 2 no longer costs any free parameters — as little as Route 1 with a bound radius. The division into three routes thereby becomes obsolete, but not entirely: a second condition remains, and it is independent of φ.

What becomes transferable

The balance of freedoms. If a structure enforces the assignment, it is no longer a stipulation, whether one calls it compactification or projection.

Route 1 with free radius	one parameter
Route 1 with $\tilde{T} \cdot m = 1$	none
Route 2 with free direction	several parameters
Route 2 with icosahedral axis	none

The distinction thus does not run between the routes, but between *enforced* and *stipulated*.

What does not become transferable

The conservation of information. A projection is by definition not injective. Even with an enforced direction, $(1, 0, 0, +1)$ and $(1, 0, 0, -1)$ merge into the same image $(1, 0, 0)$. Twelve winding vectors become six images, no matter who fixed the direction.

Under compactification this does not happen. The winding number remains distinguishable — $+1$ and -1 are different states and carry different quantum numbers, although space has three dimensions.

Where the two nevertheless meet

Under compactification the information does not vanish; it **migrates**. From position into an index. What was a fourth coordinate before is a winding number afterwards.

A projection with an enforced direction does the same — **provided the preimage pair carries a conserved index**. If it carries none, it is loss. If it carries one, it is a rebooking, and then the difference from compactification is merely one of phrasing.

The rule

Not three routes, but two conditions.

1. Is the assignment enforced by a structure or stipulated by the observer?

2. Is the lost positional information caught by a conserved index?

If both are met, it does not matter what the procedure is called. If the first fails, one has parameters. If the second fails, one has loss.

What this means for the examples

	condition 1	condition 2
Kaluza-Klein, free radius	no	yes
cut-and-project, generic	no	no
FFGFT, $\tilde{T} \cdot m = 1$	yes	yes
icosahedral projection	yes	only with an index

The last row is the interesting one. It is the case in which φ removes the stipulation but the question of the index remains open — and that question cannot be answered by geometry, only by stating which quantity carries the index.

In FFGFT it is the winding number on the mass direction. In a pure projection construction the corresponding quantity would have to be named, otherwise the loss is real.

B.6 The finding that decides the matter

D4 does not need to be reduced at all. It decomposes by itself.

Of the 24 shortest vectors:

	count	meaning
with $x_4 = 0$	**12**	lie entirely in the first three coordinates
with $x_4 \neq 0$	**12**	six with $x_4 = +1$, six with $x_4 = -1$

And the twelve with $x_4 = 0$ are not just anything:

$$\text{kissing number of D3} = \text{FCC} = 12.$$

FCC stands for *face-centered cubic* — a densest sphere packing in three dimensions and the unique densest lattice packing, stacking sequence ABCABC. A sphere in the interior touches six neighbours in its layer, three above and three below: $6 + 3 + 3 = 12$. That this is the maximum goes back to the question of Newton and Gregory and was finally proved in 1953.

Lattice and packing are the same object in two languages: the shortest vectors of D3 are $(\pm 1, \pm 1, 0)$ and permutations, exactly twelve, and they point to the twelve contact points.

Why the pyramid is fundamental in three dimensions

The twelve neighbour vectors form, as a convex hull, a **cuboctahedron** — eight triangles, six squares, all vertices at distance $\sqrt{2}$. This is the neighbourhood figure of three-dimensional space.

And the same packing has a second shape everyone knows. Placing spheres into the hollows of a square layer — the orange stack — each sphere in the interior has

$$4 \text{ neighbours below} + 4 \text{ in the plane} + 4 \text{ above} = 12$$

Exactly the kissing number. **The square pyramid is FCC**, merely stood up differently: the view runs along the cube diagonal instead of along the edge. Hexagonal layers in the sequence ABCABC and square layers with hollows describe the same lattice.

The fundamental point is uniqueness among lattices. FCC is the unique densest **lattice** packing (Gauss 1831); the maximal density $\pi/(3\sqrt{2})$ was proved by Hales in 1998 (formally verified in 2014) — it is also reached by HCP and every other stacking sequence (Barlow stacking) with the same kissing number 12; HCP, however, is not a lattice packing. The pyramid is thus, among lattices, not one building form among several, but the form the density enforces. This is why copper, gold and aluminium crystallise this way, and why the fruit stand works.

For this note two things matter in that. First, the 12 is not a conventional number but provenly maximal in 3D. Second, three-dimensional space thereby has a distinguished neighbourhood figure against which one can measure whether a construction computes in it or not: whoever works spatially with 24 does not work in the cuboctahedron.

$$24 = 12 + 12$$

Half of the D4 kissing number is three-dimensional space. The other half leads out of it — and becomes winding neighbours as soon as the fourth direction is compact, six in each direction of circulation.

This is the lattice form of $\mathrm{T}^4 = \mathrm{T}^3 \times S^1_m$: three spatial circles plus the fourth direction, and the split $12 + 12$ falls out instead of being put in.

B.7 What follows for the invariants

The invariants of D4 are **4D quantities**. They are spatial quantities only if all four directions are space.

	D4 (4D)	D3 = FCC (3D)
kissing number	24	12
packing density	$\pi^2/16 = 0.616850$	$\pi/(3\sqrt{2}) = 0.740480$

Whoever computes with 24 as a spatial kissing number thereby implicitly asserts four spatial dimensions. Whoever has three spatial dimensions plus one other direction computes spatially with 12, and the other twelve neighbours are something else — not neighbours in space, but windings.

This is no small matter. It affects every quantity formed from the neighbourhood number.

Which packing density belongs in K^{-36}?

From this a question follows for the author's own construction, and it had to be checked: the bulk exponent of the sector ladder (A270) is bound to a packing density. If only three of the four directions are space — is it then the 4D density or the 3D density?

	inverse density	required exponent	distance to $k^*/100$	to integer
D4 (4D)	$16/\pi^2 = 1.6211389$	**35.9926**	0.098 = **0.27 %**	0.0074
D3 = FCC (3D)	$3\sqrt{2}/\pi = 1.3504745$	22.3836	13.707 = **38.08 %**	0.3836

And at integer exponent:

D4: $(75/74)^{36} = 1.621301$ against 1.621139 $+0.010\,\%$
D3: $(75/74)^{22} = 1.343538$ against 1.350474 $-0.514\,\%$

(In the notation K^{-n} with $K = 74/75$; written out as $(75/74)^n$ to keep the sign of the exponent unambiguous.)

How sharp the assignment is

The five neighbouring integer exponents come out very differently — the neighbours of 36 lie roughly a hundred times further away:

n	$(75/74)^n$	against $16/\pi^2$
34	1.578354	−2.639 %
35	1.599683	−1.323 %
36	1.621301	+**0.010** %
37	1.643210	+1.361 %
38	1.665416	+2.731 %

Not a fit with a soft edge, but a window with two edges: the exponent is unambiguous.

Two qualifications belong next to this. The $0.010\,\%$ measures the rounding of 35.9926 to the integer 36, not the agreement of the two routes — their distance is 0.098 in the exponent and is covered by the stated precision of the factor 100 (register 190, R67). And the sharpness concerns the *assignment* to the integer, not the *derivation*: why a recursion depth should correspond to a packing density remains open (P35).

The result is unambiguous. The 3D density does not fit. It requires exponent 22.38 instead of 36.09, i.e. $38\,\%$ off, and it additionally lies close to a rounding boundary — 0.38 against 0.007 for the 4D density. Even as an isolated hit it would be weak. **The 4D density belongs in K^{-36}.**

Why this is not a contradiction

The bulk is T^4, not T^3. The fourth direction contributes to the packing even though it is not space — because packing happens where the torus is, and the torus has four directions.

This is admissible, but it must be said. The mass direction is a genuine torus direction with integer windings, not a caption. It counts in the packing because it is geometrically present, and it does not count in the spatial kissing number because it is not space.

The two statements thus stand side by side without contradicting each other:

spatial neighbourhood	$\longrightarrow$	12 (FCC, the $x_4 = 0$ vectors)
packing of the bulk	$\longrightarrow$	$\pi^2/16$ (all four directions)

And what follows for comparable constructions

The same number, two different justifications — and typically only one of them is spoken aloud.

In FFGFT, $\pi^2/16$ is the packing density of a four-dimensional torus of which three directions are space and one carries the mass. That is a statement that can be justified and is justified.

In a construction that computes on the same lattice but treats all four directions as spatial, $\pi^2/16$ is the packing density of a four-dimensional *space*. That is an assertion about the number of spatial dimensions — and it is rarely spoken aloud, because the number appears as a neutral lattice constant.

It is not neutral. Whoever uses it says something about the number of spatial dimensions, whether they want to or not.

B.8 Application: constructions with four spatial lattice directions

The case that occurs in practice: a construction computes on D4 with 24 offsets, has a step counter for the iteration, and nowhere declares a time direction. From this it follows necessarily: **all four coordinates are space.**

That is an assertion about the world, not merely about the model — an object in four spatial dimensions is not a particle in our space, but an entity in a space that does not exist. Even with a matching node count it would remain open what that count has to do with a mass.

And the ways out are uncomfortable:

- One direction becomes time after the fact → then **three** spatial dimensions remain, and the spatial kissing number is 12, not 24. Everything sitting on 24 would have to be recomputed.
- Time is appended as a fifth direction → then there are four spatial dimensions, one too many.
- It stays at $4+0$ → then one must say how a four-dimensional spatial object acquires a mass in our space.

This is the fundamental question for any such construction, and it precedes the numerical findings: as long as it is open, any correction to individual quantities addresses only symptoms.

It is mostly not asked. Where time appears neither in the code nor in the documents nor in the correspondence, the question does not yet exist for those involved — and that is understandable, for an iteration *feels* like time without being one.

Asked, not answered. Whoever builds such a construction knows its internals; this note reads from outside. There may be answers not visible from here.

B.9 The two conditions, applied to string theory

The rule from the chapter “Two conditions” can be applied, and the result is not sweeping. It names exactly one place.

Condition 2 is met exemplarily

The compact directions are compact. Nothing is projected, nothing merges. Winding numbers, Kaluza-Klein momentum, winding sectors, Hodge numbers, intersection

numbers — all conserved indices. The positional information of the six rolled-up directions migrates completely into quantum numbers.

This is the strength of the approach and should not be passed over. In the language of the rule: the information is rebooked, not lost.

Condition 1 is met by half

The dividing line runs through the middle of the construction.

Enforced:

$D = 10$ from anomaly freedom and the critical dimension
$N = 1$ SUSY in 4D demands SU(3) holonomy, hence Calabi-Yau
from that: $10 - 4 = 6$ compact directions

The **number** of rolled-up directions is not choosable. That is a genuine enforcement, and it is stronger than anything FFGFT can adduce for its four circles — there the choice of T^4 is marked as [STIPULATION].

Stipulated:

which Calabi-Yau manifold	($\sim 10^5$ known, total number unknown)
flux quantum numbers	($\sim 10^{500}$ combinations)
Kähler and complex structure moduli	(continuum, then stabilised)
standard embedding or not	

The number of directions is enforced. The **geometry within** is not.

What the landscape thereby is

> The landscape is exactly the unmet part of condition 1.

Not an error and not a scandal, but the place where the structure stops enforcing. A precise statement of location instead of an accusation.

And it explains why the problem is so stubborn: it cannot be fixed by more computation. Where a structure does not enforce, no computation can make it do so. An additional condition from outside is needed — and that is exactly what has been sought for decades.

The asymmetry that connects the two

What is compactified are **spatial** directions. Time stays outside. By A060 it follows: no dynamical scale in the state space. All scales must come from elsewhere, and in this construction they come from the moduli.

But moduli are exactly what condition 1 does not cover.

time outside → no scale in the structure
→ scales from moduli
→ moduli are the unenforced part
→ landscape

The landscape arises where the scales must come from. And the scales must come from there because time stands outside. It is a single choice that produces both: roll up six spatial directions and leave time open.

What this does not say

It does not say that string theory is wrong. It meets condition 2 better than anything considered here, and its dimension count is more compellingly justified than the four circles of FFGFT.

Nor does it say that the alternative would be cheaper. A rolled-up time costs causality objections, Pauli's theorem and the thermal reading — and it is nowhere accepted so far.

What it says is exclusively this: **the landscape and the necessity of attached scales are not two problems but one.** Whoever wants to solve one must start at the other.

B.10 Verification script

```
import itertools, math, collections

D4 = [v for v in itertools.product([-1,0,1], repeat=4)
      if sum(x*x for x in v) == 2]
D3 = [v for v in itertools.product([-1,0,1], repeat=3)
      if sum(x*x for x in v) == 2]

print(len(D4))                                  # 24
print(sum(1 for v in D4 if v[3] == 0))          # 12
print(sum(1 for v in D4 if v[3] != 0))          # 12
print(len(D3))                                  # 12  (FCC)
print(len(set(v[:3] for v in D4)))              # 18  (projection)
print(collections.Counter(
    v[3] for v in D4 if v[3] != 0))             # {1: 6, -1: 6}

print(math.pi**2 / 16)                          # 0.616850  D4
print(math.pi / (3 * 2**0.5))                   # 0.740480  D3
```

Extended version with all numbers of this document: `python/Dok311_Skripte/ffgft_311_vier_auf_drei.py`.

B.11 References

Internal documents: A060 (time as scale), A085 (anchor), A120 ($\theta = 2/9$, five-fold rotation), A270 (bulk exponent 36, P35), Doc. 275 (scale recursion depth k^*), Doc. 310 (resonance geometry as reference, concert-pitch anchor), Register 190 (R67: uncertainty of the factor 100 in the denominator of $k^*/100$).

On status: working note for the author's own use. The chapters on the routes, on the decomposition finding $24 = 12 + 12$ and on the packing density are computed; the chapter "Application" is a question and not an assertion.

Chapter C

Doc. 312 — The Closure Scale: Λ as a Derived Quantity of the Compact Geometry

What this is about

The cosmological constant Λ has a function in Einstein's field equations: it sets the large-scale curvature scale. FFGFT needs this function as well – but not as a free constant of the equations; rather as a **derived quantity**. This document works out the identification:

$$\Lambda \longrightarrow \Lambda^* := \frac{1}{R_H^2} \qquad \text{with} \qquad R_H = \frac{c}{H_0} = \frac{2}{\pi}\,\frac{\bar{\lambda}_e}{\xi^{10}}.$$

Three results:

1. The dimensionless form $\Lambda^* \bar{\lambda}_e^2 = (\pi/2)^2\,\xi^{20}$ is a pure number – the entire SI reference resides in the tuning anchor (*Kammerton*) $\bar{\lambda}_e$ (Doc. 310).
2. The 10^{123} fine-tuning problem dissolves into a *structural decomposition*: ~ 122 decades $= 77.5$ (from ξ^{20}) $+ 44.8$ (from $(l_P/\bar{\lambda}_e)^2$) $- 0.4$ (order-one factor). The fine-tuning question becomes the structural question "why step 10 of the ξ ladder?"
3. The inheritance from Doc. 309 stands unchanged: as long as the exponent 10 is not derived forward (P20), Λ^* is [K] *conditional* – correct by construction, not by derivation.

This document is not part of the A-series. It does not overwrite any existing documents; required follow-up changes will be registered separately in Doc. 190. References: Doc. 182 (R_H as geometric maximal scale), Doc. 190 (P20, P33), Doc. 267 (degeneracy), Doc. 279 (H_0 chain), Doc. 308 (a_0, inertial regime; A272 precursor on the Λ reading), Doc. 309 (inheritance problem), Doc. 310 (Kammerton principle).

Reader's guide: the thread of the argument

The document has grown; ten chapters and an appendix. So that the arc stays visible, here is the narrative in brief.

Starting point (Ch. A–C): Λ acquires a carrier. The field equations contain a term setting the largest curvature scale; ΛCDM *fits* its value. Here it is *identified*: FFGFT already possesses a largest length, R_H – hence $\Lambda^* = 1/R_H^2$, the closure scale, no new parameter. The 10^{123} problem thereby dissolves into the arithmetic of two structural ratios. What remains honest: everything hangs on exponent 10 (P20), hence the conditional label [K|P20] throughout.

The switch (Ch. A.1): full adoption of the field equations. Lovelock forces their form and leaves open only the G value, the Λ value and the choice of solution. FFGFT therefore does not replace the equations – it supplies what they leave open.

The price (Ch. D, appendix): the stability question. Fully valid equations mean dynamical geometry – why does the torus neither roll in nor roll out (Eddington 1930)? The appendix computes the candidate: winding against momentum modes stabilise exactly at R_H, and the required tension is again Λ^* – one scale, two roles. Open as obligation C.

The turn to time (three chapters, one message). *Time in the field equations:* time is a metric direction, not an external clock; FFGFT states what the rate is attached to – mass. *The statics of the relation:* time dilation is a static map $m(x)$; telling it dynamically forgets the mass side – and expansion vs. mass drift is a choice of frame (Wetterich), not a measurement. *Thinking in light paths:* why this is overlooked – the messenger of our cosmology carries no clock itself ($d\tau = 0$); clocks are masses, and the decisive tests are matter readings. Plus the limit at c: relational, not internal.

The sharpening (the compact time direction). If time is a metric direction and everything lives on T^4, the time direction is compact as well. Yield: time cycle 92 Gyr, quantum exactly $\hbar H_0$, temperature exactly Gibbons–Hawking without a horizon, and from it $a = cH_0$ – the a_0 anchoring from one argument instead of two. Price: the causality question, obligation D.

The grounding (relative motion). We orbit nonetheless: locally, motion *on* the map (pure kinematics); globally, the topology singles out a rest frame – and exactly one is observed, the CMB dipole (369.8 km/s exactly). Convention becomes structural statement, testable ($4.2'$ offset).

The epilogue (Einstein). He had all the ingredients for $\tilde{T} \cdot m = 1$, but the relation would have made $\hbar$ foundational – the one stone he refused to build with. FFGFT builds with it and in return adopts his equations unchanged: division of labour, not refutation. The epilogue thus closes back onto the switch of Ch. A.

In one sentence: Einstein's equations hold in full; what ΛCDM fits (Λ) and what Einstein left open (what time is attached to), the compact T^4 structure supplies from a single ladder step – at the declared price of four open obligations: A (exponent 10), B (κ), C (stability), D (causality of the compact time direction).

Conventions

Labels.

[K]	*Core derivation* – derived mathematically or numerically from the base settings; reproducible.
[K|P20]	Core derivation *conditional on* the setting of exponent 10 (P20).
[S]	*Structural statement* – motivated but not yet fully proven.
[SETZUNG]	*Setting* – declared assumption, not derived further.

Central quantities.

$\xi = 4/30000$	Single fundamental parameter [SETZUNG, P33].
$\bar{\lambda}_e$	Reduced Compton wavelength of the electron, 3.8616×10^{-13} m; Kammerton (Doc. 310).
l_P	Planck length, 1.6163×10^{-35} m.
H_0	$(\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e = 66.82$ km/s/Mpc [K|P20].
$R_H = c/H_0$	Geometric maximal scale of the compact geometry, 1.384×10^{26} m – not an expansion horizon (Doc. 182).
Λ^*	Closure scale, $:= 1/R_H^2$; carrier of the Λ function in FFGFT.
κ	Order-one factor of the observational relation $\Lambda_{\text{obs}} = \kappa \Lambda^*$; open.

C.1 The function of Λ – and what carries it

Λ sits on the left

In $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$, Λ is a geometry term of dimension $1/\text{length}^2$. Its function: to set a large-scale curvature scale against which all local curvatures are small. This function is reading-independent – it exists in the expansion reading as much as in the static one.

FFGFT adopts the field equations as valid. By the Lovelock theorem, $G_{\mu\nu} + \Lambda g_{\mu\nu}$ is, in $D = 4$, the only divergence-free second-order tensor built from the metric – the *form* of the equations is forced, and the Λ term is necessarily admitted. What the equations leave open is exactly three things: the value of G, the value of Λ, and the choice of solution applied to the cosmos. What is model-dependent is therefore not the equation but the ΛCDM triad {Λ fit, FLRW application, expansion reading of z}. Accordingly, the FFGFT programme is not to re-derive the equations but to supply what they leave open: the Λ value (this document), the G value (A145), and the justification of the choice of solution. The two-halves synthesis (Doc. 307/308) additionally shows that FFGFT reproduces the local sector *independently* – consistency, not mere compatibility.

The static reading needs no new degree of freedom

FFGFT already contains a large-scale length: R_H, the geometric maximal scale of the compact T^4/Z_3 geometry (Doc. 182). The identification

$$\Lambda^* := \frac{1}{R_H^2} \tag{C.1}$$

introduces *no* new parameter – it names an already existing quantity according to its function. [K|P20]

Demarcation from the A-series (A272): There it was shown that Λ *as a fundamental constant of nature* is an artefact of the expansion reading. That stands. This document supplies the constructive complement: the *function* of Λ is real and is carried by a derived scale. Artefact as a constant – real as a derived scale.

Naming

In FFGFT the quantity Λ^* is not called the "cosmological constant" but the **closure scale** (of the compact geometry). The name follows the same discipline as a_0 as the "transition scale of the inertial regime" (Doc. 308): it states *what* the quantity is – a step of the ξ ladder, not a new constant of nature.

C.2 The chain [K|P20]

Dimensional form

From $H_0 = (\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e$ it follows directly:

$$\Lambda^* = \frac{H_0^2}{c^2} = \left(\frac{\pi}{2}\right)^2 \frac{\xi^{20}}{\bar{\lambda}_e^2} = 5.218 \times 10^{-53}\ \mathrm{m}^{-2}. \tag{C.2}$$

Comparison with observation (Planck 2018, $\Lambda_{\mathrm{obs}} \approx 1.106 \times 10^{-52}\ \mathrm{m}^{-2}$):

$$\kappa := \Lambda_{\mathrm{obs}}\, R_H^2 = 2.12. \tag{C.3}$$

In ΛCDM this factor is $3\,\Omega_\Lambda \approx 2.07$ (with H_0^{Planck}) – there, a fit. In FFGFT, κ is **open and must be derived** (Ch. C.6). It is explicitly *not* closed here by any convenient combination of constants.

Dimensionless form

Multiplying Eq. (C.2) by $\bar{\lambda}_e^2$:

$$\boxed{\Lambda^* \bar{\lambda}_e^2 = \left(\frac{\pi}{2}\right)^2 \xi^{20} = 7.78 \times 10^{-78}} \tag{C.4}$$

A pure number. The entire SI reference resides – as with H_0/v_e in Doc. 309 – in the choice of the Kammerton $\bar{\lambda}_e$. The closure scale is thereby fully embedded in the resonance geometry (Doc. 310): dimensionless structure (ξ^{20}, $\pi/2$), a single unit anchor.

C.3 Dissolving the 10^{123} fine-tuning

The standard formulation

The fine-tuning problem is usually stated as the ratio $\rho_\Lambda/\rho_{\text{QFT}} \sim 10^{-123}$, equivalently

$$\Lambda_{\text{obs}}\, l_P^2 \sim 10^{-122}.$$

ΛCDM has no explanation: the number is set.

The structural decomposition [K|P20]

In FFGFT the same number is a product of two structural ratios:

$$\Lambda^*\, l_P^2 = \underbrace{\left(\frac{\pi}{2}\right)^2}_{+0.39\ \text{dex}} \underbrace{\xi^{20}}_{-77.5\ \text{dex}} \underbrace{\left(\frac{l_P}{\bar{\lambda}_e}\right)^2}_{-44.8\ \text{dex}} = 1.36 \times 10^{-122}. \tag{C.5}$$

Factor	Decades	Origin
ξ^{20}	-77.5	step 10 of the ξ ladder, squared
$(l_P/\bar{\lambda}_e)^2$	-44.8	Kammerton-to-Planck ratio
$(\pi/2)^2$	$+0.39$	geometry factor of the H_0 chain
Sum	-121.9	$\Lambda^* l_P^2 = 10^{-121.87}$

Interpretation: There is nothing to tune. The ~ 122 decades are not an unexplained coincidence but the arithmetic consequence of *which step of the ladder* the closure scale sits on and *where the Kammerton lies relative to the Planck scale*. The fine-tuning question ("why is Λ so small?") becomes the structural question ("why exponent 10?") – and exactly that question is already registered as P20. The decomposition thus creates no new open item; it reduces two apparently separate problems (the 10^{123} problem and P20) to *one*. [S]

What the decomposition does not deliver

It does not explain why $n = 10$. As long as P20 is open, the dissolution of the fine-tuning is *conditional* – it stands and falls with the forward derivation of the exponent. This is the same inheritance as in Doc. 309 and is not argued away.

C.4 Einstein 1917 and the stability question

The historical finding

Einstein's static universe (1917) requires exactly $\Lambda = 1/R^2$ (with R the curvature radius of the closed static solution). That the FFGFT identification $\Lambda^* = 1/R_H^2$ lands formally at

the same place is remarkable – but the classical static solution is **unstable** (Eddington 1930): a small perturbation makes it collapse or expand. This was the historical death sentence of the static reading within GR.

The stability question is open – and not evaded

Since FFGFT adopts the field equations as fully valid (Sec. C.1), the metric is dynamical – and the Eddington question therefore applies here as well, in principle. Escaping via "the geometry is not dynamical" would contradict FFGFT's own position and is explicitly *not* invoked.

The relevant difference from the classical 1917 Einstein solution lies elsewhere: Einstein's static universe is a solution on an open spatial 3-sphere geometry whose equilibrium is an unstable balance point between attraction and the Λ term. The FFGFT configuration, by contrast, is defined on the compact T^4/Z_3 topology. **The candidate for stabilisation is the topology itself**: compact cycles as a boundary condition not available to the perturbation modes that make the classical solution collapse or expand – analogous to the way winding numbers on compact cycles cannot be deformed continuously.

Honest status: this is a *candidate mechanism* [S], not a proof. The forward proof would have to carry out the perturbation analysis of the static configuration on T^4/Z_3 and show that the unstable modes of the Einstein solution are excluded or stabilised by the compact topology. Until then, the Eddington question stands as an open item in the register (Ch. C.6, item C) – posed more sharply than before, because the full validity of the field equations makes it unavoidable.

C.5 Cross-anchoring and the local sector

The same ladder step carries a_0 [K|P20]

$$c H_0 = 6.49 \times 10^{-10}\ \mathrm{m/s^2}, \qquad \frac{c H_0}{2\pi} = 1.03 \times 10^{-10}\ \mathrm{m/s^2} \quad \text{vs.} \quad a_0^{\mathrm{RAR}} \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}.$$

Closure scale (Λ^*) and transition scale of the inertial regime (a_0, Doc. 308) sit on the *same* step ξ^{10}. A pure H_0 fit would not enforce this – this is the cross-anchoring of Doc. 309, extended by one entry:

Sector	Quantity	ξ dependence	Status
Cosmology	H_0	ξ^{10}	[K|P20]
Cosmology	$\Lambda^* = 1/R_H^2$	ξ^{20}	[K|P20]
Inertial regime	a_0	ξ^{10}	[K]/[S]
Particle physics	lepton mass ladder	powers of ξ	[K]
Quantum mechanics	$K_{\mathrm{frak}} = 1 - 100\xi$	ξ	[K]

The local sector remains untouched [K]

$\Lambda^* \sim 10^{-52}$ m^{-2} is negligible against any local curvature (solar system: curvature scales $\gtrsim 10^{-27}$ m^{-2} at the solar limb; ratio $\gtrsim 10^{25}$). The Schwarzschild solution with $\Lambda = 0$, the two-halves synthesis, and all classical tests (Doc. 307/308) remain exactly valid – the closure scale automatically supplies the correct hierarchy without having to be switched off locally.

C.6 What remains to be derived

A – The exponent 10 (P20, priority)

The forward derivation of $n = 10$ from the T^4/Z_3 structure is the single point on which the entire chain hangs. Without it: everything [K|P20], inheritance per Doc. 309.

Candidate readings – explicitly marked as such:

- 10 = number of independent components of the symmetric tensor $g_{\mu\nu}$ in $D = 4$ ($= D(D+1)/2$).
- $10 = 1 + 2 + 3 + 4$ (triangular number of the dimensions).
- $10 = \binom{5}{2}$ (pair count over five elements {four cycles, one Z_3 fixed point} or similar).

Warning (R61 discipline): These readings are post-hoc. The selection space of small integers with "pretty" combinatorics is large – the same situation as with the 32 smooth 2-3-5 numbers near $1/\xi$ (A-series): a hit is structurally almost unavoidable and therefore *no* signal. Admissible is only a derivation that *forces* $n = 10$ before H_0 is looked at – ideally with a second, independent consequence of the same mechanism. None of the three readings is booked as a result here. [SETZUNG candidates, not bookable]

B – The order-one factor κ

$\kappa = 2.12$ (Eq. (C.3)) must follow from the geometry (candidate source: curvature or volume factor of the compact T^4/Z_3 structure; in the Einstein static solution the corresponding factor would be exactly 1, in ΛCDM it is $3\,\Omega_\Lambda$). Until derived, κ remains open and is **not** fitted. A later "hit" via a convenient combination of constants without a mechanism would be rejected by the R61 standard.

C – Stability of the static configuration

Perturbation analysis of the static configuration on T^4/Z_3 with fully valid field equations (Sec. C.4): proof that the unstable modes of the classical Einstein solution are excluded or stabilised by the compact topology. After the adoption of the full field equations (Sec. C.1), this is the *unavoidable* third derivation obligation – without it, the static reading is merely asserted, not secured. [S]

D – Compact time direction: KMS identification and causality

A twofold obligation from the chapter on the compact time direction: (i) forward justification that the compact real-time direction carries the thermal KMS periodicity (Gibbons–Hawking temperature without a horizon, Eq. (C.8)); (ii) proof that the field configurations close periodically on the time cycle (causality condition instead of CTC paradox) – coupled to the mode count of Appendix C.A, since the same boundary conditions (periodic/antiperiodic) decide both questions. [S]

C.7 Time in the field equations

Inside, not outside [K]

In $G_{\mu\nu} = (8\pi G/c^4)\,T_{\mu\nu}$ the indices run over $0, 1, 2, 3$ – the 0 component *is* time ($x^0 = ct$). There is no equation "for space" plus a separate time evolution: time is one of the four directions of the metric. Concretely, g_{00} carries the clock rate (gravitational time dilation), the g_{0i} carry frame dragging (Lense–Thirring), the g_{ij} carry space. The two-halves synthesis (Doc. 307/308) is exactly this split: time coupling $\hat{=}$ the g_{00} half, fractal path elongation $\hat{=}$ the g_{rr} half.

Outside the equations, by contrast, there is *no* time: no external clock parameter against which the whole evolves. The equations are covariant – every coordinate choice of "time" is equally admissible, none physically preferred. Time is not a container but is generated by the solution: only a metric yields proper times along world lines, $d\tau^2 = -g_{\mu\nu}dx^\mu dx^\nu/c^2$.

Constraints and the problem of time [K]

In the ADM 3+1 split the ten equations decompose into six genuine evolution equations and four **constraints**, among them the Hamiltonian constraint $\mathscr{H} \approx 0$: the total energy of spacetime is zero; there is no external clock. Canonical quantisation turns this into the Wheeler–DeWitt equation $\hat{H}\Psi = 0$ – a fundamental equation of the universe in which no time variable appears at all. This is the well-known "problem of time": fundamentally, time disappears from the description and must be reconstructed relationally – from correlations between fields.

Connection to FFGFT [S]

Three formulations of the same direction thus stand side by side:

- **GR:** time arises from the metric; no external parameter.
- **Wheeler–DeWitt:** fundamentally there is no time variable; time is relational.
- **FFGFT:** time is the reciprocal of mass, $\tilde{T} \cdot m = 1$, anchored in the T^4 structure with $\partial\mathrm{T}^4 = \varnothing$ (Doc. 306) – which is why "what was at $t = 0$?" is wrongly posed.

The difference can be named precisely: GR leaves open *what* the clock rate is attached to; FFGFT gives the answer – to mass. That is exactly the content of the g_{00} half of the two-halves synthesis. Adopting the full field equations (Sec. C.1) is

therefore consistent on the time side as well: FFGFT does not contradict GR's relational time structure, it concretises it. Status [S]: the correspondence $\tilde{T} \cdot m = 1 \leftrightarrow g_{00}$ sector is shown in the weak field (Doc. 307); the correspondence to the full Hamiltonian constraint is a structural statement, not a proof.

C.8 The statics of the relation: time dilation as a mass map

Half the booking [K]

$\tilde{T} \cdot m = 1$ holds fundamentally *statically*: it is a constitutive relation, not an equation of motion. Time dilation is nevertheless almost always told dynamically – "the clock in the potential is slowed down, the photon loses energy on the way up". This telling forgets the mass side of the booking.

The static counter-account is well established (Okun, Selivanov, Telegdi 2000): in a static gravitational field the photon frequency is *constant* along the entire path (time-translation invariance; the frequency is a conserved quantity). What differs are the *atoms* at different locations – their transition energies, i.e. effectively their energy/mass levels in the potential. The redshift arises in the *comparison of two different clocks*, not through anything that "happens" to the photon en route.

In FFGFT language: $\tilde{T}(x) \cdot m(x) = 1$ holds at every point; "time dilation" is the static **map** $m(x)$, not dynamics. Whoever speaks dynamically without carrying the mass side books only half the relation. The g_{00} half of the two-halves synthesis (Doc. 307) *is* this map.

The same forgetfulness carries the expansion reading [K]

On the cosmic scale a published hard result exists (Wetterich 2013, *Universe without expansion*): by field redefinition (Weyl transformation), an expanding universe with constant masses is **exactly equivalent** to a static space with growing particle masses. Only dimensionless ratios are observable; whether $a(t)$ or $m(t)$ runs is a choice of frame, not physics. The expansion reading appears without alternative only because "masses constant" is set tacitly – one forgets that the masses must change along when the frame is changed.

The degeneracy of Doc. 267 thereby receives an external foundation: the dispute "expanding or not" is undecidable in this form. What is decidable is only which frame has the *simpler structure of settings* – and that is exactly the question of this document (Λ fitted vs. closure scale derived).

Consequence for the light-curve stretching [S]

The historically hardest objection against static models is the $(1 + z)$ stretching of supernova light curves: classical tired light fails there because it changes only the photon side. In the mass reading the stretching is automatic: $\tilde{T} \propto 1/m$ – if the masses change, *all* clock rates change reciprocally with them; wavelength and clock rate are the same booking. $\tilde{T} \cdot m = 1$ structurally cannot make this mistake. [S: qualitative;

the quantitative forward proof of the four discriminators – light curves, $T(z)$, Tolman, Alcock–Paczyński – remains a programme.]

Honest limit

The Wetterich equivalence shows that the static reading is *possible* and observationally identical – not that it is *correct*. The FFGFT claim is the stronger one: that the static T^4 version is the *natural* one (no Λ setting, closure scale derived, $\partial\mathsf{T}^4 = \emptyset$), while the expansion version translates the same physics into a dynamical frame with a fitted Λ. This claim stands and falls with the open items A–D – it is asserted here, not proven.

C.9 Thinking in light paths

The messenger without proper time [K]

Almost our entire picture of time and cosmos is built on light paths: light cones, redshift, distance ladders, simultaneity – all defined or measured via photons. Yet the photon is the one messenger that *carries no clock itself*: on null geodesics $d\tau = 0$; along a light path no proper time elapses; for $m = 0$ the relation $\tilde{T} \cdot m = 1$ is not even defined. Defining the time structure of the world through the *timeless* messenger is a methodological inversion: clocks are masses ($\nu = mc^2/h$, Compton clock); light is at best the ruler, never the clock.

The legacy of operationalism [K]

The inversion has a clear historical origin: Einstein 1905 defined simultaneity *operationally via light signals* – a brilliant measurement prescription that unnoticedly became ontology. Remarkably, the SI system itself has long built in the correct division of labour: since 1983 *length* comes from light (c fixed), but the *second* comes from matter – ν_{Cs}, a transition in a massive atom (cf. Doc. 309: all units traceable to this one matter anchor). The SI books time on matter and length on light. $\tilde{T} \cdot m = 1$ makes this division explicit instead of blurring it in light-path thinking.

Consequences for cosmology [S]

All cosmological information reaches us on *one* past light cone, and light transports only dimensionless ratios. It follows necessarily: whoever thinks *only* in light paths cannot in principle decide the frame question (expansion vs. mass drift, previous chapter; Wetterich, Doc. 267) – the degeneracy is not a weakness of the data but a consequence of the timeless messenger. The decision falls where *matter clocks* can be read at a distance: the $(1 + z)$ stretching of supernova light curves is exactly that – a distant matter clock (the physics of the exploding star) whose rate we compare with ours. $T(z)$ measurements on molecular clouds likewise: thermometers made of matter. The discriminators that really bite are all matter readings – not light paths.

Placement in the two-halves synthesis [S]

The synthesis (Doc. 307) says the same locally: the deflection splits into the light-path half (fractal path elongation, Doc. 182; g_{rr}) and the matter-clock half (time coupling $\tilde{T} \cdot m = 1$; g_{00}). Einstein 1911 had only the clock half and obtained half the deflection; whoever today thinks only in light paths makes the mirror-image mistake on the other side. Only the booking of both halves is complete – and in this respect cosmology still stands where the deflection calculation stood in 1911: one half systematically underexposed.

The limit at c: relational, not internal [K/S]

As matter approaches the speed of light, clock rate and mass change together: for the resting observer the moving clock slows by γ, the energy grows by γ – $\tilde{T} \propto 1/\gamma$, $E \propto \gamma$, the product stays invariant. This is the kinetic version of $\tilde{T} \cdot m = 1$ and historically exactly de Broglie's "harmony of phases" (1924): internal clock slowed by γ, phase frequency raised by γ, the relation stays closed.

From this follows, first, the opposite of an internal limit: because *all* masses, clock rates and binding energies of an organism change together and reciprocally, it notices nothing internally – in the comoving frame the physics is exactly the familiar one. In this reading the principle of relativity is not a posit but a consequence of the co-change. There is no speed at which biology "stops cooperating".

The natural limits sit elsewhere, and all three are hard (numbers: `ffgft_312_grenze_c.py`):

1. **Acceleration, not speed.** The *transition* costs: a human sustains ~1–2 g long-term. At a constant 1 g, γ grows as $\cosh(g\tau/c)$ – about one year of proper time per factor e; the limit concerns the change, never the state.
2. **Energy.** γ diverges: a 70 kg body at $\gamma = 10$ costs $\sim 5.7 \times 10^{19}$ J – of the order of the world's annual energy consumption. Not a prohibition, a practical wall.
3. **The environment – the actual natural limit.** The co-change returns from outside: for the traveller the environment blueshifts by γ. The interstellar medium (~1 H atom/cm^3) becomes a particle beam – at $v = 0.99c$ each proton hits with several GeV, and the unshielded dose rate exceeds the lethal dose in fractions of a second; at very large γ even the CMB becomes an X-ray source. The limit is not "the organism cannot take the speed" but: *it cannot take what the speed makes of its environment.* Relational to the end – in the spirit of $\tilde{T} \cdot m = 1$: no absolute state of "fast", only relations, and the relation to the medium becomes lethal long before anything internal fails.

   The same wall hits *inanimate* structure, only shifted: electronics is not immune either. Already at $0.2c$ ($\gamma \approx 1.02$) every ISM proton hits with ~19 MeV – right in the range of maximal displacement damage in semiconductors (lattice atoms knocked off their sites, transistors degrading cumulatively); add total ionising dose and single-event effects. A nanogram of interstellar dust at $0.2c$ carries the kinetic energy of a rifle bullet (~1.8 kJ). The Starshot studies (Hoang, Loeb et al. 2017) arrive exactly there: fly edge forward, mm-thick sacrificial layer eroded in a controlled way over the flight time – and even then the survival of the electronics is one of

the open core questions of the concept, not a solved one. The difference between biology and electronics is only the dose tolerance (~5 Gy lethal vs. $\sim 10^2$–10^5 Gy function-destroying depending on hardening) and the damage mechanism – not the principle.

The limit is thus more general than an organism limit: it hits **any structure-bearing matter** – everything that holds information in ordered matter (cells, crystal lattices, memory states) is exposed to the blueshifted environment. In the language of A271: structure-bearing matter holds states (level 2), and the blueshifted environment is an erasure attack on those states – the c limit is also a limit of *information preservation in moving matter*. Truly immune is only the limiting case itself: the massless, structureless photon with $d\tau = 0$.

This closes the arc to the messenger without proper time: the photon ($m = 0$, $d\tau = 0$) is the limiting case matter *never* reaches – not because an internal barrier breaks, but because the energy diverges and the environment becomes a wall first. For masses, c is not an attainable speed but the asymptote of the relation itself ($\tilde{T} \to 0$ would mean $m \to \infty$).

Two clarifications against predictable objections from both sides: the limit *prices* the relativistic travel dream ($\gamma \gg 1$ as a vehicle of time dilation); it does *not forbid* slow interstellar travel – at 0.1–$0.2c$ dose rates are orders of magnitude milder and shielding conceivable. And it holds regardless of whom it pleases: the standard is the declared assumption ($n = 1/\mathrm{cm}^3$, unshielded, 5 Gy), not the desirability of the result – the same standard that rejects wishful results *for* bold projects also protects them from impossibility pathos without booking.

C.10 The compact time direction

The consequence [K|P20]

If time is one of the four metric directions (previous chapter) and FFGFT lives on T^4 with $\partial \mathrm{T}^4 = \emptyset$, then the **time direction is compact** as well: the field equations hold not on $\mathbb{R} \times \mathrm{T}^3$ but on the full T^4. For an isotropic torus (all four cycles of equal length, $L^* = 2\pi R_H$) the time cycle automatically has the length

$$\tau_{\text{cycle}} \;=\; \frac{L^*}{c} \;=\; \frac{2\pi}{H_0} \;=\; 2.90 \times 10^{18}\ \text{s} \;\approx\; 91.9\ \text{Gyr}. \tag{C.6}$$

The cycle length is the primary quantity – and it is prefactor-free. Written as a length rather than a rate, the $\pi/2$ of $H_0 = (\pi/2)c\xi^{10}/\bar{\lambda}_e$ disappears entirely:

$$L^* \;=\; \frac{2\pi c}{H_0} \;=\; 4\,\frac{\bar{\lambda}_e}{\xi^{10}}, \tag{C.7}$$

exactly (numerically to ten digits: $L^*/\bar{\lambda}_e = 2.252541 \times 10^{39} = 4/\xi^{10}$). In words: *the cycle is four Kammertöne times the ladder step.* The $\pi/2$ of the H_0 form is therefore not a prefactor but the conversion $H_0 = 2\pi c/L^*$ – pure notation. The objection that a freely choosable order-one prefactor would void the ladder hit does not apply: in its place stands an integer.

What remains open is the *origin of the four*. The obvious candidates – the four torus directions, the doubling outward/return (2^2), the D4 lattice – are all post-hoc and are *not* booked under R61 discipline as long as no forward derivation exists.

No new setting: the same ladder step ξ^{10}, a fourth role of the same scale (after Λ^*, T_w, $\hbar H_0$). The energy quantum of the time cycle, $2\pi\hbar/\tau_{\text{cycle}} = \hbar H_0$, is identical to the momentum quantum of the spatial cycles at the C1 minimum – isotropy of the four directions at the quantum level as well.

Compact time and temperature [S]

In field theory, periodicity in *Euclidean* time with period τ is equivalent to a temperature $T = \hbar/(k_B\tau)$ (KMS condition, Matsubara formalism). Inserting the time-cycle period – with the customary KMS factor 2π, i.e. $\tau_{\text{eucl}} = \tau_{\text{cycle}} = 2\pi/H_0$ – yields

$$\boxed{T \;=\; \frac{\hbar H_0}{2\pi k_B} \;=\; 2.63\times10^{-30}\ \text{K}} \tag{C.8}$$

This is **exactly the Gibbons–Hawking temperature** of the de Sitter horizon – here, however, without a horizon and without expansion: the temperature follows from the compactness of the time direction itself.

Honest status [S]: the KMS equivalence holds for periodic *imaginary* time (after Wick rotation). Identifying FFGFT's compact real-time direction with this thermal periodicity is a structural statement, not a proof – it belongs to the forward obligations (item D).

Unification with the a_0 chain [S]

Setting the Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ equal to the temperature (C.8) gives

$$a \;=\; cH_0 \;=\; 6.49\times10^{-10}\ \text{m/s}^2, \tag{C.9}$$

i.e. precisely the anchor of the a_0 derivation in Doc. 308 (transition scale of the inertial regime, $cH_0/2\pi = 1.03\times10^{-10}$ m/s^2 against $a_0^{\text{RAR}} \approx 1.2\times10^{-10}$). What were two separate arguments in Doc. 308 (Unruh anchoring; ξ^{10} chain) here becomes one: **the compactness of the time direction is the structural source of the Unruh anchoring.**

The price: closed timelike curves [open]

A compact time direction formally means: world lines return to their starting value of the time coordinate after τ_{cycle} – in GR the classical problem of closed timelike curves (CTCs). Two mitigations can be formulated, both [S]:

1. **Empirical:** the period of ~ 92 Gyr exceeds any observable time span; no observational conflict exists.
2. **Relational:** in the reading of the previous chapter the time coordinate is not a container one "returns" into. Periodic coordinate time is paradoxical only if the field configurations do *not* close periodically. The causality condition therefore reads: **periodicity of the field configurations on the time cycle** – on T^4 exactly

the natural boundary condition. The same choice (periodic vs. antiperiodic per field) simultaneously fixes the mode count of the tipping bound in Appendix C.A (C1′): causality and stability hang on the *same* boundary-condition structure. Neither is a proof; the CTC question enters the obligations as item D.

C.11 Relative motion and the torus rest frame

We move: the Earth around the Sun, the Sun around the Galaxy, the Local Group toward the Great Attractor. How does that fit a static compact reading? The answer has a local and a global half, and both are computed (`ffgft_312_relativbewegung.py`).

Locally: motion on the map [K]

All orbital motions are non-relativistic:

Motion	v [km/s]	$\gamma - 1$
Earth rotation (equator)	0.5	1.2×10^{-12}
Earth around Sun	29.8	4.9×10^{-9}
Sun around Galaxy	233	3.0×10^{-7}
Sun rel. CMB	369.8	7.6×10^{-7}
Local Group rel. CMB	620	2.1×10^{-6}

The clock rate of a moving body is the product of two bookings: the static map $m(x)$ (chapter "The statics of the relation") times the kinetic factor $1/\gamma$ (co-change). This is exactly how GPS computes: potential term plus velocity term. Orbital motion is motion *on* the map, not an objection to it. The hierarchy also closes vectorially: Sun-rel.-CMB minus the galactic orbit gives 552 km/s for the Milky Way relative to the CMB – literature value ~ 550.

Globally: the topology singles out a frame [K]

On compact topology, Lorentz invariance holds *locally* without restriction, but boosts are no longer global symmetries: the twin paradox has an absolute answer on the torus – the twin remaining in the torus rest frame ages the most, distinguishable by the *winding number* of the world line. Quantitatively: at $\beta = 0.99$ one circumnavigation takes 92.9 Gyr of coordinate time at 13.1 Gyr of proper time (deficit 79.8 Gyr). The distinction is structurally absolute but practically inert – every circumnavigation takes at least $\tau_{\text{cycle}} = 92$ Gyr. No aether: nothing material rubs; the distinction is topological, no local measurement can detect it, only global markers.

The CMB dipole as candidate [S]

Such a distinguished frame is observed: the CMB dipole yields, with $\Delta T/T = 1.2336{\times}10^{-3}$, exactly $v = 369.8$ km/s – the Doppler reading of a rest frame, consistent with the direct

measurement to $0.03\,\%$. In the expansion reading this is "merely" the comoving frame, a convention. In the compact reading it is the candidate for the **torus rest frame** itself: all motions of the hierarchy are velocities relative to the structure, and the dipole is their sum. The compact time direction hangs on it as well: τ_{cycle}, the $\hbar H_0$ quantum and the KMS temperature are defined in the torus frame; a moving observer reads the mode structure Doppler-shifted – again the dipole.

Honest status: the identification torus frame = CMB frame is natural and consistent, but an identification [S], not a derivation.

Topological period and light circumnavigation must be distinguished. The period of the time cycle is a *topological* quantity, $\tau_c = L^*/c = 91.9$ Gyr. A photon would take $1.35\,\%$ longer for one circumnavigation because of the fractal path elongation (Doc. 313; 93.2 Gyr). For KMS, Unruh and Gibbons–Hawking what counts is the *field periodicity*, not the photon travel time: $T = \hbar H_0/(2\pi k_B)$, the quantum $\hbar H_0$ and $a = cH_0$ therefore remain unchanged.

Test signatures and self-consistency [S]

1. **Matched-circle offset.** For compact topology near the observational bound ($L^*/D_{\text{LSS}} \approx 1.01$), circle signatures should be offset around the dipole vector by $\sim \beta$ rad $= 4.2$ arcminutes – a specific, in-principle accessible directional signature. – *provided* the scattering wall exceeds the half-cycle. With the fractal path correction (Doc. 313) it moves to $0.979\,\pi$ and no longer self-intersects; then no matched circles exist and this signature lapses. It therefore holds only in the smooth version; the remaining test surface is then the non-monotonicity at the antipode (Doc. 313).
2. **Drift.** Our motion through the structure amounts to 34.8 Mpc per time cycle ($0.12\,\%$ of L^*) and 5.2 Mpc over the recombination light-travel time ($0.019\,\%$): proper motion produces no self-contradiction of the static reading.

In short: relative motion remains locally relative (SR untouched) and becomes globally relational to the structure. What is a convention in the expansion reading (the comoving frame) is here a structural statement with a named test signature.

C.12 Why Einstein did not see $\tilde{T} \cdot m = 1$

Historical-methodological classification; no physical booking. The question is meant seriously: the ingredients of the relation come from Einstein himself.

The ingredients lay on his desk

$E = mc^2$ (1905) and $E = h\nu$ (1905, light quanta) together give

$$\nu = \frac{mc^2}{h} \qquad \Longleftrightarrow \qquad T \cdot m = \frac{h}{c^2} \ (\to 1 \text{ in natural units}). \tag{C.10}$$

Every mass *is* a frequency, hence a clock – that is $\tilde{T} \cdot m = 1$ in raw form, from two Einstein formulas. It was spelled out by **de Broglie in 1924**: the particle's "internal clock" with $\nu = mc^2/h$ was the core of his dissertation. Einstein reviewed and praised

the work ("he has lifted a corner of the great veil") – but everyone, including de Broglie himself, turned it into the *wavelength* and dropped the *clock*. The Compton time $\hbar/(mc^2)$ was henceforth treated as a derived scale, not as what time *is*.

Four reasons

1. His programme ran the other way. For Einstein, geometry was primary and mass a *source* – on the right of the equation, as $T_{\mu\nu}$. The figure of thought "mass curves the stage" blocks the figure "mass *is* the clock rate". $\tilde{T} \cdot m = 1$ makes mass constitutive of time itself – the inversion his field programme did not foresee. The irony: in 1911 he had *only* the time coupling (half the deflection – the g_{00} half of the two-halves synthesis). He stood on the T side before turning to spatial geometry.

2. The relation contains $\hbar$. Written out it reads $T = \hbar/(mc^2)$ – it weds the quantum constant and relativity at the most fundamental level. Einstein considered quantum theory provisional and wanted to *replace* it, not fuse with it. An ontology whose basic relation contains $\hbar$ was for him not a foundation but a symptom of the unfinished. The FFGFT starting stance – both theories well confirmed, both incomplete, no fundamental contradiction possible, hence dualise – is exactly the stance he could not take.

3. The clarity about units was missing. $\tilde{T} \cdot m = 1$ presupposes seeing through $\hbar$ and c as mere conversion factors – that time and mass carry no independent dimensions. This view became second nature only with the practice of natural units in particle physics, and fully consistent only in 2019, when the SI fixed all units to constants (cf. Doc. 309: all SI units traceable to ν_{Cs}). In Einstein's day the second and the gram were ontologically different things; a reciprocity between them looked like a category error.

4. It does not look like a law. Einstein sought field equations – dynamics. $\tilde{T} \cdot m = 1$ is not an equation of motion but a constitutive relation: in the terminology of *Order without Hierarchy*, a **principle** (*Grundsatz*), not a law. Such statements look trivial (everyone knew the Compton time) as long as one does not take them ontologically. The step is not the formula but the *change of status* of the formula.

Late confirmation and conclusion

The clock reading was confirmed operationally only in 2013: the Compton-clock experiment (Lan, Müller et al., Berkeley) measured a mass *as a frequency standard* – mass and clock rate operationally the same. By then the relation was 88 years old.

The honest conclusion: not blindness but fidelity to a programme. Einstein saw the half that fitted his picture (mass curves the clock rate) and not its reciprocal (the clock rate *is* the mass), because the latter would have made $\hbar$ foundational – the one stone he refused to build with. FFGFT builds with exactly this stone – and in return adopts his equations unchanged (Sec. C.1): division of labour, not refutation.

C.13 Status overview

Statement	Content	Status
Identification	$\Lambda^* = 1/R_H^2$, no new parameter	[K|P20]
Dimensionless form	$\Lambda^* \bar{\lambda}_e^2 = (\pi/2)^2 \xi^{20}$	[K|P20]
Structural decomposition	10^{-122} as $\xi^{20} \cdot (l_P/\bar{\lambda}_e)^2$	[K|P20]
Reduction	fine-tuning → P20 (one open item instead of two)	[S]
Local sector	Λ^* automatically negligible locally	[K]
Field equations	fully adopted (Lovelock); FFGFT supplies Λ value, G value, choice of solution	[S]
Time	inside (metric, g_{00}); relational; FFGFT concretises: $\tilde{T} \cdot m = 1$	[K]/[S]
Statics	time dilation as mass map $m(x)$ (Okun et al.); expansion vs. mass drift frame-equivalent (Wetterich)	[K]
Light paths	photon without proper time ($d\tau = 0$): light = ruler, mass = clock; discriminators are matter readings	[K]/[S]
Limit at c	relational: none internal (γ co-change); hard: acceleration, energy, environment	[K]/[S]
Time cycle	$\tau = 2\pi/H_0 = 91.9$ Gyr; quantum $= \hbar H_0$	[K|P20]
Temperature	$\hbar H_0/(2\pi k_B)$ = Gibbons–Hawking, without horizon; source of the Unruh/a_0 anchoring	[S]
Causality	CTC → periodicity of fields on the time cycle	open (D)
Rest frame	topology singles out a global frame; candidate: CMB dipole (369.8 km/s exactly); test signature $4.2'$ offset only without the fractal correction (Doc. 313)	[K]/[S]
Stability	open; candidate: compact topology as boundary condition	open (C)
$\kappa = 2.12$	order-one factor	open
Exponent 10	forward derivation	open (P20)

Registration: The entry of the closure scale and the open items named here into Doc. 190 (correction register, append-only), as well as any follow-up adjustments in Doc. 308/309, will be done separately and are not part of this document.

C.A Appendix: Calculations on the Stability Question (C1–C3)

What this is about

Section C.4 (item C) leaves the stability question open: with fully valid field equations the torus radius is dynamical – the Eddington question returns as a modulus question (compactified or rolled out?), structurally the same trap as the decompactification runaway in the T^6/Z_3 review (Quílez Zamora). This supplement computes three things:

- **C1** – radius stabilisation as a winding-/momentum-mode equilibrium (toy model, Brandenberger–Vafa type), computed *inversely*: which winding tension forces the minimum to $L^* = 2\pi R_H$?
- **C2** – Eddington contrast: time constant of the classical instability vs. the sign of V'' in the torus model.
- **C3** – order-of-magnitude check against CMB topology bounds.

All numbers: `ffgft_312C_stabilisierung.py`. Status throughout [K|P20] × [S] (toy model: one cycle, fixed quantum numbers). No fits – T_w is determined inversely and then *booked* on the ξ ladder; the hit on step ξ^{20} is a result, not an input.

C1 – The winding-/momentum-mode equilibrium

Model

Effective potential for the cycle length L of one T^4 cycle:

$$V(L) = \underbrace{N_w T_w L}_{\text{winding}} + \underbrace{N_k \frac{2\pi\hbar c}{L}}_{\text{momentum quanta}} \qquad (+\,\text{Casimir, small}). \tag{C.11}$$

Winding modes carry energy $\propto L$ (tension times length; topological, not continuously deformable), momentum quanta $\propto 1/L$. Their competition produces a minimum at

$$L_{\min} = \sqrt{\frac{2\pi N_k \hbar c}{N_w T_w}}, \qquad V''(L_{\min}) > 0. \tag{C.12}$$

This is the mechanism the T^6/Z_3 model lacked in the review: there, no $\propto L$ term stopped the roll-out.

Inverse calculation [K|P20]

Requiring $L_{\min} = L^* = 2\pi R_H = 4\bar{\lambda}_e/\xi^{10}$ (with $N_w = N_k = 1$) determines the winding tension:

$$\boxed{T_w = \frac{2\pi\hbar c}{L^{*2}} = \frac{\hbar c \Lambda^*}{2\pi} = \frac{\pi}{8}\frac{E_e}{\bar{\lambda}_e}\xi^{20} = 2.63 \times 10^{-79}\ \text{N}} \tag{C.13}$$

Both identities are exact (numerically confirmed to 10^{-16}). Two findings:

1. **One scale, two roles.** The tension forcing the minimum to R_H is directly proportional to the closure scale: $T_w = \hbar c\,\Lambda^*/(2\pi)$. Closure scale (Doc. 312) and stabilising tension sit on the *same* ladder step ξ^{20} – no second independent scale is needed. [S]
2. **Self-consistency at the minimum.** The momentum quantum at the minimum is exactly
$$\frac{2\pi\hbar c}{L^*} = \frac{\hbar c}{R_H} = \hbar H_0 = 2.28 \times 10^{-52}\ \text{J} = 1.4 \times 10^{-33}\ \text{eV},$$
and at the minimum equipartition holds, $E_{\text{winding}} = E_{\text{momentum}}$ (Brandenberger–Vafa property). The energy scale of the stabilised cycle *is* the Hubble energy – nothing new has to be set.

Casimir correction and tipping bound

One massless periodic bosonic degree of freedom contributes $E_C = -\pi\hbar c/(6L)$ ($|E_C|/E_{\text{momentum}} = 1/12$ per degree of freedom). Shift of the minimum:

N_{Casimir} (net)	L_{min}/L^*	finding
0	1.000	unchanged
4	0.817	minimum survives, shifted
24	–	no minimum: collapse

Honest limit: for $N_{\text{Casimir}} \geq 12\,N_k$ the model tips over. The forward obligation therefore includes the mode count of the actual T^4/Z_3 spectrum (bosons minus fermions, periodic/antiperiodic) – not only the tension derivation.

C2 – Eddington contrast [K]

The classical Einstein solution (1917) has no minimum but a balance point; perturbations grow exponentially with

$$\tau = \frac{1}{c\sqrt{\Lambda^*}} = \frac{R_H}{c} = 4.6 \times 10^{17}\ \text{s} = 14.6\ \text{Gyr}. \tag{C.14}$$

The instability is thus no academic worry: it tips the configuration on the Hubble time scale. No static model can hide behind a slow instability.

In the torus model, by contrast, $V''(L^*) > 0$: perturbations of the radius oscillate instead of running away. The oscillation frequency depends on the modulus inertia M_{mod} (open); what decides the Eddington question here is solely the *sign* of V''.

C3 – Order check: CMB topology bound [S]

$$L^* = 28.2\ \text{Gpc} \qquad \text{vs.} \qquad D_{\text{LSS}} \approx 27.8\ \text{Gpc} \qquad \Rightarrow \qquad \frac{L^*}{D_{\text{LSS}}} = 1.01\text{–}1.02. \tag{C.15}$$

With the fractal path correction. D_{LSS} is a *light path*, L^* a topological quantity – by the assignment rule (Doc. 313) $K_{\text{frak}} = 1 - 100\xi$ applies here. The ratio shifts to

$L^*/D_{\mathrm{LSS}} = 1.03$: the scattering wall lies more clearly below the half-cycle. This is more compatible with the null results of the topology searches but costs the matched-circle signature (chapter on relative motion).

Matched-circles searches (Planck) exclude cycle lengths clearly *below* the diameter of the last-scattering surface; L^* lies about 1–2 % *above* it: marginally consistent, right at the observational edge. This is a strength, not a weakness – the compact reading is thereby **falsifiable**, not arbitrary.

Caveat: distance measures must be reinterpreted in the static reading (Doc. 267 degeneracy); this is an order check, not a final test.

Result and forward obligations

	Finding	Status
C1	$T_w = \hbar c \Lambda^*/(2\pi) = (\pi/8)(E_e/\bar{\lambda}_e)\xi^{20}$; momentum quantum at minimum $= \hbar H_0$	[K|P20]×[S]
C1′	tipping bound: minimum vanishes for $N_{\mathrm{Casimir}} \geq 12\, N_k$	[K] (model)
C2	Einstein static: $\tau = R_H/c = 14.6$ Gyr; torus: $V'' > 0$	[K]
C3	$L^*/D_{\mathrm{LSS}} = 1.01$–$1.02$ (fractal 1.03) – at the edge, falsifiable	[S]

What C1 delivers and what it does not: the inverse calculation shows that stabilisation at R_H requires *no new scale* – the closure scale carries both roles, and the energy scale at the minimum is $\hbar H_0$. It does *not* show that the T^4/Z_3 winding dynamics actually supplies this tension. The forward obligations are thereby precise:

1. Derive T_w from the winding dynamics on T^4/Z_3 (target value Eq. (C.13) – sealed before the calculation, R61 discipline).
2. Mode count of the actual spectrum (check the tipping bound $N_{\mathrm{Casimir}} < 12\, N_k$).
3. Modulus inertia M_{mod} for the oscillation frequency (C2 quantitative).

Registration: entry into Doc. 190 will be done separately.

Chapter D

Appendix A to Doc. 312 — Altermagnetism as a Laboratory Analogy

Purpose of This Appendix

Doc. 312, § Lorentz (source lines 936–947), makes the statement:

> *On compact topology Lorentz invariance holds locally without restriction, but boosts are no longer global symmetries. The distinction is topological; no local measurement can detect it — only global markers.*

The concept "locally broken, globally preserved" is abstract and difficult to ground without experimental reference. Altermagnetism provides a directly measured, mathematically fully analysed example of the same structure at a completely different scale. This appendix documents the analogy and marks its limits.

Purpose: Didactic reference and terminological calibration for "local $\neq$ global". No new physical content for FFGFT. Source: Šmejkal, Sinova, Jungwirth, *Emerging Research Landscape of Altermagnetism*, Phys. Rev. X **12**, 040501 (2022).

D.1 Altermagnetism: Three-Way Classification via Spin Groups [E]

The non-relativistic spin-group classification of collinear magnets yields exactly three phases, determined by which symmetry operation connects the magnetic sublattices:

Type	**Sublattice link**	**Kramers deg.**	M_{net}
Ferromagnet	none (one spin sublattice)	lifted (Zeeman)	$\neq 0$
Antiferromagnet	translation t or inversion P	preserved	$= 0$
Altermagnet	rotation C_n ($n = 2, 4, 6$)	locally lifted	$= 0$

The decisive distinction between antiferromagnet and altermagnet: in both, the net magnetisation is zero. But in the antiferromagnet a translation or inversion connects the sublattices — this *enforces* Kramers degeneracy $E(\mathbf{k},\uparrow) = E(\mathbf{k},\downarrow)$ across the entire Brillouin zone. In the altermagnet a genuine rotation (C_4 in RuO_2) connects the sublattices — this lifts the degeneracy in a direction-dependent manner.

D.2 The Model Hamiltonian and Its Symmetries [E]

The effective single-particle Hamiltonian of the d_{xy}-wave altermagnet (Šmejkal et al., Eq. 3):

$$H(\mathbf{k}) = 2t\cos k_x \cos k_y \cdot \sigma_0 + 2t_J \sin k_x \sin k_y \cdot \sigma_z. \tag{D.1}$$

The energy eigenvalues are $E_\pm(\mathbf{k}) = 2t\cos k_x \cos k_y \pm 2t_J \sin k_x \sin k_y$ with spin splitting

$$\Delta E(\mathbf{k}) = E_+ - E_- = 4t_J \sin k_x \sin k_y. \tag{D.2}$$

Proof: C_4 maps the spin-up onto the spin-down Fermi surface [B]

The C_4 rotation in $\mathbf{k}$-space acts as $(k_x, k_y) \to (-k_y, k_x)$. Then:

$$\begin{aligned} E_{\text{dn}}(-k_y,\, k_x) &= 2t\cos\big(-k_y\big)\cos(k_x) - 2t_J \sin\big(-k_y\big)\sin(k_x) \\ &= 2t\cos\big(k_y\big)\cos(k_x) + 2t_J \sin\big(k_y\big)\sin(k_x) = E_{\text{up}}(k_x, k_y). \quad \checkmark \end{aligned} \tag{D.3}$$

The spin-up Fermi surface is mapped exactly onto the spin-down surface by C_4 (numerically confirmed, `pruef_altermagnet_1_symmetrie.py`).

Two further analytically provable statements:

1. **Global compensation:** $\frac{1}{(2\pi)^2}\int_{\text{BZ}} \Delta E(\mathbf{k})\, \mathrm{d}^2k = 0$, since $\int_{-\pi}^{\pi} \sin k_x \, \mathrm{d}k_x = 0$. No macroscopic net magnetisation. **[B]**
2. **Nodal lines:** $\Delta E = 0$ exactly at $k_x = 0$ or $k_y = 0$ (two orthogonal nodal lines, protected by mirror-plane symmetries that map one spin sublattice onto the other). **[B]**

D.3 Locally Broken, Globally Preserved — the Shared Structure

Structural comparison with Doc. 312 § Lorentz

FFGFT / Doc. 312	Altermagnet / Šmejkal 2022
Compact topology $T^4/\mathbb{Z}_3$	Crystal lattice with C_4 rotation
Lorentz invariance holds locally	T-symmetry locally broken ($\Delta E \neq 0$)
Boosts: no longer global symmetries	Spin: globally compensated ($M = 0$)
Only topological markers detectable	Only ARPES/spin-current measurable
Scale: Planck length	Scale: ångström (crystal lattice)

The structure is the same: a symmetry (time-reversal T or Lorentz boost) is locally broken in the sense that it fails at a single point (in **k**-space or for a specific direction of motion in space). But the global integral — over the entire Brillouin zone or over all directions — is zero: the overall structure is compensated. Locally this breaking is detectable only by direction-resolved measurements; an isotropic probe reveals nothing.

D.4 Limits of the Analogy [E]

Restrictions — explicit

1. **Different scales.** Altermagnetism: $\sim$ Ångström. FFGFT: Planck scale ($\sim 10^{-35}$ m). No shared observables.
2. **Different symmetry groups.** The spin group $r \times R_s$ of the altermagnet and the orbifold group $T^4/\mathbb{Z}_3$ of FFGFT are different mathematical objects.
3. **No information flow.** FFGFT does not explain solid-state effects. Altermagnetism contributes no new derivations to FFGFT.
4. **Direction of the analogy.** The analogy serves the didactic calibration of the concept "locally $\neq$ globally". It is not a derivation and must not be read as a confirmation of FFGFT.

D.5 Verification

All statements in § 2 are numerically and symbolically confirmed by `pruef_altermagnet_1_symmetrie.py`. The script is located in `2/python/Dok312_Anhang_Skripte/`.

Reference: L. Šmejkal, J. Sinova, T. Jungwirth, *Emerging Research Landscape of Altermagnetism*, Phys. Rev. X **12**, 040501 (2022). arXiv:2204.10844.

Chapter E

Doc. 313 — No Beginning: The Thermal History on the Time Cycle

What this is about

Doc. 312 ends with obligation D(ii): can the field configurations close periodically on the compact time cycle ($\tau_c = 2\pi/H_0 = 91.9$ Gyr) *while* carrying the observed thermal sequence – nucleosynthesis, recombination, structure formation? A circle has no beginning; the observations show a gradient. This document computes the question (`ffgft_313_zeitzyklus_geschichte.py`, `ffgft_313_fraktale_wegkorrektur.py`) and finds three results and one hard open core:

1. Ω_m **from the antipode condition [K|P20×E]:** the chain $H_0 \to T_0 \to \Omega_r \to$ antipode condition gives, with the measured T_0, $\Omega_m^* = 0.3139$ against Planck 0.315 ± 0.007 ($-0.16\,\sigma$) – with no parameter fitted to Ω_m (Ch. F).
2. **Antipode finding [K|P20]:** the observable history fills *one half-cycle*. With the fractal path correction (Ch. B, independently from A040) the particle horizon sits within $0.14\,\%$ of the antipode ($\theta = 0.9986\,\pi$; smooth: $1.012\,\pi$), the scattering wall at $0.979\,\pi$ – the "beginning" is the antipode of the time cycle, not an event.
3. **Smooth closure [K]:** the observed mass-drift profile reaches the antipode with slope $\sqrt{\Omega_r} \approx 0.01$ – the smoothness condition of an even periodic closure is met to $1\,\%$, and the return half is exactly the photonically unobservable half.
4. **Entropy obstruction [K]:** exact τ_c periodicity is not dynamically generic (Poincaré discrepancy $\sim 10^{10^{104}}$); it must be imposed as a boundary condition. Fine-grained admissible, coarse-grained open – the sharpened core of D(ii), which presents as a *fork*: Doc. 295 knows the non-closing case (log spiral) in which the entropy requirement lapses.

Not part of the A-series; registration in Doc. 190 separately. References: Doc. 306 ($\partial\mathsf{T}^4 = \emptyset$), 267 (degeneracy), 312 (closure scale, compact time direction, statics of the relation, torus rest frame).

Reader's guide: the thread of the argument

Nine chapters, one idea. So that the arc stays visible, here is the narrative in brief.

The starting position. Doc. 306 says: the torus has no boundary, the question about $t = 0$ is wrongly posed. Doc. 312 makes this concrete – the time direction is a cycle of 92 Gyr. What remained open is the *history*: observations show a clear thermal sequence. How does a gradient live on a circle?

Preliminary. All distances here are light paths, and space is not smooth. Before a single number is quoted, it must be settled where the fractal path correction applies: for path-through-structure yes, for path-through-path no. The rule has two control tests – and passes both, one of them (the CMB peaks) without having been made for it.

The finding. The entire observable history fills *one half-cycle*. The particle horizon sits within $0.14\,\%$ of the antipode. The "big bang" is therefore a *place* on the cycle – the region of smallest masses and slowest clocks – not an event.

The lower bound. There is no $z \to \infty$: the cycle quantum $\hbar H_0$ is a hard bound, and masses cannot run to zero. The singularity is thus not merely wrongly posed but *excluded* – and the same scale that bounds from above (Λ^*) bounds from below.

The closure and its price. The mass profile enters the antipode with slope $\sqrt{\Omega_r} \approx 0.01$ – an even periodic closure is met to $1\,\%$. But it is a *boundary condition*, not a consequence of the dynamics: the Poincaré discrepancy is 10^{104} orders of magnitude in the exponent. Fine-grained, periodicity is admissible; coarse-grained it is open – the hard core of D(ii). Doc. 295 knows the non-closing case (log spiral); D(ii) is therefore more accurately a *fork* than a gap.

Hawking. A single KMS rule tempers Unruh, Gibbons–Hawking and black holes; the membrane picture of Ch. 04 thereby gets its thermometer, and the ladder meets the cosmos at $r_s = R_H/2$. For the cycle closure this does not help: evaporation is 56 orders of magnitude too slow. Hawking fits the picture and *sharpens* D(ii).

The prediction. Inverting the antipode condition, Ω_m is no longer free: the chain $H_0 \to T_0 \to \Omega_r \to$ antipode gives, with the measured T_0, $\Omega_m^* = 0.3139$ against Planck 0.315 ± 0.007 – with no parameter fitted to Ω_m. The T_0 formula of Doc. 061 matches the measured value only in the unit eV; it is not a derivation, and the question of T_0 remains open (R70).

The test surfaces. Two: the non-monotonicity at the antipode (CνB/GW, distant) and the low-ℓ cutoff $\ell_{\min} = 3.08$ – which matches the quadrupole suppression unexplained since COBE 1992, on existing data.

In one sentence: the "beginning" is the antipode of the time cycle – a place, not an event; the observed history is one half of a map closable to $1\,\%$; and the remaining burden lies not with the fields but with coarse-grained entropy.

E.1 No boundary, no $t = 0$: the starting position

FFGFT's position on singularity and big bang stands since Doc. 306: $\partial \mathsf{T}^4 = \varnothing$ – the torus has no boundary, "what was at $t = 0$?" is wrongly posed. Doc. 312 made this concrete and gave it three supports:

Geometric: the time direction is a cycle of length 91.9 Gyr. A closed circle has no starting point – "$t = 0$" is as meaningless on it as "the beginning of the equator".

Backwards: in the mass-drift reading (Doc. 312, statics of the relation; Wetterich), running back does not send distances to zero but masses: $\tilde{T} \cdot m = 1$ then means $\tilde{T} \to \infty$ – no density divergence, but a regime of ever slower clocks. The "singularity" is where the clocks stand still, not where the density diverges.

Forwards: no collapse into a singularity – the winding/momentum minimum (appendix of Doc. 312) has $V'' > 0$.

What Doc. 312 left open is the *history*: observations show a clear thermal sequence. How does a gradient live on a circle? That is D(ii), and here it is computed, not narrated.

E.2 Preliminary: the fractal path correction

All distances in this document are *light paths*. But FFGFT space is not smooth-affine; it carries the fractal dimension $D_f = 3 - \xi$ (A040). A traversed path accumulates an elongation over it, so the smoothly computed distance is too large. Before any numbers, therefore, the question of where this factor applies.

The factor

A040 supplies it:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3}\right)^{D_f^{\text{eff}}/2} = 1 - 100\,\xi = 0.98665, \qquad D_f^{\text{eff}} \approx 2.973. \tag{E.1}$$

The value $1 - 100\,\xi$ is confirmed by comparing the two routes to E_0 (bare and fractally corrected mean of m_e and m_μ) with α (register 190, R135); the form of the factor is open (R139). It is not fitted to any cosmological result. Its uncertainty is quantified: $n = 100 \pm 2$ (register 190, R67), i.e. $\pm 0.027\,\%$ in the effect. The path elongation is $1/K_{\text{frak}} - 1 = 1.35\,\%$.

The assignment rule

A040 restricts sharply [B]: K_{frak} concerns *absolute* values, never ratios *of the same level* – there it cancels. What "same level" means is decided by the distinction of Doc. 311:

Quantity	Level	K_{frak}
Light path η_0, $\chi(z)$	unrolled (traversed)	applies
L^*, R_H, τ_c	rolled up (topological)	—
path / structure ($\eta_0/\pi R_H$)	mixed	**applies**
path / path (χ_{rec}/η_0)	same level	cancels
angles, dimensionless ratios	same level	cancels

The reason is conceptual, not numerical: a light path exists only in the *unrolled* representation – in the rolled-up view there is no "path to the antipode", only the cycle position θ. R_H, by contrast, follows from $H_0 = (\pi/2)c\xi^{10}/\bar{\lambda}_e$, a topological structural

relation. So $\eta_0/(\pi R_H)$ compares something traversed with something topological and is *not* a same-level ratio.

Control test: $\chi_{\text{rec}}/\eta_0 = 0.98024$ is identical with and without the correction – two light paths, the factor cancels as required. The rule is thus not freely applicable; it has a testable invariant.

A second, independent test: the CMB peaks. The position of the first acoustic peak is $\ell_1 = \pi D_A/r_s$ – angular distance divided by sound horizon, both *traversed paths*. By the rule K_{frak} cancels, so the peak positions remain exactly unchanged. The cross-check shows how sharp this test is: had the factor applied to D_A alone, ℓ_1 would shift by $1.35\,\%$, i.e. by three multipoles – against a measured accuracy of ± 0.5 the rule would be falsified immediately. It passes this test without having been constructed for it.

Declaration: this assignment is a conceptual stipulation, and it improves the numbers. Both belong stated side by side. In its favour: the direction of the effect is physically compelled (a winding path covers less smooth distance in the same travel time – no choice), and the level distinction of Doc. 311 was formulated independently and earlier. Nothing known speaks against it, but it is not proven. All following numbers are therefore carried in both versions.

E.3 The antipode finding [K|P20]

Where does the history sit on the cycle?

Position on the cycle: $\theta = \chi/R_H$, with χ the light path distance (the $z \to \chi$ assignment takes the measured ΛCDM map; only dimensionless ratios enter, Doc. 267). The antipode lies at $\theta = \pi$, i.e. $\chi = L^*/2 = \pi R_H = 14.09$ Gpc. Computed:

	fraction of $L^*/2$		
	smooth	fractal	
Recombination ($z = 1090$)	0.992	0.979	$-2.1\,\%$
Particle horizon (z_{max}, Ch. D)	1.012	**0.9986**	**−0.14** %

The **particle horizon** – the formal boundary of causal contact and hence the actual "beginning" – lands within $0.14\,\%$ of the antipode with the fractal correction, against $1.2\,\%$ in the smooth calculation. That is an improvement by a factor of 9 with nothing adjusted. The remainder is, however, five times the K_{frak} uncertainty ($\pm 0.027\,\%$), so not an exact hit.

The **scattering wall** thereby moves from 0.992 to 0.979: the secondary finding "recombination at the antipode" becomes *weaker*. Both statements cannot be exact simultaneously – they differ by the invariant factor $\chi_{\text{rec}}/\eta_0 = 0.980$.

z	θ/π smooth	θ/π fractal	$m(z)/m_0$
0.5	0.140	0.138	0.667
1	0.243	0.240	0.500
3	0.465	0.459	0.250
10	0.689	0.680	0.091
100	0.917	0.905	0.0099
1090	0.992	0.979	0.00092

Finding: the entire observable history – up to $z \to \infty$ – fills one *half-cycle* to $1.2\,\%$. The last-scattering surface sits at the antipode ($99.2\,\%$), the formal "beginning" $1.2\,\%$ behind it. The "big bang" is thereby a *place on the cycle* – the antipodal region, in the mass-drift reading the region of smallest masses and slowest clocks – and not an event before time.

Honest placement of the coincidence [S]

In ΛCDM language the same fact is the known quasi-coincidence $\eta_0 H_0/c \approx \pi$ (here 3.18 against 3.14, i.e. $1.2\,\%$) – there an accident of the measured Ω values. In the compact reading it becomes a structural statement: *the scattering wall sits at the antipode.* For it to be more than a re-labelled coincidence, the location would have to follow forward. The fractal path correction is a step in exactly that direction: it comes independently from A040, is not fitted to this result, and it reduces the particle horizon's distance from the antipode from $1.2\,\%$ to $0.14\,\%$. The location is not thereby fully derived – the remainder is five times the R67 uncertainty, and why the matter content fills exactly the half-cycle remains open. Status [S], noted as a derivation target.

E.4 No infinity, no zero: the lower bound

Speaking of the "particle horizon at $z \to \infty$" is a loan from standard language and, in FFGFT, strictly speaking false: there are neither infinities nor zeros. The energy quantum of the time cycle (Doc. 312) is a hard *lower bound*:

$$E_{\min} = \hbar H_0 = 2.28 \times 10^{-52}\ \text{J}, \qquad m_{\min} = \frac{\hbar H_0}{c^2} = 2.54 \times 10^{-69}\ \text{kg}. \tag{E.2}$$

Its relation to the Kammerton is exactly the ladder step itself: $\hbar H_0/(m_e c^2) = (\pi/2)\,\xi^{10}$. From this follows a *finite* highest z: the mass drift ends when the electron mass reaches the quantum,

$$1 + z_{\max} = \frac{m_e}{m_{\min}} = \frac{1}{(\pi/2)\xi^{10}} = 3.59 \times 10^{38}. \tag{E.3}$$

What this changes numerically: nothing

The light-path contribution beyond $z = 10^9$ is $1/(\sqrt{\Omega_r}(1+z)) \approx 10^{-7}\,R_H$, beyond $z_{\max}$ still $10^{-37}\,R_H$. All numbers of the preceding chapters are unaffected.

What it changes conceptually: three things

(a) The singularity is excluded, not merely wrongly posed. So far the argument was: $\partial\mathsf{T}^4 = \emptyset$, the question about $t = 0$ is wrongly posed. Now a second, stronger one is added: masses *cannot* run to zero, because below $\hbar H_0/c^2$ no state exists. The antipodal region is not a limit one approaches but a reached endpoint of the scale.

(b) One scale sets both bounds. $\Lambda^* = 1/R_H^2$ bounds from above (largest length), $\hbar H_0$ from below (smallest energy) – the same ladder step ξ^{10}, no second parameter. The economy of the construction thus reaches beyond cosmology into quantisation.

(c) Poincaré presupposes finiteness. The recurrence theorem holds only for finite phase spaces. Without infinities and without a continuum down to zero, recurrence is therefore *guaranteed* – finiteness is the precondition of the cycle closure, not its obstacle. What remains open is only the time scale (Ch. E): that recurrence exists does not mean it happens within τ_c.

E.5 The smooth closure [K]

The profile and its endpoint slope

Periodicity requires the mass drift $m(\theta)$ to close continuously on the circle. The simplest closure is an *even* profile about the antipode: minimum at $\theta = \pi$, symmetric return on the other half. It is smooth if $dm/d\theta \to 0$ at the turning point.

In the radiation era $H \simeq H_0\sqrt{\Omega_r}(1+z)^2$, hence analytically

$$\eta_0 - \chi = \frac{c}{H_0\sqrt{\Omega_r}}\,\frac{1}{1+z} \qquad \Longrightarrow \qquad \left|\frac{dm}{d\theta}\right|_{\text{end}} = \sqrt{\Omega_r} \approx 0.0096.$$

Finding: the observed profile runs into the endpoint with slope ~ 0.01 – the smoothness condition of an even periodic closure is met to $1\,\%$. The deformation required relative to the measured history is a $1\,\%$ effect at the outermost edge.

Why this is not self-deception

The return half – where the mass drift would have to rise again – is *exactly* the half behind the scattering wall: photonically unobservable in principle. That is suspiciously convenient at first sight, but it is not a choice of this document; it is a consequence of the antipode finding: the wall *lies* at $0.992\,\pi$. The closure therefore contradicts no single photon observation – and it is nevertheless not untestable (Ch. E).

The climate-zone picture from the Doc. 312 discussion thereby becomes quantitative: the gradient is the map *along* the cycle (like climate zones along a meridian circle), not an evolution *out of* a beginning – and the “hottest” point of the map is the antipode.

E.6 The entropy obstruction: the core of D(ii) [K]

Periodicity is not generic

The entropy budget of the observable volume is $S \sim 10^{104}\, k_B$ (dominated by supermassive black holes; photons: $10^{89}\, k_B$). Poincaré recurrence of a generic state takes $\sim e^{S/k_B} \sim 10^{10^{104}}$ dynamical times – against $\tau_c \sim 10^{18.5}$ s. The discrepancy is $\sim 10^{104}$ *orders of magnitude in the exponent*: exact τ_c periodicity cannot be expected dynamically.

Consequence: periodicity is a **boundary condition** – a selection of periodic solutions from the solution space, not an outcome of evolution. It belongs to the same category as the choice of the T^4 topology itself ([STIPULATION] in the sense of Doc. 311) and must be booked as such.

Fine- and coarse-grained, separated

Fine-grained [K]: under unitary evolution the fine-grained entropy is constant; nothing microscopic forbids an exactly periodic solution. The boundary condition is admissible.

The fork: Doc. 295 knows the alternative

The account so far treats periodicity as the only option with an open entropy question. That is incomplete: Doc. 295 (“Time as a log spiral”) already spells out the non-closure, and from the same factor. A winding that should advance by π per *half turn* advances, gate by gate, only by πK_{frak} and thus misses π by $100\pi\xi$ (Doc. 295). In units of a revolution the closure defect is

$$d = 100\,\xi = \tfrac{1}{75} \quad \text{per revolution.} \tag{E.4}$$

Three cases therefore stand side by side:

Case	Condition	Closure	Period
A	$d = 0$	after 1 revolution	91.9 Gyr
B	ξ frozen	after 75 revolutions	6892 Gyr
C	ξ running (recursion)	never	log spiral

Case B follows from the rational rotation number $1 - d = 74/75$; the $75 = 1/(100\xi)$ drops out exactly. That closure occurs at *exactly* 75 and not earlier follows from $\gcd(74, 75) = 1$: the smallest n with $n \cdot 74/75 \in \mathbb{Z}$ is $n = 75$. Case C arises when ξ runs through the recursion ($\xi_n \approx 1/(100(n{+}75))$): the accumulated defect becomes logarithmic and the winding turns into the equiangular spiral $\rho = 75\, e^{\beta/2\pi}$ with self-similarity ratio e per precession round – not imposed but a consequence of the $1/n$ decay.

What this means for D(ii). Case A requires the coarse-grained closure after 91.9 Gyr – the open obligation. Case B relaxes it by a factor of 75 but changes

nothing about the Poincaré discrepancy of 10^{104} dex in the exponent. **Case C removes the entropy requirement entirely**, because nothing has to return – but it costs the compact time direction in the sense of Doc. 312: τ_c, the quantum $\hbar H_0$ and the Gibbons–Hawking temperature would lose their basis.

The deciding question is thereby nameable and simpler than the entropy question itself: is the time winding bound to *one* scale step (then A or B), or does it run through the recursion (then C)? The recursion in Doc. 295 is a scale recursion, not a time recursion; whether the cosmological time direction traverses it is undecided. D(ii) is therefore more accurately carried as a *fork* than as a gap: two of the three routes are viable, but they lead to different theories.

Coarse-grained [open]: the second law is a statement about coarse-grained entropy relative to an observer. On the return half it must decrease – which is observer-relative possible (reversal of the correlation direction; no one *there* would see a violation, since the thermodynamic arrow turns with it), but unproven. **Exactly this is the sharpened core of D(ii):** not the field periodicity (which is consistent as a boundary condition, Ch. C), but the proof that a coarse-grained entropy closure is compatible with the observed structure formation. Until then: open, without embellishment.

Reach: equilibria at other places on the cycle

The place reading entails a boundary that must be booked separately, because it is easily overlooked.

Backward extrapolation presupposes what is open here. The usual computation $T(z) = T_0(1 + z)$ obtains earlier temperatures by adiabatic continuation of today's state. That presupposes *one* continuous thermodynamic chain – precisely the coarse-grained chaining which, by the foregoing, is open (D(ii), Poincaré discrepancy 10^{104} dex).

In the place reading an early epoch is not an earlier state of the same system, but a location with a *different local structure* – smaller masses, slower clocks (Ch. B) – and hence its own equilibrium, fixed locally and not by continuation of today's. An equilibrium at another place on the cycle is therefore **not derivable** from the present one; it would require its own structural derivation *at that place*.

The present computation is not affected. T_0 enters here as today's measured value, not as the terminus of a cooling history. The formula $k_B T_0 = (16/9)\,\xi$ of Doc. 061 matches this value only in the unit eV and is not a structural derivation (Ch. F.3; R70).

What this does not mean. The boundary devalues no structural derivation. The Ω_m^* chain (Ch. F) runs through ξ, H_0 and K_{frak}, not through an adiabatic back-calculation, and is unaffected. Affected are only statements of the form "how hot was it in epoch X" insofar as they are obtained by extending today's equilibrium. Such statements are **not** covered here – neither disputed nor asserted, but outside what this computation carries.

Status: [K] as a reach statement about this computation itself; the question which equilibrium holds at a given place on the cycle is thereby named as an independent, unsolved task – it does not follow from D(ii) but stands beside it.

Which quantities then remain meaningful?

The boundary above is a refusal; it becomes useful only once one says what lies *on this side* of it. The criterion has three parts: no scale bridge, no duration, same level.

Class	Examples	Status
counts	$\lvert\mathrm{Aut}(D4)\rvert = 1152$, kissing number 24, 9 fixed points	exact; theorems, not measurements
same-level ratios	α, m_μ/m_e, Koide Q	robust; testable at 10^{-10} to 10^{-5}
with scale bridge	E_0, G, H_0, T_0, absolute masses	only with an anchor; ratios thereof free again (A040)
number and duration	$\eta = n_B/n_\gamma$, window durations, time spans	not transferable

Where the light elements fall. All three BBN observables land in the lower two classes – including Y_p, which is counterintuitive enough to spell out. Y_p depends on the dimensionless quantities Q/T_f and t/τ_n; under uniform scaling both cancel. But T_f is *not measured*: it follows from a rate equation involving H, hence from the cosmological model, and t/τ_n contains a *duration*. Y_p is therefore a mixture of structural and model quantities, not a pure same-level ratio like α or m_μ/m_e. The cancellation argument is an *implication* ("if the model comes out this way, it cancels"), not a test surface.

D/H and ^{7}Li stand further out still: both depend on η and on the window duration, ^{7}Li additionally on stellar atmosphere models in the measurement itself – three model layers stacked.

Consequence for the ^{7}Li position. The corpus booking (Docs. 025/063: nucleosynthesis without a fixed time limit) addresses the window duration and remains consistent. It is, however, *no quantitative prediction*: from "no fixed limit" no numerical value for ^{7}Li/H follows. The standard picture's factor-3 discrepancy shows that something in the chain is wrong – not where.

Counter-check, lest this become a protective claim. The same classification applies to ΛCDM equally: there too Y_p is no model-free test, η comes from a multi-parameter fit, and assigning measured abundances to an early epoch runs through the same continuation assumption. Doc. 267 already records this: "none is circle-free"; the fair comparison is parsimony against anchoring. The classification therefore does not exempt FFGFT – it places both sides under the same criterion.

E.7 Hawking radiation on the time cycle

Does Hawking radiation fit this picture? Yes – better than before, because the compact time direction (Doc. 312) has quietly supplied its foundation. At the same time it sharpens the entropy core of D(ii). Both computed (`ffgft_313_hawking_zyklus.py`).

One principle, three faces [K]

The three famous "temperatures out of nothing" – Unruh, Gibbons–Hawking, Hawking – are three separate miracles with three derivations in the textbooks. Behind

them stands *one* KMS rule: where the time structure is periodic, an observer reads temperature, $T = \hbar/(k_B\tau)$:

System	Period τ	$T = \hbar/(k_B\tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18}$ s	2.63×10^{-30} K
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4}$ s	6.17×10^{-8} K
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^{8}$ s	3.98×10^{-20} K

Doc. 312 *grounded* this rule for the cosmos: the Gibbons–Hawking temperature follows from the compactness of the time direction itself, without a horizon. Hawking radiation is therefore no foreign body and no extra postulate – it is the *local special case of the same rule*.

The membrane gets its thermometer [S]

The membrane picture (narrative Ch. 04: the hole *is* its fractal membrane, the radiation comes from the oscillating surface) so far had no answer to "why exactly this temperature?". Now it has one, in the language of the mass map: mass compresses the time structure ($\tilde{T} \cdot m = 1$ – the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with $\tau = 8\pi GM/c^3$. Short period, high temperature:

Object	T_H
PBH 10^{12} kg	1.2×10^{11} K
Earth	2.1×10^{-2} K
Sun	6.2×10^{-8} K
Sgr A*	1.5×10^{-14} K
SMBH $10^9 M_\odot$	6.2×10^{-17} K

The soap bubble of Ch. 04 has a thermometer, and the thermometer measures the clock map.

The family ladder [S]

Where does the ladder meet the cosmos? $T_H = T_{\text{GH}}$ requires $M = c^3/(4GH_0) = 4.66 \times 10^{52}$ kg – and its horizon lies at

$$r_s = \frac{2GM}{c^2} = \frac{R_H}{2} \quad \text{(exactly } 0.500\text{)},$$

at the same time exactly *half* the critical Hubble mass $c^3/(2GH_0)$. The temperature ladder runs from micro holes to the whole without a break: the cosmos is the largest family member.

What Hawking cannot do [K]

The honest friction with the cycle: evaporation times against $\tau_c = 91.9$ Gyr:

Object	t_{evap}	t_{evap}/τ_c
PBH 10^{12} kg	$10^{12.4}$ yr	$10^{1.5}$
Sun	$10^{67.3}$ yr	$10^{56.4}$
SMBH $10^9 M_\odot$	$10^{94.3}$ yr	$10^{83.4}$

Only primordial holes below the threshold mass

$$M^* = \left(\frac{\tau_c\,\hbar c^4}{5120\,\pi G^2}\right)^{1/3} = 3.3 \times 10^{11}\ \text{kg} \quad (\sim \text{asteroid})$$

close via Hawking within one cycle. A solar-mass hole is 56 orders of magnitude too slow, an SMBH 83. The FFGFT correction from Ch. 04, $P = P_{\text{std}}(1 - \xi \ln(M/M_P))$, amounts to 0.6–$1.4\,\%$ and changes nothing in this finding.

Consequence for D(ii): stellar and supermassive holes – the carriers of the 10^{104} entropy budget of Ch. D – cannot disappear via Hawking on the cycle. The return half must dismantle them by another route. Hawking radiation thus fits the picture thermodynamically perfectly and at the same time *sharpens* the coarse-grained core of D(ii): it shows how hopelessly slow the only known return route would be.

Information: two languages, one statement [S]

Ch. 04 always said: the radiation correlates with the fractal membrane structure – nothing is lost ($S_{\text{BH}}(M_\odot) = 1.05 \times 10^{77}\,k_B$ of correlations). Ch. D of this document says: fine-grained entropy is constant under unitary evolution – periodicity is microscopically admissible. That is the same statement in two languages: information preservation locally (membrane) and globally (cycle) are consistent. What remains open is solely the *coarse-grained* closure – which was the core all along.

E.8 Test surfaces [S]

Photons end at the wall ($0.992\,\pi$). Two messengers originate at or behind the antipode:

Messenger	Decoupling	Position
CνB (neutrino background)	$z \sim 6 \times 10^9$	$\theta \approx 1.01\,\pi$
primordial GW	nominally $z \sim 10^{25}$	$\theta \approx 1.01\,\pi$

What the path correction costs here: fractally, the scattering wall lies at $0.979\,\pi$, clearly *below* the half-cycle – it does not intersect itself. On the one hand this makes the compact reading more compatible with the null results of the topology

searches (the bound $(L^*/2)/D_{\mathrm{LSS}}$ rises from 1.008 to 1.022); on the other it removes a test signature: without self-intersection there are no matched circles and hence no $4.2'$ offset (Doc. 312). The price of the better numbers is a lost test opportunity – to be booked honestly, not concealed.

A *non-monotonic* signature there – the reversal of the mass drift beyond the antipode – thus remains the *only* surviving test surface of the closure.

The low-ℓ cutoff: a second test surface [K|P20]

There is a second one, and it lies closer to existing data. On a torus of edge length L^* no modes with wavelength $> L^*$ exist; the smallest admissible wavenumber is $k_{\mathrm{min}} = 2\pi/L^*$. Translated into angular scale:

$$\ell_{\mathrm{min}} \;=\; k_{\mathrm{min}}\, D_{\mathrm{LSS}} \;=\; \frac{2\pi D_{\mathrm{LSS}}}{L^*} \;=\; 3.08. \tag{E.5}$$

Below $\ell \approx 3$ modes are missing: the quadrupole ($\ell = 2$) is strongly suppressed, the octupole ($\ell = 3$) sits at the edge, and from $\ell \geq 4$ the spectrum is unaffected.

No fitted parameter. $L^* = 4\bar{\lambda}_e/\xi^{10}$ follows from the chain, D_{LSS} from the same one; varying Ω_m between 0.305 and 0.325 moves ℓ_{min} only from 3.12 to 3.04. Without the fractal path correction it would be 3.12 – the statement is robust against both freedoms.

What is observed. Since COBE (1992), confirmed by WMAP and Planck: the quadrupole lies at about one sixth of the ΛCDM expectation, the octupole slightly low, and from $\ell = 4$ the spectrum is normal. In ΛCDM this has been unexplained for three decades and is attributed to cosmic variance. The compact reading predicts exactly this structure – *one* number, in the right place, with an intact spectrum above it.

What is still missing. The precise degree of suppression requires the mode summation on the torus instead of the integration over $\mathbb{R}^3$ – a finite sum, not a Boltzmann code, but not yet carried out. Only that decides whether the factor $1/6$ is matched quantitatively or only in direction. Until then the agreement counts as an *order statement* [K|P20], not a precision test. *Check script:* `ffgft_313_lowl_cutoff.py`. Practically beyond present measurement, but nameable; together with the $4.2'$ dipole offset (Doc. 312, torus rest frame) these are two concrete signatures of the compact reading.

E.9 The antipode condition as a prediction: Ω_m^* and the κ candidate

The inversion

Ch. B read the location of the scattering wall as a finding and honestly booked the coincidence $\eta_0 H_0/c \approx \pi$ as [S]. But the condition can be *inverted*: if the compact reading requires the particle horizon to close exactly at the antipode,

$$\eta_0 = \pi R_H \qquad \Longleftrightarrow \qquad \int_0^\infty \frac{dz}{E(z)} = \pi, \tag{E.6}$$

then Ω_m is no longer a free parameter. The condition solves uniquely (flat, $\Omega_r = 9.3 \times 10^{-5}$) to:

Without the fractal correction this gives $\Omega_m^* = 0.325$ ($+1.4\,\sigma$ vs Planck). With it, the condition reads $\int dz/E = \pi/K_{\text{frak}} = 3.1841$ and yields $\Omega_m^* = 0.314$ ($-0.2\,\sigma$). The R67 uncertainty ($n = 100 \pm 2$) shifts this only between 0.3137 and 0.3141. The next section assembles the chain.

The cosmic sector thereby obtains, for the first time, a *prediction for a ΛCDM parameter* rather than a reinterpretation – falsifiable: sharper Ω_m measurements (DESI, Euclid) decide whether 0.325 holds.

What the comparison quantity is: Planck $\Omega_m = 0.315 \pm 0.007$ is not a raw datum but itself the output of a six-parameter ΛCDM fit to photon count rates – including ΛCDM transfer functions and fiducial cosmology in the analysis chains. Read precisely, the $1.4\,\sigma$ comparison is therefore: *condition against pipeline, not condition against nature.* This is the third inheritance level beside the SI anchor (Doc. 309) and the $E(z)$ form (above), and like those it is declared rather than hidden. It cuts both ways: decades of ΛCDM refinement produce robustness *and* pipeline lock-in – and where the compact reading inherits the form, it simultaneously inherits its fit successes for the shared parts. This is why the independent test points of this document deliberately sit in the minimally model-laden observable class: directions and shapes (the $4.2'$ dipole offset, non-monotonicity at the antipode) rather than fit parameters.

The chain

T_0 enters the chain. Doc. 061 gives the formula $k_B T_0 = (16/9)\xi$ for it; it matches the measured value only in the unit eV and is thus a unit-dependent agreement, not a derivation (Sec. F.3; R70). The chain with this formula value:

	Step	Value
(1)	$H_0 = (\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e$	$h = 0.6682$
(2)	$k_B T_0 = (16/9)\,\xi$ (Doc. 061)	$T_0 = 2.751$ K
(3)	$\Omega_r \propto T_0^4/h^2$	9.643×10^{-5}
(4)	$K_{\text{frak}} = 1 - 100\,\xi$ (A040)	0.98667
(5)	$\int dz/E = \pi/K_{\text{frak}}$	$\Omega_m^* = 0.3136$

against Planck 0.315 ± 0.007: $-0.19\,\sigma$. What counts is the *measured* T_0; it gives $\Omega_m^* = 0.3139$ ($-0.16\,\sigma$). The prediction is practically insensitive to the $0.92\,\%$ deviation of the T_0 formula (it propagates as 0.0003). No step of the chain contains a parameter fitted to Ω_m.

T_0 and the antipode condition

The antipode condition, tied to H_0 via the Kammerton $\bar{\lambda}_e$, can also be solved for T_0. The direction of leverage matters:

Ω_m	T_0 from the antipode condition
0.3100	3.062 K
0.3136	2.754 K
0.3150	2.625 K
0.3200	2.097 K

The leverage is $dT_0/T_0 \big/ d\Omega_m/\Omega_m \approx -12$: one per cent in Ω_m propagates as twelve per cent in T_0. The antipode condition is therefore a sharp Ω_m determiner and a blunt T_0 determiner – the comparison must be made on the Ω_m side, not the T_0 side.

Read that way: the model-independently measured $T_0 = 2.7255$ K (FIRAS, $0.02\,\%$) gives $\Omega_m = 0.3139$ via the antipode condition; the formula of Doc. 061 gives $\Omega_m = 0.3136$. The closeness of the two values is no test between two independent sectors, because the formula matches T_0 only in the unit eV (R70).

Consequence for the status: the test channel runs through Ω_m: a determination to $0.3\,\%$ (DESI, Euclid) sharpens it by a factor of 2.

Declared inheritance: the calculation uses the ΛCDM *form* of $E(z)$ – including the $(1+z)^3$ term with full Ω_m. The prediction is therefore stated precisely: *the map must be shaped so that* $\int dz/E = \pi$*; in* Λ*CDM bookkeeping this reads* $\Omega_m = 0.325$. What carries the matter share of that bookkeeping is a separate question (Sec. F.3). Status hence [K|P20 $\times E(z)$ form] – the same inheritance structure as in Doc. 309, one level deeper.

Consolidation: obligation B and the coincidence become one condition

From Ω_m^* follows $\Omega_\Lambda^* = 0.675$ and hence a structural candidate for the so far open order-one factor of the closure scale (Doc. 312, obligation B):

$$\kappa^* = 3\,\Omega_\Lambda^* = 2.058 \qquad \text{against measured } \kappa = 2.12 \;\; (-2.9\,\%) \tag{E.7}$$

(smooth: 2.026, i.e. $-4.4\,\%$).

The same consolidation logic as with the 10^{123} problem (→ P20): two open items – obligation B and the [S] coincidence of Ch. B – are reduced to *one* condition. The cross-check shows this is not circular: $\kappa = 2.12$ would require $\Omega_m = 0.294$, and then $\eta_0 H_0/c = 3.268 \neq \pi$. Antipode exactness and today's Planck κ stand in a genuine $4\,\%$ tension – test contact, not a fit.

Placements and one new open item

Ω_Λ **loses its substance status:** the "dark-energy density" is the FLRW translation of the closure scale; if the antipode condition holds, it is predicted, not fitted. **Hubble tension:** the compact reading takes an unambiguous side with the CMB-anchored value ($H_0 = 66.8$ from the ξ^{10} chain against Planck 67.4); the local SH0ES excess would then have to be a map effect of the calibration [S].

Dark matter: two regimes, one open. Dark *matter* has deliberately not been mentioned so far – and the gap belongs named. *Galactically* (rotation curves, RAR) FFGFT needs no particle DM: there the transition scale a_0 carries the load (Doc. 308);

this document says nothing about that, correctly so. *At the background level*, however, the $z \to \chi$ map silently inherits the CDM share: in ΛCDM bookkeeping, $\Omega_m^* = 0.325$ consists of baryons (≈ 0.049) plus cold dark matter (≈ 0.276). What carries this share in the compact reading – actual matter, or a property of the mass-drift profile $m(\theta)$ that books as matter in FLRW translation – is unresolved and is booked as a **second new open item**. It is independent of the galactic question and must not be conflated with it.

T_0: unit-dependent agreement, derivation open. The corpus contains a formula (Doc. 061): $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi^2$ with $E_\xi = 1/\xi$, i.e. $k_B T_0 = (16/9)\,\xi$ in electronvolts $= 2.370 \times 10^{-4}$ eV against the measured 2.349×10^{-4} eV ($+0.92\,\%$). The formula yields a pure number that agrees with the measured value only in the unit eV; it is a unit-dependent agreement, not a derivation. The Casimir length $L_\xi = 100.24\ \mu$m of Doc. 279 is itself fixed by the measured CMB energy density and ξ, and therefore does not bind T_0 independently (R136). Nor does the parallel with $\alpha = \xi(E_0/1\,\mathrm{MeV})^2$ (A266) hold: there the agreement is a property of the anchor E_0, not a derivation (R135).

What follows for the status. In the Ω_m^* chain, T_0 is a measured value. Whether T_0 follows structurally from ξ is open (R70).

Register candidates (two). *(i) Notation clash:* in A085, T_0 in the four-garments identity $m_0 c^2 = E_0 = k_B T_0 = \hbar/\tilde{T}$ denotes the *calibration temperature* (8.53×10^{10} K), whereas in Docs. 061/313 it denotes the *CMB monopole* (2.7255 K) – two quantities a factor 3×10^{10} apart under the same symbol. *(ii) Precision claim:* Doc. 061 states "$0.0002\,\%$" in its summary while its own detailed calculation reports "agreement to $1\,\%$"; the correct value is $0.92\,\%$.

E.10 Status overview

Statement	Content	Status

Statement	Content	Status
Antipode finding	horizon at antipode: fractal $0.14\,\%$ (smooth $1.2\,\%$); wall $0.979\,\pi$	[K|P20]
Path correction	$K_{\mathrm{frak}} = 1 - 100\xi$ applies to path/structure, cancels for path/path (test passed)	[K]/[S]
Coincidence placement	$\eta_0 H_0/c \approx \pi$: structural statement only after forward derivation	[S]
Smooth closure	endpoint slope $\sqrt{\Omega_r}/K \approx 0.0098$; closure met to $1\,\%$	[K]
Return half	= photonically unobservable half; no conflict with photon data	[K]
Lower bound	$\hbar H_0$ as smallest energy: no $z \to \infty$; singularity excluded by quantisation	[K]
Periodicity	boundary condition (Poincaré discrepancy 10^{104} dex in the exponent), not dynamics	[K]

Statement	Content	Status
Fork	three cases (Doc. 295): $d = 0$ / 75 revolutions / spiral; C removes the entropy requirement but costs τ_c	[K]
Entropy	fine-grained admissible; coarse-grained closure open – sharpened core of D(ii)	open
Reach	equilibria at other places on the cycle are not derivable from today's; T_0 as measured value, not extrapolated; Ω_m^* chain unaffected	[K]
Robustness	criterion: no scale bridge, no duration, same level. Counts and same-level ratios yes; BBN observables (Y_p, D/H, ^{7}Li) no – holds for ΛCDM equally	[K]
Hawking	local case of the same KMS rule (GH from Doc. 312); ladder meets cosmos at $r_s = R_H/2$	[K]/[S]
Evaporation	$M^* = 3.3 \times 10^{11}$ kg; $M_\odot$: 56 dex too slow – sharpens D(ii)	[K]
Test surfaces	CνB/GW from the antipode; non-monotonicity as signature	[S]
Low-ℓ cutoff	$\ell_{\min} = 2\pi D_{\mathrm{LSS}}/L^* = 3.08$; matches the quadrupole anomaly open since COBE	[K|P20]
Ω_m^*	antipode condition with measured T_0: 0.3139 ($-0.16\,\sigma$); $E(z)$ form inherited	[K|P20×E]
κ^*	fractal 2.058 ($-2.9\,\%$), smooth 2.026: candidate for obligation B	[S]
Matched circles	wall without self-intersection: signature from Doc. 312 lapses	loss
DM (background)	$(1+z)^3$ share of the map: carrier unresolved (galactic: a_0, Doc. 308, separate)	open
T_0	formula of Doc. 061 matches the measured value only in eV (unit-dependent); derivation from ξ open (R70)	open

Balance: after this document, "no big bang" means precisely: the "beginning" is the antipode of the time cycle – a place, not an event; the observed history is one half of a map closable smoothly to $1\,\%$ (with the fractal path correction the particle horizon meets the antipode to $0.14\,\%$); and the remaining burden lies not with the fields but with coarse-grained entropy. D(ii) is thereby not settled but reduced to its hard core.

Registration: entry into Doc. 190 will be done separately.

Chapter F

Doc. 314 — The Lattice in Hilbert Space: Geometry, Spectrum and Fibre

What this is about

FFGFT computes geometrically on T^4 with the D4 lattice and quantum-mechanically on $\mathscr{H} = L^2(\mathsf{T}^4) \otimes \mathbb{C}^3$ (Docs. 230/282). The transition raises an obvious question: *do the lattices become flat?* Does anything remain of kissing number 24, packing density and triality, or is every lattice the same in Hilbert space?

The answer: the torus is flat anyway – curvature was never the difference – and the lattice structure is not lost but **migrates into the spectrum and into the fibre**. This document computes that through (three scripts, Ch. J) and finds:

1. **The spectrum separates what curvature does not [K]:** Z^4 starts at $|k|^2 = 1$ with degeneracy 8, D4 at $|k|^2 = 2$ with degeneracy 24; odd norms are entirely absent for D4. The first excitation carries exactly $24 = 12 + 12$ – FCC half and winding half separately addressable.
2. **Triality is built in, not attached [K]:** $|\mathrm{Aut}(D4)| = 1152 = |W(F_4)|$, the quotient to $W(D4)$ is $6 = |S_3|$; among the 80 order-3 elements, 16 generate exactly the T^4/Z_3 orbifold with $|\det(1 - A)| = 9$ fixed points and act on every mode orbit as a $\mathbb{C}^3$ circulant.
3. **A negative finding with theorem character [K]:** 5 does not divide $1152 = 2^7 \cdot 3^2$, so $\mathrm{Aut}(D4)$ contains *no* element of order 5. The fivefold rotation from which $\theta = 2/9$ arises (Doc. 293) is in principle incompatible with the lattice – the phase belongs to a different layer.
4. **The doubling is nevertheless prepared [K]:** $24 = 8{\times}3 = 4{\times}6$; four disjoint six-blocks with exactly the C_3 content that $3 \oplus 3'$ requires.
5. **The reciprocity is mode-wise exact – as a *place* relation [K]:** $\lambda_4 \cdot m = 2\pi$ identically for every mode (de Broglie along the mass direction; winding number = mass number). The *temporal* reading carries $1/\gamma$, a mass mixture a Jensen excess ≥ 1: the relation becomes exact only for the resting, sharp mode.

6. **The perturbation calculation is carried out [K]:** multiplet sizes follow the irrep dimensions of the respected symmetry, not the orbits – $9+8+4+2+1$ in the fully symmetric case, $8\times1+8\times2$ for Z_3, generically 24 individual levels.
7. **The closure fork is spectrally invisible [K]:** in the rolled-up state there are no fractal corrections – winding numbers are topological, the defect $d=100\xi$ accumulates only when unrolled. Cases A/B/C of Docs. 313/295 leave the mode spectrum untouched, so Chapters E–H are case-independent. The same boundary settles the fractal question (Ch. J): K_{frak} is cumulative (Doc. 133) and acts only through the scale bridge; locally only $D_f^{\text{space}}=3-\xi$ acts, with 6.7×10^{-5}.

Not part of the A-series; status: working document. References: Docs. 230/282 ($\mathscr{H}=L^2(\mathsf{T}^4)\otimes\mathbb{C}^3$), 133 (derivation of K_{frak}: cumulative, RG run), 285 (doubling $6=3\oplus3'$), 291 (channel analysis), 293 ($\theta=2/9$ icosahedral), 295/313 (closure fork), 306 (native time–energy reciprocity), 311 (Four onto Three, $24=12+12$).

Reader's guide: the thread of the argument

The question. In Hilbert space the lattice becomes the index set of the Fourier modes. It is then no longer a sphere packing – in that sense every lattice becomes "flat". The question is whether the information disappears with it.

The answer in one line. *Geometry → spectrum + fibre.* The degeneracies of the Laplacian are the theta series of the lattice; the kissing number becomes the degeneracy of the first excitation; triality becomes the orbifold and fibre structure.

What follows. First, two flat tori are distinguishable by their spectra (Ch. B). Second, the $\tilde{T}\cdot m=1$ structure becomes spectrally readable when $R_4\neq R_{123}$, and the level ordering changes at exactly two root radii (Ch. C). Third, the Z_3 need not be imposed – it is a lattice symmetry (Ch. E).

The boundary, drawn cleanly. Fivefold symmetry is impossible on the lattice (Ch. F). This does not hit the doubling, because that lives on the *fibre* and the interface of the two levels is exactly C_3 – occupied by triality (Ch. G). And the container structure holds on *every* shell, so it cannot distinguish one (Ch. H) – the same pattern-versus-occupation logic as in Doc. 293.

The caveat. The computation uses the smooth Laplacian, i.e. $D_f=4$ exactly. All numbers are therefore *bare*; Ch. I separates what that costs: counts and ratios stay invariant, $D_f^{\text{space}}=3-\xi$ is *local* and acts with 6.7×10^{-5}, while $K_{\text{frak}}=1-100\xi$ is *cumulative* (RG run across 17 orders of magnitude) and acts only through the scale bridge. Bare are therefore only the SI absolute values ($-1.33\,\%$ via the anchor); counts, ratios and containers are not. The degeneracy splitting under perturbation is computed – it follows the irrep dimensions of the perturbation symmetry, not the orbits, and lies at $\sim10^{-5}$. Ch. D2 adds the boundary between rolled-up and unrolled states, which is why the closure fork (Docs. 313/295) stays spectrally invisible.

In one sentence: in Hilbert space the lattice becomes flat but not inconsequential – what geometry carried is carried by spectrum and fibre, and the division of labour between them is exactly determinable, as long as the smooth approximation is booked as such.

Notation

The table explains the symbols of this document. Quantities appearing only once are introduced there.

Note on notation. In lattice theory Λ is the usual symbol for a lattice (Conway/Sloane). In the FFGFT corpus, however, Λ is taken – as the cosmological constant, in ΛCDM, and in particular as the closure scale $\Lambda^* = 1/R_H^2$ (Doc. 312). This document therefore uses Γ throughout for the lattice and Γ^* for the dual lattice. Likewise L^* is *not* used here, since in Doc. 312 it denotes the cycle length; the torus edge lengths are called R_{123} and R_4.

Symbol	Meaning
Lattice and torus	
Γ	lattice in $\mathbb{R}^4$; defines the torus $\mathsf{T}^4 = \mathbb{R}^4/\Gamma$ (instead of the Λ customary in the lattice literature, see note on notation)
Γ^*	dual lattice; carries the admissible wave vectors of the Fourier modes
$D4$	root lattice D_4: all integer k with $\sum k_i$ even; kissing number 24
Z^4	cubic lattice $\mathbb{Z}^4$; kissing number 8, exactly self-dual
$D4^*$	dual D4 lattice; again a D4 lattice up to scaling
R_{123}, R_4	radii of the three spatial directions and of the fourth direction
r	radius ratio $r = R_4/R_{123}$ (dimensionless)
Spectrum	
k	wave vector (mode index), $k \in \Gamma^*$
k_4	fourth component of k; $k_4 = 0$ separates the FCC half from the winding half
$\lvert k\rvert^2$	norm; also the eigenvalue of the Laplacian on the isotropic torus
$E(k)$	eigenvalue under anisotropy, $E = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$
$N(n)$	degeneracy of the shell $\lvert k\rvert^2 = n$; coefficient of the theta series
θ_{D4}	theta series of the D4 lattice, $\theta_{D4} = (\theta_3^4 + \theta_4^4)/2$
$\theta(t)$	heat-kernel theta function, $\theta(t) = \sum_k e^{-t\lvert k\rvert^2}$
$\zeta_M(s)$	Epstein zeta function of the lattice M; $s = -1/2$ is the Casimir point
Symmetries	
$\mathrm{Aut}(\Gamma)$	automorphism group of the lattice (length-preserving self-maps)
$W(F_4)$, $W(D4)$	Weyl groups; $\lvert W(F_4)\rvert = 1152$, $\lvert W(D4)\rvert = 192$

Symbol	Meaning
A	element of order 3 in Aut($D4$); half-integral, hence computed exactly via $2A$
$\det(1-A)$	determinant; $\lvert\det(1-A)\rvert = 9$ counts the orbifold fixed points
Z_3, C_3	cyclic group of order 3 (triality resp. fibre action)
S_3, A_5	symmetric group on 3 elements; alternating group on 5 elements (icosahedral)
ω	third root of unity, $\omega = e^{2\pi i/3}$; phase $+120^\circ$
χ_0, χ_1, χ_2	the three Z_3 characters; index the orbifold sectors
Hilbert space and fibre	
$\mathscr{H}$	state space, $\mathscr{H} = L^2(\mathsf{T}^4) \otimes \mathbb{C}^3$ (Docs. 230/282)
$\mathbb{C}^3$	fibre over each point; carries the Z_3 structure
3, $3'$	the two triplets of the doubling $6 = 3 \oplus 3'$ (Doc. 285)
φ	golden ratio, $\varphi = (1+\sqrt{5})/2$; occurs only in the A_5 layer
$\theta = 2/9$	phase share from Doc. 293; not a lattice invariant (Ch. F)
FFGFT references	
$\xi = 4/30000$	geometric fundamental constant of FFGFT
$\tilde{T} \cdot m = 1$	time-mass relation; readable in the spectrum as the splitting at $r \neq 1$
Status labels	
[K]	core derivation: computed and controlled
[S]	structural statement: consistent but not derived
theorem	mathematically proven, not merely checked numerically
boundary	explicitly booked limitation of scope

F.1 The starting position: flatness is not the difference

The torus $\mathsf{T}^4 = \mathbb{R}^4/\Gamma$ is *always* flat, whatever lattice Γ defines it. The D4 torus and the Z^4 torus both have zero curvature. The difference lies not in curvature but in lattice structure: kissing number 24 against 8, packing density $\pi^2/16$ against $\pi^2/32$.

On passing to $\mathscr{H} = L^2(\mathsf{T}^4)$ the lattice becomes the **index set of the Fourier modes**: the basis consists of plane waves $e^{ik\cdot x}$, and the admissible k lie on the dual lattice Γ^*. In this sense every lattice becomes flat in Hilbert space – it is no longer a sphere packing in space but a point lattice of quantum numbers.

The geometry is not lost; it migrates. The Laplacian has eigenvalues $|k|^2$, and the degeneracy of each eigenvalue is the number of lattice vectors of that length

– the theta series of the lattice. The 24 shortest D4 vectors become the 24-fold degeneracy of the first excitation, and the decomposition $24 = 12 + 12$ of Doc. 311 (FCC neighbours plus winding neighbours) remains exactly readable as a k_4 split.

Two anchors keep this translation stable:

1. **D4 is self-dual up to scaling** – $D4^*$ is again a D4 lattice. The Fourier transform does not lead out of the family. ($\mathbb{Z}^4$ is exactly self-dual.)
2. **The fibre changes nothing about the lattice.** In $\mathscr{H} = L^2(\mathbb{T}^4) \otimes \mathbb{C}^3$ the $\mathbb{C}^3$ fibre hangs at every point and carries the Z_3 structure; the mode lattice remains Γ^*.

$$\boxed{\text{geometry} \longrightarrow \text{spectrum} + \text{fibre}} \tag{F.1}$$

F.2 The spectrum separates what curvature does not [K]

Complete shell count up to $|k|^2 = 12$:

Norm $\lvert k\rvert^2$	$\mathbb{Z}^4$	D4	D4 split ($k_4{=}0$ / $k_4{\neq}0$)
1	8	0	—
2	24	24	12 / 12
3	32	0	—
4	24	24	6 / 18
5	48	0	—
6	96	96	24 / 72
7	64	0	—
8	24	24	12 / 12
9	104	0	—
10	144	144	24 / 120
11	96	0	—
12	96	96	8 / 88

Three readings:

(i) Two flat tori are spectrally distinguishable. $\mathbb{Z}^4$ starts at norm 1 with degeneracy 8, D4 at norm 2 with degeneracy 24. For D4 the odd norms are *entirely* absent – the parity condition $\sum k_i$ even excludes them. What curvature does not distinguish, the spectrum distinguishes at once.

(ii) Independent control passed. The D4 series $24, 24, 96, 24, 144, 96 \ldots$ is the known theta series $\theta_{D4} = (\theta_3^4 + \theta_4^4)/2$; the script reproduces it coefficient by coefficient.

(iii) The $12 + 12$ decomposition is spectrally addressable. The first excitation carries exactly twelve modes with $k_4 = 0$ (the spatial FCC half) and twelve with $k_4 \neq 0$ (the winding half). On higher shells the split is asymmetric ($6/18$ at norm 4, $8/88$ at norm 12): **the $12/12$ symmetry is a property of the first shell, not a general theorem.** This must be booked explicitly – Doc. 311 argues with $24 = 12 + 12$, and that statement holds exactly where it is made.

F.3 Anisotropy: where the time-mass relation becomes readable [K]

If the fourth direction is compactified with a different radius ($r = R_4/R_{123}$, eigenvalue $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$), the first shell splits:

r	E (winding, 12 modes)	E (FCC, 12 modes)
1.0	2.000	2.000 (degenerate: 24)
1.1	1.826	2.000
1.5	1.444	2.000
2.0	1.250	2.000

The FCC modes stay fixed at $E = 2$, the winding modes follow exactly

$$E_{\text{winding}} = 1 + \frac{1}{r^2}, \qquad \Delta = 1 - \frac{1}{r^2}. \tag{F.2}$$

The splitting is a direct measure of the radius ratio – **the place where the $\tilde{T} \cdot m = 1$ structure becomes readable in the spectrum**: mass direction and spatial directions carry different excitation energies as soon as they have different lengths.

The crossing map is discrete and short

All levels have the form $E = a + b/r^2$ with integer a, b. In the range $1 < r \leq 2.5$, among all levels up to norm 8 there are **exactly two** crossing radii:

$r = \sqrt{2}$	pure winding $(0, 0, 0, \pm 2)$ crosses the FCC-12
$r = \sqrt{3}$	pure winding crosses the winding-12

The level sequence therefore does not change continuously and chaotically but at two distinguished root radii. Any ordering statement about the lowest excitations falls into exactly three regimes: $r < \sqrt{2}$, $\sqrt{2} < r < \sqrt{3}$, $r > \sqrt{3}$. Beyond $\sqrt{3}$ the *ground tone* of the spectrum is a pure winding (at $r = 2$: $E = 1.00$ with 2 modes before 1.25 with 12 before 2.00 with 12).

F.4 Time from the mass circle: the reciprocity is mode-wise exact [K]

The fourth direction is the time–mass dual one; unrolling is onto time. Two check questions: how do time periods arise – and does $\tilde{T} \cdot m = 1$ hold only on average, when mass and time “distribute equally”?

Time periods as covering-space bookkeeping

Unrolling the compact fourth direction is the universal covering $\mathbb{R} \to S^1$; the deck transformation group is $\mathbb{Z}$, and the unrolled coordinate counts cycle traversals. A k_4 mode $e^{ik_4 x_4/R_4}$ thereby becomes a periodic function with period $T_k = 2\pi R_4/k_4$. Since $\pi_1(S^1) = \mathbb{Z}$ is abelian, this bookkeeping is *lossless* – the same construction that raises the faithfulness question for non-abelian fundamental groups is faithful here by structure.

Mode-wise exactness – and what exactly is exact

In the deformation spectrum (Ch. C), $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$ is literally $E^2 = p_{\text{space}}^2 + m^2$ with $m_k = k_4/R_4$ – **the winding number is the mass number**. Hence

$$T_k \cdot m_k \;=\; \frac{2\pi R_4}{k_4} \cdot \frac{k_4}{R_4} \;=\; 2\pi \qquad \text{exactly, for every mode.} \tag{F.3}$$

The reciprocity therefore does *not* hold only on average – it is satisfied mode-wise identically, because period and mass are built from the same quantum number k_4 and the same radius. The next section makes precise, however, *which* quantity is exact here: T_k is first of all an arc length on the circle, i.e. a place interval. Side result: the $12 + 12$ splitting (Ch. C) is, in this reading, the separation massless ($k_4 = 0$, FCC) versus massive (winding), and the level crossing at $r > \sqrt{3}$ means: under strong anisotropy a pure mass mode becomes the ground tone.

The period is a place first, not a duration

The quantity $T_k = 2\pi R_4/k_4$ is an *arc length on the circle*: the distance in the coordinate x_4 after which the mode repeats. It is therefore a **place interval**, not elapsed time. Properly it is called λ_4, and the relation

$$\lambda_4 \cdot m \;=\; \frac{2\pi R_4}{k_4} \cdot \frac{k_4}{R_4} \;=\; 2\pi \tag{F.4}$$

is the **de Broglie relation** along the mass direction – wavelength times wavenumber – not yet a statement about time. It is exact and always valid, but it is a statement about x_4.

Where, then, does duration come from? Only from the unrolled coordinate. A point of unrolled time is a pair

$$t \;=\; w \cdot 2\pi R_4 \;+\; x_4, \qquad w \in \mathbb{Z} \;\text{(winding)}, \;\; x_4 \in [0, 2\pi R_4), \tag{F.5}$$

that is, *number of revolutions* plus *place on the cycle*. Duration resides entirely in the counter w – the deck transformation of the covering – and place in the remainder. On the cycle itself there is no distinguished beginning; every "moment" is a place that recurs (cf. Doc. 313).

The temporal reading carries a γ factor

If one asks for the *oscillation period in unrolled time*, the relevant quantity is $T_{\text{osc}} = 2\pi/E$ with $E = \sqrt{p_{\text{space}}^2 + m^2}$. Then

$$T_{\text{osc}} \cdot m \;=\; 2\pi\,\frac{m}{E} \;=\; \frac{2\pi}{\gamma} \;\le\; 2\pi, \qquad \text{equality} \iff p_{\text{space}} = 0. \tag{F.6}$$

p_{space}/m	γ	$T_{\text{osc}} m/2\pi$	$\lambda_4 m/2\pi$
0	1.00000	1.0000000	1.0000000
0.5	1.11803	0.8944272	1.0000000
1	1.41421	0.7071068	1.0000000
2	2.23607	0.4472136	1.0000000
5	5.09902	0.1961161	1.0000000

Place and duration are not the same here. Exactly 2π is the *place* reading; the *time* reading carries the factor $1/\gamma$ and coincides with it only for the mode at rest. Carrying T_k as a "time period" and booking $T_k \cdot m = 2\pi$ as temporal reciprocity would accordingly confuse the two.

Three statements in three directions:

Reading	Relation	Equality for
place (λ_4, de Broglie)	$= 2\pi$	always
time, moving mode	$= 2\pi/\gamma \le 2\pi$	$p_{\text{space}} = 0$
mass mixture (Jensen)	$\ge 2\pi$	sharp shell

The exact relation $\tilde{T} \cdot m =$ const therefore holds for the **resting, sharp** mode – there and only there both correction factors vanish simultaneously. Motion pushes the product down, mass spread lifts it up. This is no contradiction to Doc. 306 but the Kaluza–Klein reading of its validity conditions.

For mixtures the equation becomes an inequality

If a state superposes several k_4 shells with weights p_k, then $\langle T\rangle\langle m\rangle = 2\pi\langle k\rangle\langle 1/k\rangle$, and by Jensen

$$\langle k\rangle\langle 1/k\rangle \;\ge\; 1, \qquad \text{equality} \iff \text{exactly one shell occupied.} \tag{F.7}$$

Numerically (script 4): sharp shell (any k) $\to$ exactly 1.0000000000; $k = 1, 2$ equally occupied $\to 1.125$; uniform $k = 1 \ldots 50 \to 2.295$; thermal $p_k \propto e^{-k/T} \to 1.0000227$ ($T = 0.1$) up to 2.600 ($T = 10$); 2000 random distributions: minimum 1.077, never below 1.

Booking – the conjecture inverts: the condition for $\tilde{T} \cdot m = 1$ is *not* equal distribution but **sharpness**: exactly one occupied mass shell. Equal distribution is, on the contrary, a state with a pronounced reciprocity excess (2.29 for $k \le 50$). The excess $\langle k\rangle\langle 1/k\rangle - 1$

measures the mass spread of the state (for small spread $\approx \mathrm{Var}(k)/\langle k\rangle^2$): a mixture has neither a sharp period nor a sharp mass, and the product of the means lies strictly above 2π.

Together with the γ factor the full condition follows: $\tilde{T}\cdot m$ is constant exactly when the mode *rests* (no γ) *and* is sharp (no Jensen excess). Both deviations have different signs and different causes – motion in space versus spread in mass.

Epistemic status: lattice-side consistency check of the mode structure (Kaluza–Klein reading of the fourth direction); the native FFGFT formulation of the reciprocity is Doc. 306 – nothing is newly derived here; compatibility is checked: passed, with the sharpening "exact per mode, $\geq$ for mixtures".

The closure fork is spectrally invisible

Doc. 313 (after Doc. 295) poses the closure question as a fork: the defect $d = 100\,\xi = 1/75$ per revolution admits three cases for the *unrolled* time. With the cycle length $\tau_c = 2\pi/H_0 = 91.9$ Gyr (Doc. 312):

Case	Condition	Closure	Period
A	$d = 0$	after 1 revolution	91.9 Gyr
B	ξ frozen	after 75 revolutions	6892 Gyr
C	ξ running (recursion)	never	log spiral

Case B follows from the rational rotation number $1 - d = 74/75$: since $\gcd(74, 75) = 1$, the trajectory closes after exactly 75 revolutions, i.e. $75\,\tau_c = 6892.5$ Gyr. Case C arises when ξ runs through the recursion; the accumulated defect becomes logarithmic and the winding turns into an equiangular spiral.

This table is only quoted here. It stands so that the cases are intelligible without looking them up; its *interpretation* – entropy obligation, Poincaré discrepancy, thermal history – belongs to Docs. 312/313 and is not pursued here. Two points are nevertheless needed so the numbers are not misread.

First: these are deadlines, not repetitions. 91.9 and 6892 Gyr do not say that something repeats, but within what period a recurrence would *have to* occur if the respective case held. What is closed is the *geometry*; whether the *history* returns is independent of that and, per Doc. 313, coarse-grained open. In this reading the "Big Bang" is in any case a *place* on the cycle – the region of smallest masses and slowest clocks – not an event; a closed circle has no starting point.

Second, and this bounds the reach of all time statements: states at other places on the cycle are not derivable from today's. The usual backward extrapolation ($T(z) = T_0(1+z)$ from an adiabatic chain) presupposes *one* continuous thermodynamic history. In the place reading, an early epoch is not an earlier state of the same system but a location with a *different local structure* – different masses, different clock rates – and hence its own equilibrium, fixed locally and not by continuation of today's. The corpus demonstrates this on itself: T_0 is not extrapolated but obtained structurally from ξ ($k_B T_0 = (16/9)\xi$, Docs. 061/313). This is precisely why the coarse-grained chaining remains open (Poincaré discrepancy 10^{104} dex, D(ii) in Doc. 313).

What this does *not* mean: it devalues no structural derivation. The Ω_m^* chain in Doc. 313 runs through ξ and K_{frak}, not through an adiabatic back-calculation, and is unaffected by this boundary. Affected are only statements of the form "how hot was it in epoch X" insofar as they are obtained by extending today's equilibrium – such statements would need their own structural derivation *at that place*.

It would be tempting to read the defect as a *twist* of the boundary condition ($k_4 \to k_4 + 1/75$, Scherk–Schwarz-like). That is **wrong**, and the reason is structural:

In the rolled-up state there are no fractal corrections. Single-valuedness of the mode on the circle forces $k_4 \in \mathbb{Z}$ – integer or nothing. A winding number is *topological*; it has no room for $1/75$. Corrections require something that *accumulates*, and only a traversed path can accumulate. The very phrase "defect per revolution" presupposes counting revolutions – and revolutions are counted only in the unrolled coordinate w (Ch. D1). The fractal path correction K_{frak} is a property of the *traversal*, not of the periodicity of the field.

Hence the assignment:

Quantity	State	fractal correction
theta series, degeneracies, $24 = 12 + 12$	rolled up	no
Aut($D4$), triality, orbifold	rolled up	no
circulant, containers 4×6	rolled up	no
Casimir, heat kernel, congruences	rolled up	no
$\lambda_4 \cdot m = 2\pi$ (de Broglie)	rolled up	no
cycle length, defect d, case A/B/C	unrolled	yes
revolution count w, history	unrolled	yes
T_{osc}, γ factor	unrolled	yes

Two consequences. First: *the fork has no spectral signature.* The mode spectrum is an object of the rolled-up state; whether the trajectory closes after one, after 75 or after no revolution changes nothing about the eigenvalues of the Laplacian on T^4. The first shell carries 24 modes in all three cases.

Second, and this is the real relief: **Chapters E–H are case-independent.** Orbifold, circulant and containers do not depend on which of the three cases holds – they are statements about the compact geometry, and the defect does not act there. The fork is a question of *history* (entropy, recurrence – Doc. 313), not of *structure*.

The same boundary settles the fractal question. D_f and K_{frak} are *different* things (Doc. 133): $D_f^{\text{space}} = 3 - \xi$ is local and practically without effect (6.7×10^{-5}), $K_{\text{frak}} = 1 - 100\xi$ is cumulative across the RG run and acts only where scales are *traversed*. The rolled-up spectrum carries neither appreciably – Ch. J elaborates.

Marginal note on the number. $75 = 1/(100\xi)$; its factorisation $3 \cdot 5^2$ carries *no* meaning – the five here has nothing to do with the fivefold symmetry of Ch. F.

F.5 Reorganisations: three levels and one boundary

Flat lattices can be reorganised – on three levels that cost very different things.

Change of basis (free). Any unimodular transformation from $GL(4,\mathbb{Z})$ describes the same point lattice with different generators. The spectrum does not change at all. Pure bookkeeping.

Lattice symmetries (free, structure-bearing). Reorganisations mapping the lattice onto itself – this is where the central finding sits, Ch. E.

Rebuilding into another lattice (costs physics). Discretely via glue vectors along the chain

$$D4 \subset \mathsf{Z}^4 \subset D4^* \qquad \text{(index 2 each)}, \tag{F.8}$$

or continuously by deformation in the moduli space $GL(4,\mathbb{R})/(O(4)\times GL(4,\mathbb{Z}))$. Each step shifts degeneracies; D4 is a distinguished point there (kissing number 24, provably unbeatable in 4D).

The boundary of the spectral statement. From dimension 4 on there exist pairs of flat tori that are *isospectral but not isometric* (Schiemann). The general theorem "the spectrum determines the lattice" is *false* in 4D. Uncritical for D4 itself – kissing number 24 characterises it uniquely – but it must not be booked as a rule.

F.6 Triality is built in, not attached [K]

Computed on the lattice

$\lvert\mathrm{Aut}(D4)\rvert$	$1152 = \lvert W(F_4)\rvert$
$\lvert\mathrm{Aut}(\mathsf{Z}^4)\rvert$	$384 = 2^4 \cdot 4!$
$\lvert\mathrm{Aut}(D4)\rvert/\lvert W(D4)\rvert$	$1152/192 = 6 = \lvert S_3\rvert$

Both orders were determined by brute-force enumeration of all basis tuples with identical Gram matrix and match the known values. The quotient $6 = |S_3|$ establishes triality *directly on the lattice automorphism group*, not merely on the Dynkin diagram: D4 is the only lattice of the D_n family with three equivalent Dynkin arms; for $D5, D6, \ldots$ the outer symmetry shrinks to Z_2.

The 80 order-3 elements in three classes

Computed exactly (the triality elements are *half-integral* – the glue-vector mixture $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ enters; rounding to integers destroys them, hence the computation uses $2A$):

Class	Count	Character
trace -2, half-integral	16	fixed-point free in $\mathbb{R}^4$; eigenvalues $\{\omega,\bar\omega\}$ twice
trace 1, integral	32	fixed plane in $\mathbb{R}^4$, free on the 24 minimal vectors
trace 1, half-integral	32	6 fixed vectors + 6 triple orbits

Two consequences

(a) The orbifold is a lattice symmetry, not an extra assumption. For each of the 16 elements with trace -2 one has exactly

$$|\det(1-A)| = 9, \tag{F.9}$$

so the quotient $\mathrm{T}^4/\langle A\rangle$ is a Z_3 orbifold with *exactly 9 fixed points* – the standard number 3^2. All nine lie at third-coordinates. **The T^4/Z_3 structure need not be imposed; it is already present in D4.**

(b) On every mode orbit the Z_3 acts as a circulant. The unitary action on $L^2(\mathrm{T}^4)$ permutes the three plane waves of a triple orbit cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases $0°$, $+120°$, $-120°$ – *structurally exactly the circulant diagonalisation of the $\mathbb{C}^3$ fibre* (Doc. 291). The first shell provides eight copies: $24 = 8 \times 3 = 8 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$.

F.7 The negative finding: 5 does not divide 1152 [K]

The order distribution in $\mathrm{Aut}(D4)$, determined exactly:

Order	1	2	3	4	6	8	12
Count	1	139	80	228	464	144	96

Total 1152. **Order 5: zero elements** – and that is not a computational result but a theorem: $1152 = 2^7 \cdot 3^2$, and 5 does not divide it (Lagrange).

Consequence for $\theta = 2/9$. The fivefold rotation R_5, from which $p_0 = |\langle v_0|R_5 v_e\rangle|^2 = 2/9$ follows (Doc. 293), is *in principle* incompatible with the D4 lattice structure. The test was additionally carried out directly: all invariants of the linear lattice action at the nine orbifold fixed points ($x{\cdot}x$, $k{\cdot}x$, $x{\cdot}y$, all mod 1 in exact rational arithmetic) have denominator spectrum $\{1, 2, 3, 6\}$ – **no ninths**. Notably, $x{\cdot}y$ is a multiple of one ninth, but the numerators are always divisible by 3; the D4 geometry *suppresses* ninths at the first level.

What this means – and what it does not. $2/9$ is not an invariant of the lattice action. This does not devalue Doc. 293 but delimits it: the phase belongs to a layer *above* the lattice (the A_5 structure), the lattice carries the Z_3. The division of labour is thereby exactly named rather than surmised.

F.8 The doubling is nevertheless prepared [K]

Does the negative finding contradict the doubling $6 = 3 \oplus 3'$ of Doc. 285? No – in three specifiable stages.

(1) The prohibition hits the wrong level. $5 \nmid 1152$ forbids order-5 *lattice automorphisms*. The doubling lives on the **fibre** ($\mathbb{C}^3 \to \mathbb{C}^6$); A_5 is nowhere required as a spatial symmetry. The compatibility condition is only that the interface of the two levels

exists – and that is exactly determinable: element orders in A_5 are $\{1,2,3,5\}$, in Aut$(D4)$ they are $\{1,2,3,4,6,8,12\}$; the intersection is $\{1,2,3\}$, canonically C_3. **And exactly that is provided by triality.** The interface is not merely permitted, it is the only possible one – and it is occupied.

(2) The lattice prepares the doubling. The central involution $-1 \in$ Aut$(D4)$ commutes with the triality Z_3 and pairs the eight triple orbits of the first shell into **four disjoint pairs**. No orbit is self-paired, provably: $-k = A^j k$ would require eigenvalue -1 for an element of order 3 – impossible. Hence

$$24 \;=\; 8\times 3 \;=\; 4\times 6 \qquad \text{(orbit} \oplus \text{mirror orbit)}. \tag{F.10}$$

Each six-block carries under C_3 the character content $2\times(\chi_0\oplus\chi_1\oplus\chi_2)$ – exactly the restriction of $3\oplus 3'$. The first excitation shell provides four ready containers.

(3) The boundary, cleanly booked. The *labelling* – which factor is 3 and which $3'$ – is lattice-blind. The A_5 characters of the two triplets differ only on the 5-classes (φ against $1-\varphi$); on the 3-class both are zero and the C_3 restrictions are identical. The distinction hangs exactly on the structure that does not exist in the lattice – fully consistent with Ch. F.

Balance: the lattice provides containers and C_3 content, the A_5 layer provides the $3/3'$ labels, φ and the fivefold – and hence $2/9$.

F.9 Anchor-free results [K]

All quantities below are dimensionless – ratios, counts, congruences. No Kammerton, no SI scale enters.

Two congruence theorems for the D4 theta series

All complete shells up to norm 120 checked:

$$N(n)\equiv 0 \pmod 3 \qquad \text{and} \qquad N(n)\equiv 0 \pmod 6 \qquad \text{without exception.} \tag{F.11}$$

Both are theorems, not empirics: the triality Z_3 is fixed-point free on $\Gamma\setminus\{0\}$ (eigenvalues $\omega,\bar\omega$ – no eigenvalue 1), so *every* shell decomposes into triple orbits; together with the -1 pairing without self-pairing, into six-blocks.

Character-resolved $\mathbf{T}^4/Z_3$ spectrum

Consequence: exact equidistribution on every shell.

Norm	$N(D4)$	χ_0	χ_1	χ_2
2	24	8	8	8
4	24	8	8	8
6	96	32	32	32
10	144	48	48	48
14	192	64	64	64
18	312	104	104	104

The invariant sector of the orbifold (χ_0) has exactly $N(n)/3$ states per shell; the Z_3 projection costs no asymmetry on any shell.

Honest consequence. Equidistribution and six-containers are *not* a distinction of the first shell but a theorem for all. **The structure is everywhere – and therefore cannot select anything.** This is exactly the pattern-versus-occupation logic of Doc. 293: the pattern is present scale-invariantly; which shell is physically occupied is decided by something else.

Casimir: the densest lattice loses

ζ-regularised vacuum energy of a scalar field, $E = \pi\zeta_M(-1/2)$, both tori normalised to covolume 1 (Epstein zeta via the Riemann split representation, verified three times – against direct summation at $s = 4, 5$ and against the closed form $Z_{\mathrm{Z}^4}(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1)$, agreement to 25 digits):

	$\zeta_M(-1/2)$	$E = \pi\zeta_M(-1/2)$
Z^4 torus	-0.2966893	-0.9320768
D4 torus	-0.2764778	-0.8685805
Difference		$+0.0634963$

The Z^4 torus lies lower. The reason is the spectral gap: at equal volume the densest lattice D4 has the *largest* first eigenfrequency ($\sqrt{2}$ against 1) and hence the *less* negative zero-point energy. Notable is the reversal of ordering: for $s > 0$ D4 minimises the Epstein zeta (the known extremality of the densest lattice), at $s = 0$ all lattices agree ($\zeta(0) = -1$ universally), and at $s = -1/2$ the ordering is inverted.

Interpretation boundary, booked: this holds for the scalar, periodic field. Different field content or different boundary conditions can change the sign of the preference – "nature chooses Z^4" does *not* follow.

Heat kernel: the observability limit quantified

For both tori $t^2\theta(t) \to 1$ (identical Weyl term at equal volume); the lattice difference is exponentially small:

t	relative difference
0.5	1.2×10^{-2}
0.2	1.2×10^{-6}
0.1	1.8×10^{-13}
0.05	4.1×10^{-27}

Any observable resting only on the leading heat-kernel coefficients is lattice-blind. The D4/Z^4 difference sits exclusively in the exponentially suppressed theta-series fine structure – or directly in the spectral gap.

F.10 What is still bare here – and what is not [K]

This document computes with the *smooth* Laplacian on $\mathrm{T}^4 = \mathbb{R}^4/\Gamma$. What has to be settled is what FFGFT holds against this in fractal structure – and here **two quantities must be kept apart** that are easily confused:

Quantity	Character	Effect
$D_f^{\text{space}} = 3 - \xi$ (Doc. 133)	*local*	6.67×10^{-5}
$K_{\text{frak}} = 1 - 100\xi$ (Doc. 133, A040)	*cumulative*	1.33×10^{-2}

Doc. 133 explicitly derives K_{frak} *not* from the local dimension but from the "cumulative effect" across the RG run from the Planck to the electroweak scale (17 orders of magnitude); the factor 100 is RG amplification (Doc. 190). The local dimension alone yields, per Doc. 133, "a vanishingly small correction – practically one", computed $(D/3)^{D/2} = 0.99993$. The ratio of the two effects is exactly 200.

On notation, because this is easily slipped: the fractal dimension of FFGFT is $D_f^{\text{space}} = 3 - \xi$ – the *spatial* one, not a four-dimensional quantity. The exponent $3/2$ in $K_{\text{frak}}^{3/2}$ (Doc. 018) is likewise justified as half the *spatial* dimension. It is three dimensions, not four; a quantity $D_f = 4 - \xi$ is unknown to the corpus.

Hence the assignment – and it coincides with the boundary of Ch. D2 (rolled up versus unrolled): a Laplacian on the compact T^4 is *local*. It cannot carry a cumulative quantity; K_{frak} does not enter through the operator but through the **bridge constants** (E_0, v) in scale conversion – exactly as R72 shows for the two E_0 values (ratio $1/\sqrt{K_{\text{frak}}}$). The consequence falls into three very unequal parts.

Untouched: everything anchor-free

By the assignment rule (register R70, Doc. 313), K_{frak} applies to absolute quantities and cancels for ratios of the same level. Untouched therefore:

- all **counts** – 24, 8, $N(n)$, 8×3, 4×6: integer combinatorics, no length;
- the **congruences** $N(n) \equiv 0 \bmod 3$ and $\bmod 6$: group theory, no metric content;
- the **character split** $N/3$ per sector;
- all **ratios** E/E_{min} and the crossing radii $\sqrt{2}$, $\sqrt{3}$ – structure over structure, the factor cancels.

This is the real reason why Ch. H is called "anchor-free results": those statements are fractal-invariant.

One factor: all absolute energies

With an anchor, $E = \hbar H_0 |k|$ (ground mode $= \hbar H_0$, see Ch. B together with Doc. 312). These values are bare and carry the factor:

	bare	with K_{frak}
first D4 shell ($\lvert k\rvert = \sqrt{2}$)	3.230×10^{-52} J	3.187×10^{-52} J
pure winding ($\lvert k\rvert = 2$)	4.567×10^{-52} J	4.507×10^{-52} J

Throughout $-1.33\,\%$. The same holds for the Casimir energies of Ch. H and for the spectral gap itself. Bookkeeping, settled; not structurally interesting.

Important for the assignment: the factor sits *not* in the spectrum but in the *anchor*. The dimensionless eigenvalues $\lvert k\rvert^2$ remain unchanged; what is corrected is the conversion to SI via $\hbar H_0$ – a scale bridge, that is, precisely the route along which a cumulative quantity may act. Anyone looking only at ratios sees none of it (A040 rule, R72).

The operator, not the conversion: the perturbation calculation [K]

The third part cannot be settled by multiplication. At $D_f^{\text{space}} = 3 - \xi$ the operator is *no longer an exactly smooth Laplacian* – and exact symmetry is precisely what carries the degeneracies. Only the *local* effect is relevant here: K_{frak} is cumulative per Doc. 133 and therefore does not belong on a local operator, so $O(100\xi)$ is ruled out as a perturbation strength.

Perturbation strength	$\Delta E/E$	ΔE at $E = \hbar H_0$
local, $(D/3)^{D/2}$ (Doc. 133)	6.7×10^{-5}	1.5×10^{-56} J
$O(\xi)$ (upper bound)	1.3×10^{-4}	3.0×10^{-56} J

The perturbation thus lies **two orders of magnitude below** what an estimate via K_{frak} would give.

The 24-fold degeneracy would then no longer be a degeneracy but a multiplet with fine structure. Which sub-multiplets it decomposes into is decided by the *symmetry of the perturbation* – and, as the calculation shows, via *representation theory*, not via the orbit structure:

Perturbation respects	Multiplet structure of the 24 (computed)
full Aut($D4$) (c_g norm-dependent only)	$9 + 8 + 4 + 2 + 1$
triality (Z_3) and -1	$8 \times 1 \;+\; 8 \times 2$
only Z_3	$8 \times 1 \;+\; 8 \times 2$
only -1	24×1
neither	24×1

Computed (script 4): first-order perturbation theory on the 24×24 matrix $\langle k|V|k'\rangle = c_{k-k'}$, symmetry classes via orbit averaging of the coefficients, three independent random perturbations per class – the multiplet structure is seed-independent and identical in all cases.

The obvious expectation is wrong. One assumes the orbits stay together (Z_3 and -1: the 24 as a whole; only Z_3: 8 triple orbits; only -1: 4 six-blocks). In fact multiplet sizes follow the *irreducible representations* of the respected symmetry group, not the orbits. Concretely: (i) every position-dependent perturbation breaks the translation invariance of the torus – and *that* carried the 24-fold degeneracy, not the point symmetry. (ii) Z_3 is abelian, all irreps one-dimensional; the doublets are the antiunitary pairing of the conjugate characters $\chi_1 \leftrightarrow \chi_2$, while the χ_0 sector decomposes into 8 singlets. (iii) -1 alone enforces nothing (free Z_2 action, one-dimensional characters). (iv) Even the maximally symmetric perturbation does not keep the 24 together: $24 = 9 \oplus 8 \oplus 4 \oplus 2 \oplus 1$, consistent with the irrep dimensions of $W(F_4)$ on the root shell [K numerically; the irrep assignment is B, without a character computation].

This closes the open point – and defuses it: qualitatively it still holds that only as much of the 24-fold degeneracy survives as the perturbation symmetry provides in irrep dimension (at best $9+8+4+2+1$, generically nothing). Quantitatively, however, the relevant perturbation is $\sim 10^{-5}$ relative, not 10^{-2}: the splitting would be fine structure far below any readability, and the container structure (4×6, Ch. G) remains practically intact.

Chapter balance. *Bare* are only the SI absolute values, and via the anchor, not via the operator. The rolled-up quantities of this document – degeneracies, ratios, congruences, containers, Casimir and heat-kernel ratios – are **not** bare but correct: K_{frak} does not act there structurally. This is the same boundary as in Ch. D2, read from the other side.

F.11 Status overview

Statement	Content	Status
Translation	geometry → spectrum + fibre; lattice becomes mode index, degeneracies = theta series	[K]
Distinguishability	Z^4 from norm 1 (8), D4 from norm 2 (24); odd norms empty for D4	[K]
$12 + 12$	first shell exactly FCC + winding; higher shells asymmetric ($6/18$, $8/88$)	[K]
Anisotropy	$E_{\text{wind}} = 1 + 1/r^2$; crossings only at $r = \sqrt{2}$ and $\sqrt{3}$; three regimes	[K]
Triality	$\lvert\text{Aut}(D4)\rvert = 1152$, quotient $6 = \lvert S_3\rvert$; 80 order-3 elements in three classes	[K]
Orbifold	16 elements with $\lvert\det(1 - A)\rvert = 9$: T^4/Z_3 is a lattice symmetry, not an extra assumption	[K]
Circulant	Z_3 acts on every triple orbit with $\{1, \omega, \omega^2\} = \mathbb{C}^3$ fibre structure	[K]
$5 \nmid 1152$	no order-5 automorphisms; $2/9$ is not a lattice invariant (denominators $\{1, 2, 3, 6\}$)	[K] theorem

Statement	Content	Status
Doubling	interface $A_5 \cap \mathrm{Aut}(D4) = C_3$, occupied; $24 = 4 \times 6$ as containers	[K]
Labelling	3 versus $3'$ is lattice-blind (difference only on 5-classes)	[K] boundary
Congruences	$N(n) \equiv 0 \bmod 3$ and mod6 on all 60 shells checked, with proof ground	[K] theorem
Selection	containers and equidistribution are generic – the structure cannot distinguish any shell	[K]
Casimir	$E(\mathrm{Z}^4) = -0.9321$ against $E(D4) = -0.8686$; ordering reversed relative to $s > 0$	[K], field-content dependent
Heat kernel	Weyl term identical, difference 4×10^{-27} at $t = 0.05$: leading order is lattice-blind	[K]
Spectral theorem	"spectrum determines lattice" false in 4D (Schiemann); uncritical for D4, but not a rule	[K] boundary
Reciprocity	$\lambda_4 \cdot m = 2\pi$ exact as a place relation (de Broglie); temporally $2\pi/\gamma \leq 2\pi$; for mixtures $\geq 2\pi$ (Jensen) – exact only at rest and sharp	[K]
Closure case	rolled up, no fractal correction (k_4 topological); fork A/B/C spectrally invisible, Chs. E–H case-independent	[K]
Perturbation multiplets	splitting follows irrep dimensions, not orbits: $9{+}8{+}4{+}2{+}1$ (full), $8{\times}1{+}8{\times}2$ (Z_3), $24{\times}1$ (generic)	[K]
Fractal status	$D_f^{\text{space}} = 3 - \xi$ local ($6.7{\times}10^{-5}$) versus K_{frak} cumulative ($1.33\,\%$, RG run, Doc. 133): bare are only SI absolute values via the anchor, not the rolled-up spectrum	[K]

Balance: in Hilbert space the lattice becomes flat but not inconsequential. The kissing number becomes the degeneracy, triality becomes the orbifold and the fibre, the radius ratio becomes the level splitting. The *division of labour* is determined just as sharply: fivefold symmetry – and with it φ and $2/9$ – cannot be carried by the lattice *itself*, and that by theorem ($5 \nmid 1152$), not for want of computational effort. This is a statement about the *layer*, not about the quantity: on the fibre, $2/9$ is explicitly admissible, since the interface $A_5 \cap \mathrm{Aut}(D4) = C_3$ is occupied by triality (Ch. G), and with $24 = 4 \times 6$ the lattice even provides the containers for the doubling $3 \oplus 3'$. What is excluded is therefore not $2/9$ but only its derivation *from* lattice invariants – the phase demonstrably enters one layer higher. And the container structure is generic: it provides, it does not select.

F.12 Check scripts

Three scripts, all deterministic, all controls passed:

- `d4_skript_1_spektrum_deformation.py` – theta series D4/Z^4, k_4 split, deformation r, exact crossing radii (standard library only).
- `d4_skript_2_trialitaet_orbifold_phasen.py` – Aut$(D4) = 1152$, order-3 classification (computed exactly with $2A$), $|\det(1 - A)| = 9$, circulant, phase test $2/9$, order distribution, $24 = 4 \times 6$ (numpy).
- `d4_skript_4_stoerung_reziprozitaet.py` – first-order perturbation theory on the 24 shell (multiplet structure per symmetry class, 3 seeds each) and reciprocity $T \cdot m$ (mode-wise; Jensen for mixtures) (numpy, mpmath).
- `d4_skript_3_schalen_casimir.py` – shell theorems, Epstein zeta at $s = -1/2$, heat kernel (numpy, mpmath).

Controls against known values: θ_{D4}, $|W(F_4)| = 1152$, $2^4 \cdot 4! = 384$, $|W(D4)| = 192$, orbifold fixed-point count $3^2 = 9$, Jacobi identity for Z_{Z^4}.

Registration: entry into Doc. 190 will be done separately, should the delimitation against Doc. 293 be booked there.

Chapter G

Doc. 315 — The Form of K_{frak}: Additive or Multiplicative?

G.1 What this is about

The fractal correction factor appears throughout the corpus in the additive form

$$K_{\text{frak}}^{\text{add}} \;=\; 1 - 100\,\xi \;=\; \frac{74}{75} \;=\; 0.98\overline{6},$$

while a multiplicative–cumulative form would be conceivable:

$$K_{\text{frak}}^{\text{mul}} \;=\; (1-\xi)^{100} \;=\; 0.9867543\,\ldots$$

The relative distance between the two forms is $8.88{\cdot}10^{-5}$. The corpus confirms the *value* $74/75$ several times independently (A040, A130, A270); whether this also discriminates the *form* – additive against multiplicative – had remained unexamined. This document supplies the examination and answers two questions: Which corpus location can resolve the two forms? And – as a structural question – *why* should one form be the right one?

Euler's musical spiral serves as the control case. It is the classic example of a cycle that *never* closes and thereby provides the counter-probe to the ξ cycle, which closes *exactly* after 75 revolutions. The comparison of the two cases supplies, in the end, the conceptual axis on which the form question hangs: rolled-up versus unrolled description.

Preliminary clarification (avoiding a category error): The value $74/75$ is **[K]** by the A040 derivation; the question of this document concerns exclusively whether independent witnesses can *measure* the additive against the multiplicative form – i.e. the status of the form as a measured rather than merely derived statement.

G.2 Control case I: Euler's spiral never closes

The impossibility argument

[Q] Euler's Tonnetz (*Tentamen novae theoriae musicae*, 1739) spans the intervals of just intonation as a lattice of fifths ($3/2$) and major thirds ($5/4$); the octave (2) is

divided out. The question "after how many repetitions does the spiral close?" has an exact answer: *never*. A closure would require

$$2^x \cdot 3^y \cdot 5^z = 1 \qquad \text{with integers } (x, y, z) \neq (0, 0, 0),$$

contradicting the uniqueness of prime factorisation. Equivalently: $\log 2$, $\log 3$, $\log 5$ are linearly independent over $\mathbb{Q}$. The orbit lies dense on the octave circle but never meets its starting point.

Near-closures in the 5-limit

[Q] What exists instead are near-closures whose deficit is in each case a named comma:

Path on the Tonnetz	Residual (comma)	Ratio	Cents
8 fifths + 1 maj. third vs. 5 octaves	**schisma**	$32805/32768$	≈ 1.95
4 fifths vs. third + 2 octaves	syntonic comma	$81/80$	≈ 21.51
12 fifths vs. 7 octaves	Pythagorean comma	$531441/524288$	≈ 23.46
3 maj. thirds vs. 1 octave	lesser diesis	$128/125$	≈ 41.06

The practically best closure in the 5-limit is the schisma (≈ 2 cents, below the hearing threshold); the best-known is the circle of fifths after 12 fifths with the Pythagorean comma as residual.

Near-closures in the 7-limit (script result)

[K] Adding the natural seventh $7/4$ as a third axis (lattice $\mathbb{Z}^3$ over $3, 5, 7$), the same impossibility argument applies. The systematic search (`euler_spirale_7limit.py`, exact fraction arithmetic, search space $|b| \leq 12$ fifths, $|c| \leq 6$ thirds, $|d| \leq 4$ sevenths; an assertion confirms along the way that no exact closure ever occurs) yields near-closures an order of magnitude better than anything in the 5-limit:

Cents	Comma	Path $3^b\,5^c\,7^d$	Name
0.325	$250047/250000$	$b = -6,\ c = 6,\ d = -3$	landscape comma
0.396	$4375/4374$	$b = -7,\ c = 4,\ d = 1$	**ragisma**
0.721	$2401/2400$	$b = -1,\ c = -2,\ d = 4$	breedsma
1.954	$32805/32768$	$b = -8,\ c = -1,\ d = 0$	schisma (5-limit, for comparison)
5.758	$5120/5103$	$b = -6,\ c = 1,\ d = -1$	hemifamity
7.712	$225/224$	$b = -2,\ c = -2,\ d = 1$	septimal kleisma

The ragisma is the simplest sub-cent closure (7 fifths down, 4 thirds and 1 seventh up miss the starting point by 0.4 cents); the coarser 7-limit counterpart of the syntonic comma is Archytas' comma $64/63$ (≈ 27 cents). More prime axes make the lattice denser – the spiral approaches itself faster, but never closes in principle.

How the spiral is closed after all: temperament

[B] A closure only arises when commas are *tempered out*, i.e. declared equal to 1. Twelve-tone equal temperament sets the fifth by decree to exactly $7/12$ of the octave – it *rationalises* the step size, and only thereby does the circle of fifths close. The closed circle is not a property of the pure intervals but an artefact of rationalisation. This sentence becomes load-bearing in Section G.6.

G.3 Control case II: the xi cycle closes exactly

[B] The ξ cycle stands on the other side of the same dichotomy. The step size is

$$100\,\xi = \frac{400}{30000} = \frac{1}{75} \qquad \text{– exactly rational,}$$

because $\xi = 4/30000$ is a rational stipulation (A020). A step of $1/75$ of a cycle closes after exactly 75 repetitions with zero residual; $K_{\text{frak}} = 1 - 1/75 = 74/75$ is the closed path minus one segment (A040, A050). This is the general torus dichotomy: rational winding number ⇒ closed orbit; irrational ⇒ dense, never-closing orbit.

	Musical spiral (5/7-limit)	ξ cycle (FFGFT)
Step size	$\log_2(3/2),\ \log_2(5/4), \ldots$ irrational	$1/75$ rational
Reason	primes mult. independent	ξ stipulated rational
Behaviour	dense, never meets itself	closes exactly after 75
“Near-closure”	commas (schisma, ragisma, ...)	none needed
Verification	numerical only (cents)	exact fraction arithmetic

The closure after 75 is thus no numerical accident but a direct consequence of the rationality of ξ – the same rationalisation that music must impose after the fact through temperament is stipulated in the framework from the outset.

G.4 The form question

[B] The two forms are identical to first order; their entire difference is the second binomial term:

$$(1-\xi)^{100} = 1 - 100\xi + \binom{100}{2}\xi^2 - \ldots, \qquad K^{\text{mul}} - K^{\text{add}} = 4950\,\xi^2 - O(\xi^3) = 8.76 \cdot 10^{-5}.$$

Numerically verified: the binomial term $4950\xi^2 = 8.8 \cdot 10^{-5}$ covers $99.6\,\%$ of the full distance; the remainder is higher-order. A corpus location can decide the form only if its own uncertainty is smaller than this distance – for power-law locations: smaller than the correspondingly amplified distance.

G.5 Three witnesses

Witness A: the α point (A130)

[K] A130 leads to the scale E_0 by two closely related routes: route 1 as the bare geometric mean $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV, route 2 as the same mean with the fractal correction, $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.397$ MeV. They differ only by K_{frak}. The measured α fixes this factor without using it:

$$K_{\text{mess}} = \xi m_e m_\mu \alpha^{-1} = 0.98650.$$

[K] Finding: the residual against the additive form is $1.7 \cdot 10^{-4}$, against the multiplicative $2.6 \cdot 10^{-4}$. Both lie far above the form distance $8.9 \cdot 10^{-5}$; the location has a residual of its own that is larger than the distance between the forms. Witness A confirms the *value* $74/75$ to $1.7 \cdot 10^{-4}$ but makes *no form statement*.

[K] Register status (R72): The register books the two E_0 values as *the same value bare and fractally corrected* (7.348 bare, 7.398 α-anchored; ratio $1/\sqrt{K_{\text{frak}}}$ by the A040 rule). A determination of the *exponent* follows from the ratio: $n = 101.2$, alongside $n = 100 \pm 2$ (R67) and $n = 100.27$ (sector ladder A270). In this n language the form question reads: additive $K = 1 - n\xi$ versus multiplicative $(1-\xi)^n$ differ by $\Delta n_{\text{eff}} = n(n-1)\xi/2 \approx 0.66$ – none of the three n determinations currently reaches this sharpness.

Witness B: the high-power location (A270)

[K] A270 carries the identity $K_{\text{frak}}^{-36} \approx 16/\pi^2$ with documented agreement $0.010\,\%$. The exponent 36 amplifies the form distance to $36 \cdot 8.9 \cdot 10^{-5} = 0.32\,\%$ – far above this tolerance. The location is therefore resolution-capable:

Expression	Value	rel. dev. from $16/\pi^2 = 1.6211389$
$(74/75)^{-36}$ (additive)	1.6213007	$+\mathbf{1.0 \cdot 10^{-4}}$
$\left((1-\xi)^{100}\right)^{-36}$ (multipl.)	1.6161261	$-3.1 \cdot 10^{-3}$

Verdict: 31:1 in favour of additive – conditional on the assumption that $16/\pi^2$ is the correct reference. In A270 this is carried as a coincidence without forward derivation (P35), but Monte-Carlo secured.

[K] Upgrade of the reference through Doc. 314: The constant is identifiable corpus-natively. $\pi^2/16$ is the *packing density of the D4 lattice* (Doc. 314: "kissing number 24 against 8, packing density $\pi^2/16$ against $\pi^2/32$") – hence

$$\frac{16}{\pi^2} = \frac{1}{\Delta(\text{D4})}.$$

The A270 identity thus reads $K_{\text{frak}}^{-36} = 1/\Delta(\text{D4})$: the reciprocal packing density of precisely the lattice whose structure Doc. 314 verifies in Hilbert space. The reference is therefore no longer a free-floating pretty number but a lattice invariant of the

framework; P35 narrows from “where does the constant come from?” to “why does the exponent 36 couple to the packing density”. The condition of witness B becomes considerably weaker thereby, but does not vanish.

Witness C: the power form (A040)

[K] The power form $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2}$ with $D_f^{\text{eff}} = 2.973$ yields 0.986651: residual additive $1.6 \cdot 10^{-5}$, multiplicative $1.0 \cdot 10^{-4}$ – tends additive. However, the four-digit rounding of D_f^{eff} ($\pm 2.5 \cdot 10^{-4}$ on K) exceeds the form distance: not an independent witness, but consistent with B.

G.6 Rolled up and unrolled: why the additive form has a reason

Section G.5 treats the form question as a measurement problem. This section treats it as a *structural problem*: the two forms are not two arbitrary computational ansätze but the exact bookkeepings of *two different representations of the same geometry*. The form question thereby becomes a precise physical question: In which domain is the correction exactly defined – rolled up or unrolled?

The two domains

[B] *Rolled up (T^4, winding picture).* On the compact cycle the natural quantity is the phase or arc length, and phases *add*. A correction that shortens each winding by the segment ε sums linearly over N windings. The closure condition from A040,

$$75\,(1-\varepsilon) = 74 \quad \Rightarrow \quad \varepsilon = \frac{1}{75} = 100\xi,$$

is *exact* in this domain – as a bookkeeping identity of the winding, not as an approximation: 75 shortened revolutions close precisely when one full revolution is missing in total. The additive form $K = 1 - 100\xi$ is therefore not a linearisation of anything; it is the native, closed bookkeeping of the rolled-up geometry – representable in exact fraction arithmetic because ξ is rational.

[B] *Unrolled (scale flow, measurement domain).* Unrolling the cycle in the scale direction turns the winding count into a run over orders of magnitude (the RG flow from A040, ~ 17 decades). In this representation corrections act stepwise on amplitudes, and stepwise amplitude corrections compose multiplicatively: $(1-\xi)^{100} \approx e^{-100\xi}$. Were the unrolled description fundamental, *this* form would be exact and the additive only its first order.

The form distance is the translation error

[B] By Section G.4 the entire form distance is the second-order binomial term, $4950\xi^2$. It is thus nothing other than the error of the translation between the domains at second order. The form question reads sharply: Which domain carries the exact

definition, which the approximation? Precisely this – and only this – is what the witnesses measure; currently only witness B resolves it.

The Euler parallel makes the assignment unambiguous

[B] The musical spiral provides the control case in which the domain assignment is *uncontested* – because there, measurement takes place. What ear and instrument reach are frequencies, i.e. quantities of the unrolled scale domain; intervals compose multiplicatively there ($3/2 \cdot 3/2 \cdot \ldots$), and only the logarithm rolls them onto the octave circle – whereby the irrationality enters and prevents closure in principle. The ξ cycle, conversely, is *defined* in the rolled-up domain: from the outset a circle structure with rational winding number, without two multiplicatively defined objects that would first have to be made comparable logarithmically.

	Music (Euler)	FFGFT (ξ cycle)
Measured quantity	frequency (unrolled)	— (defined rolled up)
Composition	intervals multiply	winding phases add
Bridge to the circle	logarithm (forced)	none needed
Step size	$\log_2(3/2)$ irrational	$1/75$ rational
Closure	never (commas only)	exact after 75

The assignment is complete: multiplicative composition belongs to the unrolled measurement/scale domain, additive composition to the rolled-up winding domain.

The corpus already contains the fork: case B against case C

[K] The assignment does not stand beside the corpus but within it. The closure fork (Docs. 295/313, cited in Doc. 314) carries three cases: case A ($d = 0$, closure after one revolution), case B (ξ frozen: rational rotation number $1 - d = 74/75$; because $\gcd(74, 75) = 1$, closure after exactly 75 revolutions) and case C (ξ runs via the recursion: "the cumulated defect becomes logarithmic and the winding an equiangular spiral" – no closure). This is precisely the form question as physics: case B *is* the additive bookkeeping (linear defect, rational winding, exact closure), case C *is* the multiplicative–logarithmic composition – and the equiangular spiral is the same mathematical picture as the Euler spiral of control case I. The two forms of K_{frak} are thus the bookkeepings of the two branches of the fork; the question "additive or multiplicative?" is the question of which branch carries the correction.

Doc. 314 (Ch. D2) additionally sharpens the domain boundary in the terminology used here: *in the rolled-up state there are no fractal defects; the defect* $d = 100\xi$ *accumulates only unrolled*, and K_{frak} is *a property of the traversal*, entering via the bridge constants (E_0, ν), not via the local operator. The additive form is accordingly the bookkeeping of the *closed* traversal with frozen ξ (case B); the multiplicative would be the bookkeeping of the *running* accumulation (case C). All corpus uses of K_{frak} (A040 rule, R70 assignment, R72) book case B.

What the witnesses then mean

[B] In this reading, witness B does not measure a computational detail but the domain membership of the correction – and it answers: the exact bookkeeping is the *rolled-up* one; witness C tends the same way. The multiplicative form is the approximation a naive observer of the unrolled scale narrative would write down – correct to first order, wrong from $4950\xi^2$ on.

This also resolves the apparent tension within A040: the document contains both – the closure comma (rolled-up, exact reading) and the cumulative RG flow (unrolled derivation of the factor 100). Both are consistent because they agree to first order; by the witness evidence, the rolled-up one is exact. The RG flow derives *why* the correction has magnitude 100ξ; the winding geometry fixes *how* it composes. The naive RG argument for the multiplicative form is thereby placed rather than refuted: it is the correct argument of the unrolled domain – and that domain, by the witness evidence, is not the defining one.

Connection to the foundational architecture

[B] The assignment "rolled up $=$ exact" is not an ad-hoc rescue of the additive form but congruent with the foundational architecture of the framework: the compact geometry (T^4, rational ξ stipulation) is primary; scale-flow narratives are derived descriptions. That the exact arithmetic of the corpus works throughout with fractions is the same statement in methodological form: exact fractions exist only in the rolled-up domain – the unrolled one knows only approximations with ξ^2 residuals. In this respect a consistently rolled-up framework *must* carry the additive form; were the measurement to come out multiplicative, not only K_{frak} would be affected but the primacy claim of the geometry itself.

G.7 Status and falsification line

Bookkeeping status

$K_{\text{frak}} = 1 - 100\xi = 74/75$: value **[K]** (A040). Form additive: **[B]**, once conditionally confirmed – the condition ($16/\pi^2$ reference, witness B) is substantially weakened by Doc. 314, since $16/\pi^2 = 1/\Delta(\text{D4})$ is identified corpus-natively and only the coupling to the exponent 36 remains open (P35 narrowed). Structurally the additive form is the bookkeeping of fork branch case B (frozen ξ, rational winding), which all corpus uses book; the multiplicative alternative $(1-\xi)^{100}$ is the bookkeeping of the spiral branch case C, is disfavoured by the conditional witness B and supported by no witness.

[B] *Why nevertheless not [K]-measured:* The only sharp discrimination (witness B) hangs on the status of $16/\pi^2$ as reference (P35, narrowed). If this premise later turns out correct, the form statement becomes unconditional. Conversely, the multiplicative form remains logically possible as long as it is open: then the $16/\pi^2$ proximity would

be a coincidence – and the naive RG argument of the unrolled domain would have been right.

[B] *Falsification line.* The question would be decidable unconditionally at a location where K stands at high power against a *measurement* rather than a coincidence constant. The candidate already stands in the corpus: the baryon prediction from A270 (sector exponent $+2$, i.e. K^{38} level, form distance there $\approx 0.34\,\%$). A baryon mass check of better accuracy would decide additive against multiplicative without conditions – and at the same time test the domain assignment of Section G.6: were it to come out multiplicative, not only the form choice would be corrected; the correction would live in the scale domain, against the winding reading of the closure comma. The structural argument thus raises the stakes; it binds the form question to the primacy question.

G.8 Open items

- **P-315-2:** The residual of the α point: α requires $K_{\text{frak}} = 0.98650$, i.e. $1.7 \cdot 10^{-4}$ from $74/75$ (in n language 101.2 instead of 100). A correction term of the theory or a pole-mass effect? Consistent with R72.
- **P35 (narrowed by Doc. 314):** With $16/\pi^2 = 1/\Delta(\text{D4})$ the constant is identified as the reciprocal D4 packing density; what remains open is the forward derivation of the coupling $K_{\text{frak}}^{-36} = 1/\Delta(\text{D4})$, in particular the role of the bulk exponent 36.
- **P-315-3 (n sharpness):** In n language, form discrimination requires $\Delta n \leq 0.66$ ($= n(n-1)\xi/2$). The three n routes from R72 (101.2; 100 ± 2; 100.27) do not reach this; the sector ladder (100.27) is the most promising candidate for sharpening.
- **Sharpening A040:** state D_f^{eff} with more digits (needed: better than $\pm 1 \cdot 10^{-4}$) so that witness C becomes resolution-capable.
- **Baryon location (A270):** mass check at the K^{38} level as unconditional form test – simultaneously a test of the fork assignment case B/C.

G.9 Verification

All numbers of this document are reproducible by the following scripts (pure standard library; exact fraction arithmetic where possible):

- `python/315_Skripte/euler_spirale_7limit.py` – comma search in the 5/7-limit, exact; assertion against exact closure.
- `python/315_Skripte/pruefrechnung_kfrak_form.py` – witnesses A/B/C with verdict logic (witness A via the K_{frak} required by α; uncertainty against the effective, possibly amplified form distance).

Inputs: $\xi = 4/30000$ exact; $m_e = 0.51099895069$ MeV, $m_\mu = 105.6583755$ MeV (CODATA 2022).

Chapter H

Doc. 316 — FFGFT and the Riemann Zeta Function

Preliminary: the answer in short

Result

The Riemann hypothesis is an **arithmetic** matter. The torus and ξ contribute nothing to its solution. The result of this document is a *negative* one – three obvious geometric approaches are excluded with reasons given.

The blanket formulation would be too coarse in one respect, hence the refinement.

What remains purely a matter of numbers. Where the zeros lie, how they stand relative to one another, how one grasps them individually – that is carried by arithmetic, by the primes. Three geometric approaches were tested and all are negative: no compact geometry can carry the zeros as its spectrum (Section H.3, at theorem level), ξ does not shift the critical line (Section H.2), and the ξ ladder discretises nothing (Section H.5, out-of-sample test negative).

What is nevertheless not merely a matter of numbers. Two things. First, the Riemann zeta is a *factor* in the spectrum of the $\mathbb{Z}^4$ torus – an identity, not a picture (Section H.2). The Riemann hypothesis is thereby a statement about a geometric object of the framework; only one that the geometry cannot answer. Second, the form of the zero density says *where* the question belongs: $N(T)$ grows like $T \ln T$, and no Laplacian on any compact manifold can deliver that – it requires a dilation structure, hence the unrolled domain. The geometry indicates the address without supplying the answer.

The striking thing is precisely that both occur together: the numbers demonstrably sit in the torus spectrum, but the torus cannot generate them.

The yield. The dead ends are closed, with reasons rather than conjecture – and four positive findings stand alongside them: two identities of the framework (the factorisation of the torus spectral zeta and the symmetry position of the Casimir interval) and two measurements on the prime structure (the bicoherence-measured cross-term situation and its physical evidence in the harmonic series).

H.1 Starting point and claim

[Q] The corpus mentions the Riemann hypothesis in exactly one place: Doc. 024 notes, in an AI/network context, that the "number space" has dimensions "modulated by the Riemann hypothesis and ζ functions". That is a conceptual marginal remark without an epistemic marker and without derivation. A worked-out connection does not exist in the corpus.

[K] At the same time Doc. 314 contains material that touches the ζ function directly: Epstein zeta, Casimir point, heat kernel on T^4. This document examines systematically what follows.

Claim. It proves nothing about the Riemann hypothesis and does not claim to. Three things are clarified: which connections hold as identities **[K]**, which path carries constructively **[B]**, and which obvious paths are *demonstrably* not passable **[X]**. The exclusion is the most valuable part – it saves work on dead ends.

H.2 Two findings that hold as identities

The Riemann zeta is a factor of the torus spectral zeta

[K] Doc. 314 carries the closed form of the $\mathbb{Z}^4$ spectral zeta (verified threefold there; confirmed here against the direct sum with analytic tail correction to 10^{-7} through 10^{-12}):

$$Z_{\mathbb{Z}^4}(s) \;=\; 8\left(1-4^{1-s}\right)\zeta(s)\,\zeta(s-1).$$

This is not an analogy but an identity. The zeros of the torus spectral zeta therefore have three sources:

Source	Location of the zeros
$\zeta(s)=0$	$\Re(s)=1/2$ (non-trivial – the RH)
$\zeta(s-1)=0$	$\Re(s)=3/2$ (shifted by 1)
$1-4^{1-s}=0$	$s=1+2\pi i k/\ln 4$ (lattice factor)

On this reading the Riemann hypothesis is a statement about the spectrum of the $\mathbb{Z}^4$ torus – about a geometric object of the framework, not merely about a number series.

The Casimir point lies symmetric to the critical line

[K] Doc. 314 computes the ζ-regularised vacuum energy $E=\pi\zeta_M(-1/2)$ and finds the order reversal: D4 minimises the Epstein zeta for $s>0$, at $s=0$ all lattices are equal, and at $s=-1/2$ the $\mathbb{Z}^4$ torus lies lower (-0.9321 against -0.8686).

The Casimir point $s=-1/2$ has, under the functional equation $\zeta(s)=\chi(s)\zeta(1-s)$, the mirror point $s=3/2$. The axis of symmetry:

$$\frac{-1/2+3/2}{2} \;=\; \frac{1}{2} \qquad \text{– exactly the critical line.}$$

The physically occupied interval $[-1/2,\, 3/2]$ has the critical line as its midpoint: consistency between physics and arithmetic.

[X] No proof follows from this – the critical line is fixed by the functional equation anyway. Nor can ξ shift it: a perturbation of order $\xi \sim 10^{-4}$ (even with the RG amplifier 100ξ) does not reach an axis fixed by a functional identity.

H.3 The decisive obstruction: Weyl against Riemann-von Mangoldt

The obvious move would be to seek an operator whose eigenvalues are the zeros (Hilbert-Pólya programme), and to deform $\mathrm{T}^4/\mathbb{Z}_3$ suitably for that purpose. **This entire branch is excluded – at theorem level, not as intuition.**

[Q] *Weyl's law.* For *every* compact Riemannian manifold of any dimension d, asymptotically $N(\lambda) \sim C_d \,\mathrm{Vol}\, \lambda^{d/2}$ – a pure power. Curvature, boundary and orbifold points change only lower orders, not the leading term.

[Q] *Riemann-von Mangoldt (proved).*

$$N(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O(1/T).$$

[K] The comparison via the local exponent $\mathrm{d}\ln N/\mathrm{d}\ln T$:

T	$N(T)$	local exponent $d/2$
10^3	$6.49 \cdot 10^2$	1.2457
10^5	$1.38 \cdot 10^5$	1.1153
10^8	$2.48 \cdot 10^8$	1.0642
10^{12}	$3.95 \cdot 10^{12}$	1.0403
10^{20}	—	1.0231

The exponent is $1 + 1/(\ln(T/2\pi) - 1)$ – it falls monotonically towards 1 but is *never constant*. Range over 17 decades: 0.22. The best power fit in the window $[10^3, 10^6]$ deviates systematically by up to $11\,\%$.

Exclusion at theorem level

[X] There is no compact Riemannian manifold – of whatever dimension, curvature, topology or orbifold structure – whose Laplace spectrum is the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

Consequence. The branch "deform $\mathrm{T}^4/\mathbb{Z}_3$ suitably" is settled completely, not only in the flat case. The extra $\ln T$ is the signature of a **scale degree of freedom**: a dilation direction with logarithmically growing volume. That is not a Laplacian on a manifold.

Not a curvature problem. What generates logarithmic lengths is a dilation structure – the multiplicative group $\mathbb{R}_+^*$ – not curvature; the Berry-Keating operator $H = xp$ is the generator of dilation, not of a hyperbolic flow. The Weyl obstruction therefore applies independently of any metric and renders the curvature question moot.

H.4 The constructive link – and why it does not carry

If the zero density requires a scale degree of freedom, the responsible domain is the **unrolled** one (Doc. 315) – multiplicative, logarithmic, with dilation as its natural operation.

The approach

[Q] *Berry-Keating (1999)* [1]. The semiclassical count of the dilation generator $H = (xp + px)/2$ in a phase space with a minimal cell:

$$N(E) = \frac{1}{2\pi}\big[\, E\ln\big(E/(l_x l_p)\big) - E + l_x l_p \,\big].$$

With $l_x l_p = 2\pi$ this becomes exactly the Riemann-von Mangoldt formula (verified: difference constant $0.125 = 1 - 7/8$, a pure boundary term, over $T = 15 \ldots 5000$).

The obvious chain would then be: Doc. 314 Ch. D1 carries $\lambda_4 \cdot m = 2\pi$ as an exact position relation; in the unrolled domain the quantity conjugate to scale is the dilation momentum; hence FFGFT supplies the cell condition.

Why the chain does not carry

[X] Stage 1 – the cell condition is empty of content. $l_x l_p = 2\pi\hbar$ is, in natural units, the Planck cell, and follows from the canonical quantisation of *every* conjugate pair. That $\lambda_4 m = 2\pi$ has the same form is no independent finding – the de Broglie relation *is* the statement $\hbar = 1$. Moreover the count is logarithmically insensitive to the cell (a factor 2 wrong $\Rightarrow$ N changes by $\sim 1\,\%$); a "hit" would not have been a test in any case.

[X] Stage 2 – the actual content fails quantitatively. Berry-Keating requires something sharper than the cell: two *independent* cutoffs $x > l_x$ and $p > l_p$ whose product is the Planck cell. The FFGFT double cutoff makes this testable:

Quantity	Value
UV: $L_0 = \xi\,\ell_P$	$2.155 \cdot 10^{-39}$ m
IR: $L^* = 4\bar{\lambda}_e/\xi^{10}$	$8.698 \cdot 10^{26}$ m
Ratio L^*/L_0	$4.04 \cdot 10^{65}$
required	$2\pi \approx 6.28$
Shortfall	**65 decades**

[X] Stage 3 – a finite window against infinitely many zeros. $\ln(L^*/L_0) = 151.06$, that is $16.93\ \xi$ ladder rungs; the Berry-Keating count within the window gives ~ 1816 modes. A finite scale window carries finitely many modes; the zeros are countably infinite. Even if stages 1 and 2 held, at most an initial segment would result – and an initial segment is not a spectral identity.

Marginal observation, explicitly not booked: 16.93 lies close to 17. Both cutoffs carry their own uncertainties – the prefactor 4 is open per R73, ℓ_P depends on G. Without an error budget this is not assessable under P35.

What the chain achieves

[B] What does carry is a structural statement, and it is not little: that the counting function has the form $(T/2\pi)\ln(T/2\pi)$ at all requires a **dilation generator**. A Laplacian on a compact manifold cannot deliver it (Section H.3). The substantive statement is the *identification of the structure* – the unrolled domain carries the dilation – not the numerical value 2π.

H.5 The missing boundary condition – test negative

$H = xp$ has *continuous* spectrum. The counting function fixes the mean density, not the individual zeros. What is missing is the boundary condition that turns the continuum into a discrete spectrum.

The natural candidate. The ξ ladder generates a discrete subgroup $\Gamma_\xi = \{\xi^n\} < \mathbb{R}_+^*$; the quotient $\mathbb{R}_+^*/\Gamma_\xi$ is a *circle* of circumference $L = \ln(1/\xi) = 8.9227$. Admissible dilation frequencies would then be multiples of $2\pi/L = 0.7042$.

Test discipline

[K] Proximity arguments require three controls (P35): a null distribution over random periods in the *same* range; artefact control, because periods larger than the data range automatically produce small scores; and an out-of-sample check with look-elsewhere correction.

The artefact control is mandatory: periods above the data range automatically produce low scores, because all quotients round to the same integer. At period $2\pi/|\ln(74/75)| = 468$, for instance, all zeros lie below 102; a score near zero means nothing there. The null distribution must cover the same range of periods as the candidates, or spurious signals arise.

Result

[K] Candidates within the admissible range (null distribution: mean 0.2502):

Candidate	Period	Score	p value
$2\pi/\ln(1/\xi)$	0.7042	0.2812	0.92
$2\pi/\ln 75$	1.4553	0.2043	**0.005**
$\ln(1/\xi)$	8.9227	0.2891	0.94
2π	6.2832	0.2366	0.18

The borderline candidate $2\pi/\ln 75$ was checked out of sample (zeros 31–50): score 0.2686, $p = 0.81$; look-elsewhere corrected $p = 1.0$.

Boundary condition

[X] The ξ ladder does not supply the missing boundary condition. No ξ-derived period candidate survives the out-of-sample check.

Why this was structurally to be expected

[K] A single-period compactification produces an *equidistant* lattice. The zeros, however, show GUE statistics with level repulsion (Montgomery-Odlyzko [11, 10]): mean normalised spacing 0.9815 (should be ~ 1), fraction of spacings < 0.3 equal to 0.000 – against ~ 0.04 for GUE and ~ 0.26 for a lattice-like spectrum. A circle cannot deliver GUE in principle.

The missing boundary condition is carried by arithmetic (the primes), not by geometry.

H.6 The harmonic approach: ξ as a Tonnetz point

The preceding sections treat ξ as a *magnitude* – a parameter whose smallness decides its effect. A second approach asks instead about its *harmonic structure*, that is, about the position of ξ in the prime lattice. It leads to a different point of contact with the theory and yields a sharper exclusion.

The starting observation

[K] The prime factorisation

$$\frac{1}{\xi} \;=\; 7500 \;=\; 2^2 \cdot 3 \cdot 5^4$$

places ξ as a point in Euler's Tonnetz with coordinates $(a_2, a_3, a_5) = (-2, -1, -4)$. This is the same lattice structure as in the control case of the musical spiral: the primes 2, 3, 5 as axes of the 5-limit space. Equivalently in logarithmic form:

$$\ln(1/\xi) \;=\; 2\ln 2 + \ln 3 + 4\ln 5 \;=\; 8.922658\ldots$$

The harmonic reading suggests a different point of contact than the Laplace operator. A Tonnetz carries three incommensurable logarithmic frequencies $\ln 2$, $\ln 3$, $\ln 5$; three independent frequencies generate a quasi-periodic, precisely not equidistant spectrum. The approach therefore touches the arithmetic of the ζ function directly.

The responsible point of contact: Weil's formula

[Q] Weil's explicit formula couples zeros and primes:

$$\sum_{\rho} h(\gamma) \;=\; (\text{archimedean term}) \;-\; 2\sum_{p,k} \frac{\ln p}{p^{k/2}}\, \hat{h}(k \ln p).$$

The dual *length spectrum* of the zeros is therefore

$$\mathscr{L} \;=\; \{\, k \ln p \;:\; p \text{ prime},\; k = 1, 2, 3, \dots \},$$

that is, multiples of logarithms of **single** primes. Whether ξ appears there is directly testable: the Fourier transform of the zero density must show peaks at the lengths in $\mathscr{L}$.

Measurement

[K] Tested is $F(u) = \left|\sum_n e^{i\gamma_n u}\right|/N$ for the first 100 zeros; the background is 3000 random u in the same range (median 0.0304, 95 % threshold 0.2005, 99 % threshold 0.2572).

u	$F(u)$	Assignment	
1.09861	0.2321	$\ln 3$	(peak, 95 %)
1.60944	0.2651	$\ln 5$	(peak, 99 %)
1.94591	0.2625	$\ln 7$	(peak, 99 %)
2.39790	0.2610	$\ln 11$	(peak, 99 %)
2.56495	0.2676	$\ln 13$	(peak, 99 %)

The primes are visible; the method works. The same measurement for ξ-derived lengths:

u	$F(u)$	Quantity
8.92266	0.1775	$\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5$
4.46133	0.0777	$\ln(1/\xi)/2$
4.31749	0.0384	$\ln 75 = \ln 3 + 2\ln 5$
2.23066	0.0189	$\ln(7500)/4$
0.70418	0.1313	$2\pi/\ln(1/\xi)$

None reaches the 95 % threshold.

Resolution limit. The measurement becomes assessable only once u spreads the phases over the full circle, that is $u \gg 2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.02824$. For $u \to 0$ all phases coincide and $F(u) \to 1$, independently of any structure. Candidates below this limit – for instance $\ln(75/74) = 0.01342$ – are *not measurable* with 100 zeros; they yield neither confirmation nor refutation.

The structural reason

Combination against factorisation

Tonnetz and Euler product share the prime axes but use them in **opposite** ways. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ logarithmises to a sum over prime powers: each prime stands on its own, without cross terms – hence $\mathscr{L}$ contains only $k \ln p$. The Tonnetz lives on the opposite: an interval is $2^a 3^b 5^c$ with several

non-zero exponents, and it is precisely the mixing that generates the commas. ξ is such a mixed point.

[X] $\ln(1/\xi)$ is therefore a *sum* of lengths and not a length; sums do not appear as peaks in the trace formula. The nearest genuine length is $3\ln 19 = 8.83332$ at distance 0.08934 – without significance, since proximity within the dense length spectrum says nothing.

Sharpness of the exclusion. It is structural, not a matter of magnitude: it holds unchanged if ξ were of order 1. What matters is not the value of ξ but that $1/\xi$ is composite. This section therefore carries a stronger reason than any estimate based on the smallness of ξ.

Remaining test surface. The *individual* axes $\ln 2$, $\ln 3$, $\ln 5$ are lengths in the Weil spectrum – but they are so for any theory containing primes at all; the factorisation of 7500 selects nothing. A statement would arise only if the *exponents* $(2, 1, 4)$ produced a weighting within the spectrum. The following subsection tests this.

Does the 5-limit include the higher ones?

The limit extension of Test B raises the converse question: can it be shown that the 5-limit already contains the higher limits? The answer comes in five gradations, and the gradation repeats the pattern of the whole document (`z9_limit_einschluss.py`).

[K] (1) Exactly: no, provably. $\ln 7 = a\ln 2 + b\ln 3 + c\ln 5$ with rational a, b, c would mean, after clearing denominators, $7^q = 2^A 3^B 5^C$ – contradicting unique factorisation. This holds for every prime above the limit, hence for infinitely many.

[K] (2) As a dense approximation: yes, arbitrarily well. Since $\ln 2, \ln 3, \ln 5$ are linearly independent over $\mathbb{Q}$, the lattice lies dense in $\mathbb{R}$:

N	Error	Cents	Exponents	Tenney
2	$2.82 \cdot 10^{-2}$	48.77	$(2, 2, -1)$	$5.6 \cdot 10^{-3}$
4	$7.97 \cdot 10^{-3}$	13.80	$(-1, -2, 3)$	$4.4 \cdot 10^{-4}$
6	$4.45 \cdot 10^{-3}$	7.71	$(-5, 2, 2)$	$1.4 \cdot 10^{-4}$
8	$2.29 \cdot 10^{-4}$	0.40	$(1, 7, -4)$	$3.7 \cdot 10^{-7}$

The error of the best approximation $(1, 7, -4)$ is exactly $\ln(4375/4374)$ – **the ragisma**. The coarse approximation $(6, -2, 0) = \ln(64/9)$ has error $\ln(64/63)$, **Archytas' comma**. Both are known from the Euler control case: there as near-closures in the 7-limit, here as the error of the 5-limit approximation to the 7-axis. It is the same quantity read from the opposite direction.

[K] (3) At finite resolution: yes, indistinguishable. The peak width of the measurement in Section H.6 is $2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.0021$; the ragisma, at $2.29 \cdot 10^{-4}$, is only 0.11 peak widths:

Location	u	Power
$\ln 7$ exact	1.945910	2.144
approximation $(1, 7, -4)$	1.945682	2.144
coarse $(6, -2, 0) = \ln(64/9)$	1.961659	1.417
control $\ln 7 + 0.05$	1.995910	0.042

The fine approximation is indistinguishable from $\ln 7$, the coarse one falls off, and the control point lies in the noise. At this resolution the 5-limit does effectively contain the 7-axis – a statement about the measuring procedure, not about the arithmetic.

[K] (4) Amplitude-weighted: no, and clearly so. The Tenney weight of the approximation $(1, 7, -4)$ is $3.7 \cdot 10^{-7}$ against $1/7$ for $\ln 7$ as an axis of its own – a factor $3.9 \cdot 10^{5}$. This also explains *why* the approximation can be so accurate: it requires large exponents, and large exponents mean small weight. Accuracy and weight run opposite to one another.

[B] (5) By tempering: yes, but by stipulation. Declaring the ragisma equal to 1 makes $7 = 2 \cdot 3^7/5^4$ exact in the tempered system; likewise, tempering out Archytas' comma makes the approximation $64/9$ exact.

Inclusion does not arise from structure

The 5-limit does not include the higher limits – unless approximation, limited resolution or tempering is allowed to count as inclusion. This is the same pattern as everywhere in this document: inclusion arises through *rationalisation* (tempering) or through *blur* (resolution), never through structure. Likewise Euler's spiral closes only by tempering, and likewise coupling in the Weil spectrum sits only at exact prime powers (Section H.6).

For ξ nothing new follows, but a confirmation: that ξ is a pure 5-limit point cannot be cured by any extension – and the approximations that might replace an extension carry no weight.

Measurement instead of assertion: the bicoherence test

The statement that the Euler product has no cross terms was up to this point argued from the *form* of the product. It can be measured – with the procedure responsible for it and with built-in controls.

Setup. The sequence of zeros is taken as a point measure; its transform in the length variable is $F(u) = \sum_n e^{i\gamma_n u}$. Averaged over height blocks, bispectrum and bicoherence [6] follow as in Section H.10. The zeros are computed here via the Riemann-Siegel formula: 2469 of them up to $t = 3000$ (Riemann-von Mangoldt expects 2468.6), residual error $7.5 \cdot 10^{-3}$ in γ, corresponding to 0.02 rad of phase error at $u \approx 2.6$.

Test plan. The Weil spectrum contains $k \ln p$. This yields predictions with positive *and* negative controls: if the sum of two lengths is again a length – that is, a prime power – coupling must appear; if it leads to a composite number, none may appear.

Pre-check. The power of $F(u)$ shows the individual lengths $\ln 2 \dots \ln 13$ all above the 99 % threshold. The background must be cleaned of the 13 genuine lengths in

the window – random u hit one of them with about 2 % probability and otherwise form the outlier tail (median 0.055 against a $99\,\%$ threshold of 0.881 after cleaning).

Result (24 height blocks of 102 zeros each; critical value from 2000 phase-randomised realisations per pair):

Pair	Sum	b^2	$b_c(99\%)$	Verdict	expected
$(\ln 2, \ln 2)$	$\ln 4 = 2^2$	**0.8088**	0.2612	coupling	coupling
$(\ln 3, \ln 3)$	$\ln 9 = 3^2$	**0.8462**	0.2214	coupling	coupling
$(\ln 2, 2\ln 2)$	$\ln 8 = 2^3$	**0.7586**	0.2432	coupling	coupling
$(\ln 2, \ln 3)$	$\ln 6$ comp.	0.0253	0.1832	none	none
$(\ln 2, \ln 5)$	$\ln 10$ comp.	0.0252	0.1912	none	none
$(\ln 3, \ln 5)$	$\ln 15$ comp.	0.0175	0.1718	none	none

Cross terms: measured

Six of six predictions met, with a contrast of a factor 30 to 50 between prime powers and composite numbers. Coupling occurs exactly where the sum of two lengths is again a length, and nowhere else. **[K]** The cross-term statement is thereby measured, not merely argued structurally.

Consequence for ξ. $\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5$ is a sum over *three different* primes and leads to the composite number 7500. By the pattern measured here, no coupling can sit there – which the independent Fourier finding of Section H.6 ($F = 0.1775$, below every threshold) confirms. Open item Z-4 is thereby closed, and the exclusion of Section H.6 now rests on a measurement rather than on an argument from form.

Can a weighting exist? Three tests

The remaining test surface read: a statement would arise only if the exponents $(2, 1, 4)$ produced a weighting in the Weil spectrum. Three independent tests decide this question negatively (`z7_gewichtung_unmoeglich.py`).

[K] Test A – weighting strength. Higher harmonics have smaller amplitude; the standard measure is Tenney height, weight $\sim 1/(nd)$. An exponent makes the strength free:

$$w = \left(2^{|a|}\, 3^{|b|}\, 5^{|c|}\right)^{-\alpha},$$

with $\alpha = 0$ unweighted and $\alpha = 1$ Tenney. The weight sum has the closed form $\prod_p [1 + 2p^{-\alpha}/(1 - p^{-\alpha})]$ and converges for every $\alpha > 0$:

α	Weight sum	eff. point count	Character
0.10	4,284	23,843	blindly dense
0.30	245.8	1,767	middle
0.50	56.9	360	middle
1.00	9.0	36	axes only

Both extremes are unsuitable: unweighted, the lattice hits everything and selects nothing; at Tenney weighting only the axes remain. What matters is the middle around $\alpha \approx 0.3$ with several hundred to a thousand carrying points. The selection test across the whole range:

α	M(Weil)	M(random)	Ratio	p
0.15	2.1205	2.0943	1.013	0.27
0.30	0.5344	0.5210	1.026	0.31
0.50	0.1414	0.1493	0.947	0.64
1.00	0.0114	0.0226	0.503	0.93

No α produces selection; the ratio stays at or below one. The weighting strength is not the problem.

[K] Test B – limit extension. The 5-limit is the simple form of the Tonnetz; extended Tonnetze take in further prime axes, 7 being the next. Tested at $\alpha = 0.3$; primes up to the respective limit are axes and hence excluded as trivial.

Limit	Axes	Points	Ratio	p
5	3	2,197	1.049	0.28
7	4	28,561	1.015	0.38
11	5	371,293	1.048	0.11

Weightier than the p values is the structural point: every extension merely shifts the boundary, it closes no gap.

Limit	trivially covered	entirely absent
5	$2, 3, 5$	all $p \geq 7$
7	$2, 3, 5, 7$	all $p \geq 11$
13	$2 \ldots 13$	all $p \geq 17$

A Tonnetz with limit L covers the finitely many primes $p \leq L$ by definition and the infinitely many $p > L$ not at all – there is nothing in between. Weil's formula, however, sums over all primes. The limiting case settles it: a Tonnetz with infinite limit would no longer be a structure; it would contain every prime as an axis, that is, the Euler product itself – without additional content and without any relation to ξ.

In addition, $\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$ has exponent zero on all axes from 7 onwards. An extension enlarges the lattice but does not change the position of ξ within it; ξ remains a pure 5-limit point.

[K] Test C – rigidity. This is the actual reason, and it holds independently of any numerical computation. The weights of the explicit formula are not free parameters but a consequence of the Euler product:

$$\log \zeta(s) = \sum_{p,k} \frac{1}{k}\, p^{-ks},$$

from which $\ln p/p^{k/2}$ follows by differentiation and shifting to $s = 1/2$. There is no place at which an additional factor could be inserted without altering the Euler product itself. Setting $\prod_p (1 - g\,p^{-s})^{-1}$ with $g \neq 1$ by way of trial produces a different Dirichlet series with a shifted abscissa of convergence and *without* a functional equation.

Why no weighting is possible

The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is precisely what defines the critical line. A weighting meant to explain the location of the zeros thereby destroys the object whose zeros it seeks to explain. **[X]**

Test C carries on its own; A and B show in addition that the numerical routes yield nothing either – neither through the choice of weighting nor through extended Tonnetze. Open item Z-1 is therefore not unanswered but **decided in the negative**.

H.7 Balance

Statement	Status
ζ is a factor of the torus spectral zeta	**[K]** verified
Casimir interval has critical line as axis	**[K]**
Zero density requires dilation generator	**[B]** structure
Cell condition as an independent finding	**[X]** only $\hbar = 1$
Double cutoff as BK truncation	**[X]** 65 decades
Compact manifold with zero spectrum	**[X]** Weyl
ξ ladder discretises the zeros	**[X]** test negative
$\ln(1/\xi)$ in the Weil length spectrum	**[X]** a sum, not a length
Cross terms in the Euler product	**[K]** measured by bicoherence
5-limit contains higher limits	**[X]** only approximated/tempered
K_{frak} as tempering instrument	**[X]** no comma in the residue
Harmonic series: primes as generators	**[K]** physical evidence
Weighting by the exponents $(2, 1, 4)$	**[X]** Z-1 decided
ξ shifts the critical line	**[X]** functionally fixed
Relaxation circumvents the limit	**[X]** potential must carry it
FFGFT solves the Riemann hypothesis	**[X]**

The most honest sentence. Four statements hold as **[K]**, and they fall into two pairs. Two are *identities* of the framework: the factorisation of the torus spectral zeta and the symmetry position of the Casimir interval. Two are *measurements* on the prime structure: the bicoherence-measured cross-term situation and its physical evidence in the harmonic series. None of the four says anything about the *location* of the zeros; they say how the primes compose and where the zeta stands in the spectrum of the framework.

The constructive link, by contrast, reduces to $\hbar = 1$ and carries nothing specific. What remains there is a structural statement: the zero density requires a dilation generator, and that lives in the unrolled domain.

What a viable approach would have to achieve: a non-trivial boundary condition on the unrolled scale direction that (a) carries infinitely many modes – hence is not a finite cutoff window – and (b) produces GUE statistics rather than equidistance. Both requirements point in the same direction: the structure must be arithmetic, not geometric.

H.8 The harmonic series as physical evidence

The control case of the musical spiral has so far been treated purely arithmetically. It has a physical counterpart that confirms the measured structure independently: the natural harmonic series of a valveless wind instrument. An alphorn has neither valves nor finger holes; it plays the series $n\,f_0$ unaltered, as a direct image of the natural numbers (`z11_alphorn.py`).

Primes as generators

[K] Among the natural tones, exactly the prime orders introduce tonal qualities that cannot be composed from lower ones. The composite orders are combinations of what is already present: $4 = 2 \cdot 2$ is the octave of the octave, $6 = 2 \cdot 3$ the octave of the fifth, $9 = 3 \cdot 3$ the fifth of the fifth.

n	octave-reduced	Cents	Role
2	1.00000	0.0	prime – new axis
3	1.50000	702.0	prime – new axis
4	1.00000	0.0	$2 \cdot 2$ – combination
5	1.25000	386.3	prime – new axis
6	1.50000	702.0	$2 \cdot 3$ – combination
7	1.75000	968.8	prime – new axis
9	1.12500	203.9	$3 \cdot 3$ – combination
11	1.37500	551.3	prime – new axis

This is the Euler product structure, made audible: each prime stands on its own, the composites are products. **The bicoherence test measures the same structure** (Section H.6) in the Weil spectrum – there coupling only at prime powers, here new tonal quality only at primes. The acoustic and the number-theoretic finding are independent of one another and agree.

Amplitude decay as an acoustic Tenney analogue

[K] High natural tones are harder to sound and carry less amplitude – with rising order the moved air mass decreases. Formally this corresponds to the Tenney height of Section H.6, which weights a ratio by $1/(nd)$: both scales fall like $1/n$. The physical reason (decreasing air mass) and the number-theoretic one (increasing harmonic distance) lead to the same form. The high primes colour the timbre but do not carry the foundation – exactly as the heavy points of the weighted Tonnetz lie on the axes while the distant combinations carry no weight.

The same structure in the mass sector

The harmonic series indexes its modes by integers n; the mass formula of the corpus does the same. Doc. 006 sets

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i), \qquad E_{\text{char},i} = \frac{1}{\xi_i},$$

with quantum numbers n_i, l_i, j_i from the spatial wave equation and rational factors. Here $f(n,l,j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$; elsewhere in this document f stands for a frequency. There the particles are explicitly conceived as *discrete resonance modes* – the same language as for the natural tones, and the same indexing by integers.

[K] The factors can therefore be read like Tonnetz coordinates. For the leptons, by way of example:

Particle	r_i	Factorisation	Limit
Electron	$4/3$	$2^2/3$	3
Muon	$16/5$	$2^4/5$	5
Tau	$25/9$	$5^2/3^2$	5

The lepton generation thus lies in the same 5-limit lattice as ξ itself ($\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$). In the quark sector higher primes occur occasionally; whether these are genuine axes or artefacts of the factor fitting is a question of the mass sector and is to be treated there.

[B] Reference. The mass formula and the assignment of quantum numbers are given in Doc. 006 [13]. The factors r_i are fitted to the measured values there and remain stipulations; their derivation from an (n,l,j) formula does not hold, and the precision tables are therefore not evidence (R138). The present document refers to it only to record the structural parallel: integer indexing, rational factors and an octave respectively generation hierarchy are the same form in acoustics and in the mass sector. No statement about the Riemann zeta follows from this – here the primes appear as *building blocks* of ratios, not as lengths of a spectrum.

Winding number, not frequency ratio

[K] For orbits on the torus what counts is the *winding number* – the fraction of the cycle, hence the logarithm to the octave base – not the frequency ratio itself. The two must be distinguished:

Fifth	Ratio	Kind	Winding number	Kind
pure $3/2$	1.5000000	rational	0.5849625	**irrational**
tempered	1.4983071	irrational	0.5833333	**rational** $= 7/12$

The check decides: twelve pure fifths, octave-reduced, end at 1.013643 – the residue is the Pythagorean comma (23.46 cents). Twelve tempered fifths end at exactly 1.000000000000.

The assignment

Pure natural tones have irrational winding and a dense, open orbit. Tempered tones have rational winding and a closed orbit. The closed circle of fifths is an artefact of tempering; in pure intonation it is a spiral. This is the same dichotomy as in Section H.2: rational winding ⇒ closure.

Cent bookkeeping

[B] The theoretical deviation of a natural tone from the tempered scale and the instrument-induced shift are two different quantities; only their sum is the net deviation. The separation is necessary because the two are easily conflated:

n	Ratio	theor. dev.	Instrument	Net
4	1.00000	0.0	0.0	0.0
7	1.75000	−31.2	−12.5	−43.7
11	1.37500	−48.7	+20.0	−28.7

The 7th natural tone already deviates by -31.2 cents from the tempered seventh; the bell geometry pulls it further down. The 11th lies -48.7 cents below the tritone in theory but is raised again by the cone and the player's embouchure. The instrument values are order-of-magnitude figures from instrument acoustics, given here only to illustrate the bookkeeping.

H.9 Averaging, tempering, resolution: where does the instrument sit?

Euler's spiral closes only by tempering (Section H.2), and the 5-limit contains the higher limits only by approximation, resolution or tempering (Section H.6). In both cases closure arises not from the structure but from something added from outside. It is worth determining what that addition is – for in three domains it sits in three different places.

Three domains, three locations

For hearing, the averaging sits in the *receiver*: critical bands, finite frequency resolution, fusion of closely spaced tones. The measuring instrument becomes part of the result. Tempering here is not an arbitrary stipulation but follows from a property of the listener.

In pure mathematics there is no receiver. The ratios exist regardless of whether anyone distinguishes them. If something is to close that does not close of itself, tempering must be inserted as a *stipulation* – a deliberate intervention, documented in the comma that thereby disappears.

In nature it is open whether there is a receiver. The question therefore arises whether the fractal correction plays that role.

The testable form

[B] A tempering always leaves a residue, and that residue is a *comma*: a ratio of smooth numbers, that is, of products of small primes. The corpus knows one residue – the residue $1.7 \cdot 10^{-4}$ between the value required by α via the two E_0 routes (0.98650) and the stipulated $K_{\text{frak}} = 74/75$ (P-315-2), about 7700 times the relative uncertainty of $m_e m_\mu/\alpha$. The question is thus testable: does this residue have comma structure?

[K] It does not. The best rational approximations to $1 + 1.68 \cdot 10^{-4}$ are $5941/5940$, $11881/11879$ and $17822/17819$; none is 5- or 7-limit smooth. The control probe shows the test's discriminating power: all six commas examined – Pythagorean, syntonic, schisma, breedsma, ragisma, landscape – are recognised as 5- or 7-limit smooth.

On test conduct: the smoothness test must be coupled to a condition on approximation quality. Otherwise the trivial approximation $1/1$ counts as smooth although it misses the target value by 100 %.

Where the fractal correction stands

K_{frak} is not the tempering instrument

If K_{frak} were a tempering, it would have to leave a comma-like residue. The residue present does not have that structure. **[X]** The residue is not the by-product of a rationalisation; the two routes match the comma value $100\,\xi$ to within about 1 % (α requires $n = 101.2$).

[B] A further clarification is easily overlooked: in FFGFT the stipulation sits not at K_{frak} but one level below – at $\xi = 4/30000$ itself. That is the rational intervention; $K_{\text{frak}} = 74/75$ is its *consequence*, not the act. In this respect the fractal correction is not a tempering but the bookkeeping of an already closed cycle.

[B] Nevertheless it sits structurally where an instrument would sit: Doc. 314 Ch. D2 carries K_{frak} as a *property of the traversal*, entering via the bridge constants rather than the local operator. That is literally "the measuring instrument becomes part of the result": K_{frak} lies not in the field but in the traversing. It resembles a measurement effect in *location*, not in *character* – it is exact, not averaging.

The actual candidate: cutting off rather than averaging

[B] If nature has an instrument, it is more likely the double cutoff. It generates a resolution limit, and precisely there the procedure becomes part of the result – without anything being tempered. Section H.6 shows it in a concrete case: at a peak width of 0.0021 the 5-limit approximation to $\ln 7$ is no longer distinguishable from the exact value, although the ragisma between them remains *present*.

The difference in one sentence

A tempering *alters* the values and leaves a comma. A resolution limit alters nothing and leaves nothing – it merely makes differences invisible. Both make distinctions disappear; only the first produces residues.

Hence the tripartition. Hearing averages *actively* and produces commas. Mathematics has no receiver; there tempering must be done by stipulation. Nature, if the framework is right, cuts off *passively*: distinctions vanish without rationalisation – and the fractal correction is not involved in this; the scale limit is.

[B] *Status.* The comma test is an exclusion and rules out one particular reading of K_{frak}. It does not show that no averaging mechanism acts in nature at all – only that the fractal correction is not one. Whether the double cutoff is more than an analogy as an instrument remains open; it depends in turn on quantities the corpus carries as open (the prefactor 4 in L^*, R73).

H.10 Sources and methodological placement

The procedures used in this document are standard, and so are their pitfalls. This section places the usable literature and states where the reach of the procedures employed is bounded.

Load-bearing sources

Berry & Keating (1999) [1]. The basis of Section H.4. Comparison of the counting functions suggests that the t_n are eigenvalues of an unknown Hermitian operator obtained by quantising a classical system with Hamiltonian $H_{cl} = xp$. The work also states the requirement on such a system: the "Riemann dynamics" would have to be chaotic and possess periodic orbits *whose periods are multiples of logarithms of the primes*. That is precisely the length spectrum of Section H.6 – and the condition on which the Tonnetz fails, since it yields sums rather than multiples.

Sierra (2019) [2] **and Sierra & Townsend (2008)** [3]. These show the state of the Hilbert-Pólya programme: spectral realisations via xp with boundary interaction, via Landau levels, and most recently via a massless Dirac fermion in Rindler spacetime with delta potentials on the square-free integers. Notable for the present question: all these constructions encode the arithmetic *explicitly* (square-free numbers, modular forms, boundary functions from the Riemann-Siegel formula). None derives it from a geometry. This supports the finding of Section H.3 from the other side: where the zeros are realised spectrally, it happens by building in number theory, not by choosing a space.

Pápics (2012) [4]. The work methodologically closest to Section H.6. It examines whether detected frequencies stand in harmonic relation to one another – the same problem in asteroseismology. Central result: combination frequencies occur in random data with high probability, and as data quality rises the distinction between genuine and spurious combinations becomes *harder*, not easier. The usable discriminator is not the existence of individual combinations but the *number* of lowest-order

combinations compared against simulations with matching properties. Exactly this approach – comparison against a simulated null distribution rather than evaluation of individual hits – underlies the permutation tests of this document.

Pólóskei et al. (2018) [5]. Treats the procedure actually responsible for the question of harmonic coupling. Bicoherence

$$b^2(f_1, f_2) = \frac{|B(f_1, f_2)|^2}{A\{|X(f_1)X(f_2)|^2\}\, A\{|X(f_1+f_2)|^2\}}$$

measures not amplitudes but *phase coupling*: whether $f_3 = f_1 + f_2$ occurs with a fixed phase relation $\varphi_3 = \varphi_1 + \varphi_2 +$ const. Decisive for the present use is the warning about false positives: for non-stationary or short signals, classical bicoherence yields high values *without* any coupling. The proposed remedy is a Monte Carlo procedure using the measured Fourier amplitudes with *randomised phases*, from which the null distribution $\rho(b)$ and a critical value $b_c(\alpha)$ are obtained. The trade-off is also stated: the number of averages can be increased only at the cost of frequency resolution ($\Delta f = f_s N/M$).

Further usable procedures

The following methods fall below the Fourier resolution limit by assuming a model (a sum of exponentials) instead of transforming blindly. They have not been employed here but would be the next stage:

- **Subspace methods:** MUSIC (Schmidt), ESPRIT (Roy/Kailath), Min-Norm. They decompose the covariance matrix into signal and noise subspaces.
- **Matrix pencil** [7] for damped and undamped sinusoids.
- **Iterative methods:** RELAX, NOMP; and Volterra-series approaches for the self-consistent removal of combination frequencies (Lares-Martiz et al., MNRAS 2020).
- **Multitaper** (Thomson) with an F-test for line components; **multichannel singular spectrum analysis** for extracting dynamical modes without a Fourier basis.

On the magnitude of the gain: in a published comparison of three frequencies at 440/445/450 Hz over 100 ms, ESPRIT attains a frequency RMSE of 0.087 Hz against 2.88 Hz for spectral MUSIC – the difference between model classes thus exceeds the difference from Fourier analysis.

Reach of the procedures employed

[B] The scripts z6 and z7 use the bare Fourier sum $F(u) = |\sum_n e^{i\gamma_n u}|/N$ – the simplest tool available. Three limitations follow:

1. *No coupling measurement.* $F(u)$ measures whether structure sits at length u at all, not whether two lengths are *coupled*. The structural claim that the Euler product has no cross terms is argued in this document from the form of the product and measured in Section H.6; the remaining tests of this document manage with the simpler procedure.
2. *Statistics.* Sections H.6 and H.6 compute with 100 zeros from the literature; for bispectra that is too few. Section H.6 therefore computes 2469 zeros itself.

3. *Not a time signal.* The sequence of zeros is not a measured signal; stationarity is undefined, and the standard tests for non-stationarity do not apply directly.

These limitations do not touch the **[X]** findings of this document: the Weyl obstruction (Section H.3) and the rigidity of the Weil weights (Section H.6, Test C) are structural arguments and independent of the choice of spectral method. They concern only the numerical tests, whose force is thereby bounded from below – which does not weaken the negative overall finding but merely qualifies its numerical component.

H.11 Closing reflection: why one cannot compute backwards

At the end stands an image that draws the individual findings together, and an interpretation that goes beyond them. The two must be kept apart.

What the calculations show

The primes are axes – in the Tonnetz as in the Euler product. The same numbers, the same lattice structure. For simple ratios the inference back to the underlying harmony is unambiguous: $\ln 2$, $\ln 3$, $\ln 5$ appear as clear lines in the spectrum of the zeros, each in its place, each assignable.

With composition this unambiguity is lost, for two reasons, both visible in the tests.

The first is *ambiguity*. A simple ratio has few possible decompositions. A composite one has many – and the further the lattice is spanned, the denser the possibilities lie, until in the end every number fits somehow. It is not the rarity of a combination that creates the difficulty but its multiplicity of readings: where everything is reachable, reaching says nothing.

The second reason weighs heavier. The Euler product treats each prime *on its own*, without cross terms; its spectrum knows only $k \ln p$, multiples of individual prime logarithms. As soon as a ratio mixes several primes it is no longer a single harmony but a superposition – and superpositions do not appear in the spectrum as lines of their own. $\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$ is such a superposition.

An image from practice captures it exactly: from the sound of a single reed one can infer the fundamental. In a full chord across several registers one hears the mixture – but the fundamental of the mixture does not exist. There are only the fundamentals of the individual reeds. ξ is a chord, not a note.

The direction that works – and the one that does not

This allows a precise statement of what is actually closed here. It is not the connection between harmony and structure; it is its *reversibility*.

Forwards it carries: from simple ratios the composite can be built, step by step, each step traceable. This is how intervals arise from fifths and thirds, and how ξ arises from 2, 3 and 5.

Backwards it does not carry. Composition is not invertible because it destroys information: many paths lead to the same place, and the path taken can no longer be read off the result. This is not a weakness of method that better techniques could remedy, but a property of the matter itself. The tests of this document confirmed it at three different points: at the weighting strength, at the limit extension, and at the rigidity of the Weil weights.

Interpretation

Interpretation, not a finding

It is natural to suppose that complex formations in nature rest on such harmonies – that the composite is built from simple ratios, as a chord from notes. This conjecture is old and has been fruitful in many respects; it also stands behind the construction of ξ.
What this document shows is something else and more modest: even if that is so, the path cannot be walked backwards. Reconstructing the underlying harmonies from the finished formation is not difficult but impossible – composition has erased the assignment.

The first sentence is an interpretation and remains one; nothing in this document supports it and nothing refutes it. The second is computed.

Between the two lies the actual yield: whoever suspects a harmonic order behind complex structures must establish it *forwards* – by construction and prediction. The way back, inferring the order from the finished structure, is not open. This holds for the Riemann hypothesis, and it holds more generally.

H.12 The Fourier transform as a projection – and why that settles it

This document has applied the Fourier transform to 100 and 2469 zeros respectively (Sections H.6 and H.6) and made prime lengths visible in the process. It is worth understanding that operation precisely – for it reveals why neither classical FFT, nor quantum Fourier transform, nor a photonic system can solve the Riemann hypothesis.

What a Fourier transform does and does not do

[K] A Fourier transform – classical, optical, or as a quantum Fourier transform (QFT) – is a *projection*: it translates one representation into another without generating new information. What is not contained in the input data does not appear in the output.

This is visible concretely in the measurements of this document. The Fourier transform of the zeros shows peaks at $\ln 2$, $\ln 3$, $\ln 5$, $\ln 7$, $\ln 11$, $\ln 13$ – because this information is *already contained in the zeros*, via the Euler product. It shows no peak at $\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5$ – because that information is not there. The transform

invents no structure; it *projects* the existing structure onto a different coordinate system.

Consequence: A Fourier analysis of the zeros cannot solve the Riemann hypothesis, because it does not generate the arithmetic information – it can only find it if it is already there. And the decisive information (exactly where the zeros lie) must be obtained by computation, not by projection.

The structure of Shor's algorithm

[K] Shor's algorithm decomposes factorisation into two steps. The first – evaluating $f(x) = a^x \bmod N$ – is classical overhead that cannot be parallelised; it grows with bit size. The second – the QFT for period finding – is the actual quantum operation.

But the QFT too *generates* no new information: it finds the period r already present in the prepared states. *Obtaining* those states is the limiting step. The same holds for the zeros: the Fourier analysis finds what the Riemann-Siegel formula has already computed. The QFT does not solve the RH because it does not compute the zeros – it takes them as given.

The topological obstruction

[K] Section H.3 shows the decisive obstruction: the density of zeros grows like $T \ln T$, while every compact manifold (and every finite qubit register) has eigenvalue density $N(\lambda) \sim C\lambda^{d/2}$. That is not a power – and hence structurally incompatible. A quantum computer with n qubits spans a compact, finite state space; it cannot carry the $T \ln T$ density as an eigenvalue spectrum. This is not a technical but a structural obstacle.

The connection to Shor: Shor's algorithm factorises concrete numbers efficiently on quantum hardware – a real achievement for a specific problem. But a general exponential advantage of quantum computers over classical methods is not established, and for the RH the Weyl obstruction applies without qualification.

Projection and counting

The Fourier transform operates in the additive space of logarithms; the Riemann hypothesis is a statement about the multiplicative structure of the Euler product. A projection translates the representation – it does not change the question.

The core of the separation

The Fourier transform is a projection of multiplicative structure onto an additive basis. It can *find* the prime structure when it is present in the input data – as the bicoherence test demonstrates. It cannot *generate* it. The Riemann hypothesis is a statement about the consistency of factorisation – and that question cannot be answered by any projection, whether classical, optical, or as a quantum Fourier transform.

Note on the Shor documents in the corpus. Older documents (Docs. 075, 076, 147, 176) draw a connection between FFGFT and Shor's algorithm. That connection is associative, not substantive: factorising a particular number is structurally distinct from a statement about all primes. There is no FFGFT factorisation method distinguishable from Shor's (R65).

H.13 Relaxation as a third mechanism

The preceding sections treat procedures that *transform*: Fourier transform, quantum Fourier transform, optical interference. All are projections and generate no information (Section H.12). There is a third mechanism that does not transform but *settles*: physical relaxation into an energy minimum.

The mechanism

[Q] An Ising system minimises

$$H_{\text{Ising}} = -\tfrac{1}{2}\sum_{ij} J_{ij}\sigma_i\sigma_j - \sum_i h_i\sigma_i, \tag{H.1}$$

and every combinatorial optimisation problem maps polynomially onto this form. Whoever finds the ground state has solved the problem. Analog Ising machines – for instance coherent optical setups of degenerate parametric oscillators – realise this as physical relaxation rather than algorithmic iteration. The corpus treats this approach in Doc. 161 (T0 Ising machine) and Doc. 162 (comparison of optimisation frameworks).

The method is established. Programmable setups with 100 spins and all-to-all coupling, as well as networks with 2000 nodes, are published (McMahon et al., *Science* 354, 2016; Inagaki et al., *Science* 354, 2016).

Noise as a tool

[Q] The central difficulty of analog Ising machines is getting trapped in local minima. Noise resolves this: it forces spin flips even where they are energetically unfavourable, opening a wider region of phase space. This is not conjecture but measured practice with a quantitative law.

Decisive is an *optimal* noise strength: too little leaves the system trapped, too much drives it into a Boltzmann sampling regime in which the ground state is likewise missed. The optimal strength follows a scaling law

$$\gamma \sim \beta\sqrt{C}, \tag{H.2}$$

with coupling strength β and mean number C of connections per spin; the fit reaches $R^2 = 0.97$ for small problems (Mys, Verschaffelt & Van der Sande, arXiv:2508.19107, 2025). This allows the quality to be predicted without a parameter scan – the times to solution obtained lie at about twice the optimum.

What relaxation achieves and what it does not

[K] Relaxation is *not* a projection. It does not transform between representations but settles into a state. The exclusion of Section H.12 therefore does not apply directly; the mechanism must be examined in its own right.

[X] It fails at the same point, but for a different reason. The potential J_{ij} must *already* carry the sought structure: the optimisation problem is encoded into the coupling matrix, and relaxation finds what is written there. An arithmetic structure not contained in J_{ij} arises no more from settling than from transforming.

There is also a practical limit that the sources themselves state: not all problems reach high success rates. In the study cited, five of the forty Gset instances yield, for *none* of the models examined, a value within $99.5\,\%$ of the best known cut. Analog Ising machines are capable heuristics, not exact solvers – the ground-state problem remains NP-hard.

Three mechanisms, one limit

Transformation (Fourier, QFT, optics) projects and generates no information. Relaxation settles and finds what is in the potential. Both presuppose that the sought structure is already present – in the input data or in the coupling matrix respectively. The Riemann hypothesis is a statement about that structure itself.

The question, properly posed

This allows the initial question to be put more sharply. “Can a quantum computer solve the Riemann hypothesis?” confuses the tool with the object – and forces every technology to be examined separately.

The tenable formulation reads: *Is the Riemann hypothesis a question that can be answered by transformation or relaxation?* The answer is no, and it holds independently of technology: both mechanisms presuppose the arithmetic structure rather than generating it. The limit lies not in the tool but in the object.

H.14 Settled items

The questions raised in the course of this document are all decided. They are recorded here so that they are not taken up again.

- **Z-1 – weighting by the exponents** $(2, 1, 4)$**:** decided in the negative. Neither a weighting strength nor a limit extension produces selection; above all, the weights of the explicit formula are not free parameters (Section H.6).
- **Z-2 – meaning of the lattice factor** $(1 - 4^{1-s})$**:** clarified. The factor is the Dirichlet signature of Jacobi’s four-square theorem, $r_4(n) = 8 \sum_{d|n,\, 4\nmid d} d$, and encodes the divisibility condition “4 does not divide d”. The associated ladder of zeros $s = 1 + 2\pi i k/\ln 4$ with spacing $\pi/\ln 2 = 4.5324$ is a property of the sum of four squares; the nearest quantity of the framework (the prefactor 4 from L^*, R73) is $12\,\%$ away. The 4

is indeed the number of squares and hence the dimension, but it acts arithmetically here, not geometrically (`z12_gitterfaktor_waermekern.py`).

- **Z-3 – heat kernel and theta-series fine structure:** void. The difference between D4 and $\mathbb{Z}^4$ rests on a parity condition – D4 occupies only even norms, and at those the two lattices agree exactly. No test surface follows: at small t the Weyl term dominates both series equally, at large t both are exponentially small, and the informative location is the Casimir point $s = -1/2$, whose evaluation is given in Doc. 314.
- **Z-4 – bicoherence test:** carried out and confirming in six of six controls (Section H.6).

Not recommended: further search for ξ periods in the zeros. Section H.5 excludes this structurally via the GUE argument, not merely numerically.

H.15 Verification

All numbers are reproducible by twelve scripts (standard library only, target values as assertions, seed 20780458):

- `z1_weyl_obstruktion.py` – Weyl's law against Riemann-von Mangoldt, local exponent, power fit.
- `z2_zellbedingung.py` – Berry-Keating count, equivalence to the Riemann form, cross-check against Weyl.
- `z3_randbedingung_test.py` – ξ ladder test with artefact control, null distribution, out-of-sample, GUE control.
- `z4_epstein_casimir.py` – factorisation against the direct sum, Casimir symmetry point.
- `z5_zellbedingung_kritik.py` – examination of the cell argument in three stages.
- `z6_tonnetz_weil.py` – Tonnetz coordinates of ξ, Fourier detection of the Weil length spectrum, resolution limit.
- `z12_gitterfaktor_waermekern.py` – Jacobi's four-square theorem as the origin of the lattice factor; shell occupation of D4 against $\mathbb{Z}^4$.
- `z11_alphorn.py` – harmonic series, winding number versus frequency ratio, cent bookkeeping.
- `z10_kommatest.py` – does the residue $1.7 \cdot 10^{-4}$ of P-315-2 (0.98650 versus $74/75$) have comma structure? Smoothness test with control probe.
- `z9_limit_einschluss.py` – five readings of limit inclusion; ragisma and Archytas' comma as approximation errors.
- `z8_bikohaerenz.py` – zeros via Riemann-Siegel, bicoherence with phase randomisation, positive and negative controls.
- `z7_gewichtung_unmoeglich.py` – weighting strength α, limit extension (5-, 7-, 11-limit), permutation tests, rigidity of the Weil weights.

Inputs: $\xi = 4/30000$ exact; $\ell_P = 1.616255 \cdot 10^{-35}$ m, $\bar{\lambda}_e = 3.8615926796 \cdot 10^{-13}$ m (CODATA 2022); zero values (first 50) from the literature.

[1] Berry, M. V., & Keating, J. P. (1999). *The Riemann Zeros and Eigenvalue Asymptotics*. SIAM Review, 41(2), 236–266. https://doi.org/10.1137/S0036144598347497

[2] Sierra, G. (2019). *The Riemann Zeros as Spectrum and the Riemann Hypothesis*. Symmetry, 11(4), 494. https://doi.org/10.3390/sym11040494

[3] Sierra, G., & Townsend, P. K. (2008). *Landau Levels and Riemann Zeros*. Physical Review Letters, 101, 110201. https://doi.org/10.1103/PhysRevLett.101.110201

[4] Pápics, P. I. (2012). *The puzzle of combination frequencies found in heat-driven pulsators*. Astronomische Nachrichten, 333(10), 1053–1056. https://doi.org/10.1002/asna.201211815

[5] Pólóskei, P. Z., Papp, G., Por, G., Horvath, L., & Pokol, G. I. (2018). *Bicoherence analysis of nonstationary and nonlinear processes*. arXiv:1811.02973 [eess.SP]. https://arxiv.org/abs/1811.02973

[6] Kim, Y. C., & Powers, E. J. (1979). *Digital bispectral analysis and its applications to nonlinear wave interactions*. IEEE Transactions on Plasma Science, 7(2), 120–131.

[7] Hua, Y., & Sarkar, T. K. (1990). *Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise*. IEEE Transactions on Acoustics, Speech and Signal Processing, 38(5), 814–824.

[8] Lares-Martiz, M., Garrido, R., & Suárez, J. C. (2020). *Self-consistent method to extract non-linearities from pulsating star light curves – I. Combination frequencies*. Monthly Notices of the Royal Astronomical Society, 498(4), 6069–6086. https://doi.org/10.1093/mnras/staa2762

[9] Euler, L. (1739). *Tentamen novae theoriae musicae ex certissimis harmoniae principiis dilucide expositae*. Petropoli: Ex Typographia Academiae Scientiarum.

[10] Odlyzko, A. M. (1987). *On the distribution of spacings between zeros of the zeta function*. Mathematics of Computation, 48(177), 273–308.

[11] Montgomery, H. L. (1973). *The pair correlation of zeros of the zeta function*. Analytic Number Theory, Proc. Sympos. Pure Math. XXIV, 181–193. American Mathematical Society.

[12] Tiesinga, E., Mohr, P. J., Newell, D. B., & Taylor, B. N. (2024). *CODATA Recommended Values of the Fundamental Physical Constants: 2022*. Reviews of Modern Physics, 97, 025002. https://physics.nist.gov/cuu/Constants/

[13] Pascher, J. (2026). *FFGFT – Fundamental Fractal Geometric Field Theory*. Corpus, in particular Doc. 006 (particle masses, quantum numbers), Docs. 230/231/232 (Hilbert space), Docs. 295/313 (closure fork), Doc. 314 (D4 lattice, Epstein zeta, Casimir), Doc. 315 (rolled up/unrolled), Doc. 190 (correction register). https://github.com/jpascher/T0-Time-Mass-Duality

[14] Mys, L., Verschaffelt, G., & Van der Sande, G. (2025). *Predicting the optimal noise strength for solving optimization problems with analog Ising machines*. arXiv:2508.19107 [physics.app-ph]. https://arxiv.org/abs/2508.19107

[15] McMahon, P. L., et al. (2016). *A fully programmable 100-spin coherent Ising machine with all-to-all connections*. Science, 354(6312), 614–617.

[16] Inagaki, T., et al. (2016). *A coherent Ising machine for 2000-node optimization problems*. Science, 354(6312), 603–606.

Part II

Part II — Leptons, Spectral Theory and Gauge Structure

Chapter I

Doc. 317 — Topological Origin of Lepton Generations: KSAU and FFGFT

Summary

The generation problem – why exactly three particle families exist, differing only in their masses – belongs to the unsolved puzzles of the Standard Model. This study compares two independent approaches deriving masses from fundamental geometry: the *Knot-Synchronization-Adhesion Unified Theory (KSAU)* and the *Fundamental Fractal Geometric Field Theory (FFGFT)*. The synthesis reveals a deep structural correspondence: the seemingly abstract FFGFT parameters receive a direct physical interpretation through the knot topology of KSAU.

Status markers

All statements carry FFGFT status markers:
[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

I.1 Point of Departure: The Generation Problem

The experimental mass ratios of the charged leptons:

$$m_e : m_\mu : m_\tau \approx 1 : 206.77 : 3477.4 \tag{I.1}$$

are described in the Standard Model by free Yukawa couplings. Neither their values nor the number of generations follow from the framework.

I.2 The FFGFT Framework

The FFGFT derives all physical constants from the single dimensionless parameter

$$\xi = \frac{4}{30000} = 1.\overline{3} \cdot 10^{-4} \tag{I.2}$$

on a compact 4D torus T^4. The fundamental relation

$$\tilde{T} \cdot m = 1 \tag{I.3}$$

follows from the torus geometry. Field modes exist as standing waves with quantised winding numbers (n_θ, n_ϕ).

Resonance ladder of lepton masses

[K] The lepton masses follow the geometric ladder

$$m_i = r_i \cdot \xi^{p_i} \cdot v \tag{I.4}$$

with Higgs VEV $v = 246$ GeV and rational quantum numbers (r_i, p_i) (Table I.1). The exponents p_i are winding numbers on a logarithmic torus with the steps $\Delta p = 1/2$ ($e \to \mu$) and $\Delta p = 1/3$ ($\mu \to \tau$), common grid $1/6$.

Table I.1: FFGFT quantum numbers of the charged leptons

Particle	Generation	n_θ	n_ϕ	$r_i = n_\phi^2/n_\theta$	p_i
Electron e	1	3	2	$4/3$	$3/2$
Muon μ	2	5	4	$16/5$	1
Tau τ	3	9	5	$25/9$	$2/3$

[K] The mass ratios are parameter-free (without v):

$$\frac{m_\mu}{m_e} = \frac{12}{5} \cdot \xi^{-1/2} \approx 207.85 \quad (\text{PDG: } 206.77,\ +0.52\,\%) \tag{I.5}$$

$$\frac{m_\tau}{m_\mu} = \frac{125}{144} \cdot \xi^{-1/3} \approx 16.992 \quad (\text{PDG: } 16.817,\ +1.04\,\%) \tag{I.6}$$

$$\frac{m_\tau}{m_e} = \frac{25}{12} \cdot \xi^{-5/6} \approx 3532 \quad (\text{PDG: } 3477,\ +1.57\,\%) \tag{I.7}$$

Table I.2: Comparison of ladder predictions with PDG values

Ratio	Ladder (FFGFT)	PDG	Deviation
m_μ/m_e	207.85	206.77	+0.52 %
m_τ/m_μ	16.992	16.817	+1.04 %
m_τ/m_e	3532	3477	+1.57 %

[S] Falsification test: the prediction $m_\tau = 1776.97$ MeV will be decided by Belle-II (PDG: 1776.86 ± 0.12 MeV).

I.3 The KSAU Theory: Topological Knots

Scientific status of the KSAU theory

[SETZUNG] The KSAU theory (Yui, Zenodo 2026) is an unpublished preprint without peer review. The knot assignment to lepton generations is a heuristic hypothesis.

- Peer-reviewed foundations: Möbius energy (Freedman et al. 1994), Avrin 2012 (*Symmetry*, MDPI), Blatt et al. 2025 (arXiv)
- Not peer-reviewed: KSAU knot assignment and mass formula

Doc. 317 uses KSAU as a **heuristic impulse**, not as established physics. All KSAU-specific statements carry **[S]**.

[S] The KSAU theory models charged leptons as stable topological knot excitations in the quantum vacuum.

Mathematical foundation: Möbius energy

[K] The Möbius energy is mathematically rigorously defined (Freedman, He & Wang 1994):

$$E_{\text{Möb}}(\gamma) = \iint \left(\frac{1}{|\gamma(s) - \gamma(t)|^2} - \frac{1}{d_\gamma(\gamma(s), \gamma(t))^2} \right) |\gamma'(s)||\gamma'(t)| \, \mathrm{d}s \, \mathrm{d}t \tag{I.8}$$

It is conformally invariant and possesses minimisers in every knot class. Blatt et al. (2025) prove: in every torus knot class $T(a, b)$ for co-prime a, b there exist at least two distinct critical points of the Möbius energy.

[K] Torus-knot status of the lepton assignments:

Knot	Type	Torus notation	Möbius-critical
3_1 (trefoil)	torus knot	$T(2,3)$	**yes** (Blatt et al.)
6_3 (amphicheiral)	not a torus knot	—	unknown
7_1	torus knot	$T(2,7)$	**yes** (Blatt et al.)

Important restriction: Exact analytical values $E(K)$ for 3_1, 6_3, 7_1 are not known; only numerical approximations (Kim & Kusner 1993). The KSAU mass formula requires concrete $E(K)$ values. The status of the mass prediction is therefore **[S]**.

[S] The knot structure of the leptons according to KSAU:

Table I.3: KSAU knot structure of the leptons

Lepton	Knot	Knot type	Genus g	Span(Δ_K)	$E(K)/E_0$
Electron e	3_1	Trefoil	1	2	1.000
Muon μ	6_3	amphicheiral	2	4	$e^{\gamma \cdot \Delta E}$
Tau τ	7_1	torus knot	3	6	$e^{2\gamma \cdot \Delta E}$

Placement in related literature

[K] Jeon et al. (arXiv:2407.11731, 2024) demonstrate the existence of stable knot solitons in a BSM extension of the Standard Model ($U(1)_{PQ} \times U(1)_{B-L}$, Chern-Simons coupling). Stability arises from the linking of two strings, not from Möbius energy. There is *no* direct connection to lepton masses or generations. The paper establishes that knot solitons can in principle exist in high-energy physics – nothing more.

The Faddeev-Skyrme model (Faddeev & Niemi 1997) permits knot solitons in a non-renormalisable sigma model. It is not a phenomenological model of the Standard Model.

The KSAU mass assignment is a hypothesis (Zenodo 2026) that has not appeared in any peer-reviewed publication. Doc. 317 treats it as a comparison framework with status **[SETZUNG]**, not as established physics.

The mass follows according to KSAU from the Möbius energy functional:

$$m(K) = m_0 \cdot \exp(\gamma \cdot (E(K) - E_0)), \quad \gamma = 1/e \approx 0.3679 \tag{I.9}$$

[K] The Alexander polynomial links the knot directly to the generation number: $\text{Span}(\Delta_K) = 2 \cdot n_g$ (Table I.3).

I.4 Synthesis: Structural Correspondence

Table I.4: Structural correspondence KSAU ↔ FFGFT

Parameter	KSAU	FFGFT	Synthesis
Basic object	Vacuum knots $3_1, 6_3, 7_1$ **[S]**	Resonances on log. torus **[K]**	Knots on torus topology **[S]**
Generation scaling	Seifert genus $g = 1, 2, 3$	$\Delta p = 1/2,\ 1/3$	Topological step height
Form factor	Möbius energy $E(K)$	$r_i = n_\phi^2/n_\theta$	Geom. resonance of winding
Symmetry pivot	6_3 is amphicheiral **[S]**	$p_\mu = 1$ (integer) **[K]**	Symmetry of 2nd generation **[S]**
Chirality	Inner/outer projection **[S]**	Projector P_+ **[B]**	Origin of $V\!-\!A$ structure **[S]**

Torus-knot bridge theorem [S]

Let $\mathcal{T}$ be a compact 4D torus with winding numbers (n_θ, n_ϕ) and $\xi = 4/30000$. Then:

1. The three lepton generations correspond to torus knots with genus $g = 1, 2, 3$.
2. The mass ratios follow the ladder $m_i = r_i \cdot \xi^{p_i} \cdot v$ with $p_g = \frac{3}{2}\left(\frac{2}{3}\right)^{g-1}$ and $r_i = n_\phi^2/n_\theta$; the geometric sequence with ratio $2/3$ (Doc. 158) reproduces the table values $3/2,\ 1,\ 2/3$ exactly.
3. The chiral projector P_+ selects the outer state ($g_L \neq 0$, $g_R = 0$).

Status: **[S]** – the KSAU side is external (Zenodo 2026) and not verified here.

Table I.5: Quantum numbers of the charged leptons: FFGFT ladder and KSAU topology

Lepton	Gen.	n_θ	n_ϕ	r_i	p_i	Knot	g	Span
e	1	3	2	$4/3$	$3/2$	3_1	1	2
μ	2	5	4	$16/5$	1	6_3	2	4
τ	3	9	5	$25/9$	$2/3$	7_1	3	6

I.5 Complete Quantum Number Table

The table contains only the charged leptons. Carrying the same (n_θ, n_ϕ) structure with elevated exponents over to the neutrinos (Doc. 006) does not hold, because the ladder $m = r\xi^p \nu$ would then give neutrino masses of ≈ 67 eV, ≈ 1.9 eV and ≈ 12 eV, all above the KATRIN bound of 0.45 eV. Doc. 340 is authoritative for the neutrino sector. KSAU knot assignments for the neutrinos are open.

I.6 Relation to the Corpus

The lepton quantum numbers (r_i, p_i) are fully documented in Doc. 006 (particle masses, Tables 1–3) and verified by Python scripts with exact arithmetic. The FFGFT fundamental relations ($\tilde{T} \cdot m = 1$, $\xi = 4/30000$) are derived in A-series documents A080, A100, A110. The chiral projection (P_+, $g_R = 0$) is treated in A192.

[SETZUNG] The KSAU knot assignment is an external theory (Yui, Zenodo 2026), used here as a comparison framework. The structural correspondence is **[S]**.

I.7 ξ as a Galois Number: Derivation from GF(3^n)

The winding numbers (n_θ, n_ϕ) of the three lepton generations (Table I.5) are fully derivable from the GF(3^n) structure **[K]**. From this, $\xi = 4/30000$ follows as a Galois number.

Winding numbers from GF(3^n)

Quantity	Value	Galois origin	
$n_{\theta,1}$	3	char(GF(3^n))	axiomatic
$n_{\phi,1}$	2	$\lvert\text{GF}(3)^*\rvert$	mult. group of prime field
$n_{\theta,2}$	5	$n_{\theta,1} + n_{\phi,1}$	recursion; also min. prime factor of $\lvert\text{GF}(81)^*\rvert = 80$
$n_{\phi,2}$	4	$\varphi(5)$	Euler-φ of min. prime factor
$n_{\theta,3}$	9	$n_{\theta,2} + n_{\phi,2}$	recursion; also char^2
$n_{\phi,3}$	5	min. prime factor of $\lvert\text{GF}(81)^*\rvert$	

Recursion law: $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$ **[K]**. The initial condition $(3, 2)$ follows from GF(3) alone. GF$(3^4)^* = 80 = 2^4 \cdot 5$ supplies the first prime factor > 3 not present at any earlier level.

Core identity and derivation of ξ

The FFGFT mass-ratio formula (Doc. 006) and the Galois group orders yield the same number:

$$\frac{(r_\mu/r_e)^2}{\xi} = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \tag{I.10}$$

These are not two computations of the same value but a single algebraic equation from which ξ follows:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{1}{7500} = \frac{4}{30000} \quad [\mathrm{K}] \tag{I.11}$$

All factors are Galois orders: $8 = |\mathrm{GF}(9)^*|$, $27 = |\mathrm{GF}(27)|$, $5 = \mathrm{min.prime}(|\mathrm{GF}(81)^*|)$. Verification script: `pruef_331_xi_aus_galois.py`.

I.8 Open Points

1. **Covariant derivatives D_μ (A192):** The chiral projection explains $SU(2)_L$ coupling qualitatively; the covariant derivatives are not yet written out explicitly.
2. **Correction $N_0 \approx 38.6$ (A105):** Candidate for the generation-linear correction of ladder deviations; not derived from ξ. Concerns only residual deviations.
3. **Neutrino knots:** KSAU assignments for neutrino generations not yet established.

[1] Pascher, J. (2026). *T0-Time-Mass-Duality – FFGFT v1.3.0*. Zenodo. https://zenodo.org/records/21838986

[2] Yui / Protocol Project (2026). *Knot-Synchronization-Adhesion Unified Theory (KSAU) v1.6.1*. Zenodo.

[3] Avrin, J. S. (2012).

Knots on a Torus: A Model of the Elementary Particles. *Symmetry* (MDPI), **4**(1), 39–115. DOI: 10.3390/sym4010039

[4] Faddeev, L., & Niemi, A. J. (1997). Knots and particles. *Nature*, 387, 372–374.

[5] Freedman, M. H., He, Z.-X., & Wang, Z. (1994). Möbius energy of knots. *Annals of Mathematics*, 139(1), 1–50.

[6] Jeon et al. (2025). Tying Knots in Particle Physics. *Physical Review*.

[7] Blatt, S., Gilsbach, A., Reiter, Ph., & von der Mosel, H. (2025). Stability of knot equivalence at low regularity, and symmetric critical knots for the Möbius energy. arXiv:2512.02998.

[8] Kim, D., & Kusner, R. (1993). Torus knots extremizing the Möbius energy. *Experimental Mathematics*, 2(1), 1–9.

[9] Khanna, F. C., Malbouisson, A. P. C., Malbouisson, J. M. C., & Santana, A. E. (2014). *Quantum field theory on toroidal topology: algebraic structure and applications*. arXiv:1409.1245 [hep-th]. Submitted to *Physics Reports*. *Content:* Mathematical foundation of QFT on torus topology ($\Gamma_D^d = (S^1)^d \times \mathbb{R}^{D-d}$); treats thermal field theory, Casimir effect, superconductivity, and hadronic phase structure. No connection to lepton masses.

Chapter J

Doc. 318 — Scope of Mass Derivations in FFGFT: Rest Masses, Hadron Dynamics, and Open Bridges

List of symbols

Symbol	Meaning	Value / unit
M_Z	Mass of the Z boson; reference scale for α_s	91.19 GeV
Λ_{QCD}	QCD confinement scale	≈ 200–217 MeV
$\alpha_s(\mu)$	Strong coupling at scale μ	$\alpha_s(M_Z) \approx 0.118$; $\alpha_s(m_\tau) \approx 0.33$
$D = N_c$	Number of colour degrees of freedom (Doc. 160)	$D = 3$
y_e	Yukawa coupling of the electron	$m_e/v \approx 2.08 \cdot 10^{-6}$
v	Higgs VEV	246 GeV
m_p, m_n	Proton / neutron mass	938.3 / 939.6 MeV
RGE	Renormalisation group equation	describes μ-dependence of α_s
K_{frak}	Fractal interface factor (rolled-up → SI)	$1 - 100\xi \approx 0.987$
m^{bare} / m^{SI}	Geometric / measured mass	nat. units / MeV
ξ	FFGFT geometry parameter	$4/30000 \approx 1.333 \cdot 10^{-4}$

Summary

The FFGFT derives particle masses from the geometry parameter $\xi = 4/30000$ on the compact 4D torus $T^4/\mathbb{Z}_3$. This scope is precise for rest masses from the Higgs-Yukawa sector (leptons, quark rest masses). Hadron masses such as the proton mass contain a dominant non-perturbative QCD contribution that requires its own framework. This document draws the boundary explicitly and grounds it physically in the relativistic energy relation $E^2 = (m_0c^2)^2 + (pc)^2$ and the picture of massless gluons inside the proton.

Status markers

[K] FFGFT derives rest masses (leptons, quark Yukawa) from ξ-geometry. **Hadron masses** have additionally a non-perturbative QCD contribution: **open bridge**.

J.1 Starting point: the complete energy relation

The popular $E = m_0c^2$ is the rest energy, i.e. the special case $p = 0$ of the complete relation below; as such it is exact. Linearising the Lorentz factor

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} \qquad (v \ll c) \tag{J.1}$$

gives, for small velocities, $E = \gamma m_0c^2 \approx m_0c^2 + \frac{1}{2}m_0v^2$. The complete, exact relation is:

$$\boxed{E^2 = (m_0c^2)^2 + (pc)^2} \tag{J.2}$$

This equation is the actual core – not the simplified formula. The second term pc does not vanish when a particle is massless. For massless particles ($m_0 = 0$):

$$E = pc, \tag{J.3}$$

so a massless particle moving at the speed of light still contributes to the total energy – and thus to the measurable mass of a system.

J.2 The proton: mass as emergent property

The proton mass $m_p \approx 938.3$ MeV is the clearest example of this emergence:

Table J.1: Energy contributions to the proton mass

Contribution	Value	Fraction	Mechanism
u and d quark rest masses	≈ 10 MeV	$\approx 1\,\%$	Higgs-Yukawa coupling
QCD energy (quark kinetic energy, gluon field energy, trace anomaly)	≈ 928 MeV	$\approx 99\,\%$	QCD confinement
Proton mass	938.3 MeV	100 %	

According to the lattice QCD decomposition (Yang et al., PRL 121, 212001 (2018)) the proton mass splits into quark kinetic energy ($\sim 32\,\%$), gluon field energy ($\sim 36\,\%$), trace anomaly ($\sim 23\,\%$) and quark mass term ($\sim 9\,\%$); the $\approx 1\,\%$ row counts only the valence current-quark masses.

The proton consists of three quarks (uud), whose rest masses together amount to about 10 MeV. The remainder – 928 MeV, around 99 % – is QCD energy: kinetic energy of

the quarks, field energy of the gluons, the carriers of the strong force, and the trace anomaly (decomposition below Table J.1). Gluons are massless ($m_0 = 0$). They move at the speed of light. By Eq. (J.2) they nevertheless contribute to the total energy of the system through their momentum.

Timelessness inside the proton

For massless particles moving at the speed of light, time dilation is maximal: the proper time of these particles is zero. Gluons inside the proton exist in a state where no time passes. For them, the spatial extent of the world is zero – length contraction is maximal.

And yet: these timeless, massless constituents account for the dominant part of what is measured externally as the proton mass. Mass is a property at the surface of a system, not an intrinsic property of its components.

[S] This observation connects directly to the FFGFT fundamental relation. $\tilde{T} \cdot m = 1$ states: intrinsic time and mass are dual. The interior of the proton is precisely the timeless level – and this constitutes the majority of the externally apparent mass. Rest mass arises through projection from the timeless to the time-bearing domain. The proton illustrates $\tilde{T} \cdot m = 1$ inadvertently but precisely. This connection is conceptually robust but not yet a formal derivation; it is noted for a future document.

J.3 Two sectors – two mechanisms

The ratio $m_p/m_e \approx 1836$ connects two physically distinct sectors:

Quantity	Sector	Scale
m_e, m_μ, m_τ	Higgs-Yukawa	$\xi^{p_i} \cdot r_i \cdot v$
$m_{\nu_e}, m_{\nu_\mu}, m_{\nu_\tau}$	neutrino sector	own structure (Doc. 340)
m_u, m_d, m_s (rest)	Higgs-Yukawa	(open, Doc. 006)
m_p, m_n (total)	QCD confinement + Higgs-Yukawa	$\Lambda_{\text{QCD}} + m_{\text{quarks}}$

[B] Proton mass and electron mass arise through entirely different physics. m_e is a rest mass from the Higgs-Yukawa sector; m_p consists to $99\,\%$ of QCD confinement energy. A ratio of these two quantities connects two independent scales (Λ_{QCD} and $y_e \cdot v$). There is no structural reason why the geometry of a single parameter ξ should simultaneously fix both scales.

Consequence: m_p/m_e is not a derivation target of the current FFGFT corpus. An apparently exact five-digit agreement would be a warning signal, not a success criterion – because QCD contributions carry percentage-level theory uncertainties.

J.4 Neutrinos: the other limiting case

In contrast to the proton, neutrinos are the opposite limit: extremely small rest masses lying entirely within the Higgs-Yukawa sector.

In FFGFT, neutrinos are treated as “nearly massless photons” (Doc. 007). Carrying the (n_θ, n_ϕ) winding structure of the charged leptons with elevated ξ exponents over to the neutrinos (Doc. 006) does not hold: the exponent $25/9$ of the ν_τ row is the prefactor r_τ, copied into the exponent column, and the rows give masses above the KATRIN bound (Doc. 317). The authoritative treatment of the neutrino sector is Doc. 340. Like the charged leptons, neutrinos concern rest masses; the proton lies outside this domain.

J.5 Rolled up and unrolled: the structural reason for the two-sector separation

Doc. 311 distinguishes three ways in which the fourth spatial direction of the 4D torus T^4 becomes the three-dimensional world:

Way	Mechanism	Problem
Compactification (Kaluza-Klein)	fourth direction becomes a circle, windings = modes	radius R is a free modulus
Projection (cut-and-project)	4D lattice → 3D cut	loses lattice invariants
Fourth is not space	carries mass, time, internal index	without compactness no mode tower

[K] FFGFT combines ways 1 and 3, bound together by $\tilde{T} \cdot m = 1$: the fourth direction is *rolled up* (compact) *and bound* (it carries mass). The radius R is therefore not free – it is the mass. This is the only case in which a scale arises without bringing in an additional parameter.

Rolled up: rest masses from the Yukawa sector

[K] Quark and lepton rest masses arise from *rolled-up* modes: winding numbers (n_θ, n_ϕ) on the compact torus generate quantised masses $m_i = r_i \cdot \xi^{p_i} \cdot v$ (Docs. 006, 317). Integer winding numbers are forced by the compactness of the torus. This is the geometric origin of the Yukawa couplings: not free parameters, but topological invariants.

Unrolled: gluons as dynamic fields in 3D space

[S] Gluons are massless vector bosons propagating in the unrolled, non-compact 3D space. They are not winding modes on the torus but dynamic degrees of freedom of the gauge field in the unfolded space. Their energy does not come from ξ-windings but from motion and QCD confinement.

The proton in this language:

Contribution	Origin	Sector
Quark rest masses ($\sim 1\,\%$)	rolled-up modes, Yukawa	FFGFT, **[K]**
QCD energy ($\sim 99\,\%$: quark kinetic energy, gluon field energy, trace anomaly)	unrolled modes, QCD dynamics	open bridge, **[S]**

Hadron masses have contributions from *both* regimes – that is the structural reason why a unified ξ-derivation does not directly apply. The boundary between rolled-up and unrolled is the boundary of the scope.

J.6 The fractal correction K_{frak}: interface factor in the transition rolled-up → unrolled

[K] K_{frak} is the factor that appears in the transition from the geometrically rolled-up (bare) description to the measured (unrolled, SI) description (Docs. 012, 133):

$$m_i^{\text{SI}} = K_{\text{frak}} \cdot C_{\text{conv}} \cdot m_i^{\text{bare}}, \qquad K_{\text{frak}} = 1 - 100\xi \approx 0.9867 \tag{J.4}$$

The value $1 - 100\xi$ is confirmed by comparing the two E_0 routes with α (Doc. 190, R135); the form of the factor is open **[S]**, and the factor 100 is set in Doc. 133, not derived. The physical interpretation is the fractal porosity of spacetime at Planck scales ($D_{\text{frak}} = 2.94$, Doc. 003), which reduces the effective interaction strength relative to a smooth space. In the ladder $m_i = r_i\, \xi^{p_i}\, v$, K_{frak} does not appear separately: there the correction is absorbed into the bridge constant v (Doc. 190, P40).

The key point: K_{frak} cancels in ratios

[K] In mass ratios K_{frak} drops out completely (Doc. 012):

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}} \tag{J.5}$$

This is structurally significant: the mass ratios that FFGFT derives in the lepton sector (m_μ/m_e, m_τ/m_μ, Doc. 317) are purely geometric and do *not* require K_{frak}. K_{frak} appears only when absolute SI values are needed – for the gravitational constant G, the fine-structure constant α, and absolute mass values in MeV or kg.

Scope and limitation for hadrons

K_{frak} is the interface factor between two levels:

Level	Description	K_{frak}
Rolled-up (bare)	m^{bare}, torus geometry, ξ-powers	not needed
Transition	$m^{\text{SI}} = K_{\text{frak}} \cdot C_{\text{conv}} \cdot m^{\text{bare}}$	**here**
Unrolled (SI)	measurable mass in MeV, kg	already included

For the **hadron sector**, K_{frak} would appear in the transition $\Lambda_{\text{QCD}}^{\text{bare}} \to \Lambda_{\text{QCD}}^{\text{SI}}$ – but $\Lambda_{\text{QCD}}^{\text{bare}}$ itself is not derived from ξ. K_{frak} as an interface factor does not resolve the underlying problem: the 928 MeV contribution is missing not because of a lacking conversion, but because the underlying QCD mechanism is not yet treated in the FFGFT corpus.

A135: settability as a diagnostic

[K] A135 states the settability criterion: what can be norm-set to one was a property of the measurement system. $K_{\text{frak}} = 1 - 100\xi$ cannot be set to 1 without changing ξ – it is bound to something physical, not a pure measurement-system artefact. For precision predictions Doc. 133 applies.

J.7 Scope of the ξ mass derivations

Table J.2: Complete scope of mass derivations in FFGFT

Particle	Dominant contribution	Status	Doc.
e, μ, τ	Higgs-Yukawa, rest mass	**[K]**	006, 317
ν_e, ν_μ, ν_τ	neutrino sector, own structure	**[S]**	340
u, d, s, c, b, t (rest)	Higgs-Yukawa	**[S]** open	006
W, Z	Higgs mechanism	**[S]** scale correct	—
Higgs h	Higgs self-coupling	**[K]** ξ_{Higgs}	A-series
p, n (total)	QCD + Higgs-Yukawa	**open bridge**	—
m_p/m_e	sector-mixed	**not claimed**	—

J.8 Three scenarios for the QCD bridge

Whether ξ also captures the QCD sector is an open research question. Three scenarios are conceivable:

1. **ξ fixes Λ_{QCD} via orbifold geometry.** The $T^4/\mathbb{Z}_3$ structure generates nine orbifold fixed points. Whether these force a natural QCD scale is open. Doc. 041 lists the ansatz $\Lambda_{\text{QCD}} = v \cdot \xi^{1/3}$; evaluated, it gives $v \cdot \xi^{1/3} \approx 12.6$ GeV and misses $\Lambda_{\text{QCD}} \sim 217$ MeV (factor ≈ 58; $246\,000$ MeV $\cdot\, \xi^{1/3} = 246\,000 \cdot 0.05109$ MeV $= 12\,568$ MeV). A derivation of the QCD scale from ξ is therefore not available. **[S]**
2. **Trace anomaly on compact spaces.** The renormalisation group for the strong coupling $\alpha_s(\mu)$ on compact tori generates a natural infrared scale. Khanna et al. (2014, arXiv:1409.1245) provide the mathematical framework; whether ξ enters there is unexplored. **[S]**
3. **Hadron sector with SM inputs (Doc. 005) and geometric α_s derivation (Doc. 160).** Doc. 005 contains a complete proton calculation (0.938100 GeV, deviation $0.02\,\%$) using the extended fractal formula. It uses $\alpha_s = 0.118$ and $\Lambda_{\text{QCD}} = 0.217$ GeV as Standard Model inputs. The claim "without free parameters" means: no additional parameters beyond the SM inputs.

   Essential precision: Doc. 160 (lepton lifetimes) derives α_s geometrically:

$$\alpha_s(m_\tau) = D \cdot \xi^{1/(D+1)} = 3 \cdot \xi^{1/4} \approx 0.322 \quad (\text{exp: } 0.330,\ -2.3\,\%) \tag{J.6}$$

   $D = 3$ colour degrees of freedom, $D+1 = 4$ spacetime dimensions. This is α_s at scale m_τ, not at M_Z. The strong coupling runs with energy: $\alpha_s(m_\tau) \approx 0.33$ and $\alpha_s(M_Z) \approx 0.118$ are different values of the *same* running coupling. The FFGFT derivation targets the correct scale (Doc. 160) and uses 0.118 for the proton calculation in Doc. 005 because M_Z is the relevant scale there. **[K]**

   Open point: the complete RGE bridge from $\alpha_s(m_\tau)$ to $\alpha_s(M_Z)$ within FFGFT is not yet carried out. **[S]**

Scenario 3 is the current position. Scenarios 1 and 2 are marked research questions. The approach in Doc. 041 ($\Lambda_{\text{QCD}} = v \cdot \xi^{1/3}$) does not support Scenario 1, since it yields ≈ 12.6 GeV instead of ~ 217 MeV.

J.9 Relation to the corpus

Structural classification of the early documents

[K] The early documents (Docs. 001–041) primarily worked out the *structural connections* between ξ and the physical constants – not precision derivations. The formulas in these documents are order-of-magnitude approaches: they show *that* and *how* a coupling constant depends on ξ, not necessarily the exact prefactor.

Document	Content	Character
Doc. 041	$\alpha_s = \xi^{-1/3}$, $\alpha_W = \xi^{1/2}$ (nat. units)	structural sketch **[S]**
Doc. 041	$\Lambda_{\text{QCD}} = v \cdot \xi^{1/3} \approx 12.6$ GeV	misses Λ_{QCD} **[S]**
Doc. 160	$\alpha_s(m_\tau) = 3\xi^{1/4} \approx 0.322$	precision derivation **[K]**
Doc. 011	$\alpha = \xi E_0^2$	property of the anchor E_0, no prediction
Doc. 006	lepton masses $m_i = r_i \xi^{p_i} v$	ladder structure with reference point **[K]**

In particular: Doc. 041 lists $\alpha_s \approx 0.118$ (experiment) and $\xi^{-1/3} = 19.57$ (natural units) side by side without unit conversion. This is not an error in the physics but a missing bridge – the conversion from natural to SI units is absent there. Doc. 160 provides the cleaner derivation.

Corpus references

- **Doc. 005** (tm-extension): proton calculation with extended fractal formula; $\alpha_s = 0.118$ and $\Lambda_{\text{QCD}} = 0.217$ GeV as SM inputs (not ξ-derivations).
- **Doc. 006** (particle masses): lepton rest masses [K]; hadron masses not treated.
- **Doc. 007** (neutrinos): neutrinos as nearly massless photons.
- **Doc. 340** (neutrinos): neutrino sector (authoritative).
- **Doc. 041** (parameter derivation): structural connections of all couplings; order-of-magnitude estimates, not SI precision.
- **Doc. 160** (lepton lifetimes): geometric derivation $\alpha_s(m_\tau) = 3\xi^{1/4}$, precision $-2.3\,\%$ [K].
- **Doc. 317** (KSAU-FFGFT): quantum number table of the charged leptons.
- **A-series**: ξ_{Higgs}, Koide relation.

J.10 Entry in Doc. 190 (R76)

This clarification is declared as register entry R76 in Doc. 190: scope of mass derivations explicitly delimited; hadron sector documented as open bridge; m_p/m_e not

claimed as a derivation target of the current corpus; three scenarios for the QCD bridge marked as research questions.

Chapter K

Doc. 319 — The Proton as a Vibrating Torus: Geometric Foundations of Mass Emergence in FFGFT

Summary

In FFGFT the proton is not described as a collection of quarks and gluons in a volume, but as a *vibrating torus* $T^4/\mathbb{Z}_3$. Quarks and gluons are different mode types of this structure. The proton mass is the vibrational energy of the torus. This document develops the geometric picture, anchors it in the corpus, and marks all statements by the epistemic standard [K]/[S]/[SETZUNG].

Reading note: inner perspective

[K] We describe the proton torus *from inside* – as part of the same T^4 system (Doc. 248). There is no outside perspective. What "confinement" or "freedom" means is relational: it describes the relation between the observer and the respective topological system, not an intrinsic property of the observed field.

K.1 The Geometric Picture

[K] FFGFT starts from a compact 4D torus T^4 with $\mathbb{Z}_3$ orbifold symmetry. The fundamental relation is (Doc. 003):

$$\tilde{T} \cdot m = 1 \tag{K.1}$$

Intrinsic time and mass are dual. This has immediate consequences for all modes on the torus.

[K] The triality of the D4 lattice is built into the lattice geometry, not postulated (Doc. 314): the order-3 elements in Aut$(D4)$ are *half-integer* – the glue-vector mixture $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ enters. The $T^4/\mathbb{Z}_3$ structure with exactly 9 fixed points follows from the D4

geometry. **[S]** Reading these half-integer elements as the geometric origin of the fermionic statistics of the quark modes is an interpretation: no calculation connects half-integer matrix entries with spin $\frac{1}{2}$ or antisymmetry.

K.2 Two Mode Types: Quarks and Gluons

[K] The compact $\mathbb{Z}_3$ fibre carries the discrete, mass-bearing spectrum – not the unrolled axis (Doc. 285):

> "The discrete, mass-bearing spectrum sits on the compact $\mathbb{Z}_3$ fibre, not on the unrolled axis."

Table K.1: Quark and gluon modes on the torus [K]

Mode	Spectrum	Mass	Proper time $\tilde{T}$	Doc.
Quark modes	discrete, $\mathbb{Z}_3$ fibre	$m > 0$	$\tilde{T} < \infty$	006, 285, 314
Gluon modes	continuous, locally flat	$m = 0$	$\tilde{T} = \infty$	003, 270

[K] Gluons are massless ($m = 0$), hence $\tilde{T} = \infty$ – they perform no measurement act (Doc. 270). Locally, flatness holds: "in the spatial limit, local flatness" (Doc. 270). This means: locally the massless gluon sees $\mathbb{R}^3$, but that is a local homeomorphism, not a global statement.

K.3 The Relational Nature of Confinement and Freedom

[K] Unrolling is a measurement act of the embedded observer (Doc. 270): "The measurement chooses one circle as time and unrolls it; a physical act, not a mathematical limit."

[K] We (observers) are part of the T^4 system – not outside (Doc. 248): "We are part of the system. We always describe from inside, never from outside."

This yields a fundamental insight for the photon-gluon distinction:

Confinement is relational [K/S]

Photon and gluon are both massless ($m = 0$), both timeless ($\tilde{T} = \infty$), both see $\mathbb{R}^3$ locally. **Both carry boundary conditions.**
The difference is not intrinsic but *relational*:

- **Photon:** We (observers) are part of the same T^4 system. The photon inhabits the same topological sector. Its boundary condition is our boundary condition. It therefore appears to us as freely propagating.
- **Gluon:** We face the $\mathbb{Z}_3$ substructure of the proton torus from outside. The gluon inhabits a different topological sector. It appears to us as confined.

"Confinement" is not a property of the gluon alone – it is a property of the relation observer ↔ system (Doc. 248, [K]). **[S]** The complete topological derivation of why photon and gluon inhabit different sectors remains open.

K.4 Massless Field Modes: Locally Pure Energy

[K] For massless fields ($m = 0$), $\tilde{T} \cdot m = 1$ is not defined – not as a limit, but exactly: on null geodesics $d\tau = 0$; along a light path no proper time elapses. There is no rest frame (Doc. 312):

> "Masslessness means $v = c$ exactly – the saturation of the inequality $v \leq c$. There is no $c^*(\lambda)$; $dc/d\lambda \equiv 0$."

[K] What remains locally is exclusively $E = pc$ – pure energy (Docs. 134, 312). This holds for photon field modes and gluon field modes *equally*. From a local point of view, every massless field mode is pure energy. There is no "gluon here" or "photon there" in an absolute sense – only energy density in space.

[K] "Particles" are not fundamental in FFGFT (Docs. 049, 156): "There are no particles at all! What we call particles are different excitation patterns in the single field $\delta m(x,t)$." Photon and gluon are field modes – not objects.

[S] The difference photon/gluon lies not in the local description (both: $E = pc$, both: $d\tau = 0$, both: locally flat), but in the *global topological embedding* of the field mode:

Field mode	Topology	Global behaviour
Photon	$U(1)$ flux, extended	propagates in T^4 (our system)
Gluon	$\mathbb{Z}_3/SU(3)$ linking	standing wave in proton torus

[S] The gluon field is not localisable: as a non-Abelian $SU(3)$ field mode (Doc. 145: colour-anticolour linking of three flux threads) it interacts with itself. There is no bounded "gluon object" – only a global field configuration of the proton torus. The eight gluon modes correspond to the eight non-trivial linking states of the three flux threads = eight generators of $SU(3)_c$ (Doc. 145, [S]; algebraically confirmed by Doc. 321 [B]).

K.5 Topological Phase Closure as Confinement Mechanism

[K] On the closed torus a wave must complete an integer phase shift over the circumference:

$$\oint k \, \mathrm{d}l = 2\pi n, \quad n \in \mathbb{Z} \tag{K.2}$$

[K] The $\mathbb{Z}_3$ action on $L^2(T^4)$ permutes the mode orbits cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases $0°$, $+120°$, $-120°$ – structurally exactly the circulant diagonalisation of the $\mathbb{C}^3$ fibre (Doc. 314). This enforces discrete modes without walls.

[S] The step from this mode structure to full QCD confinement dynamics has not yet been formally carried out.

K.6 The Duality $\tilde{T} \cdot m = 1$ in the Proton

[K] The complete energy relation $E^2 = (m_0 c^2)^2 + (pc)^2$ (Doc. 318) shows: massless gluons carry energy $E = pc$ despite $m_0 = 0$ – together with the kinetic energy of the quarks and the trace anomaly, their field energy forms the QCD energy, $\sim 99\,\%$ of the proton mass (Doc. 318).

[K] This is not a paradox. A standing wave has no proper time ($\tilde{T} = \infty$), but its energy $E = \hbar\omega$ is real and measurable. The duality $\tilde{T} \cdot m = 1$ states: timelessness and masslessness are equivalent. The *system energy* of the bound state – not the eigen-energy of individual modes – is the proton mass.

Table K.2: Time-bearing vs. timeless oscillation [K]

Property	Quark (time-bearing)	Gluon (timeless)
Proper time	$\tilde{T} > 0$	$\tilde{T} = \infty$
Energy	Rest mass + motion	Pure field energy $E = pc$
Spectrum	Discrete ($\mathbb{Z}_3$ fibre)	Locally continuous
Statistics	Fermion (D4 triality, interpretation [S])	Boson

K.7 Spin Statistics from D4 Triality

[K] Lattice facts (Doc. 314): the 80 order-3 elements in $\mathrm{Aut}(D4) = W(F_4)$ (1152 elements) fall into three classes: 16 with trace -2 (*half-integer*, fixed-point-free in $\mathbb{R}^4$, eigenvalues $\omega, \bar{\omega}$ each twice), 32 with trace 1 integer and 32 with trace 1 half-integer. **[S]** Reading this half-integrality as the geometric foundation of the fermion statistics of the quark modes is an interpretation: the class decomposition has been computed explicitly, whereas a connection between half-integer matrix entries and spin $\frac{1}{2}$ or antisymmetry has not.

[S] The draft approach "$(n_\theta + n_\phi)$ odd $\rightarrow$ fermion" captures the result but not the path. Correct is: D4 triality delivers half-integer representations, independently of the parity of the winding numbers.

K.8 Colour Charge and $\mathbb{Z}_3$ Structure

[K] The $\mathbb{Z}_3$ triality of the D4 lattice is built in, not postulated (Doc. 314). $\mathbb{Z}_3$ is the centre of $SU(3)$.

[B] The complete algebraic derivation of the $SU(3)_c$ gauge structure from $\mathbb{Z}_3$ triality is carried out in Doc. 321 (§3–4). The eight bilinear transition operators between the

three eigensectors $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$ of $L^2(T^4)$ satisfy the $\mathfrak{su}(3)$ commutation relations with canonical structure constants. $N_c = 3$ is algebraically necessary, not postulated.

[K] Doc. 049: "Nodes cannot exist alone, must form neutral combinations." Algebraically sharpened by Doc. 321: only states with triality $T_R = 0$ are asymptotic states – confinement as triality selection.

Status of the colour charge thesis – closed by Doc. 321 [B]

The $SU(3)_c$ emergence from $\mathbb{Z}_3$ triality is algebraically proved (Doc. 321). Colour charge is geometrically fundamental: the three $\mathbb{Z}_3$ eigensectors are the three colour degrees of freedom. Confinement as $T_R = 0$ selection is the algebraic form of Doc. 049.

K.9 Λ_{QCD} and Minimal Frequency

[S] The minimal frequency on a torus with radius $R \approx 1$ fm:

$$\omega_{\text{min}} = \frac{1}{R} \approx 197 \text{ MeV} \approx \Lambda_{\text{QCD}} \tag{K.3}$$

($1 \text{ fm}^{-1} = 197.3$ MeV in natural units). Not a complete derivation; Doc. 041 gives $v \cdot \xi^{1/3} = 246.22 \text{ GeV} \cdot 0.05109 = 12.58$ GeV, 64 times ω_{min}, which is not a value for Λ_{QCD}.

K.10 The Proton as a Whole

[K] The proton is not a container with particles but a *system quantity*: the quark modes (discrete, $\mathbb{Z}_3$ fibre, massive) define the boundary condition. The gluon modes (locally continuous, massless) fill the system with field energy. The proton mass is the total energy of this coupled system.

[S] The geometric picture "quarks at the boundary, gluons in the volume" is conceptually robust but not yet a formal derivation from the torus geometry.

K.11 Relation to the Corpus

- **Doc. 003**: $\tilde{T} \cdot m = 1$, $D_{\text{frak}} = 2.94$, K_{frak} [K]
- **Doc. 248**: Embedded observer, inner perspective [K]
- **Doc. 270**: Measurement as unrolling, local flatness [K]
- **Doc. 285**: Discrete spectrum on $\mathbb{Z}_3$ fibre [K]
- **Doc. 311**: Rolled-up/unrolled, three ways [K]
- **Doc. 314**: D4 triality, half-integer, $T^4/\mathbb{Z}_3$ from D4 [K]
- **Doc. 318**: Proton as open bridge, $E^2 = (m_0c^2)^2 + (pc)^2$, scope [K]
- **Doc. 041**: $v \cdot \xi^{1/3} = 12.6$ GeV, does not match Λ_{QCD}

- **Doc. 049**: Confinement formulation – algebraically sharpened by Doc. 321 [K]
- **Doc. 321**: Algebraic derivation of $SU(3)_c$ from $\mathbb{Z}_3$ triality; $N_c = 3$; confinement as $T_R = 0$ selection [B]
- **Doc. 322**: Hilbert-space embedding; fixed-point boundary conditions on $T^4/\mathbb{Z}_3$ [K]

K.12 Open Questions [S]

1. $SU(3)_c$ **emergence from** $\mathbb{Z}_3$ **triality.** [**B** – closed by Doc. 321] Eight Gell-Mann operators derived algebraically from $\mathbb{Z}_3$ eigensectors; $N_c = 3$ necessary.
2. **Why photon and gluon inhabit different sectors.** The relational explanation (Doc. 248) is [K]. The formal topological derivation remains [S].
3. $\Lambda_{\mathbf{QCD}}$ **from torus geometry.** Doc. 041 gives no matching value (see the section on Λ_{QCD}). Complete derivation open.
4. **RGE bridge** $\alpha_s(m_\tau) \to \alpha_s(M_Z)$**.** Doc. 160: $\alpha_s(m_\tau) = 3\xi^{1/4}$ [K]. Bridge open [S].
5. **Formal boundary conditions on** $T^4/\mathbb{Z}_3$**.** [**K** – worked out in Doc. 322] Fixed-point boundary conditions and Hilbert-space embedding complete.

Chapter L

Doc. 320 — Detailed Field-Geometric Spectral Theory

List of Symbols

Symbol	Meaning	Value / Unit
ξ	FFGFT geometry parameter	$4/30000 = 1.3\overline{3} \times 10^{-4}$
v	Higgs vacuum expectation value	246.0 GeV
m_e	electron mass (neutrino sector ref.)	0.511 MeV
r_i	topological winding ratio	n_ϕ^2/n_θ
p_i	exponent in mass formula	dimensionless
η_{D4}	D_4 projection coefficient	$81/640$
F_{D4}	D_4 volume element	$27/1\,600\,000$
$[K]$	derivable from ξ	status marker
$[B]$	algebraically proved	status marker
$[S]$	sketched / plausible	status marker

Summary

The FFGFT derives all particle masses of the leptonic sector from the single dimensionless geometry parameter $\xi = 4/30000$ and the topology of the $T^4/\mathbb{Z}_3$ orbifold. This document follows the derivation explicitly in every step: from the sub-Planck lattice, through the D_4 projection and topological winding numbers, to the measurable masses. The mathematical Hilbert-space foundation is provided in Doc. 322.

Status markers

All statements carry FFGFT status markers:
[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

L.1 Starting Point: the Sub-Planck Geometry

Why a sub-Planck level?

The Standard Model describes particle masses through Yukawa couplings – free parameters measured experimentally but not explained within the framework itself. FFGFT goes one level deeper: masses do not originate from free couplings but from the geometry of space-time on scales below the Planck length ℓ_P.

[SETZUNG] At the sub-Planck scale t_0 space-time has a discrete packing structure. The most efficient regular lattice in four dimensions is the D_4 lattice – the 4D analogue of the densest sphere packing in 3D.

The D_4 lattice and its symmetry

Properties of the D_4 lattice:

- **4 orthogonal dimensions** with identical scaling.
- **Automorphism group** $\mathrm{Aut}(D_4) \cong W(F_4)$ with $|W(F_4)| = 1152$ elements.
- **24 nearest neighbours** per lattice point.
- The torus $T^4/\mathbb{Z}_3$ is the natural compactification, with the $\mathbb{Z}_3$ identification reflecting the three quark colour charges.

[K] The fundamental FFGFT relation $\tilde{T} \cdot m = 1$ follows from the duality between torus radius $R \propto \tilde{T}$ and mass $m \propto 1/R$: on a compact torus the smallest mode spacing is inverse to the radius.

L.2 The Geometry Parameter ξ and its Origin

Definition and numerical value

[K]

$$\xi = \frac{4}{30000} = \frac{1}{7500} = 1.3\overline{3} \times 10^{-4}. \tag{L.1}$$

Numerator 4 equals the number of orthogonal dimensions of the D_4 lattice – topologically fixed, not freely chosen.

Denominator 30000 equals the phase volume of the fractal T^4 compactification: product of the $\mathbb{Z}_3$ orbifold order (3) and the number of fundamental Fourier modes on the torus (10^4, the fourth power of the first non-trivial mode).

Connection to Higgs geometry

The Higgs mass provides a consistency check of ξ (Doc. 354, A190, Doc. 385). With the Standard Model self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$ one has $\xi_{\mathrm{EFT}} = \lambda_h^2 v^2/(16\pi^3 m_h^2) = m_h^2/(64\pi^3 v^2) \approx 1.30 \cdot 10^{-4}$, about $2.3\,\%$ below $4/30000$. The Higgs geometry is thus a

consistency check of ξ to within a few per cent, not an exact derivation and not a second independent route to the same value. Inserting $\lambda_h = m_h/v$ does not hold: it would give $1/(16\pi^3) = 2.016 \cdot 10^{-3}$, 15 times $4/30000$, and an iteration $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ does not lead from there to $4/30000$ but towards 0 (its only fixed point).

The D_4 projection coefficient η_{D4}

The projection coefficient used is

$$\eta_{D4} = \frac{81}{640} = 0.1265625; \tag{L.2}$$

[S] its origin is open: a counting of the Lie-algebra modes ($N_c^2 \cdot 9$ over $N_g \cdot 8 \cdot 8$ with $N_g = 80$) gives $81/5120$, one eighth of this value. The D_4 volume element:

$$F_{D4} = \xi \cdot \eta_{D4} = \frac{27}{1\,600\,000} = 1.6875 \times 10^{-5}. \tag{L.3}$$

Numerical check: $4 \cdot 81 = 324$, $30000 \cdot 640 = 19\,200\,000$, $324/19\,200\,000 = 27/1\,600\,000$. ✓

L.3 Charged Leptons: Derivation from Winding Numbers

Why winding numbers?

Field modes on a compact torus T^2 are standing waves. The periodicity condition $\psi(x + 2\pi R) = \psi(x)$ permits only integer winding numbers $n \in \mathbb{Z}$. On the T^2 torus two independent circles carry winding numbers n_θ and n_ϕ.

[K] The three generations of charged leptons correspond to the three topologically distinguished (n_θ, n_ϕ) states forming the simplest rational ratios $r_i = n_\phi^2/n_\theta$:

Table L.1: FFGFT quantum numbers of the charged leptons

Particle	n_θ	n_ϕ	$r_i = n_\phi^2/n_\theta$	p_i
Electron e	3	2	$4/3$	$3/2$
Muon μ	5	4	$16/5$	1
Tau τ	9	5	$25/9$	$2/3$

The exponent ladder

[K] The exponents p_i form a geometric sequence with ratio $2/3$ per generation, $p_g = \frac{3}{2}\left(\frac{2}{3}\right)^{g-1}$:

$$p_e = \tfrac{3}{2}, \quad p_\mu = 1, \quad p_\tau = \tfrac{2}{3}. \tag{L.4}$$

The steps $3/2 \to 1 \to 2/3$ are $-1/2$ and $-1/3$; the constant quantity is the ratio $2/3$. The general mass formula:

$$m_i = r_i \cdot \xi^{p_i} \cdot v, \qquad v = 246.0 \text{ GeV}. \tag{L.5}$$

Step-by-step mass calculation

Pre-computed ξ powers:

$$\xi^{3/2} = 1.5396 \times 10^{-6}, \qquad \xi^{1} = 1.3\overline{3} \times 10^{-4}, \qquad \xi^{2/3} = 2.6099 \times 10^{-3}. \tag{L.6}$$

[K]

$$m_e = \tfrac{4}{3} \cdot 1.5396 \times 10^{-6} \cdot 246.0 \times 10^{3}\ \text{MeV} = \mathbf{0.50499\ MeV} \quad (\text{PDG: } 0.511\ \text{MeV},\ -1.18\,\%), \tag{L.7}$$

$$m_\mu = \tfrac{16}{5} \cdot 1.3\overline{3} \times 10^{-4} \cdot 246.0 \times 10^{3}\ \text{MeV} = \mathbf{104.96\ MeV} \quad (\text{PDG: } 105.66\ \text{MeV}), \tag{L.8}$$

$$m_\tau = \tfrac{25}{9} \cdot 2.6099 \times 10^{-3} \cdot 246.0 \times 10^{3}\ \text{MeV} = \mathbf{1783.4\ MeV} \quad (\text{PDG: } 1776.86\ \text{MeV}). \tag{L.9}$$

Parameter-free mass ratios

[K] The VEV v cancels in ratios:

$$m_\mu/m_e = \tfrac{12}{5} \cdot \xi^{-1/2} = 207.85 \quad (\text{PDG: } 206.77,\ +0.52\,\%), \tag{L.10}$$

$$m_\tau/m_\mu = \tfrac{125}{144} \cdot \xi^{-1/3} = 16.99 \quad (\text{PDG: } 16.82,\ +1.04\,\%), \tag{L.11}$$

$$m_\tau/m_e = \tfrac{25}{12} \cdot \xi^{-5/6} = 3532 \quad (\text{PDG: } 3477,\ +1.57\,\%). \tag{L.12}$$

Table L.2: Lepton masses and ratios: FFGFT vs. PDG

Quantity	Formula	Theory	PDG	Dev.	Status
m_e	$\frac{4}{3}\xi^{3/2}v$	0.50499 MeV	0.511 MeV	$-1.18\,\%$	[K]
m_μ	$\frac{16}{5}\xi^{1}v$	104.96 MeV	105.66 MeV	$-0.66\,\%$	[K]
m_τ	$\frac{25}{9}\xi^{2/3}v$	1783.4 MeV	1776.86 MeV	$+0.37\,\%$	[K]
m_μ/m_e	$\frac{12}{5}\xi^{-1/2}$	207.85	206.77	$+0.52\,\%$	[K]
m_τ/m_μ	$\frac{125}{144}\xi^{-1/3}$	16.99	16.82	$+1.04\,\%$	[K]

L.4 Neutrino Sector: Fixed-Point Localisation on the Orbifold

This section derives neutrino masses from orbifold fixed points (F_2, F_5, F_7, exponents $\{9/4, 2, 9/5\}$) and achieves Δm^2_{atm} with a deviation of $+16.0\,\%$ (with K_{frak} $+12.8\,\%$). Doc. 339/340 derives both mass differences from the $GF(27)^*$ Frobenius orbit structure and closes the atmospheric discrepancy at $1.0\,\%$ deviation [K]. The fixed-point derivation is a development step; the valid derivation is Doc. 340.

According to Doc. 343 (Theorem D‴), the quadratic character mod 13 (self-inverseness of the Frobenius orbit under $k \mapsto k^{-1}$) determines Majorana/Dirac for the Galois masses in Doc. 340: ν_1 (orbit O_1) and ν_2 (orbit O_3) are Majorana, ν_3 (orbit O_4) is a **Dirac neutrino [B]**. The fixed-point assignment F_2, F_5, F_7 in this section provides no Majorana/Dirac character; this follows only from the Galois algebra of Doc. 340/343.

Structural difference from charged leptons

Charged leptons arise as winding modes on the T^2 torus and couple to the Higgs VEV v. Neutrinos are fundamentally different: they are localised at the *geometric fixed points* of the $T^4/\mathbb{Z}_3$ orbifold – points x_0 satisfying $\tau(x_0) = x_0$. This topological distinction produces a double ξ-suppression compared to the charged leptons. For the formal Hilbert-space treatment of the fixed-point boundary conditions see Doc. 322, Eq. (15).

The neutrino mass formula

[K]

$$m_{\nu_i} = m_e \cdot \xi^{p_i}, \qquad m_e = 0.511 \text{ MeV}. \tag{L.13}$$

Table L.3: Fixed-point localisation of neutrino flavours (Doc. 322)

Fixed pt.	Position	Neutrino	Exponent p_i	Mass
F_2	$\frac{2\pi R}{3}(1,1,1,0)$	ν_1	$9/4$	0.976 meV
F_5	$\frac{4\pi R}{3}(1,1,1,0)$	ν_2	2	9.084 meV
F_7	$\frac{4\pi R}{3}(1,1,1,1)$	ν_3	$9/5$	54.11 meV

Step-by-step neutrino mass calculation

[K] Pre-computed ξ powers and masses:

$$\xi^{9/4} = 1.9103 \times 10^{-9}, \quad m_{\nu_1} = 511\,000 \text{ eV} \cdot 1.9103 \times 10^{-9} = \mathbf{0.976 \text{ meV}}, \tag{L.14}$$

$$\xi^{2} = 1.\overline{7} \times 10^{-8}, \quad m_{\nu_2} = 511\,000 \text{ eV} \cdot 1.\overline{7} \times 10^{-8} = \mathbf{9.084 \text{ meV}}, \tag{L.15}$$

$$\xi^{9/5} = 1.059 \times 10^{-7}, \quad m_{\nu_3} = 511\,000 \text{ eV} \cdot 1.059 \times 10^{-7} = \mathbf{54.11 \text{ meV}}. \tag{L.16}$$

Mass-squared differences and oscillation dynamics

Solar mass-squared difference Δm^2_{21}

$$\Delta m^2_{21} = (9.084 \text{ meV})^2 - (0.976 \text{ meV})^2 = 82.52 - 0.95 = 81.57 \text{ meV}^2 = \mathbf{8.16 \times 10^{-5}} \text{ eV}^2. \tag{L.17}$$

PDG 2023: 7.53×10^{-5} eV^2. Deviation: $+8.3\,\%$.

Atmospheric mass-squared difference Δm^2_{32}

$$\Delta m^2_{32} = (54.11 \text{ meV})^2 - (9.084 \text{ meV})^2 = 2927.9 - 82.5 = 2845.4 \text{ meV}^2 = \mathbf{2.845 \times 10^{-3}} \text{ eV}^2. \tag{L.18}$$

PDG (NuFIT 5.3): 2.453×10^{-3} eV^2. Deviation: $+16.0\,\%$.

Open point: atmospheric mass-squared difference

Δm^2_{21} agrees to $+8.3\,\%$, within the typical FFGFT range. Δm^2_{32} deviates by $+16.0\,\%$ – about twice as much and with the same sign: m_{ν_3} is too large. Equal sign and comparable magnitude point to a common cause rather than two independent problems.

The fixed-point exponents p_i are secured [K]; the source of the discrepancy is an open point. Three candidates were examined:

- K_{frak} **correction at fixed point** F_7 (Doc. 315): F_7 is the only fixed point in the full $(1,1,1,1)$ direction and thus carries a mass-circle component. With $K_{\text{frak}} = 1 - 100\xi = 74/75$ one obtains $m_3 = 53.39$ meV and $\Delta m^2_{32} = 2.768 \times 10^{-3}$ eV², i.e. $+12.8\,\%$ – an improvement, but not sufficient.
- **Mixing term** F_5–F_7: the fixed points are not fully orthogonal. An off-diagonal term in the mass matrix on the orbifold would shift both eigenvalues and would carry the right sign. This requires the complete mass matrix, not worked out here – the most promising candidate.
- **A different exponent** – unlikely: the exponent required for the PDG value is $p_3 = 1.8081$, only $0.45\,\%$ above $9/5 = 1.8000$. No rational fraction with small numerators lies closer ($16/9 \to +74\,\%$, $11/6 \to -37.5\,\%$, $20/11 \to -17.1\,\%$). The exponent $9/5$ is the best rational candidate and is grounded in the D_4 coordination.

The deviation therefore most likely does not stem from a wrong exponent but from an effect correcting $m_{\nu_i} = m_e \xi^{p_i}$ multiplicatively. Status: **[S]**.

Cosmological mass sum

$$\sum_i m_{\nu_i} = 0.976 + 9.084 + 54.11 \text{ meV} = 64.17 \text{ meV} = 0.0642 \text{ eV}. \tag{L.19}$$

Planck 2018 limit: $\sum m_\nu < 0.12$ eV. ✓

Table L.4: Neutrino spectrum: FFGFT prediction vs. experiment

Quantity	Formula	Prediction	Experiment	Status
m_{ν_1}	$m_e \xi^{9/4}$	0.976 meV	< 0.8 eV	[K]
m_{ν_2}	$m_e \xi^{2}$	9.084 meV	~ 8.7 meV	[K]
m_{ν_3}	$m_e \xi^{9/5}$	54.11 meV	~ 50 meV	[K]
Δm^2_{21}	$m^2_{\nu_2} - m^2_{\nu_1}$	8.16×10^{-5} eV²	7.53×10^{-5} eV²	[K] $+8.3\,\%$
Δm^2_{32}	$m^2_{\nu_3} - m^2_{\nu_2}$	2.845×10^{-3} eV²	2.453×10^{-3} eV²	[S] $+16.0\,\%$
$\sum m_\nu$	$\sum m_{\nu_i}$	0.0642 eV	< 0.12 eV	[K]

L.5 Symmetry Breaking and Gauge Structure

$SU(3)_c$: colour charges from $\mathbb{Z}_3$ fibration

[K] The $\mathbb{Z}_3$ symmetry generates three discrete complex-conjugate phase levels $\omega = e^{2\pi i/3}$, corresponding to the three colour charges Red, Green, Blue. The eight generators of $\mathfrak{su}(3)$ follow from the eight real degrees of freedom of the $\mathbb{Z}_3$ projection. **[B]** Full algebraic derivation of the Gell-Mann matrices from orbifold modes is in Doc. 321.

Electroweak bosons ($W^{\pm}$, Z^0)

[B]

$$m_W = \frac{1}{2}\, g\, v \approx 80.38 \text{ GeV}, \qquad m_Z = m_W/\cos\theta_W \approx 91.19 \text{ GeV}. \tag{L.20}$$

The numerical values insert the measured g and θ_W and are inputs, not a prediction: with $g = 0.6517$, $v = 246$ GeV one gets $m_W = 80.16$ GeV, 80.38 GeV requires $g = 0.6535$; $m_Z = m_W/\cos\theta_W$ with $\cos\theta_W = m_W/m_Z$ is circular.

L.6 Complete Overview of All Spectra

Table L.5: Systematic overview of all FFGFT spectra, Doc. 320

Group	Particle	Prediction	PDG	Status
Leptons	Electron e	0.50499 MeV	0.511 MeV	[K]
	Muon μ	104.96 MeV	105.66 MeV	[K]
	Tau τ	1783.4 MeV	1776.86 MeV	[K]
Neutrinos	ν_1 (F_2)	0.976 meV	< 0.8 eV	[K]
	ν_2 (F_5)	9.084 meV	~ 8.7 meV	[K]
	ν_3 (F_7)	54.11 meV	~ 50 meV	[K]
Oscillations	Δm^2_{21}	8.16×10^{-5} eV^2	7.53×10^{-5} eV^2	[K] +8.3 %
	Δm^2_{32}	2.845×10^{-3} eV^2	2.453×10^{-3} eV^2	[S] +16.0 %
	$\sum m_\nu$	0.0642 eV	< 0.12 eV	[K]
Bosons	γ, g	0	0	[K]
	$W^{\pm}$	80.38 GeV (input)	80.38 GeV	input
	Z^0	91.19 GeV (input)	91.19 GeV	input

The $W^{\pm}$ and Z^0 values insert the measured g and θ_W and are inputs; θ_W is not derived from ξ in this document (RG running from ξ: Doc. 323). For the neutrino rows the derivation in Doc. 340 is authoritative (see the neutrino-sector section).

L.7 Cross-references and Open Bridges

Table L.6: Derivation chain Doc. 320 and cross-references

Step	Doc.	Status
$\xi = 4/30000$ from D_4 lattice	Docs. 001, 205	[K]
Higgs consistency check of ξ ($\approx 2.3\,\%$)	Doc. 354, A190	[K]
K_{frac} interface factor	Docs. 012, 133	[K]
$\tilde{T} \cdot m = 1$	Doc. 003	[SETZUNG]
Winding numbers (n_θ, n_ϕ)	Doc. 317	[K]
Lepton masses m_e, m_μ, m_τ	Docs. 006, 317	[K]
Hilbert-space embedding, operator $\hat{F}$	Doc. 322	[K/S]
Fixed-point positions F_2, F_5, F_7	Doc. 322	[K]
Δm^2_{21} ($+8\,\%$)	this Doc.	[K]
Δm^2_{32} ($+16.0\,\%$)	this Doc.	[S] open
$SU(3)_c$ from $\mathbb{Z}_3$	Doc. 321	[B]
Weinberg angle θ_W from ξ	Doc. 323	[K]
Quark sector (hadrons)	Docs. 318, 319	open bridge

Chapter M

Doc. 321 — Algebraic Derivation of the $SU(3)_c$ Gauge Structure

List of Symbols

Symbol	Meaning	Value / Unit
$\mathbb{Z}_3$	cyclic group of order 3	$\{e, g, g^2\}$, $g^3 = e$
ω	primitive $\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
τ	generator of the $\mathbb{Z}_3$ action on T^4	unitary
P_k	projectors onto ω^k eigensectors	$k = 0, 1, 2$
$\mathfrak{su}(3)$	Lie algebra of $SU(3)$	dim 8
λ_a	Gell-Mann matrices, $a = 1, \ldots, 8$	Hermitian, traceless
f_{abc}	$\mathfrak{su}(3)$ structure constants	totally antisymm.
T_R	triality representation index	$T_R \in \{0, 1, 2\}$ mod 3
$\mathbf{3}, \bar{\mathbf{3}}, \mathbf{8}$	irreducible $SU(3)$ repr.	quark, antiquark, gluon
N_c	number of colour degrees of freedom	$N_c = 3$
α_s	strong coupling constant	$3\xi^{1/4}$ at m_τ (Doc. 160)
ξ	FFGFT geometry parameter	$4/30000$
$[K]$	derivable from ξ	status marker
$[B]$	algebraically proved	status marker
$[S]$	sketched / plausible	status marker

Summary

Doc. 321 carries out the algebraic derivation marked [S] open in Docs. 319, 320 and 322: the Gell-Mann matrices and the complete $SU(3)_c$ Lie algebra emerge from the $\mathbb{Z}_3$ triality of the $T^4/\mathbb{Z}_3$ orbifold. The proof proceeds in four steps: (1) the $\mathbb{Z}_3$ projectors on $L^2(T^4)$ yield a decomposition into three eigensectors; (2) the bilinear transitions between these sectors generate eight linearly independent operators; (3) these eight operators satisfy the $\mathfrak{su}(3)$ commutation relations; (4) the triality index distinguishes quarks ($T_R = 1, 2$) from gluons ($T_R = 0$) and explains confinement structurally.

Status markers

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

M.1 Starting Point: $\mathbb{Z}_3$ as the Centre of $SU(3)$

The centre of a Lie group

The centre $Z(G)$ of a group G consists of all elements that commute with every other element: $Z(G) = \{z \in G : zg = gz \ \forall g \in G\}$.

[K] For $SU(3)$:

$$Z(SU(3)) = \mathbb{Z}_3 = \{\mathbf{1},\ \omega\mathbf{1},\ \omega^2\mathbf{1}\}, \quad \omega = e^{2\pi i/3}. \tag{M.1}$$

[K] This is not a coincidence: $\mathbb{Z}_3 = Z(SU(3))$ means that any $\mathbb{Z}_3$ symmetry embeddable into the unitary group leads automatically into the $SU(3)$ structure. The $\mathbb{Z}_3$ action on $T^4/\mathbb{Z}_3$ is therefore not an additional FFGFT assumption but the minimal unitary extension of the $\mathbb{Z}_3$ centre to a full gauge group.

Triality: the $\mathbb{Z}_3$ charge of $SU(3)$ representations

[B] Every irreducible representation of $SU(3)$ has a *triality* $T_R \in \{0, 1, 2\}$, defined by the $\mathbb{Z}_3 = Z(SU(3))$ action:

$$\omega\mathbf{1} \xrightarrow{\text{repr. } R} \omega^{T_R} \cdot \mathbf{1}_R. \tag{M.2}$$

Representation	Dimension	Triality T_R	Physical role
$\mathbf{1}$	1	0	singlet (colour-neutral)
$\mathbf{8}$	8	0	gluons (adjoint)
$\mathbf{3}$	3	1	quarks
$\bar{\mathbf{3}}$	3	2	antiquarks

[K] Confinement is equivalent to requiring all observable states to have triality $T_R = 0$ (Doc. 049: “nodes cannot exist alone, must form neutral combinations”).

M.2 The $\mathbb{Z}_3$ Sector Decomposition on $L^2(T^4)$

Three projectors from one $\mathbb{Z}_3$ action

[K] The $\mathbb{Z}_3$ group $\{e, g, g^2\}$ acts on T^4 through the unitary operator τ ($\tau^3 = \mathrm{id}$). On $L^2(T^4)$ this induces three orthogonal projectors:

$$P_k = \frac{1}{3}\sum_{j=0}^{2} \omega^{-jk}\tau^j, \quad k = 0, 1, 2. \tag{M.3}$$

[B] These projectors are pairwise orthogonal and complete:

$$P_k^2 = P_k, \quad P_k P_l = \delta_{kl} P_k, \quad P_0 + P_1 + P_2 = \mathrm{id}. \tag{M.4}$$

Proof of idempotency: $P_k^2 = \frac{1}{9}\sum_{j,j'} \omega^{-k(j+j')}\tau^{j+j'} = \frac{1}{9}\sum_n 3\,\omega^{-kn}\tau^n = P_k$. ✓

The three eigensectors and their modes

[K] In the Fourier basis $e^{in\cdot x/R}$ the $\mathbb{Z}_3$ action $(x_1, x_2, x_3, x_4) \mapsto (x_2, x_3, x_1, x_4)$ maps (n_1, n_2, n_3) to (n_3, n_1, n_2). The sum $n_1 + n_2 + n_3$ is unchanged under τ and therefore does not fix the sector. A single Fourier mode is an eigenvector of τ only for $n_1 = n_2 = n_3$ (then in $\mathscr{H}_0$). The eigenstates are orbit combinations

$$\psi_{n,k} = \sum_{j=0}^{2} \omega^{-jk}\,\tau^j\, e^{in\cdot x/R} \in \mathscr{H}_k, \qquad \tau\psi_{n,k} = \omega^k\,\psi_{n,k}; \tag{M.5}$$

each three-element orbit $\{n, \tau n, \tau^2 n\}$ yields one state each in $\mathscr{H}_0$, $\mathscr{H}_1$ and $\mathscr{H}_2$. Thus the elements of the orbit of $(1, 0, 0)$ all have sum 1, and the orbit still yields one state in each of the three sectors:

- $\mathscr{H}_0$: colour-neutral (gluons, singlets).
- $\mathscr{H}_1$: quarks.
- $\mathscr{H}_2$: antiquarks.

M.3 Algebraic Construction of the Eight Gell-Mann Operators

Bilinear transitions between sectors

[B] The diagonal basis (e_0, e_1, e_2) with $\tau e_k = \omega^k e_k$ corresponds to the three colour states (Red $= e_0$, Green $= e_1$, Blue $= e_2$). The $3^2 - 1 = 8$ traceless Hermitian operators on $\mathbb{C}^3$ are constructed as:

Transitions $\mathscr{H}_0 \leftrightarrow \mathscr{H}_1$: $T_1 = \frac{1}{2}(|e_0\rangle\langle e_1| + |e_1\rangle\langle e_0|)$, $T_2 = \frac{i}{2}(|e_1\rangle\langle e_0| - |e_0\rangle\langle e_1|)$.

Transitions $\mathscr{H}_0 \leftrightarrow \mathscr{H}_2$: $T_4 = \frac{1}{2}(|e_0\rangle\langle e_2| + |e_2\rangle\langle e_0|)$, $T_5 = \frac{i}{2}(|e_2\rangle\langle e_0| - |e_0\rangle\langle e_2|)$.

Transitions $\mathscr{H}_1 \leftrightarrow \mathscr{H}_2$: $T_6 = \frac{1}{2}(|e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|)$, $T_7 = \frac{i}{2}(|e_2\rangle\langle e_1| - |e_1\rangle\langle e_2|)$.

Diagonal operators: $T_3 = \frac{1}{2}(|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|)$, $T_8 = \frac{1}{2\sqrt{3}}(|e_0\rangle\langle e_0| + |e_1\rangle\langle e_1| - 2|e_2\rangle\langle e_2|)$.

Identification with the Gell-Mann matrices

[B] $T_a = \frac{1}{2}\lambda_a$ with:

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$
$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{M.6}$$

M.4 Proof of the $\mathfrak{su}(3)$ Commutation Relations

Normalisation condition

[B] Probe for $a = b = 1$:

$$\mathrm{tr}(T_1 T_1) = \tfrac{1}{4}\mathrm{tr}(\lambda_1^2) = \tfrac{1}{4}\mathrm{tr}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \tfrac{2}{4} = \tfrac{1}{2}. \quad \checkmark \tag{M.7}$$

Selected commutators

[B]

$[T_1, T_2] = iT_3$:

$$[T_1, T_2] = \tfrac{1}{4}[\lambda_1, \lambda_2] = \tfrac{i}{2}\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = iT_3. \quad \checkmark \tag{M.8}$$

$[T_1, T_4] = \frac{i}{2}T_7$: T_1 connects $\mathscr{H}_0 \leftrightarrow \mathscr{H}_1$, T_4 connects $\mathscr{H}_0 \leftrightarrow \mathscr{H}_2$, so their commutator falls in the $\mathscr{H}_1 \leftrightarrow \mathscr{H}_2$ sector. The imaginary transition T_7 appears (not T_6):

$$[T_1, T_4] = \tfrac{1}{4}[\lambda_1, \lambda_4] = \tfrac{i}{2}T_7. \quad \checkmark \tag{M.9}$$

Structure constant: $f_{147} = 1/2$ (canonical $\mathfrak{su}(3)$ value).

$[T_3, T_8] = 0$: both diagonal; diagonal matrices commute. ✓

T_3 and T_8 generate the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{su}(3)$ (maximum abelian subalgebra, dimension 2).

Main result Doc. 321 [B]

The eight bilinear transition operators between the three $\mathbb{Z}_3$ eigensectors of $L^2(T^4)$ satisfy the $\mathfrak{su}(3)$ Lie algebra relations $[T_a, T_b] = if_{abc}T_c$ with the canonical structure constants and the normalisation condition $\mathrm{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$. The $SU(3)_c$

gauge group emerges algebraically necessarily from the $\mathbb{Z}_3$ triality of the $T^4/\mathbb{Z}_3$ orbifold.

M.5 Physical Interpretation

Eight gluons as transition operators

[K] The eight gluons correspond to the eight bilinear transition operators. Gluons have triality $T_R = 0$ (adjoint representation): the centre element $\omega\mathbf{1}$ acts trivially on the adjoint. In the sector decomposition only the diagonal operators T_3, T_8 have $\Delta k = 0$; the off-diagonal generators $T_1, T_2, T_4, \ldots, T_7$ shift k by $\pm 1 \pmod 3$ (e.g. T_1 connects $\mathscr{H}_0$ and $\mathscr{H}_1$). Triality 0 of the gluons is a property of the representation and does not follow from $\Delta k = 0$ in this decomposition. A quark (triality 1) emitting a gluon (triality 0) remains a quark (triality 1).

Confinement as triality selection

[K]

State	Composition	Triality	Observable?
Meson ($q\bar{q}$)	$T_R = 1 + 2$	$0 \pmod 3$	Yes
Baryon (qqq)	$T_R = 1 + 1 + 1$	$0 \pmod 3$	Yes
Single quark	$T_R = 1$	1	No

[B] A single quark has $T_R = 1 \neq 0$ and is therefore not a $\mathbb{Z}_3$ singlet. Its state function is not localised in $\mathscr{H}_0$ and cannot propagate as a free asymptotic state – the geometric confinement criterion (Doc. 049 in algebraic form).

M.6 Connection to the Strong Coupling α_s

[K] Doc. 160 derives the strong coupling from orbifold geometry. With $N_c = 3$ now algebraically proved (three $\mathbb{Z}_3$ eigensectors):

$$\alpha_s(m_\tau) = N_c \cdot \xi^{1/(N_c+1)} = 3 \cdot \xi^{1/4} = 3 \cdot \left(\frac{4}{30000}\right)^{1/4} \approx 0.322. \tag{M.10}$$

PDG: $\alpha_s(m_\tau) \approx 0.330$. Deviation: $-2.3\,\%$.

[B] The factor $N_c = 3$ in Eq. (M.10) was an input in Doc. 160. Doc. 321 derives it algebraically: N_c is the order of the $\mathbb{Z}_3$ group, i.e. the number of eigensectors of the $\mathbb{Z}_3$ action on $L^2(T^4)$. Eq. (M.10) is now fully parameter-free.

[K] Why $N_c = 3$ and not $N_c = 2$ or $N_c = 4$:

- $N_c = 3$ is the order of $\mathbb{Z}_3 = Z(SU(3))$ – the centre of the smallest simple compact non-abelian Lie group whose centre contains $\mathbb{Z}_3$; E_6 (centre $\mathbb{Z}_3$) and $SU(3k)$ (centre $\mathbb{Z}_{3k}$) also contain $\mathbb{Z}_3$ in their centre (the order of the centre is the determinant of the Cartan matrix: A_2: 3, A_5: 6, A_8: 9, E_6: 3).

- D_4 triality (Doc. 314) enforces exactly three classes of spin representations.
- For $\mathbb{Z}_n$ with $n > 3$ the $T^4/\mathbb{Z}_n$ orbifold has fewer independent fixed points, destroying the neutrino sector (Doc. 320).

M.7 $SU(2)_L$ and $U(1)_Y$ from the Orbifold Sector Structure

Remaining torus degrees of freedom after $\mathbb{Z}_3$ projection

[K] The $\mathbb{Z}_3$ projection acts on the first three torus directions (x_1, x_2, x_3). The fourth direction x_4 is $\mathbb{Z}_3$-invariant and remains unfolded:

$$T^4/\mathbb{Z}_3 = \underbrace{T^3/\mathbb{Z}_3}_{SU(3)_c} \times \underbrace{T^1}_{U(1)_Y} . \tag{M.11}$$

Within the $\mathbb{Z}_3$-invariant sector $\mathscr{H}_0$ a $\mathbb{Z}_2$ symmetry between the ω- and ω^2-sectors ($\mathscr{H}_1$ and $\mathscr{H}_2$) provides the source of $SU(2)_L$.

$U(1)_Y$ from the fourth torus direction

[B] The fourth torus direction carries a global $U(1)$ phase symmetry $x_4 \mapsto x_4 + \alpha$ with generator $\hat{Y} = \partial/\partial\alpha|_{\alpha=0}$. It acts as $\hat{Y}|n_4\rangle = n_4|n_4\rangle$, quantising hypercharge by the fourth winding number. Compatibility with $\mathbb{Z}_3$ forces $Y \in \frac{1}{3}\mathbb{Z}$.

$SU(2)_L$ from the $\mathbb{Z}_2$ pairing of sectors

[B] Complex conjugation $\mathscr{C} : \mathscr{H}_1 \leftrightarrow \mathscr{H}_2$ is a $\mathbb{Z}_2$ symmetry. In the reduced 2×2 sub-algebra of the $\mathscr{H}_1$-$\mathscr{H}_2$ transitions it generates the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ with $[\sigma_i/2, \sigma_j/2] = i\varepsilon_{ijk}\sigma_k/2$ – the $\mathfrak{su}(2)$ algebra, yielding $SU(2)_L$.

Weinberg angle from the trace formula of the five orbifold states

[B] The five FFGFT orbifold states and their quantum numbers:

State	Sector	T_3	Q	$Y = 2(Q - T_3)$
d_R, d_G, d_B	$\mathscr{H}_1$ (triality 1)	0	$-1/3$	$-2/3$
e^-	$\mathscr{H}_2$ (triality 2)	$-1/2$	-1	-1
ν_e	$\mathscr{H}_0$ (triality 0)	$+1/2$	0	-1

The Weinberg angle follows from the trace formula in the 5-plet:

$$\sin^2\theta_W = \frac{\mathrm{tr}[T_3^2]}{\mathrm{tr}[Q^2]} = \frac{3 \cdot 0 + (1/2)^2 + (1/2)^2}{3 \cdot (1/3)^2 + 1^2 + 0^2} = \frac{1/2}{4/3} = \boxed{\frac{3}{8} = 0.375.} \tag{M.12}$$

[B] This is a direct algebraic consequence of the orbifold sector structure with $N_c = 3$ and coincides with the classical $SU(5)$-GUT result of Georgi and Glashow (1974), now derived from FFGFT geometry.

RG running to M_Z

[K] The measured value $\sin^2\theta_W(M_Z) = 0.2312$ (PDG) differs from the GUT-point value by $\Delta\sin^2\theta_W = 3/8 - 0.2312 = 0.1438$, which is the well-known RG running from M_{GUT} to M_Z.

[K] The formal derivation of the dependence of $\Delta\sin^2\theta_W$ on ξ is carried out in Doc. 323. The result $\sin^2\theta_W(M_Z) = 3/8 - \frac{55\alpha_{\text{em}}}{24\pi}[\ln(m_{\text{Pl}}/M_Z) + \frac{19}{12}\ln\xi] = 0.2308$ deviates $-0.19\,\%$ from PDG.

Status balance $SU(2)_L \times U(1)_Y$

$U(1)_Y$: **[B]** from the fourth torus direction.
$SU(2)_L$: **[B]** from the $\mathbb{Z}_2$ pairing $\mathscr{H}_1 \leftrightarrow \mathscr{H}_2$.
$\sin^2\theta_W|_{\text{GUT}} = 3/8$: **[B]** from the trace formula of the five orbifold states.
$\sin^2\theta_W(M_Z) = 0.2308$: **[K]** RG running derived from ξ in Doc. 323 ($-0.19\,\%$).

Complete gauge group picture

Factor	Origin	Generators	Status
$SU(3)_c$	$\mathbb{Z}_3$ triality on $T^3 \subset T^4$	8	[B] §3–4
$SU(2)_L$	$\mathbb{Z}_2$ pairing $\mathscr{H}_1 \leftrightarrow \mathscr{H}_2$	3	[B] §7.3
$U(1)_Y$	4th torus direction T^1	1	[B] §7.2
$\sin^2\theta_W\|_{\text{GUT}} = 3/8$	trace formula	—	[B] §7.4
$\sin^2\theta_W(M_Z)$	RG running	—	[K] Doc. 323
Full SM gauge group	$SU(3)_c \times SU(2)_L \times U(1)_Y$	12	[B]/[K]

M.8 Cross-References and Corpus

Overall status Doc. 321

The complete algebraic derivation of the $SU(3)_c$ gauge structure is complete **[B]**. $N_c = 3$ is algebraically necessary; $\alpha_s(m_\tau) = 3\xi^{1/4}$ is fully parameter-free. $U(1)_Y$ (fourth torus direction) and $SU(2)_L$ ($\mathbb{Z}_2$ sector pairing) are algebraically derived **[B]**. The Weinberg angle at the GUT point $\sin^2\theta_W = 3/8$ follows from the trace formula of the five orbifold states **[B]**. The full SM gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ is algebraically founded. Doc. 323 derives the RG running $\sin^2\theta_W(\text{GUT}) \to \sin^2\theta_W(M_Z)$ from ξ **[K]**, deviation $-0.19\,\%$.

Table M.1: Derivation chain Doc. 321 and corpus references

Statement	Doc.	Status
$\mathbb{Z}_3 = Z(SU(3))$ (centre)	group theory	[B]
Triality as $\mathbb{Z}_3$ charge	repr. theory	[B]
$\mathbb{Z}_3$ projectors on $L^2(T^4)$	this Doc.	[B]
Three eigensectors $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$	this Doc.	[K]
Eight bilinear transition operators	this Doc.	[B]
$\mathfrak{su}(3)$ commutation relations	this Doc.	[B]
Gell-Mann matrices from sector transitions	this Doc.	[B]
$N_c = 3$ algebraically derived	this Doc.	[B]
$\alpha_s(m_\tau) = 3\xi^{1/4}$ (now complete)	Doc. 160 + this	[K]
Confinement as triality selection	Doc. 049, this	[K]
D_4 triality, $\mathbb{Z}_3$ centre	Doc. 314	[K]
Gluon topology (flux tubes)	Docs. 145, 319	[S]
$U(1)_Y$ from 4th torus direction	this Doc.	[B]
$SU(2)_L$ from $\mathbb{Z}_2$ sector pairing	this Doc.	[B]
$\sin^2\theta_W\vert_{\mathrm{GUT}} = 3/8$ (trace formula)	this Doc.	[B]
$\sin^2\theta_W(M_Z)$ RG running from ξ	Doc. 323	[K]

Chapter N

Doc. 322 — Spectral Theory and Hilbert-Space Embedding of the FFGFT Tori

List of Symbols

Symbol	Meaning	Value / Unit
$\mathscr{H}$	Hilbert space of physical states	separable
$\hat{F}$	fundamental fractal spectral operator	self-adjoint on P_0
$\hat{F}_{D_4}$	D_4 sub-operator	eigenvalue ξ
$\sigma(\hat{F})$	spectrum of $\hat{F}$	$\subset \mathbb{R}$ on P_0
$E(\lambda)$	spectral projection measure	$E(\lambda)^2 = E(\lambda)$
$\mathscr{D}(\hat{F})$	domain of $\hat{F}$	dense in $\mathscr{H}$
$P_{\mathbb{Z}_3}$	$\mathbb{Z}_3$ projection operator	idempotent
$\hat{\Delta}_{T^4}$	Laplacian on $T^4/\mathbb{Z}_3$	essentially self-adj.
$\hat{W}$	winding-number operator	eigenvalues (n_θ, n_ϕ)
D_f	fractal dimension of the orbifold	$3 - \xi$
D_f^{frac}	approximate fractal dimension	2.94 (no ξ-derivation)
K_{frac}	interface factor rolled → SI	$1 - 100\xi = 74/75$ (Doc. 133)
ξ	FFGFT geometry parameter	$4/30000$
ω	$\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
$[K]$	derivable from ξ	status marker
$[B]$	algebraically proved	status marker
$[S]$	sketched / plausible	status marker
[SETZUNG]	axiomatic setting	status marker

Summary

Doc. 322 develops the mathematical foundation of FFGFT spectral theory. The mass formulae of Doc. 320 receive here their rigorous Hilbert-space embedding: the mass

operator $\hat{M}_f$ on the state space $\mathscr{H}_{\text{FFGFT}}$ has the particle masses as eigenvalues; the geometry parameter $\xi = 4/30000$ is the smallest eigenvalue of the D_4 sub-operator of $\hat{F}$. The mass formulae of Docs. 006, 317 and 320 are not phenomenological ansätze but eigenvalue equations of a self-adjoint operator.

Status markers

All statements carry FFGFT status markers:
[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

N.1 Foundations of Spectral Theory in Hilbert Spaces

Motivation: from matrices to operators

In linear algebra a symmetric $n \times n$ matrix A possesses exactly n real eigenvalues and an orthonormal eigenbasis, enabling the diagonal decomposition $A = UDU^T$.

In quantum mechanics – and in FFGFT – operators act on *infinite-dimensional* Hilbert spaces. The momentum operator, the Hamiltonian, and the FFGFT spectral operator have no finite-dimensional analogue. Spectral theory generalises the eigenvalue concept to this case.

The Hilbert space: definition and physical meaning

Definition (Hilbert space). A Hilbert space $\mathscr{H}$ is a complete inner-product vector space. Its elements have finite norm $\|\psi\| = \sqrt{\langle\psi,\psi\rangle} < \infty$, which in position space means

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx < \infty. \tag{N.1}$$

Only such square-integrable wave functions are physically realisable – they describe normalisable probability densities.

Note: Eigenstates of operators with *continuous* spectrum – e.g. position eigenstates $|y\rangle$ with $\langle x|y\rangle = \delta(x-y)$ – are not normalisable and lie outside $\mathscr{H}$ itself, in the extended distributional space. In FFGFT this issue arises in the IR limit; the sub-Planck regime has a purely discrete spectrum.

The spectrum of a linear operator

For a linear operator T on a Hilbert space the spectrum $\sigma(T)$ generalises the set of all eigenvalues:

$$\sigma(T) = \{\lambda \in \mathbb{C} : (T - \lambda I) \text{ not invertible with bounded inverse}\}. \tag{N.2}$$

One distinguishes:

- **Point spectrum** $\sigma_p(T)$**:** "true" eigenvalues with $T\psi = \lambda\psi$, $\psi \neq 0$.

- **Continuous spectrum $\sigma_c(T)$:** $(T - \lambda I)$ injective but not surjective; no normalised eigenvector.
- **Absolutely continuous and singular spectrum:** important for Schrödinger operators on fractal spaces.

The spectral theorem: centrepiece of the theory

[B] For self-adjoint operators ($T = T^\dagger$ with suitable domain) the spectral theorem holds: there exists a unique *spectral measure* E – a projection-valued function $\lambda \mapsto E(\lambda)$ – such that

$$T = \int_{\sigma(T)} \lambda \, dE(\lambda). \tag{N.3}$$

For compact operators Eq. (N.3) reduces to the familiar discrete sum $T = \sum_i \lambda_i \langle \cdot, e_i \rangle e_i$. Self-adjointness guarantees $\sigma(T) \subset \mathbb{R}$ – real spectra, the necessary condition for measurable observable values.

N.2 The Fundamental Fractal Operator $\hat{F}$

Definition and Hilbert-space structure

[SETZUNG] FFGFT postulates a quantum-logical fractal operator $\hat{F}$ acting on a separable Hilbert space $\mathscr{H}$, formally encoding the fractal self-similarity of the sub-Planck geometry:

$$\hat{F} \equiv \bigoplus_{n=1}^{\infty} S_n \circ L_n, \tag{N.4}$$

with scale operators $S_n = r_n U_n$ ($r_n \in (0,1)$ contraction rates, U_n unitary) and logical projectors L_n satisfying $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$.

The domain

$$\mathscr{D}(\hat{F}) = \left\{ \psi \in \mathscr{H} \,:\, \sum_{n=1}^{\infty} \| S_n (L_n \psi) \|^2 < \infty \right\} \tag{N.5}$$

is dense in $\mathscr{H}$. **[B]** On the $\mathbb{Z}_3$-invariant sector P_0, $\hat{F}$ is self-adjoint: since $L_0 = \xi \cdot \ell_P$ (Doc. 180) renders the scale ladder finite, $\sum_n r_n < \infty$ and hence $\hat{F}$ is bounded with $\mathscr{D}(\hat{F}) = \mathscr{H}$; symmetry follows from the $\mathbb{Z}_3$ pairing $(k, -k)$ on the $\mathbb{Z}_3$-invariant sector P_0. The deficiency indices are $(0,0)$ – there is exactly one self-adjoint realisation (Doc. 327). On the full space $\hat{F}$ is normal under the pairing, $\varphi_{-k} = \bar{\varphi}_k$ (Doc. 327, Theorem 3); only $\mathrm{Re}\,\hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint there, and its use is a choice.

ξ as smallest eigenvalue of the D_4 sub-operator

[K] The geometry parameter is not a free parameter but a spectral value:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}. \tag{N.6}$$

This explains structurally why ξ appears as the base of all mass formulae: it is not a chosen parameter but the lowest mode of the geometry operator.

N.3 The FFGFT State Space

Tensor-product structure

[K] The physical state space:

$$\mathscr{H}_{\text{FFGFT}} = \mathscr{H}_{\text{geom}} \otimes \mathscr{H}_{\text{spin}} \otimes \mathscr{H}_{\text{flavor}}, \tag{N.7}$$

with:

1. $\mathscr{H}_{\text{geom}} = L^2(T^4/\mathbb{Z}_3, d\mu_f)$: square-integrable functions on the fractal orbifold.
2. $\mathscr{H}_{\text{spin}} = \mathbb{C}^{2\lceil D_f \rceil}$: spinor space; $\lceil D_f \rceil = 3$.
3. $\mathscr{H}_{\text{flavor}} = \ell^2(\mathbb{N}^2)$: flavour space with winding numbers (n_θ, n_ϕ).

The fractal measure $d\mu_f$ with 100-fold recursion

[K] The fractal measure on $T^4/\mathbb{Z}_3$ is built by 100 successive fractalisations of the Lebesgue measure:

$$d\mu_f^{(0)}(x) = dx, \tag{N.8}$$

$$d\mu_f^{(k+1)}(x) = \prod_{i=1}^{4} \left(\sum_{n=1}^{100} c_n^{(k)}\, \delta\!\left(x_i - x_i^{(n)}\right) \right) d\mu_f^{(k)}(x), \tag{N.9}$$

with $c_n^{(k)} = n^{-(D_f^{(k)}-3)}$, $D_f^{(k)} = 3 - \xi^{(k)}$, $\xi^{(k)} = (4/30000)(1 - k/100)$. By this scheme $\xi^{(100)} = 0$ and hence $D_f^{(100)} = 3$. The product of the coefficients $c_n^{(k)} = n^{\xi^{(k)}}$ over $k = 0, \dots, 99$ gives $n^{\xi \sum_k (1-k/100)} = n^{50.5\,\xi}$; after 100 steps the measure reads

$$\boxed{d\mu_f(x) = \prod_{i=1}^{4} \left(\sum_{n=1}^{100} n^{50.5\,\xi}\, \delta\!\left(x_i - x_i^{(n)}\right) \right) dx_i}, \quad \xi = \tfrac{4}{30000}, \tag{N.10}$$

with $n^{50.5\,\xi} \approx 1 + 0.0067 \ln n$ (about 1.031 at $n = 100$). The first step $k = 0$ alone has the coefficients n^{ξ}. $D_f = 3 - \xi$ as a limit does not follow from the recursion in this form.

N.4 The Torus T^4 in Hilbert Space

Fourier decomposition on T^4

[B]

$$L^2(T^4) \cong \bigoplus_{n \in \mathbb{Z}^4} \mathbb{C} \cdot e^{i n \cdot x / R}, \qquad \Delta_{T^4}\, e^{i n \cdot x/R} = -\tfrac{|n|^2}{R^2}\, e^{i n \cdot x/R}. \tag{N.11}$$

The $\mathbb{Z}_3$ orbifold projection

[K]

$$P_{\mathbb{Z}_3} = \tfrac{1}{3} \sum_{k=0}^{2} \tau^k, \qquad \mathscr{H}(T^4/\mathbb{Z}_3) = \mathsf{Fix}(P_{\mathbb{Z}_3}) \subset L^2(T^4). \tag{N.12}$$

A state lies in the orbifold space exactly when $\tau\psi = \psi$.

Fixed points and neutrino localisation

[K] The boundary condition at a fixed point x_0:

$$\psi_\chi(x_0 + y) = \chi \cdot \psi_\chi(x_0 + g_*(y)). \tag{N.13}$$

The three neutrino flavours are localised at fixed points (Doc. 320):

$$\nu_1 \leftrightarrow F_2 = \tfrac{2\pi R}{3}(1,1,1,0), \quad \nu_2 \leftrightarrow F_5 = \tfrac{4\pi R}{3}(1,1,1,0), \quad \nu_3 \leftrightarrow F_7 = \tfrac{4\pi R}{3}(1,1,1,1). \tag{N.14}$$

N.5 The Complete Spectral Operator

Tensor-product form

[K]

$$\hat{F} = \hat{\Delta}_{T^4} \otimes \hat{S} \otimes \hat{W}, \qquad \hat{M}_f = v \cdot \xi^{\hat{p}} \otimes \hat{r}. \tag{N.15}$$

Commutator conditions and MASA

[K]

$$[\hat{F}, \hat{S}] = 0, \qquad [\hat{n}_\theta, \hat{n}_\phi] = 0. \tag{N.16}$$

The MASA $\mathscr{A} = \mathrm{span}\{\hat{n}_\theta, \hat{n}_\phi, \hat{p}\}$ yields the complete spectral labelling (n_θ, n_ϕ, p).

N.6 Generalised Fourier Transform on $T^4/\mathbb{Z}_3$

[K] Projection kernel and GFT:

$$K_\lambda^{(p)}(x) = \sum_{n \in \mathbb{Z}^4} \frac{\delta_{\lambda, \Lambda_n^{(p)}}}{|n|^{D_f/2}}\, e^{in\cdot x/R}, \qquad \Lambda_n^{(p)} = \frac{|n|^2}{R^2} \cdot \xi^p. \tag{N.17}$$

$$\tilde{\psi}(\lambda) = \langle K_\lambda^{(p)}, \psi \rangle_{\mathscr{H}} = \int_{T^4/\mathbb{Z}_3} \overline{K_\lambda^{(p)}(x)}\, \psi(x)\, d\mu_f(x). \tag{N.18}$$

N.7 Spectral Derivation of Particle Masses

Charged leptons as torus excitations

[K] The mass eigenvalue equation $\hat{M}_{e,\mu,\tau}|n_\theta, n_\phi, p\rangle = m_i |n_\theta, n_\phi, p\rangle$ gives:

$$m_i = \frac{n_\phi^2}{n_\theta} \cdot \xi^{p_i} \cdot v. \tag{N.19}$$

Particle	n_θ	n_ϕ	p_i	m_i
Electron e	3	2	$3/2$	0.505 MeV
Muon μ	5	4	1	104.96 MeV
Tau τ	9	5	$2/3$	1783.4 MeV

The table values are formula values with $v = 246$ GeV. The electron formula value $(4/3)\xi^{3/2} \cdot 246$ GeV $= 0.5050$ MeV (with $v = 246.22$ GeV: 0.5054 MeV) lies $1.2\,\%$ below the measured 0.511 MeV.

Neutrinos as fixed-point eigenstates

[K]

$$P_{\mathbb{Z}_3}|\nu_i\rangle = |\nu_i\rangle, \qquad m_{\nu_i} = m_e \cdot \xi^{p_i}. \tag{N.20}$$

Neutrino	Fixed point	$\dim_{\text{loc}}$	p_i	m_{ν_i}
ν_1	F_2	$2 + 1/4$	$9/4$	0.976 meV
ν_2	F_5	2	2	9.084 meV
ν_3	F_7	$2 - 1/5$	$9/5$	54.11 meV

Spectral selection rule for the exponent ladder

[S] The commutator condition

$$\hat{F} \cdot \hat{\xi}^{1/3} = \hat{\xi}^{1/3} \cdot \hat{F} \tag{N.21}$$

makes the spectrum invariant under scaling by $\xi^{1/3}$. Exponents on the same orbit differ by multiples of $1/3$:

$$\Delta p = \frac{k}{3}, \quad k \in \mathbb{Z} \quad \text{(within one orbit)}, \tag{N.22}$$

as for $\mu \to \tau$ ($\Delta p = 1/3 = 1/D$, Doc. 158), with ν_2 on the same orbit. The exponent ladder spans several orbits: p_e, p_{ν_1} and p_{ν_3} do not lie on the $1/3$ grid of the muon orbit. The rule therefore describes an *orbit structure* of the operator $\hat{F}$; the commutator condition itself is not derived.

N.8 Fractal Dimension and K_{frac}

[K] Exact FFGFT value (it does not follow from the recursion above in its present form):

$$\boxed{D_f = 3 - \xi = 3 - \frac{4}{30000} = \frac{22499}{7500} \approx 2.9998\overline{6}}. \tag{N.23}$$

The interface factor from Doc. 133:

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9866\overline{6}. \tag{N.24}$$

In Doc. 133 the factor 100 is set, not derived; the value $1 - 100\xi$ is confirmed by comparing the two E_0 routes with α, the form of the factor is open **[S]**. In mass ratios K_{frac} cancels exactly (Doc. 012).

Table N.1: All spectral states of the FFGFT and their Hilbert-space components (Doc. 322)

Sector	State	Eigenvalue	Formula	$\mathscr{H}$-component	Status
Leptons	$\|e\rangle$	0.505 MeV	$\frac{4}{3}\xi^{3/2}v$	$\mathscr{H}_{\text{fl}}(3,2)$	[K]
	$\|\mu\rangle$	104.96 MeV	$\frac{16}{5}\xi^{1}v$	$\mathscr{H}_{\text{fl}}(5,4)$	[K]
	$\|\tau\rangle$	1783.4 MeV	$\frac{25}{9}\xi^{2/3}v$	$\mathscr{H}_{\text{fl}}(9,5)$	[K]
Neutrinos	$\|\nu_1\rangle$	0.976 meV	$m_e\xi^{9/4}$	$\text{Fix}(P)\cap F_2$	[K]
	$\|\nu_2\rangle$	9.084 meV	$m_e\xi^{2}$	$\text{Fix}(P)\cap F_5$	[K]
	$\|\nu_3\rangle$	54.11 meV	$m_e\xi^{9/5}$	$\text{Fix}(P)\cap F_7$	[K]
Bosons	$\|g_a\rangle$	0	$\hat{F}\|g_a\rangle = 0$	$\text{Fix}(P_{\mathbb{Z}_3})$	[K]
	$\|W^{\pm}\rangle$	80.38 GeV	$\frac{1}{2}g_2 v$	$\mathscr{H}_{\text{spin}}$	[B]
	$\|Z^{0}\rangle$	91.19 GeV	$m_W/\cos\theta_W$	$\mathscr{H}_{\text{spin}}$	[B]

N.9 Complete Spectral Correspondence Table

Overall status Doc. 322

The Hilbert-space embedding of FFGFT is structurally complete: $\hat{F}$ is defined, the state space is constructed, all mass formulae from Doc. 320 are identified as eigenvalue equations, and the self-adjointness of $\hat{F}$ is proved on the $\mathbb{Z}_3$-invariant sector P_0 (Doc. 327): $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is thereby well-defined as a spectral value, provided the D_4 sector lies in P_0. The particle spectrum follows from spectral theory, not from free parameters.

Chapter O

Doc. 323 — Derivation of the Weinberg Angle at M_Z from FFGFT Geometry

List of Symbols

Symbol	Meaning	Value
ξ	FFGFT geometry parameter	$4/30000$
m_{Pl}	Planck mass	1.221×10^{19} GeV
m_τ	tau mass (FFGFT)	$\frac{25}{9}\xi^{2/3} v = 1783$ MeV
M_Z	Z^0 mass	91.19 GeV
M_{GUT}	GUT scale (FFGFT)	$m_{\mathrm{Pl}} \cdot \xi^{19/12}$
$\alpha_{\mathrm{em}}(M_Z)$	running EM coupling at M_Z	$1/128$
$\alpha_s(m_\tau)$	strong coupling (FFGFT, Doc. 160)	$3\xi^{1/4}$
N_c	colour degrees of freedom (Doc. 321)	3
n_f	active quark flavours (m_τ–M_Z)	5
β_3	1-loop coeff. $SU(3)_c$	$(11N_c - 2n_f)/3 = 23/3$
β_2	1-loop coeff. $SU(2)_L$	$19/6$
p	FFGFT GUT-scale exponent	$19/12$
$[K]$	derivable from ξ	status marker
$[B]$	algebraically proved	status marker

Summary

Doc. 321 derives $\sin^2\theta_W|_{\mathrm{GUT}} = 3/8$ algebraically from the trace formula of the five orbifold states **[B]**. The open bridge to the measured value $\sin^2\theta_W(M_Z) = 0.2312$ (PDG) is closed here.

The complete FFGFT formula is:

$$\boxed{\sin^2\theta_W(M_Z) = \frac{3}{8} - \frac{55\,\alpha_{\mathrm{em}}(M_Z)}{24\pi}\left[\ln\frac{m_{\mathrm{Pl}}}{M_Z} + \frac{19}{12}\ln\xi\right] = 0.2308,} \tag{O.1}$$

at $-0.19\,\%$ deviation from PDG. All inputs come from the FFGFT corpus. The exponent $p = 19/12 = 3/2 + 1/(4N_c)$ connects the electron mass exponent ($p_e = 3/2$, Doc. 320) with the D_4-lattice phase correction.

Status markers

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible

O.1 Starting Point and Strategy

[B] Doc. 321 derives $\sin^2\theta_W|_{\mathrm{GUT}} = 3/8$ from the trace formula at the unification point. The measured value at M_Z differs by

$$\Delta \sin^2\theta_W = 3/8 - 0.2312 = 0.1438, \tag{O.2}$$

which is the RG running between M_{GUT} and M_Z. The derivation proceeds in three steps:

1. $\alpha_s(M_Z)$ from the 1-loop RG run of $\alpha_s(m_\tau)$ – fully from FFGFT.
2. GUT scale $M_{\mathrm{GUT}} = m_{\mathrm{Pl}} \cdot \xi^{19/12}$ from FFGFT geometry.
3. $\sin^2\theta_W(M_Z)$ from the Georgi-Quinn-Weinberg formula.

O.2 Step 1: $\alpha_s(M_Z)$ from FFGFT Running

[B] With $N_c = 3$ (Doc. 321) and $n_f = 5$ active quark flavours between m_τ and M_Z:

$$\beta_3 = \frac{11N_c - 2n_f}{3} = \frac{23}{3}. \tag{O.3}$$

[K] Integrating the 1-loop RGE:

$$\frac{1}{\alpha_s(M_Z)} = \frac{1}{\alpha_s(m_\tau)} + \frac{\beta_3}{2\pi}\ln\frac{M_Z}{m_\tau}. \tag{O.4}$$

Inserting FFGFT values ($\alpha_s(m_\tau) = 3\xi^{1/4} = 0.3224$, $m_\tau = (25/9)\xi^{2/3}v = 1.783$ GeV, $M_Z = 91.19$ GeV):

$$\frac{1}{\alpha_s(M_Z)} = 3.102 + \frac{23/3}{2\pi} \times 3.934 = 3.102 + 4.800 = 7.902. \tag{O.5}$$

$$\boxed{\alpha_s(M_Z) = 0.1265} \quad (\text{PDG: } 0.1181,\ +7.1\,\%). \tag{O.6}$$

The $+7.1\,\%$ discrepancy is the known 1-loop error; 2-loop corrections and quark-threshold effects (particularly the b-quark threshold) bring it to ≈ 0.118. The Weinberg-angle derivation depends on $\alpha_s(M_Z)$ only indirectly via M_{GUT} and is much less sensitive to this error.

O.3 Step 2: The GUT Scale from FFGFT Geometry

[K] The GUT scale in FFGFT:

$$M_{\mathrm{GUT}} = m_{\mathrm{Pl}} \cdot \xi^{p}, \quad p = \frac{19}{12} = \frac{3}{2} + \frac{1}{4N_c}. \tag{O.7}$$

The exponent $p = 19/12$ has a geometric derivation:

- $p_e = 3/2$: electron mass exponent (Doc. 320, Eq. 5), the fundamental sub-Planck suppression of the first lepton generation.
- $1/(4N_c) = 1/12$: discrete phase correction of the four-dimensional $\mathbb{Z}_3$ fibration on T^4. The factor $4N_c = 12$ counts the product of torus dimensions (4) and colour degrees of freedom ($N_c = 3$, Doc. 321).

[K] Numerically:

$$M_{\mathrm{GUT}} = 1.221 \times 10^{19}\ \mathrm{GeV} \cdot \xi^{19/12} = 8.94 \times 10^{12}\ \mathrm{GeV}, \tag{O.8}$$

in the range 10^{12}–10^{15} GeV typical for non-supersymmetric GUT models. M_{GUT} serves here only as the running parameter of the GQW formula; as the mass of $SU(5)$ X bosons this value is excluded by proton decay: $\tau_p \sim M_X^4/(\alpha_G^2 m_p^5) \approx 3 \times 10^{23}$ yr at 8.94×10^{12} GeV, whereas $\tau(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr requires $M_X \gtrsim 5 \times 10^{15}$ GeV.

O.4 Step 3: $\sin^2\theta_W(M_Z)$ from the GQW Formula

[B] The 1-loop $SU(5)$ result (Georgi, Quinn, Weinberg 1974):

$$\sin^2\theta_W(\mu) = \frac{3}{8} - \frac{55\,\alpha_{\mathrm{em}}(\mu)}{24\pi} \ln\frac{M_{\mathrm{GUT}}}{\mu}. \tag{O.9}$$

With $M_{\mathrm{GUT}} = m_{\mathrm{Pl}} \cdot \xi^{19/12}$:

$$\begin{aligned} \ln\frac{M_{\mathrm{GUT}}}{M_Z} &= \ln\frac{m_{\mathrm{Pl}}}{M_Z} + \frac{19}{12}\ln\xi \\ &= 39.44 + \tfrac{19}{12} \times (-8.923) = 39.44 - 14.13 = 25.31. \end{aligned} \tag{O.10}$$

[K] Result:

$$\begin{aligned} \sin^2\theta_W(M_Z) &= \frac{3}{8} - \frac{55}{24\pi} \cdot \frac{1}{128} \cdot 25.31 \\ &= 0.3750 - 0.1442 = \mathbf{0.2308}. \end{aligned} \tag{O.11}$$

$$\boxed{\sin^2\theta_W(M_Z)_{\mathrm{FFGFT}} = \frac{3}{8} - \frac{55\,\alpha_{\mathrm{em}}(M_Z)}{24\pi}\left[\ln\frac{m_{\mathrm{Pl}}}{M_Z} + \frac{19}{12}\ln\xi\right] = 0.2308} \tag{O.12}$$

PDG: 0.2312. Deviation: $-0.19\,\%$.

O.5 Summary and Status

Table O.1: FFGFT inputs to Doc. 323 and their origin

Quantity	FFGFT formula	Value	Doc.
ξ	$4/30000$	1.333×10^{-4}	basis
N_c	from $\mathbb{Z}_3$ triality	3	Doc. 321 [B]
n_f	quark thresholds	5	Doc. 320
$\alpha_s(m_\tau)$	$3\xi^{1/4}$	0.3224	Doc. 160 [K]
m_τ	$(25/9)\xi^{2/3}v$	1.783 GeV	Doc. 320 [K]
M_Z	$m_W/\cos\theta_W$	91.19 GeV	Doc. 320 [K]
m_{Pl}	$1/\sqrt{G_N}$	1.221×10^{19} GeV	external
$\alpha_{\text{em}}(M_Z)$	$1/128$	7.81×10^{-3}	external
$p = 19/12$	$p_e + 1/(4N_c)$	1.5833	this Doc. [K]

Table O.2: FFGFT predictions vs. PDG

Quantity	FFGFT	PDG	Dev.
$\sin^2\theta_W\vert_{\text{GUT}}$	$3/8 = 0.3750$	—	[B] exact
M_{GUT}	8.94×10^{12} GeV	$\sim 10^{13}$ GeV	$\sim 10\,\%$
$\alpha_s(M_Z)$	0.1265	0.1181	$+7.1\,\%$
$\sin^2\theta_W(M_Z)$	0.2308	0.2312	$-0.19\,\%$

Overall status Doc. 323 [K]

The open bridge from Doc. 321 is closed. The RG running $\sin^2\theta_W\vert_{\text{GUT}} = 3/8 \to \sin^2\theta_W(M_Z) = 0.2308$ is derived fully from FFGFT quantities. The $-0.19\,\%$ deviation lies well within the 1-loop uncertainty. Two external inputs remain (m_{Pl}, α_{em}); their derivation from ξ is an explicit open bridge **[S]**.

Part III

Part III — Hyperbits, the Hawking Mechanism, Coupling and the Continuum

Chapter P

Doc. 324 — $G(6, \mathbb{Z}_3\mathbb{C})$ Vacuum Structure and FFGFT

List of Symbols

G(6,$\mathbb{Z}_3\mathbb{C}$) Symbols after Matzke (2026)

Symbol	Meaning	Remark
$G(6, \mathbb{Z}_3\mathbb{C})$	Clifford algebra, 6 dimensions, coefficient field $\{-1, 0, \pm 1, \pm i\}$	$2 \equiv -1 \pmod 3$
$e_1, \ldots, e_6$	Basis vectors, $e_k^2 = +1$	
$N_k^{\pm}$	Nilpotent vectors, $(N_k^{\pm})^2 = 0$	$N_k^{\pm} = e_a \pm i\, e_b$
P_{Pk}	Idempotents, $P_{Pk}^2 = P_{Pk}$	$P_{Pk} = (\mathbf{1} + i\, e_a e_b)/2$
V	Vacuum operator, $V = N_1^+ N_2^+ N_3^+ N_1^- N_2^- N_3^-$	involution in $\mathbb{Z}_3\mathbb{C}$: $V^2 \equiv -V \pmod 3$
P_{Nk}	Anti-particle idempotents, $P_{Nk}^2 = P_{Nk}$	$V = -64 \cdot P_{N1} P_{N2} P_{N3}$
T_k	Trine operator, $T_k^3 = \mathbf{1}$	$T_k = \mathbf{1} + (\omega - 1) P_{Pk}$
ω	Primitive $\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
I, J, K	Commutative quaternions (grade 4), $I^2 = J^2 = K^2 = +1$	encode 3 spatial dimensions
iE_7	Complex volume form, $(iE_7)^2 = +1$	universal phase operator

FFGFT Symbols after Pascher (2023–2026)

Symbol	Meaning	Value
ξ	FFGFT geometry parameter	$4/30000$
$T^4/\mathbb{Z}_3$	4D torus with $\mathbb{Z}_3$ orbifold projection	
τ	Generator of the $\mathbb{Z}_3$ action on T^4	unitary
P_k	$\mathbb{Z}_3$ projectors on $L^2(T^4)$	$\frac{1}{3}\sum_j \omega^{-jk} \tau^j$, $k = 0, 1, 2$
$\mathscr{H}_k$	Eigensectors: $\mathscr{H}_k = P_k L^2(T^4)$	three colour sectors
$\hat{F}_{D_4}$	D_4 sub-operator of the spectral operator	$\xi = \lambda_{\min}(\hat{F}_{D_4})$
$C_2(R)$	Quadratic Casimir operator (Lie algebra)	$C_2(SU(3)_{\text{fund}}) = 4/3$
N_{Fourier}	Fourier mode count of the T^4 torus	$10^4 = 30000/\lvert\mathbb{Z}_3\rvert$
N_c	Colour degrees of freedom, algebraically necessary	3
v	Higgs vacuum expectation value	246 GeV

FFGFT Document Numbering

This document references the FFGFT corpus (Johann Pascher, 2023–2026, ORCID 0009-0000-6518-4064):

- **Doc. XXX**: Document No. XXX of the FFGFT corpus. Full list: `github.com/jpascher/T0-Time-Mass-Duality`
- **RXXX**: Correction/addendum No. XXX in the register Doc. 190.
- **[B]**: algebraically proved **[K]**: derivable from ξ **[S]**: sketched/plausible

Documents relevant to this comparison: Doc. 009 (Casimir effect, factor $4/3$), Doc. 049 (confinement), Doc. 160 (α_s), Doc. 314 (D_4 lattice), Doc. 321 ($SU(3)_c$ derivation), Doc. 322 (Hilbert space), Doc. 323 (Weinberg angle).

Summary

This document compares the vacuum structure of $G(6, \mathbb{Z}_3\mathbb{C})$ after Matzke [1] with FFGFT spectral theory and numerically investigates the open connection question: is $\xi = 4/30000$ the smallest spectral value of the vacuum operator V in G(6)?

Results of the numerical investigation (Python script `324_G6_FFGFT_verify.py`):

- The Trine theorem is verified: $T_k^3 = \mathbf{1}$ to 6.5×10^{-16} **[B]**.
- Matzke's vacuum operator $V = N_1^+ N_2^+ N_3^+ N_1^- N_2^- N_3^- = -64 \cdot P_{N1} P_{N2} P_{N3}$ is an involution in $\mathbb{Z}_3\mathbb{C}$: $V^2 = -64V$, i.e. $V^2 \equiv -V \pmod 3$ **[B]**. The complementary Pp-product $P_{P1} P_{P2} P_{P3}$ is a rank-1 projector with eigenvalues $\{0^{(7)}, 1^{(1)}\}$, but lies on a different ray ($|\langle \text{vac}_V | \text{vac}_{Pp} \rangle| = 0$).
- ξ is *not* a direct spectral value of V. The connection is *structural*, not spectral.
- The trine product $T_1 T_2 T_3$ is a global $\mathbb{Z}_3$ operator: $(T_1 T_2 T_3)^3 = \mathbf{1}$, eigenvalues $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$ **[B]**.
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4$ algebraically derived – R84 **[K]**.

Status markers

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible

P.1 Starting Point: Two Frameworks, One Physics

Matzke's G(6,$\mathbb{Z}_3\mathbb{C}$) Vacuum Structure

[S] Matzke constructs the physical vacuum state V as an algebraic object in $G(6, \mathbb{Z}_3\mathbb{C})$, a Clifford algebra over the coefficient field $\mathbb{Z}_3\mathbb{C} = \{-1, 0, +1, \pm i\}$. The central vacuum relation is

$$V^2 = -V \quad (\text{in } \mathbb{Z}_3\mathbb{C} \text{ arithmetic}), \tag{P.1}$$

where $2 \equiv -1 \pmod 3$. Three nilpotent vector pairs $N_k^{\pm} = e_a \pm i\, e_b$ with $(N_k^{\pm})^2 = 0$ decompose the $\mathbb{C}^6$ spinor space into three eigensectors. The Trine theorem states: for every nilpotent N with $N^2 = 0$,

$$T = \mathbf{1} + (\omega - 1)N \quad \Rightarrow \quad T^3 = \mathbf{1}. \tag{P.2}$$

FFGFT Spectral Theory

[K] In FFGFT, $\xi = 4/30000$ is the smallest eigenvalue of the D_4 sub-operator $\hat{F}_{D_4}$ (Doc. 322):

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}. \tag{P.3}$$

The $\mathbb{Z}_3$ projection $P_k = \frac{1}{3}\sum_{j=0}^{2} \omega^{-jk}\tau^j$ on $L^2(T^4)$ decomposes the Hilbert space into three eigensectors $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$. Their bilinear transitions generate the $\mathfrak{su}(3)$ algebra (Doc. 321, **[B]**).

P.2 Numerical Verification of the Trine Theorem

Construction of the Vacuum Structure

We use the 8-dimensional spinor representation of G(6) over $\mathbb{C}$, given by Γ matrices as tensor products of Pauli matrices:

$$\begin{aligned} &\Gamma_1 = \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad \Gamma_2 = \sigma_y \otimes \mathbf{1} \otimes \mathbf{1}, \quad \Gamma_3 = \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \\ &\Gamma_4 = \sigma_z \otimes \sigma_y \otimes \mathbf{1}, \quad \Gamma_5 = \sigma_z \otimes \sigma_z \otimes \sigma_x, \quad \Gamma_6 = \sigma_z \otimes \sigma_z \otimes \sigma_y. \end{aligned} \tag{P.4}$$

The Clifford relation $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}\mathbf{1}_8$ holds (numerically to $< 10^{-14}$).

Nilpotents over three Witt pairs $(0,1)$, $(2,3)$, $(4,5)$:

$$N_k^{+} = \Gamma_{2k} + i\,\Gamma_{2k+1}, \quad N_k^{-} = \Gamma_{2k} - i\,\Gamma_{2k+1}, \quad (N_k^{\pm})^2 = 0 \quad \textbf{[B]}. \tag{P.5}$$

Idempotents:

$$P_{Pk} = \frac{\mathbf{1} + i\,\Gamma_{2k}\Gamma_{2k+1}}{2}, \qquad P_{Pk}^2 = P_{Pk} \quad \text{(numerically to } 0\text{)} \quad \textbf{[B]}. \tag{P.6}$$

Anti-particle idempotents and vacuum operator:

$$P_{Nk} = \frac{\mathbf{1} - i\,\Gamma_{2k}\Gamma_{2k+1}}{2}, \qquad P_{Nk}^2 = P_{Nk}, \quad P_{Pk}P_{Nk} = 0, \quad P_{Pk} + P_{Nk} = \mathbf{1} \quad \textbf{[B]}. \tag{P.7}$$

$$V = N_1^{+}N_2^{+}N_3^{+}N_1^{-}N_2^{-}N_3^{-} = -64 \cdot P_{N1}P_{N2}P_{N3}. \tag{P.8}$$

Trine Operators and Their Spectrum

With $\omega = e^{2\pi i/3}$:

$$T_k = \mathbf{1} + (\omega - 1)\, P_{Pk}. \tag{P.9}$$

[B] Numerical result:

$$T_k^3 = \mathbf{1} \quad \text{to } 6.5 \times 10^{-16}. \tag{P.10}$$

The Trine theorem is exactly verified in the 8-dimensional spinor representation of G(6).

The trine product:

$$(T_1T_2T_3)^3 = \mathbf{1} \quad \text{to } 1.9 \times 10^{-15}. \tag{P.11}$$

Eigenvalues of $T_1T_2T_3$: $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$ – a global $\mathbb{Z}_3$ phase operator on the 8-dimensional spinor space.

P.3 Spectrum of the Vacuum Operator V

[B] Matzke's vacuum operator $V = -64 \cdot P_{N1}P_{N2}P_{N3}$ has in the 8-dimensional spinor representation over $\mathbb{C}$:

$$\mathrm{Tr}(V) = -64, \quad \mathrm{rank}(V) = 1, \quad V^2 = -64\,V. \tag{P.12}$$

In $\mathbb{Z}_3\mathbb{C}$ arithmetic, $-64 \equiv -1 \pmod 3$, hence:

$$V^2 \equiv -V \pmod 3 \qquad \text{(involution — Matzke's relation confirmed \textbf{[B]})}. \tag{P.13}$$

The Pp-product $P_{P1}P_{P2}P_{P3}$ is a rank-1 projector with $\sigma = \{0^{(7)}, 1^{(1)}\}$, but lies on a different ray: $|\langle \mathrm{vac}_V | \mathrm{vac}_{Pp}\rangle| = 0$ **[B]**. Numerically checked with `pruef_324_vakuum_involution.py`.

[K] $\xi = 4/30000$ is *not* a spectral value of V. The connection is structural:

Table P.1: Structural comparison $G(6, \mathbb{Z}_3\mathbb{C})$ vs. FFGFT $T^4/\mathbb{Z}_3$

Quantity	$G(6, \mathbb{Z}_3\mathbb{C})$	FFGFT $T^4/\mathbb{Z}_3$
$\mathbb{Z}_3$ symmetry	Witt decomposition of $\mathbb{C}^6$	Orbifold projection on T^4
Three sectors	three nilpotent pairs $N_k^{\pm}$	$\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$
$N_c = 3$	number of Witt pairs	order of $\mathbb{Z}_3$
Confinement	$T_R = 0$ (triality)	$T_R = 0$, algebraic [B]
Vacuum	involution $V = -64 \cdot P_{N1}P_{N2}P_{N3}$, $V^2 \equiv -V$	spectral value $\xi = \lambda_{\min}(\hat{F}_{D_4})$
$SU(3)_c$	bilinear transitions [S]	proved [B] (Doc. 321)
Fermion masses	nilpotents as directions	eigenvalues $m_i = r_i\xi^{p_i}v$ [K]
Numerator 4 in ξ	4 Witt pairs in $G(8)$	4 torus dimensions
Denominator 30000	$\lvert\mathbb{Z}_3\rvert \times 10^4$ (modes)	$\lvert\mathbb{Z}_3\rvert \times 10^4$ (modes)

P.4 Agreements and Divergences

[B] The following structures are identical in both frameworks and numerically verified:

1. **Trine theorem:** $T_k^3 = \mathbf{1}$ from nilpotents $N_k^{\pm}$ – identical with the $\mathbb{Z}_3$ phase structure of FFGFT projectors $P_k = \frac{1}{3}\sum \omega^{-jk}\tau^j$.
2. **Three sectors from one symmetry:** G(6) Witt decomposition ↔ FFGFT $\mathbb{Z}_3$ eigenspace decomposition. Both yield exactly three sectors; their bilinear transitions generate $\mathfrak{su}(3)$.

3. **Confinement as $T_R = 0$:** Matzke's complement condition $V(V+1) = 0$ corresponds to the triality selection in Doc. 321.
4. $N_c = 3$ **algebraically forced:** Neither G(6) nor FFGFT sets N_c as a free parameter.

Table P.2: Open and closed bridges between $G(6, \mathbb{Z}_3\mathbb{C})$ and FFGFT

Point	$G(6, \mathbb{Z}_3\mathbb{C})$	FFGFT	Character
Electroweak group	$SU(2)_L$ open [S]	$SU(2)_L$, $U(1)_Y$ [B]	G(6)-internal
Weinberg angle	not computed	0.2308 ($-0.19\,\%$) [K]	G(6)-internal
Lepton masses	not derived	$m_e, m_\mu, m_\tau < 1\,\%$ [K]	G(6)-internal
Neutrino masses	not derived	m_{ν_i} from fixed points [K]	G(6)-internal
ξ connection	Casimir quotient [K]	$\xi = C_2/N_F$ [K]	**closed**

P.5 The Actual Connection Question

The question "is ξ the smallest spectral value of V?" is fully answered by the numerical investigation and R84:

[K] ξ is not a spectral value of any G(6) operator – a closed negative finding, not an open bridge. The connection between G(6) and FFGFT lies on a different level:

$$\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{30000} \tag{P.14}$$

connects the *algebraic* structure of G(6) (Casimir invariant $C_2 = 4/3$, from $N_c = 3$ [B]) with the *geometric* scale of FFGFT (torus mode count $N_F = 10^4$). This is not a G(6)-internal statement, but the bridge formula between both frameworks – R84.

Note on the electroweak group: $SU(2)_L \times U(1)_Y$ is algebraically derived in *FFGFT* (Doc. 321 [B]). Whether Matzke's $G(6, \mathbb{Z}_3\mathbb{C})$ reproduces the same derivation internally is a *G(6)-internal* problem – not an open FFGFT bridge.

P.6 ξ from the Casimir Operator of $SU(3)$

The Quadratic Casimir Operator

[B] The quadratic Casimir operator of a Lie algebra $\mathfrak{g}$ is the group invariant

$$C_2(R) = \sum_{a=1}^{\dim \mathfrak{g}} T_a^{(R)} T_a^{(R)}, \tag{P.15}$$

where $T_a^{(R)}$ are the generators in representation R. For $SU(N_c)$ in the *fundamental* representation:

$$C_2(SU(N_c)_{\text{fund}}) = \frac{N_c^2 - 1}{2N_c}. \tag{P.16}$$

With $N_c = 3$ (algebraically necessary, Doc. 321, **[B]**):

$$C_2(SU(3)_{\text{fund}}) = \frac{9-1}{6} = \frac{8}{6} = \frac{4}{3}. \tag{P.17}$$

ξ as the Casimir Value per Fourier Mode

[K] The Fourier mode count of the T^4 torus:

$$N_{\text{Fourier}} = \frac{N_{\text{modes}}}{|\mathbb{Z}_3|} = \frac{30000}{3} = 10^4. \tag{P.18}$$

The FFGFT geometry parameter follows:

$$\boxed{\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{3 \times 10^4} = \frac{4}{30000}} \tag{P.19}$$

[B] This formula is exact. Each quantity has a geometric or algebraic justification:

- $C_2(SU(3)_{\text{fund}}) = 4/3$: algebraic from $N_c = 3$, which is algebraically necessary from the $\mathbb{Z}_3$ structure (Doc. 321).
- $|\mathbb{Z}_3| = 3$: order of the orbifold group (Doc. 321).
- $N_{\text{Fourier}} = 10^4$: Fourier mode count of the T^4 torus, from 4-dimensional topology and resolution scale.

Connection to Doc. 009: Casimir Effect and the Factor 4/3

[S] Doc. 009 relates the physical Casimir effect (vacuum energy between plates) to the factor $4/3$. The QFT Casimir energy per area between ideal plates reads

$$\frac{E_{\text{Casimir}}}{A} = -\frac{\pi^2 \hbar c}{720\, a^3} \tag{P.20}$$

without a factor $4/3$; the pressure is $-\pi^2\hbar c/(240\, a^4)$ and the energy density $-\pi^2\hbar c/(720\, a^4)$, so the prefactor ratio pressure/energy density is 3, not $4/3$ (check: $|P| = 1.30$ mPa at $a = 1\,\mu$m). The factor $4/3$ is appended in Doc. 009, not obtained from QFT. Identifying it with $C_2(SU(3)_{\text{fund}}) = 4/3$ is an FFGFT interpretation **[S]**, not a QFT statement: in this interpretation $4/3$ counts the degree-of-freedom weighting of an $SU(3)$-invariant system.

Table P.3: Factor 4/3 in ξ: algebraic derivation and Casimir interpretation

Route	Source	Factor
Casimir effect (interpretation)	Doc. 009	$4/3$ appended, FFGFT interpretation [S]
Casimir operator (algebraic)	$SU(3)$, Doc. 321	$C_2 = (N_c^2 - 1)/(2N_c)$
Result	algebraic route	$4/3 \to \xi = 4/30000$

Casimir derivation of ξ [K]

$\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}}$ derives the numerator 4 in $\xi = 4/30000$ algebraically from the quadratic Casimir operator. $N_c = 3$ is algebraically necessary (Doc. 321 [B]). The connection to the physical Casimir effect (Doc. 009) is an interpretation [S]: the QFT Casimir energy does not contain the factor $4/3$.

P.7 Conclusion

Overall status Doc. 324 – $G(6, \mathbb{Z}_3\mathbb{C}) \leftrightarrow$ FFGFT

Verified [B]:

- Trine theorem: $T_k^3 = \mathbf{1}$ in 8-dim spinor representation
- $V = -64 \cdot P_{N1}P_{N2}P_{N3}$ is an involution in $\mathbb{Z}_3\mathbb{C}$: $V^2 \equiv -V \pmod 3$, rank $= 1$ **[B]**
- Trine product $(T_1T_2T_3)^3 = \mathbf{1}$ with $\mathbb{Z}_3$ spectrum
- Three sectors, bilinear transitions, confinement as $T_R = 0$
- $C_2(SU(3)_{\text{fund}}) = 4/3$ algebraically from $N_c = 3$

Closed [K]:

- ξ is not a spectral value of V (numerically excluded) [K]
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4 = 4/30000$ algebraically derived – R84 [K]
- The Laplace operator on the vacuum ray gives $\Delta = 6 \cdot \mathbf{1}$ (trivial, since $\Gamma_a^2 = \mathbf{1}$) – no connection to ξ as an eigenvalue; the question is closed as incorrectly posed [K]

Open [S]:

- $SU(2)_L \times U(1)_Y$ in $G(6, \mathbb{Z}_3\mathbb{C})$ not yet derived – **G(6)-internal problem, not an FFGFT problem**. FFGFT has derived $SU(2)_L$ and $U(1)_Y$ from torus geometry (Doc. 321 [B]); the question is only whether Matzke's $G(6, \mathbb{Z}_3\mathbb{C})$ reproduces the same derivation internally.

Remark on the Laplace point: The original question "is there a natural G(6) operator with $\lambda_{\text{min}} = \xi$?" is answered by R84: ξ does not arise from a spectrum within G(6), but from the Casimir quotient C_2/N_{Fourier} – a bridge between algebraic structure (G(6)) and geometric scale (FFGFT torus). These are two different levels; no G(6) operator alone can have ξ as an eigenvalue.

References

[1] Matzke, D. (2026). *How the Vacuum Builds Space: Nilpotents, Quaternions, and the Planck Voxel*. ANPA Conference 2026, 13 July 2026. Website: `http://www.QuantumDoug.com`

[2] Furey, C. (2012). Unified theory of ideals. *Physical Review D* 86, 025024.

[3] Furey, C. (2018). Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra. *Physics Letters B* 785, 84–89.

[4] Pascher, J. (2023–2026). *FFGFT Corpus* (Doc. 001–324). ORCID: 0009-0000-6518-4064.
`https://github.com/jpascher/T0-Time-Mass-Duality`

[5] Pascher, J. (2026). *Doc. 321: Algebraic derivation of the $SU(3)_c$ gauge structure*. FFGFT Corpus.

[6] Pascher, J. (2026). *Doc. 323: Derivation of the Weinberg angle at M_Z*. FFGFT Corpus.

[7] Particle Data Group (2024). *Review of Particle Physics*. *Physical Review D* 110, 030001.

Chapter Q

Doc. 325 — The FFGFT Hawking Mechanism

Abstract

Doc. 313 (Ch. G) established the thermodynamic frame of Hawking radiation in FFGFT: a single KMS rule tempers Unruh, Gibbons–Hawking and black holes jointly **[K]**, the membrane picture receives its thermometer **[S]**, and the family ladder reaches the cosmos at $r_s = R_H/2$ **[S]**. What remained open was the *microscopic emission mechanism*: which modes escape, what the quantum carries, why massive quanta fall back. This document works it out completely from FFGFT-native building blocks:

1. **Temperature**: The KMS rule $T = \hbar/(k_B\tau)$ with membrane period $\tau = 8\pi GM/c^3$ follows from $\tilde{T}\cdot m = 1$ (clock compression at the membrane, Doc. 313 G.2). **[K]**
2. **Quantum**: The Hawking quantum is the bit of scale $4\pi r_s$: $k_B T_H = \hbar c/(4\pi r_s)$ exactly — the system-dependent bit energy $E_{\text{bit}} = \hbar c/L$ (Doc. 257) evaluated at the horizon scale. No universal bit value is needed anywhere. **[K]**
3. **Selection**: Only triality-0 modes ($T_R = 0$) can be emitted. Confinement (Doc. 321) and Hawking selection are *one* principle; absorption of the partner quantum is the $\mathbb{Z}_3$ projector orthogonality $P_j P_k = 0$. **[B]**
4. **Carrier**: The massless carrier is the $n_4 = 0$ torus mode (spatial winding without mass-circle winding). Massive modes ($n_4 \neq 0$) have finite $\tilde{T}$ and fall back. **[K]**
5. **Information**: The emitted quantum carries the sector pair $(k, -k)$ as an orthogonally readable quantum number ($\log_2 3$ bits) — plus the thermal 1-nat area bookkeeping. Fine-grained entropy remains constant (unitarity, Doc. 322). No information paradox. **[K]**

Two verification scripts, all assertions passed: `325_hawking_ffgft_mechanismus.py` (thermodynamic chain), `325_hawking_z3_selektion.py` ($\mathbb{Z}_3$ algebra).

Q.1 Starting point: what Doc. 313 left open

Doc. 313 Ch. G established three results: (a) the single KMS rule $T = \hbar/(k_B\tau)$ for Unruh, Gibbons–Hawking and black holes **[K]**; (b) the membrane thermometer from $\tilde{T} \cdot m = 1$ **[S]**; (c) the family ladder up to the cosmos ($r_s = R_H/2$ exactly) **[S]**. Likewise the limit: evaporation is 56 (stellar) resp. 83 (supermassive) orders of magnitude too slow for the time cycle — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it **[K]**.

What was missing was the *microstructure* of emission:

- Which modes of the $T^4/\mathbb{Z}_3$ spectrum can leave the horizon — and which principle selects them?
- What is the emitted quantum, and how is its energy related to the information unit of the framework (Doc. 257)?
- Why do massive quanta fall back — as a consequence of the geometry, not as an extra assumption?
- How is information conservation realized microscopically?

These four questions are answered here — each from an already established FFGFT building block: $\tilde{T} \cdot m = 1$ (Docs. 077, 180), system-dependent bit energy (Docs. 257, 302, A271–A273), $\mathbb{Z}_3$ triality (Doc. 321), unitarity of the spectral operator (Doc. 322).

Q.2 Building block 1: the KMS temperature from $\tilde{T} \cdot m = 1$

Wherever the local time structure is periodic with period τ, an observer reads off temperature:

$$T = \frac{\hbar}{k_B\,\tau} \tag{Q.1}$$

System	Period τ	$T = \hbar/(k_B\tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18}$ s	2.66×10^{-30} K
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4}$ s	6.17×10^{-8} K
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^{8}$ s	3.98×10^{-20} K

The membrane period $\tau = 8\pi GM/c^3$ is not an extra assumption but follows from time-mass duality: mass compresses the time structure ($\tilde{T} \cdot m = 1$ — the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with exactly this period (Doc. 313 G.2). **[K]**

The temperature is thus not the result of a Bogoliubov computation on a curved background, but a read-off rule of the clock map — the same rule for all three systems.

Q.3 Building block 2: the Hawking quantum as a system-dependent bit

The central new result of this document. The system-dependent bit energy of FFGFT (Doc. 257) reads $E_{\text{bit}}(L) = \hbar c/L$. Evaluated at the horizon scale:

Core result [K]

$$k_B T_{\mathrm{H}} = \frac{\hbar c}{4\pi r_{\mathrm{s}}} = E_{\mathrm{bit}}(L = 4\pi r_{\mathrm{s}}) \tag{Q.2}$$

The thermal Hawking quantum *is* the bit of scale $L = 4\pi r_{\mathrm{s}}$ (twice the horizon circumference). Numerically exact (script 1): $\hbar c/(k_B T_{\mathrm{H}}) = 4\pi r_{\mathrm{s}}$ to machine precision.

The system dependence of bit values (Docs. 257, 302, A271–A273) is therefore no obstacle to Hawking physics but *its explanation*: the Hawking temperature is the bit temperature of the horizon scale. A universal bit value — say, a fixed bit mass from a Landauer evaluation at the Planck temperature — is needed nowhere; every system carries its own bit energy, and that of the black hole at its horizon is exactly $k_B T_{\mathrm{H}}$.

Area bookkeeping

Emission of a quantum of energy $k_B T_{\mathrm{H}}$ carries, by Clausius, exactly $\mathrm{d}S = E/(T \cdot k_B) = 1$ nat, hence

$$\mathrm{d}A = -4\ell_{\mathrm{P}}^2 \quad \text{per nat} \tag{Q.3}$$

(script 1). The area loss per emission thus follows from Bekenstein + Clausius alone — no further geometric assumptions. **[B]**

Q.4 Building block 3: $\mathbb{Z}_3$ triality selection

The algebraic core of the mechanism. The horizon carries the $\mathbb{Z}_3$ orbifold structure of $T^4/\mathbb{Z}_3$ (Docs. 321, 322). The three $\mathbb{Z}_3$ projectors

$$P_k = \frac{1}{3}\sum_{j=0}^{2} \omega^{-jk}\, \tau^j, \qquad \omega = e^{2\pi i/3} \tag{Q.4}$$

are complete, idempotent and orthogonal (script 2):

$$P_0 + P_1 + P_2 = \mathbb{1}, \qquad P_j P_k = \delta_{jk} P_j. \tag{Q.5}$$

Absorption [B]

The infalling partner quantum erases the horizon voxel exactly when it lies in the orthogonal $\mathbb{Z}_3$ sector:

$$P_j\left(P_k \psi\right) = 0 \quad (j \neq k) \tag{Q.6}$$

— exact, no tunnelling (script 2: $\|P_2(P_1\psi)\| < 10^{-15}$). Pair creation at the membrane separates a $(k, -k)$ sector pair; the inward-falling partner is absorbed by projector orthogonality, the outward-running one escapes provided it satisfies the triality condition.

Who escapes: $T_R = 0$

The triality of a mode is its $\mathbb{Z}_3$ charge k. Emittable (asymptotically free) are only combinations with $T_R = \sum k_i \equiv 0 \pmod 3$:

Combination	T_R	emittable
Single mode $k = 0$ (photon analogue)	0	yes
Single mode $k = 1$ (quark)	1	no
Bilinear $(1, 2)$ (meson)	0	yes
Bilinear $(1, 1)$	2	no
Trilinear $(1, 1, 1)$ (baryon)	0	yes

Confinement = Hawking selection [B]

Doc. 321 identified confinement as triality selection $T_R = 0$. The same principle acts at the horizon: only colourless objects escape. Confinement and Hawking selection are *one* principle of the $\mathbb{Z}_3$ structure — not two. The $\mathbb{Z}_3$ charge is exactly conserved in every emission (script 2).

Q.5 Building block 4: the massless carrier

On $T^4 = T^3_{\text{space}} \times S^1_m$ the mass of a mode is carried by the winding n_4 on the mass circle ($\tilde{T} \cdot m = 1$). The emittable carrier is the mode with

$$n_4 = 0 : \qquad \text{spatial winding without mass winding.} \tag{Q.7}$$

A massive quantum ($n_4 \neq 0$) has finite $\tilde{T}$; in the compressed clock map at the membrane (Doc. 313 G.2) it falls back — as a consequence of the geometry, not as an extra assumption. For $|n_i| \leq 1$ there are $3^3 - 1 = 26$ massless neighbouring modes against 54 massive ones (script 2). **[K]**

The Hawking spectrum thus consists of $n_4 = 0$ modes with $T_R = 0$: massless, colourless, thermal at T_{H} — the photon-like degrees of freedom of the torus.

Q.6 Building block 5: information conservation

What the quantum carries

The emitted $T_R = 0$ quantum carries the *sector pair* $(k, -k)$ as an internal quantum number. The three possible pair states are exactly orthogonal (Gram matrix $= \mathbb{1}_3$, script 2) — which sector was withdrawn is readable from the quantum: $\log_2 3 \approx 1.585$ bits of sector information per quantum, in addition to the thermal 1-nat area bookkeeping.

Fine-grained constancy

Doc. 313 G.5 phrased it as “two languages, one statement”: locally, the radiation is correlated with the fractal membrane structure ($S_{\text{BH}}(M_\odot) = 1.05 \times 10^{77}\, k_B$ of correlations — nothing is lost); globally, fine-grained entropy is constant under unitary

evolution (unitarity of the spectral operator, Doc. 322 **[K]**). No information paradox: the membrane is a clock-map object, not a causal abyss.

Quantitative information budget [B]

Building blocks 1–5 supply all ingredients for a fully computable information budget over the entire lifetime of a black hole.

Endpoint of evaporation. From Doc. 329 **[B]**: the collapse threshold $M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ follows from pure geometry (Compton = Schwarzschild). The Schwarzschild radius at the endpoint is

$$r_{\text{s}}(M_{\text{coll}}) = \sqrt{2}\,\ell_{\text{P}}, \tag{Q.8}$$

exactly one Planck length times $\sqrt{2}$ — the smallest horizon on which a T^4 mode still fits.

Residual entropy at the endpoint — exact identity [B]:

Main result R93 [B]

$$S_{\text{BH}}(M_{\text{coll}}) = \left.\frac{4\pi r_{\text{s}}^2}{4\,\ell_{\text{P}}^2}\right|_{r_{\text{s}}=\sqrt{2}\ell_{\text{P}}} = 2\pi \text{ nat} = \frac{2\pi}{\ln 2} \text{ bit} \approx 9.065 \text{ bit}. \tag{Q.9}$$

This is $S_{\text{BH}}(M_{\text{coll}}) = 2\pi/\ln 2$ bit $= L_{\text{Matzke}}/\ell_{\text{P}}$; Matzke's stability threshold is $n_{\text{thr}} = S_{\text{BH}}/\sqrt{2} = 6.410$, where $\sqrt{2} = r_{\text{s}}(M_{\text{coll}})/\ell_{\text{P}}$ (Doc. 329 **[B]**) — the same geometry (Compton = Schwarzschild) in two languages.

Total emission volume. For a black hole of initial mass $M_{\text{init}} > M_{\text{coll}}$:

$$N_{\text{emit}} = S_{\text{BH}}(M_{\text{init}}) - 2\pi \text{ nat}. \tag{Q.10}$$

Example $M_\odot$: $N_{\text{emit}} \approx 1.05 \times 10^{77}$ nat.

Structural constant of the $\mathbb{Z}_3$ geometry [B]. The ratio of sector information to thermal information is identical for every mass:

$$\frac{I_{\text{sector}}}{I_{\text{thermal}}} = \frac{\log_2 3 \text{ bit}}{1 \text{ nat}} = \frac{\log_2 3}{1/\ln 2} = \ln 3 \approx 1.099, \tag{Q.11}$$

since each emitted quantum carries exactly $\log_2 3$ bits of sector information (§Q.6) and 1 nat $= 1/\ln 2$ bits of thermal area bookkeeping. The ratio is a structural constant of the $T^4/\mathbb{Z}_3$ geometry, independent of mass and scale. **[R94 [B]]**

Verification script: `325_informationsbilanz.py` — 9 assertions, all passed.

Q.7 The FFGFT correction and its limits

Doc. 313 gives the fractal correction of the Hawking power:

$$P = P_{\text{std}}\left(1 - \xi \ln(M/m_{\text{P}})\right), \tag{Q.12}$$

numerically 0.60 % (PBH 10^{12} kg) to 1.44 % (SMBH $10^9\,M_\odot$) — script 1. It does not change the evaporation findings: only primordial holes below

$$M_* = \left(\frac{\tau_c\,\hbar\,c^4}{5120\,\pi G^2}\right)^{1/3} = 3.3\times10^{11}\ \text{kg} \tag{Q.13}$$

close via Hawking within one time cycle $\tau_c = 91.9$ Gyr. The Sun is 56, an SMBH 83 orders of magnitude too slow (script 1) — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it (Doc. 313 G.4). **[K]**

The family ladder runs seamlessly up to the cosmos: $T_\mathrm{H} = T_\mathrm{GH}$ requires $M = c^3/(4GH_0) = 4.66\times10^{52}$ kg with $r_\mathrm{s} = R_H/2$ exactly (script 1). **[K]**

Placement of the bit result

$k_B T_\mathrm{H} = E_\mathrm{bit}(4\pi r_\mathrm{s})$ shows: Hawking physics can be formulated entirely with the system-dependent bit energy. Any construction that instead postulates a universal bit value (say, a fixed bit mass from $k_B T\ln 2$ at a privileged temperature) adds a parametrization that the temperature physics does not require — consistent with A271–A273: Landauer's bound is a statement about carrier operations, not about abstract information.

Q.8 Status overview

Element	Status
KMS rule tempers Unruh/GH/BH jointly	**[K]** (Doc. 313)
Membrane period from $\tilde{T}\cdot m = 1$	**[K]**
$k_B T_\mathrm{H} = E_\mathrm{bit}(4\pi r_\mathrm{s})$ — quantum = horizon bit	**[K]** (new)
Area quantum $-4\ell_\mathrm{P}^2$/nat from Bekenstein+Clausius	**[B]** (new)
$\mathbb{Z}_3$ absorption $P_j P_k = 0$	**[B]** (new)
$T_R = 0$ selection; confinement = Hawking selection	**[B]** (new)
Carrier = n_4-free torus mode	**[K]** (new)
Sector info $\log_2 3$ bits orthogonally readable	**[B]** (new)
Information conservation (membrane + unitarity)	**[K]** (313/322)
FFGFT power correction 0.6–1.4 %	**[K]** (313)
Greybody factors, mode-spectrum detail	**[S]** (open)
Back-reaction on orbifold fixed points	**[S]** (open)

The Hawking mechanism in FFGFT thus stands at **[K]**/**[B]**; open remain only the greybody factors (standard QFT computation on the FFGFT background) and the back-reaction of the emission on the orbifold fixed-point structure.

Verification scripts

- `python/Dok325_Skripte/325_hawking_ffgft_mechanismus.py` — thermodynamic chain: KMS, membrane thermometer, horizon bit, area bookkeeping, correction, evaporation, family ladder, information side.

- `python/Dok325_Skripte/325_hawking_z3_selektion.py` — $\mathbb{Z}_3$ algebra: projectors, absorption, triality selection, charge conservation, sector information, masslessness.

References

1. J. Pascher, Doc. 313 *No Beginning*, Ch. G (Hawking on the time cycle), FFGFT corpus, 2026.
2. J. Pascher, Doc. 321 *SU(3)$_c$ from $\mathbb{Z}_3$ triality*, FFGFT corpus, 2026.
3. J. Pascher, Doc. 322 *Spectral theory and Hilbert-space embedding*, FFGFT corpus, 2026.
4. J. Pascher, Docs. 257, 302, A271–A273 (bit energy, Landauer), FFGFT corpus.
5. J. Pascher, Doc. 180 *Derivation of L_0*, FFGFT corpus, 2026.
6. S. W. Hawking, Commun. Math. Phys. **43**, 199, 1975.
7. J. Bekenstein, Phys. Rev. D **7**, 2333, 1973.
8. W. G. Unruh, Phys. Rev. D **14**, 870, 1976.

© 2026 Johann Pascher · CC BY 4.0

Chapter R

Doc. 326 — Black Holes from Hyperbits and from FFGFT

Legend

[K] core result **[B]** algebraically proven **[S]** sketch **[Q]** source **[X]** excluded **[POSTULATE]** axiom

Abstract

In "Black Holes from Hyperbits" (IPI 2026) Douglas Matzke derives the entire physics of black holes from the Clifford-algebra axiom $e_i^2 = +1$, without general relativity and without free parameters. Central to this: the mass of a bit follows from Landauer's principle ($E_{\text{bit}} = k_B(T_P/2\pi)\ln 2$, evaluated at the KMS temperature of the Planck scale) plus $E = mc^2$ and is therefore a *universal* number — $m_{\text{bit}} = m_P \ln 2/(2\pi) \approx 0.11\, m_P$, independent of the physical system. "Landauer at T_P" refers below to this Planck scale.

FFGFT (Fundamental Fractal Geometric Field Theory) arrives at a fundamentally different insight: bit values are *system-dependent*. The minimal information unit is the winding quantum $\Delta w = 1$ on T^4 (Doc. 257), and its energy is $E_{\text{bit}} = \hbar c/L$ — dependent on the physical scale L of the system. Landauer's $k_B T \ln 2$ is the special case of one particular scale and temperature, not a universal law (A271, A272, A273, Doc. 302).

This document compares both frameworks and works out the difference in the Landauer interpretation as theoretically essential.

R.1 Why Matzke's algebra looks so simple

Before the technical comparisons, a narrative placement is worthwhile: where does the aesthetic appeal of the hyperbit approach come from – and where is it paid for? The two belong together.

The starting point: a sign

Matzke begins with $e_i^2 = +1$. That is a *bit* – something whose square is one, hence something with two states. No space, no time, no metric. Only the statement: there are distinguishable things.

The real trick is that the Clifford algebra immediately starts working on this. Take several such e_i, let them anticommute, and out of nothing arises a structure with dimension, orientation and rotations. Nothing was put in except "there are bits, and they are independent of one another" – and out comes geometry. That is the first reason for the beauty: **geometry as bookkeeping over distinguishability**, not as a presupposition.

Why $\mathrm{Cl}(6)$ in particular

The second beautiful step: $\mathrm{Cl}(6)$ decomposes into exactly three Witt pairs – three packets, each with one creator and one annihilator. Following Furey, this is precisely the structure of one fermion generation: three colours, one lepton. The three is not chosen, it *falls out*. Anyone who has tried to explain why there are three colours knows how satisfying that is.

The black-hole part

Now the actual coup. Bekenstein says: a black hole has entropy proportional to *area*, one bit per four Planck areas. Matzke turns this around: if a bit is an area, then a collection of bits is a volume – and the question "at what point does this collapse?" becomes a question about numbers, not about field equations. He sets the Compton wavelength equal to the Schwarzschild radius (the classical point where quantum mechanics and gravity meet) and obtains the Schwarzschild radius – **without ever using the Einstein equations**. For $20\,M_\odot$ he states 59.1 km; with $S_{\mathrm{BH}} = A/(4\ell_{\mathrm{P}}^2)$, however, his formula gives twice this value (section on the Schwarzschild radius).

And then the number that carries the appeal: the stability threshold sits at 6.41 bits. $\mathrm{Cl}(6)$ has 6.000 bits. The universe consists of fermions sitting exactly 0.41 bits below the collapse threshold. Matter is stable because the smallest possible fermion algebra just barely fails to collapse. That is a fine story – terse, dramatic, with a concrete number.

Where the elegance is paid for

This 6.41 hangs on a postulate. To get from bits to masses, Matzke needs an exchange rate – what does a bit weigh? He takes Landauer ($E = k_B T \ln 2$) and evaluates it at the KMS temperature of the Planck scale, $T_P/(2\pi)$. Out comes $m_{\mathrm{bit}} = m_P \ln 2/(2\pi)$, a *universal* bit mass.

However: Landauer is a statement about thermal carrier ensembles, not about abstract information (A271–A273). And the Planck temperature is not the temperature of any physical system – it is an energy scale. The choice $T = T_P/(2\pi)$ is a postulate that does not justify itself. The number 6.41 is therefore exactly as universal as that

postulate – and Doc. 325 shows that the Hawking temperature does entirely without it.

The shortest version

Matzke's algebra is beautiful because it shows how far pure combinatorics can go: from "there are bits" follow dimension, three colours, one fermion generation, the Schwarzschild radius. That is genuine reduction, not relabelling.

It is *simple* because it stops asking at three places: Bekenstein, Compton = Schwarzschild and Landauer at T_P are posited as bridges, not derived. Three borrowed bridges to a structure that is internally very frugal.

The contrast with FFGFT lies exactly there: FFGFT pays the price up front (deriving ξ, L_0, justifying the torus) and afterwards needs no bridges – but the entry is harder. Matzke pays almost nothing up front and borrows three times later. Both accounts balance; they simply look elegant at different places.

R.2 Starting points

Matzke

Single axiom: $e_i^2 = +1$ (hyperbit). **[POSTULATE]**
Physical units via three external bridges: Bekenstein ($4\ell_{\mathrm{P}}^2$ per bit), Compton–Schwarzschild (ℓ_{P} as length scale), Landauer–Einstein ($m_{\mathrm{bit}} = m_P \ln 2/(2\pi)$).

FFGFT

Single parameter: $\xi = 4/30000$, derivable as smallest eigenvalue of the spectral operator $\hat{F}$ on $T^4/\mathbb{Z}_3$ **[K]** (Doc. 322). From ξ follow G, ℓ_{P}, $L_0 = \xi \cdot \ell_{\mathrm{P}}$, $\alpha^{-1} = 137.06$ (with $m_e m_\mu$ and K_{frak}, R72; the agreement with the measured value is a property of the anchor E_0), all lepton masses — without bridge axioms **[K]** (Docs. 180, 011, 006). Minimal information unit: winding quantum $\Delta w = 1$, energy $E_{\mathrm{bit}} = \hbar c/L$ (*scale-dependent*) (Doc. 257, A271).

R.3 The Landauer question: universal vs. system-dependent bit values

This is the theoretically deepest difference between the two frameworks.

Matzke's position: a universal bit value

Matzke fixes the Landauer energy scale at the KMS temperature of the Planck scale, $T_P/(2\pi)$:

$$E_{\mathrm{bit}} = k_B \frac{T_P}{2\pi} \ln 2 \quad \Rightarrow \quad m_{\mathrm{bit}} = \frac{m_P \ln 2}{2\pi} \approx 0.110\, m_P \qquad \text{(Matzke, Ch. 6)}$$

With T_P itself, $k_B T_P = m_P c^2$ would give only $m_{\text{bit}} = m_P \ln 2 \approx 0.693\, m_P$; the value $0.110\, m_P$, and hence $n_{\text{thresh}} = 6.41$, presuppose $T_P/(2\pi)$ (section on the stability threshold). This value is *universal*: every bit carries the same mass, independent of the system (electron, black hole, vacuum). From it follow the stability threshold $n_{\text{thresh}} = 2\pi/(\sqrt{2}\ln 2) = 6.41$ and the Schwarzschild relation directly.

FFGFT position: system-dependent bit values

FFGFT understands Landauer's principle as a statement about *phase-space regions* of a thermal carrier ensemble, not about abstract information (A271):

> *Counting proceeds by distinguishability (orthogonality), not by energy. Thermodynamics counts regions and does not read them — there is no thermodynamic truth detector. Sufficient for heat production is the* ***non-bijectivity of the region map****, not logical irreversibility.*

The fundamental bit energy in FFGFT is (Docs. 257, 302):

$$E_{\text{bit}} = \frac{\hbar c}{L} \qquad \text{(FFGFT, Doc. 257)}$$

where L is the physical scale of the system. Landauer's $k_B T \ln 2$ is the limiting case at one particular scale and bath temperature (A273: "two floors — the thermal $k_B T$ and the quantum-geometric $\hbar c/L$; the larger one governs").

The bit energy accordingly varies between scales:

System	Scale L	$E_{\text{bit}} = \hbar c/L$
Planck scale	$\ell_P = 1.62 \times 10^{-35}$ m	$m_P c^2 = 1.22 \times 10^{19}$ GeV
FFGFT floor	$L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39}$ m	$m_P c^2/\xi \approx 9.16 \times 10^{22}$ GeV
Characteristic	$L_e = \hbar c/E_0 \approx 26.67$ fm	$E_0 = 7.398$ MeV
Nuclear physics	~ 1 fm	~ 197 MeV
Room-temperature bit	$L_T \sim \hbar c/(k_B T) \approx 7.6\,\mu$m	$\sim 3 \times 10^{-21}$ J $= 18$ meV (Landauer: $k_B T \ln 2$)

FFGFT critique of Matzke's Landauer bridge

Matzke sets $T = T_P/(2\pi)$ in $E_{\text{bit}} = k_B T \ln 2$. This yields a *universal* bit value. FFGFT objects: Landauer's bound holds for a thermal equilibrium ensemble at bath temperature T — a statement about the *carrier*, not about abstract information (A272). The Planck temperature T_P is not the temperature of the physical system (electron, proton, black hole in the cosmos); it is merely an energy scale. Landauer holds because of mechanics (Liouville), not because of a privileged scale.

Consequence: Matzke's universal stability threshold $n_{\text{thresh}} = 6.41$ hinges essentially on the choice $T = T_P/(2\pi)$. With a system-dependent bit energy, n_{thresh} would itself be system-dependent. **[S]**

Two floors, one principle

FFGFT distinguishes (A273, Doc. 290):

Static Bit ≡ winding quantum $\Delta w = 1$ on T^4: scale-universal, energy-independent as a counting quantity; the energy $E_{\text{bit}} = \hbar c/L$ is scale-specific.

Thermodynamic Irreversible erasure: $Q \geq k_B T \ln 2$ — cost in joules, dependent on the bath temperature T. This price has nothing to do with the information-theoretic quantity (A272: "no energy can be assigned to an information bit, only to its carrier").

Kinematic Reversible bit-flip time: $\tilde{T} = \hbar/E_{\text{bit}}$; Margolus–Levitin $\tau \geq \pi\hbar/(2E)$; energy sets the clock without loss (Doc. 290).

The four guises of a bit energy at the characteristic scale $E_0 = 7.398$ MeV:

$$m_0 c^2 = E_0 = k_B T_0 = \hbar/\tilde{T} \quad \Rightarrow \quad T_0 = E_0/k_B \approx 8.59 \times 10^{10}\ \text{K} \tag{R.1}$$

Landauer at room temperature gives $Q \approx 3 \times 10^{-21}$ J ≈ 18 meV — eight orders of magnitude below E_0. The statements "what the bit is" ($E_0 = 7.398$ MeV) and "what erasure costs" ($k_B T \ln 2 = 18$ meV at 300 K) belong to different layers (Doc. 290, A271).

R.4 Bekenstein entropy and Planck voxels

Matzke

1 bit $=$ Cl$(6) = 3$ Witt pairs $= 1$ Planck voxel $4\ell_P^3$. **[B]**
$S_{\text{BH}} = A/(4\ell_P^2) = V_{\text{shell}}/(4\ell_P^3)$; Bekbits as base unit, $\Omega = 2^b$.

FFGFT

Minimal information unit: winding quantum $\Delta w = 1$ on $\pi_1(T^4) \cong \mathbb{Z}^4$. Discreteness is topological (no half winding), not from a tiling (Doc. 302).
Planck voxels exist implicitly as orbifold fixed points. Bekenstein: $S_{\text{BH}} = A/(4\ell_P^2)$ as a consequence of the Bekenstein bound on the saturation surface **[B]**.
Holographic direction: N-dimensional → 3D (as Matzke), not $3D \to 2D$.

Convergence: fermion generation = algebraic base unit

Matzke: Cl$(6) = 3$ Witt pairs = Planck voxel = one fermion generation (Furey). **[B]**
FFGFT: $T^4/\mathbb{Z}_3$ orbifold, three $\mathbb{Z}_3$ sectors → SU$(3)_c$ → $N_c = 3$ algebraically necessary → one fermion generation. **[B]** (Doc. 321)
Both find: threefold structure, algebraically enforced, no free parameter.

Divergence: depth of derivation

Matzke presupposes $\mathrm{Cl}(6)$ as physical unit (**[POSTULATE]** via Furey identification). FFGFT derives the threefold structure from $\mathbb{Z}_3$ trialization: $N_c = 3$ follows with algebraic necessity **[B]** (Doc. 321, R79–R80).

R.5 Schwarzschild radius and Bekenstein entropy

Matzke's Novel Result #1:

$$r_s = \frac{2}{\sqrt{\pi}} \sqrt{S_{\mathrm{BH}}}\, \ell_{\mathrm{P}} \qquad \text{(Matzke)}$$

The factor $2/\sqrt{\pi}$ comes from the transition hypercube → sphere ("rounding the square").

FFGFT:

$$G = \frac{\xi^2}{4m_e} C_{\mathrm{conv}} K_{\mathrm{frak}}, \quad \ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad r_s = \frac{2GM}{c^2} \qquad \text{(FFGFT, Doc. 180)}$$

Bekenstein: consequence of the bound $S \le 2\pi RE/(\hbar c)$ on the saturation surface **[B]**.

Numerically: 20 $M_\odot$: $r_s = 2GM/c^2 = 59.1$ km; with $S_{\mathrm{BH}} = A/(4\ell_{\mathrm{P}}^2)$, Matzke's formula gives twice this value, 118.1 km. Agreement would hold only for $r_s = \sqrt{S_{\mathrm{BH}}/\pi}\,\ell_{\mathrm{P}}$ or for a different normalization of S, which the document does not state.

Convergence

Both derive r_s without the GR field equations. Both show: ℓ_{P} is enforced by self-consistency, not set axiomatically. Numerically identical up to a factor of 2 in Matzke's normalization.

Divergence

Matzke's $2/\sqrt{\pi}$ follows from hypertetrahedron geometry. In FFGFT no such form factor appears; $r_s = 2GM/c^2$ is the standard relation, and G comes from ξ.

R.6 Stability threshold

Matzke

$n_{\mathrm{thresh}} = 2\pi/(\sqrt{2}\ln 2) = 6.41$ bits. **[B]**
$\mathrm{Cl}(6)$ fermion: 6.000 bits, 0.41 bits below threshold.
A stable universe, because the minimal fermion algebra lies algebraically below the collapse threshold.

Dependence: n_{thresh} hinges essentially on $m_{\mathrm{bit}} = m_P \ln 2/(2\pi)$, i.e. on the Landauer bridge with $T = T_P/(2\pi)$.

FFGFT

Stable matter follows from $\tilde{T} \cdot m = 1$: lepton masses from rational invariants (r_i, p_i) of the torus space **[K]** (Docs. 006, 046, 320).
No direct bit-counting threshold; stability from the orbifold fixed-point structure. FFGFT does not know n_{thresh} as a bit count — and it should be system-dependent if bit values are system-dependent.

Settled: the threshold is scale-dependent [B] (Doc. 329)

Matzke's $n_{\text{thresh}} = 6.41$ is elegant but hinges on the postulate $m_{\text{bit}} = m_P \ln 2/(2\pi)$ – Landauer evaluated at the KMS temperature of the Planck scale, $T_P/(2\pi)$. Doc. 329 settles the question numerically.
The collapse condition itself is *free of any bit value*: Compton = Schwarzschild yields $M_{\text{coll}} = m_P/\sqrt{2}$ purely geometrically **[B]**. Everything non-geometric in $n_{\text{thresh}} = M_{\text{coll}}/m_{\text{bit}}$ sits in the denominator. Inserting the FFGFT bit energy $E_{\text{bit}}(L) = \hbar c/L$ there gives exactly

$$n_{\text{thresh}}(L) = \frac{L}{\sqrt{2}\,\ell_{\text{P}}}, \tag{R.2}$$

a **linear function of the scale**, not an invariant number. Across the scales of the corpus ($L_0 = \xi\ell_{\text{P}}$ to 1 nm) it varies by a factor of 4.6×10^{29}.

Matzke's scale is reconstructible: from $m_{\text{bit}} = \hbar/(cL)$ one gets $L_{\text{Matzke}} = 2\pi\ell_{\text{P}}/\ln 2 = 9.0647\,\ell_{\text{P}}$, and there the formula returns exactly 6.4097 **[B]**. The "universal" bit value *is* the FFGFT bit energy at about nine Planck lengths.

$n_{\text{thresh}} = 6.41$ as a universal number: **[X]** (Doc. 329, R88). Untouched: Compton = Schwarzschild, r_s without field equations, and the Cl(6) structure **[B]**.

R.7 Hawking radiation

Matzke

Hawking quanta are exclusively massless vacuum states $\{V, V_s\}$ of Cl(6). Annihilation: $P_k^+ \cdot P_-^{\text{vac}} = 0$ (exact in $\mathbb{Z}_3$). **[B]**
Information conserved: each V encodes the orientation of the destroyed Bekenstein voxel.
$T_H = T_P/(4\,n_{\text{mass}} \ln 2)$. **[B]**
No formation radiation (formation = time reversal of annihilation).

FFGFT (Doc. 325)

Complete mechanism from FFGFT-native building blocks:
Temperature: KMS rule $T_H = \hbar/(k_B\tau)$ with $\tau = 8\pi GM/c^3$ from $\tilde{T} \cdot m = 1$ **[K]** (Docs. 313 G, 325).

Quantum: $k_B T_H = \hbar c/(4\pi r_s)$ exactly — the Hawking quantum is the system-dependent bit of the horizon scale **[K]** (Doc. 325).
Selection: only $T_R = 0$ escapes; absorption $P_j P_k = 0$; confinement = Hawking selection **[B]** (Docs. 321, 325).
Carrier: massless $n_4 = 0$ torus mode; massive modes fall back (clock compression) **[K]** (Doc. 325).
Information: sector pair $(k, -k)$ orthogonally readable, unitarity of the spectral operator **[K]** (Docs. 322, 325).

Convergence — two complete mechanisms

Both frameworks possess an algebraically complete mechanism. The correspondences are structurally exact (Doc. 325): Matzke's $P_k^+ \cdot V_s = 0 \leftrightarrow$ FFGFT's $P_j P_k = 0$; Matzke's massless $V \leftrightarrow$ FFGFT's $n_4 = 0$ torus mode; Matzke's voxel orientation $\leftrightarrow$ FFGFT's sector pair $(k, -k)$.

Divergence — the bit-value foundation

Matzke's mechanism rests on the universal bit value $m_{\text{bit}} = m_P \ln 2/(2\pi)$; the FFGFT route (KMS from $\tilde{T} \cdot m = 1$ plus system-dependent bit energy) does without it and shows: it is a reparametrization of the KMS relation, not an independent physical input (Doc. 325). In addition FFGFT delivers the power correction $P = P_{\text{std}}(1 - \xi \ln(M/m_P))$ = 0.6–1.4 %, for which Matzke has no counterpart.

R.8 Comparison table

Quantity	Matzke/hyperbit	FFGFT
Fundamental object	$e_i^2 = +1$ **[POSTULATE]**	$\xi = 4/30000$ from spectrum **[K]**
Bit energy	$E_{\text{bit}} = k_B(T_P/2\pi)\ln 2$ (universal) **[POSTULATE]**	$E_{\text{bit}} = \hbar c/L$ (system-dependent) **[K]**
Role of Landauer	physical bridge: $m_{\text{bit}} = m_P \ln 2/(2\pi)$	carrier operation, not abstract info (A271)
Bit value	universal, $\approx 0.11\, m_P$	system-dependent, scale-L-dependent
Planck length	Compton = Schwarzschild **[B]**	$G = \xi^2/(4m_e) \cdot C_{\text{conv}} K_{\text{frak}}$, $\ell_{\text{P}} = \sqrt{\hbar G/c^3}$ **[K]**
Minimal length	ℓ_{P}	$L_0 = \xi \cdot \ell_{\text{P}} \approx \ell_{\text{P}}/7500$ **[K]**
Stability threshold	$n_{\text{thresh}} = 6.41$ bits **[B]**	orbifold fixed-point structure, no bit count
r_s without GR	$(2/\sqrt{\pi})\sqrt{S_{\text{BH}}}\,\ell_{\text{P}}$ **[B]**; with $S = A/(4\ell_{\text{P}}^2)$: $2r_s$	$2GM/c^2$ with G from ξ **[K]**

Quantity	Matzke/hyperbit	FFGFT
α^{-1}	not derived	137.06 from ξ, $m_e m_\mu$, K_{frak} **[K]**
Lepton masses	not derived	m_e, m_μ, m_τ from (r_i, p_i) **[K]**
$\sin^2\theta_W$	not derived	0.2308 **[K]** (Doc. 323)
Hawking mechanism	V/V_s annihilation **[B]**	KMS + $\mathbb{Z}_3$ selection **[K]/[B]** (Doc. 325)
$N_c = 3$ algebraically	Furey identification **[POSTULATE]**	$\mathbb{Z}_3$ trialization **[B]** (Doc. 321)
Holography	N-dim $\to$ 3D **[B]**	$T^4/\mathbb{Z}_3$ projection **[K]**
Information conservation	V encoding of voxel orientation **[B]**	unitarity of the spectral operator **[K]**

R.9 FFGFT objections to Matzke's Landauer bridge

Three concrete objections from the FFGFT corpus:

1. **Landauer applies to carrier operations, not to abstract information** (A272, A273). What Landauer 1961 proves is a lower bound for *carrier operations on a thermal equilibrium ensemble* — not a universal statement about abstract information. The identification $m_{\text{bit}} = m_P \ln 2/(2\pi)$ treats the bit as a thermodynamic object at Planck temperature. That is a realization assumption, not a law.
2. **The bit energy is scale-specific** (Docs. 302, 257). In FFGFT the minimal information unit is the winding quantum $\Delta w = 1$ on T^4 — scale-universal as a *counting quantity*, but with scale-specific energy $E_{\text{bit}} = \hbar c/L$. Landauer's $k_B T \ln 2$ is the special case at one particular scale and bath temperature. The phrase "the bit energy is $m_P \ln 2/(2\pi)$" implies a scale that is not distinguished in FFGFT.
3. **The stability threshold is not a purely algebraic statement** (derived from points 1 and 2). $n_{\text{thresh}} = 2\pi/(\sqrt{2}\ln 2) = 6.41$ follows from the Compton–Schwarzschild equation plus $m_{\text{bit}} = m_P \ln 2/(2\pi)$. The first piece (Compton = Schwarzschild) is geometric **[B]**. The second (Landauer at T_P) is a postulate. The apparent elegance of the result (Cl(6) exactly 0.41 bits below threshold) hinges on this postulate. A system-dependent bit energy would shift the threshold.

R.10 Open bridges

Points with potential for mutual cross-fertilization

[S] n_{thresh} **from the orbifold structure**: Doc. 325 shows that Matzke's bit value is a reparametrization of the KMS relation. The constructive counter-question remains open: can the FFGFT fixed-point condition for fermion stability be

translated into an effective bit-counting statement — and which system-specific bit energy then enters?

[S] **Casimir connection** (Doc. 324, R84): $\xi = C_2(\mathrm{SU}(3)_{\mathrm{fund}})/N_{\mathrm{Fourier}} = (4/3)/10^4$. This links the algebraic structure of Cl(6) (Casimir invariant $C_2 = 4/3$, from $N_c = 3$ **[B]**) with the geometric scale of FFGFT (torus mode number $N_F = 10^4$). This bridge is closed **[K]** (Doc. 324).

[S] $\sin^2\theta_W$ **from** Cl(6): FFGFT derives $\sin^2\theta_W(M_Z) = 0.2308$ **[K]** (Doc. 323). Can Matzke's Cl(6) structure produce an analogous value?

R.11 Status balance

Result	Matzke	FFGFT
r_s without GR	**[B]**	**[K]**
ℓ_{P} from self-consistency	**[B]**	**[K]**
Cl(6) = fermion generation	**[B]** (Furey)	**[B]** (orbifold)
$N_c = 3$ algebraically necessary	**[POSTULATE]**	**[B]**
Bit energy system-dependent	**[X]**	**[K]**
$\alpha^{-1} = 137.06$ (+0.017 %)	**[X]**	**[K]** (anchor E_0)
Lepton masses	**[X]**	**[K]**
$\sin^2\theta_W$	**[X]**	**[K]**
Hawking mechanism worked out	**[B]**	**[K]/[B]** (Doc. 325)
n_{thresh} (bit count, universal)	**[POSTULATE]**	**[X]** (Doc. 329: scale-dep.)
Information conservation	**[B]**	**[K]**
Holography $\neq 3D \to 2D$	**[B]**	**[K]**

Conclusion

Matzke's hyperbit framework and FFGFT are complementary structures with remarkably similar structural findings: fermion generation = algebraic base unit, $N_c = 3$ enforced, Schwarzschild radius without GR.

The deepest difference lies in the treatment of bit values. Matzke posits a universal bit value via the Landauer bridge at Planck temperature — an elegant but, by FFGFT standards, inadmissible universalization. FFGFT holds: bit values are system-dependent ($E_{\mathrm{bit}} = \hbar c/L$), Landauer applies to carrier operations of a thermal ensemble, and the stability threshold n_{thresh} would then not be a universal number.

This is no objection to the algebraic beauty of Matzke's approach, but a theoretical difference in foundations. Doc. 325 makes it constructive: the complete Hawking physics — temperature, selection, carrier, information conservation — follows from $\tilde{T} \cdot m = 1$, the $\mathbb{Z}_3$ structure and the system-dependent bit energy, without a universal bit value. Matzke's $n_{\mathrm{thresh}} = 6.41$ thus demonstrably hinges on the bit-value parametrization, not on the temperature physics.

Main finding

Matzke and FFGFT converge on **fermion generation = fundamental algebraic unit**. They diverge on **bit values**: universal (Matzke, Landauer at T_P) vs. system-dependent (FFGFT, $E_{\text{bit}} = \hbar c/L$). This difference is not cosmetic — it concerns the stability threshold and the mass of a bit. The Hawking temperature formula itself is untouched by it: both routes give the same number, but only the FFGFT route (Doc. 325) does without a universal bit value.

Relation to the FFGFT corpus

Topic	Status	Doc.
Landauer: phase-space bookkeeping	**[K]**	A271
Carrier vs. information (Landauer critique)	**[K]**	A272
Bit energy system-dependent (computing sphere)	**[K]**	A273
Information unit $\Delta w = 1$, $E_{\text{bit}} = \hbar c/L$	**[K]**	Doc. 257
Elementary cells, bit capacity, partition invariance	**[K]**	Doc. 302
$G = \xi^2/(4m_e) \cdot C_{\text{conv}} K_{\text{frak}}$, ℓ_P, L_0	**[K]**	Doc. 180
$\text{SU}(3)_c$ from $\mathbb{Z}_3$ trialization	**[B]**	Doc. 321
$\sin^2\theta_W(M_Z) = 0.2308$	**[K]**	Doc. 323
$\xi = C_2(\text{SU}(3))/N_{\text{Fourier}}$	**[K]**	Doc. 324, R84
Hawking mechanism (KMS, $\mathbb{Z}_3$, n_4=0)	**[K]/[B]**	Doc. 325
Spectral theory, Hilbert-space embedding	**[K]**	Doc. 322
Lepton masses from (r_i, p_i)	**[K]**	Docs. 006, 046, 320
$E = mc^2 = E = m$ (unit identity)	**[K]**	Doc. 077

References

1. D. Matzke, *Black Holes from Hyperbits*, IPI Annual Conference, August 2026.
2. J. Pascher, A271 *Landauer and phase-space bookkeeping*, FFGFT corpus, 2026.
3. J. Pascher, A272 *Carrier and information*, FFGFT corpus, 2026.
4. J. Pascher, A273 *The computing sphere*, FFGFT corpus, 2026.
5. J. Pascher, Doc. 257 *Information unit on* T^4, FFGFT corpus.
6. J. Pascher, Doc. 302 *Elementary cells and bit capacity*, FFGFT corpus, 2026.
7. J. Pascher, Doc. 180 *Derivation of* L_0, FFGFT corpus, 2026.
8. J. Pascher, Docs. 321/322/323/324, FFGFT corpus, 2026.
9. J. Pascher, Doc. 325 *The FFGFT Hawking mechanism*, FFGFT corpus, 2026.
10. R. Landauer, *Irreversibility and heat generation*, IBM J. Res. Dev. **5**, 183, 1961.
11. C. Furey, *Standard Model Physics from an Algebra?*, Diss., Waterloo, 2015.
12. J. Bekenstein, Phys. Rev. D **7**, 2333, 1973.

© 2026 Johann Pascher · CC BY 4.0

Chapter S

Doc. 327 — Self-Adjointness of the Fundamental Fractal Operator $\hat{F}$

Status markers: **[B]** algebraically proved; **[K]** derivable from ξ or numerically checked; **[S]** sketched or postulated, not proved (open).

Abstract

Doc. 322 defines the fundamental fractal operator

$$\hat{F} = \bigoplus_{n=1}^{N} S_n \circ L_n, \qquad S_n = r_n U_n, \quad r_n \in (0,1), \quad U_n \text{ unitary}, \tag{S.1}$$

with logical projectors $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$, and books self-adjointness as an open sketch **[S]** (R82).

The upper bound N in (S.1) is *finite*: $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) is the minimal length scale of FFGFT — below L_0 the concept of "distance" loses physical meaning. The $\bigoplus_{n=1}^{\infty}$ notation in Doc. 322 is a functional-analytic convention, not a physical infinity.

Main result [B] on P_0, [S] on all of $\mathscr{H}$

The R82 gap is **closed on the $\mathbb{Z}_3$-invariant sector** P_0; on the whole space it remains open **[S]**. Because L_0 makes the number of rungs finite, $\hat{F}$ is bounded:

$$\|\hat{F}\| \;\le\; \sum_{n=1}^{N} r_n \;<\; \infty.$$

A bounded symmetric operator is automatically self-adjoint; the deficiency indices are $(0,0)$; there is exactly one self-adjoint realisation. The $\mathbb{Z}_3$ pairing $(k,-k)$ makes $\hat{F}$ normal; on P_0, $\hat{F} = \hat{F}^\dagger$. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$-mirrored $\hat{F}$; the self-adjoint operator there is $\mathrm{Re}\,\hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ — a postulate **[S]**. $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is well-defined provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector **[B]**.

Verification script: `327_selbstadjungiertheit_F_verify.py` — 26 assertions, all passed.

S.1 Starting point

R82 (Doc. 190) records: *"Open: complete self-adjointness proof for $\hat{F}$* ***[S].*** *"* Without this proof, the central claim

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}$$

is formally unsecured: "smallest eigenvalue" requires a real spectrum, and a real spectrum requires self-adjointness.

The definition in Doc. 322 contains a genuine tension: $S_n L_n = r_n U_n L_n$ is in general *not* symmetric for arbitrary unitary U_n — $(U_n L_n)^\dagger = L_n U_n^\dagger \neq U_n L_n$. The self-adjointness must come from the *structure*. On the $\mathbb{Z}_3$-invariant sector P_0, this is what the $\mathbb{Z}_3$ pairing delivers.

S.2 Three observations from the corpus

The following points introduce no new axioms — they read from already-established corpus results what is needed for the proof.

(F1) Finite number of rungs. $L_0 = \xi \cdot \ell_P$ is the minimal length scale (Doc. 180 **[K]**). The scale ladder ends at L_0; N is finite; $\sum_{n=1}^{N} r_n < \infty$ follows trivially. No convergence condition needs to be imposed — there are no infinities in FFGFT.

(F2) $\mathbb{Z}_3$ phase pairing. U_n acts sector-wise as a phase rotation $e^{i\theta_{n,k}}$ (Doc. 322: "phase rotations on the scale sectors"), and the phases respect the orbifold symmetry: $\theta_{n,-k} = -\theta_{n,k}$ for $\mathbb{Z}_3$-charged pairs $(k, -k)$; on null sectors ($q = 0$): $\theta \in \{0, \pi\}$. The same pairing carries the Hawking sector information in Doc. 325.

(F3) Finite fractal recursion. The fractal measure arises from a finite 100-step recursion (Doc. 322), hence $0 < c \leq w \leq C$ for the weight w — bounded and strictly positive.

S.3 Proof chain

Lemma 1 — Filtration realises the Doc.-322 axioms [B]

Under (F1), the L_n satisfy exactly $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$: the "fractal inclusion structure" of Doc. 322 is a scale filtration. *(Script [1]: exact to machine precision.)*

Theorem 2 — Diagonal form and boundedness [B]

Under (F1)+(F2), $\hat{F}$ is block-diagonal:

$$\hat{F}\big|_{\mathcal{H}_k} = \varphi_k \cdot \mathbb{1}, \qquad \varphi_k = \sum_{n \geq k}^{N} r_n \, e^{i\theta_{n,k}}.$$

By the finite rung count (F1): $\|\hat{F}\| \leq \sum_{n=1}^{N} r_n < \infty$. $\hat{F}$ is **bounded**, $\mathscr{D}(\hat{F}) = \mathscr{H}$. *(Script [2]+[3].)*

Theorem 3 — $\mathbb{Z}_3$ pairing: normality [B], symmetry on P_0 [B]

Under (F2), $\hat{F}$ is normal (block-diagonal with scalars φ_k); $\varphi_{-k} = \bar{\varphi}_k$ holds when the sectors k and $-k$ carry the same scale rungs n. On the null sectors ($\theta \in \{0, \pi\}$), $\hat{F}$ is self-adjoint. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$-mirrored $\hat{F}$, $\|\hat{F} - \hat{F}^\dagger\| = 2|\mathrm{Im}\,\varphi_k|$; the self-adjoint operator there is

$$\mathrm{Re}\,\hat{F} = \tfrac{1}{2}(\hat{F} + \hat{F}^\dagger), \qquad \varphi_k^{\mathrm{cov}} = \sum_{n \geq k}^{N} r_n \cos\theta_{n,k} \in \mathbb{R}$$

(cosine form) — a postulate **[S]**. *(Script [4]: normality, $\|\hat{F} - \hat{F}^\dagger\| = 0.019$ on charged pairs, hermiticity on P_0.)*

Corollary 4 — Self-adjointness, deficiency indices $(0, 0)$ [B] on P_0

Bounded (Thm. 2) + symmetric (Thm. 3) ⇒ self-adjoint: for $\hat{F}$ on $P_0\mathscr{H}$, for $\mathrm{Re}\,\hat{F}$ on all of $\mathscr{H}$ (there a postulate **[S]**). Deficiency indices: $\ker(A \mp i) = \{0\}$ for $A = \hat{F}|_{P_0\mathscr{H}}$ or $A = \mathrm{Re}\,\hat{F}$, since $\|(A \mp i)^{-1}\| \leq 1$ for bounded self-adjoint A. **Exactly one** self-adjoint realisation — no extension choice. *(Script [5]: smallest singular value exactly 1, for $\mathrm{Re}\,\hat{F}$.)*

Theorem 5 — Tensor factor $\hat{\Delta}_{T^4}$ with fractal measure [B]

1. Δ on compact T^4 (no boundary): essentially self-adjoint, Fourier eigenbasis with eigenvalues $|n|^2 \geq 0$. (Standard.)
2. Fractal measure (F3): $0 < c \leq w \leq C$, so $\psi \mapsto \sqrt{w}\psi$ is a unitary isomorphism $L^2(d\mu_f) \to L^2(d\mu)$. The fractal measure *preserves* self-adjointness. *(Script [6].)*

Theorem 6 — $\mathbb{Z}_3$ restriction and χ-twists [B]

1. $[\hat{F}, P_0] = 0$ ($\mathbb{Z}_3$ covariance from (F2)) ⇒ restriction to the orbifold sector is self-adjoint. *(Script [7].)*
2. Fixed-point boundary conditions $\chi^3 = 1$ (Docs. 322/320): unitarily implemented twists; spectrum $(n + \alpha)^2 \geq 0$ real in all three twist classes — each carries a self-adjoint realisation. *(Script [8].)*

Corollary 7 — ξ as well-defined minimum [B]

$\hat{F}_{D_4}$ (restriction to the D_4 sector, Doc. 314) is self-adjoint, provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector $P_0\mathscr{H}$ (otherwise the statement holds for Re $\hat{F}$, postulate **[S]**). By the min-max principle:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \min_{\|\psi\|=1,\,\psi\in\mathscr{H}_{D_4}} \langle\psi, \hat{F}\psi\rangle$$

well-defined and truncation-stable: the lowest mode lies in every truncation space that contains it. *(Script [9]: ξ is inserted in the model; what is checked is existence and truncation stability of the minimum.)*

S.4 Status table

Statement	Status
$L_nL_m = L_{\min}$ realised by filtration, exactly	**[B]**
$\hat{F}$ bounded by finite rung count (L_0)	**[B]**
$\hat{F}$ normal; symmetry on P_0 as $\mathbb{Z}_3$ consequence	**[B]**
Symmetry on the whole space (Re $\hat{F}$ as postulate)	**[S]**
Self-adjointness, deficiency indices $(0,0)$, uniqueness on P_0	**[B]**
Fractal measure (finite recursion) preserves self-adjointness	**[B]**
$\mathbb{Z}_3$ restriction and all three χ-twist classes s.a.	**[B]**
$\xi = \lambda_{\min}(\hat{F}_{D_4})$ well-defined, truncation-stable (D_4 in P_0)	**[B]**
Remaining cases	whole space open **[S]**

The R82 gap is closed on the $\mathbb{Z}_3$-invariant sector; on the whole space it remains open **[S]**. There are no infinities in FFGFT; $\sum r_n < \infty$ was never an extra requirement — it follows directly from $L_0 = \xi \cdot \ell_P$.

Relation to the corpus

Building block	Doc.
Minimal length scale $L_0 = \xi \cdot \ell_P$	180
Definition of $\hat{F}$, $\mathscr{D}(\hat{F})$, R82 gap	322
D_4 lattice in Hilbert space	314
Fixed-point boundary conditions, neutrino localisation	320, 322
$\mathbb{Z}_3$ triality, sector structure	321
Sector pairing $(k,-k)$ — same as in Hawking mechanism	325
Fractal measure, 100-step recursion, $D_f = 3 - \xi$	322

References

1. J. Pascher, Docs. 180, 314, 320, 321, 322, 325, FFGFT corpus, 2026.
2. M. Reed, B. Simon, *Methods of Modern Mathematical Physics I/II*, Academic Press.
3. J. von Neumann, *Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren*, Math. Ann. 102, 1929.

© 2026 Johann Pascher · CC BY 4.0

Chapter T

Doc. 328 — Coupling Regimes, Resonance, and Synchronisation

Abstract

Communications engineering has a quantitative three-way classification of coupling regimes (undercritical, critical, overcritical) relative to the damping of the coupled subsystems. This document shows that this classification is an overlooked structural principle: the same mathematical structure recurs in quantum optics, particle physics, and synchronisation theory. The connection to FFGFT is established through ten verification scripts with explicit symbolic and numerical results.

Main results:

- **[B]** Dictionary two-circuit ↔ 2×2 mixing (Hermitian + non-Hermitian, exceptional point $g_{\mathrm{EP}} = |\gamma_1 - \gamma_2|/4$)
- **[B]** Exact duality $D^2 + V^2 = 1$ for all pure pointers; measurement transition as one-parameter continuum $\kappa = d/(2\sigma)$
- **[K]** $\kappa_{\mathrm{mix}}(\theta_W)$ derived from ξ by composition (Doc. 323); screening negative (no simple constant)
- **[B]** GST texture $H_{12} = \sqrt{m_1 m_2}$ proven symbolically (Doc. 041 formulas at leading order)
- **[K]** $K_{\mathrm{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}$ from Doc. 007/008; φ quantitatively outside all Arnold tongues
- Four check-points identified in Doc. 041 (author decision required)

T.1 Motivation and Thesis

Theoretical physics treats couplings primarily via the magnitude of coupling constants. Communications engineering, by contrast, has an older and structurally richer classification: couplings are judged not absolutely but *relative to the damping of the coupled subsystems*. This yields a three-way split — undercritical, critical, overcritical

— distinguishing not merely quantitative shifts but **qualitative changes of spectral structure**.

Thesis

This regime classification is an overlooked ordering principle. It recurs in quantum optics (strong coupling), particle physics (mixing, avoided crossing), and synchronisation theory (Kuramoto threshold, Arnold tongues) under different names. For FFGFT, in which particles are resonance states, it provides the natural language for when one state becomes two — and which frequency ratios lock or remain stably incommensurate.

T.2 The Two-Circuit Formalism [E]

Two resonators with eigenfrequencies ω_1, ω_2, quality factors Q_1, Q_2 and coupling coefficient k. Reference quantity: **critical coupling**

$$k_{\text{crit}} = \frac{1}{\sqrt{Q_1 \cdot Q_2}}. \tag{T.1}$$

Normalised coupling factor $\kappa = k/k_{\text{crit}} = k \cdot Q$ (symmetric case):

- **Undercritical** ($\kappa < 1$): single peak at ω_0; incomplete energy transfer.
- **Critical** ($\kappa = 1$): maximally flat pass-band (Butterworth); optimal power transfer.
- **Overcritical** ($\kappa > 1$): two peaks at $\omega_\pm \approx \omega_0\left(1 \pm \frac{1}{2}\sqrt{k^2 - k_{\text{crit}}^2}\right)$.

Two structural statements [E]

(2.1) Coupling shifts frequencies before splitting them (frequency pulling); the "bare" eigenfrequency is in principle unobservable.
(2.2) The threshold is a ratio: whether a system splits depends on k relative to the loss rates, not on k alone.

Numerically verified by `pruef_328_2_bandfilter.py`: peak formula to $< 10^{-5}$ relative, Butterworth curvature exactly zero at $\kappa = 1$.

T.3 Recurrence in Quantum Physics [E]

The correspondence is mathematically exact, not merely analogical:

Communications eng.	Quantum physics	Condition
Overcrit. peak splitting	Rabi splitting, avoided crossing	$g > (\kappa_c + \gamma)/2$
Critical coupling	Critical coupling (resonator QED)	coupling = loss rate
Frequency pulling	Mass/energy renormalisation, Lamb shift	coupling to continuum
Envelope interference	ρ–ω mixing	two poles, shared channel

T.4 Measurement as Coupling — the κ-Continuum [B]

Every measurement couples the readout system to the object. Complete decoupling $\equiv$ zero information flow.

Symbolic Proof (von Neumann pointer model)

System $a|+\rangle+b|-\rangle$, Gaussian pointer of width σ, shift $\pm d/2$; coupling parameter $\kappa \equiv d/(2\sigma)$. Exact results:

Exact duality — [B] (Scripts 5 and 9A)

$$V(\kappa) = \langle \phi_+ | \phi_- \rangle = e^{-\kappa^2/2} \quad \text{(back-action / residual coherence)}, \tag{T.2}$$

$$D(\kappa) = \sqrt{1 - e^{-\kappa^2}} \quad \text{(Helstrom distinguishability)}, \tag{T.3}$$

$$D^2 + V^2 = 1 \quad \text{for } \textit{every} \text{ pure pointer.} \tag{T.4}$$

Limits: $\kappa \to 0$: $(V, D) = (1, 0)$; $\kappa \to \infty$: $(V, D) = (0, 1)$.
Weak regime: $D \approx \kappa$ (linear), $1 - V \approx \kappa^2/2$ (quadratic).

Not closable — [AXIOM]

"Every measurement apparatus reduces to a pointer model" is a physical model assumption, not provable, only falsifiable. Declared as **[AXIOM]** (model framework).

All quantities V, D, κ reside on the state-transition level of the A271 three-level separation (hardware / state transition / description).

T.5 Synchronisation and Locking [E]

Adler Equation

$\dot{\phi} = \Delta\omega - \varepsilon \sin\phi$. Lock-in precisely for $|\Delta\omega| \leq \varepsilon$; lock range $\propto$ coupling ε. Verified numerically (`pruef_328_3_einrasten.py`, deviation $< 0.1\,\%$).

Kuramoto Model

Below the critical coupling K_c the ensemble remains incoherent ($r = 0$); above K_c the order parameter $r \propto \sqrt{K - K_c}$ grows continuously — a second-order phase transition. This is **not** a three-way split in the sense of Sec. 2: the Kuramoto transition separates two structural regimes (disorder / growing order), not three. The closer parallel to the band-filter three-way classification is the Adler model (Sec. 5.1): there the lock-range threshold is exactly the coupling amplitude, sharing the same threshold structure as the band-filter critical coupling.

Noise Minimum

The locked state exports entropy and is the least noisy available state — the foundation of every frequency synthesis (PLL).

Thermodynamic Clarification

Phase locking occurs exclusively in open, driven, dissipative systems. "Natural systems tend toward an optimum, not toward uniform distribution" holds for nonequilibrium steady states; the second law remains intact.

T.6 Arnold Tongues, KAM, and the Golden Ratio

Qualitative stability [E]

Tongue widths $\Delta\Omega(p/q) \propto K^q$ (Script 9B: slope 1.98 for $q = 2$, 2.69 for $q = 3$). The most irrationally approximable ratio is $\varphi = [1; 1, 1, \ldots]$ (KAM). Connects to Doc. 172/188/275 of the corpus.

Quantitative FFGFT connection [K]

From Doc. 008: $m(n) = m_0 e^{\xi n}$, coupling term $g \sim \xi\omega$, hence $g/\omega = \xi$ (n-independent). Effective K in the circle map:

$$K_{\text{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}. \tag{T.5}$$

The ξ^2 suppression (Doc. 007: $\sigma_\nu \sim \xi_0^2 G_F$) corresponds to the K^2 term in the tongue width. At $K_{\text{eff}} \ll 1$: $\Delta\Omega(1/2) \approx 5.6\times10^{-8}$. φ lies entirely outside every tongue. **[K]** (Orbifold derivation outstanding.)

T.7 Mixing Angles as Coupling Grades — Largely Closed

Dictionary two-circuit ↔ 2×2 mixing [B]

Hermitian case (Script 1):

$$\lambda_\pm = \tfrac{H_{11}+H_{22}}{2} \pm \sqrt{\left(\tfrac{H_{11}-H_{22}}{2}\right)^2 + H_{12}^2}, \quad \tan 2\theta = \frac{2H_{12}}{H_{11} - H_{22}}. \tag{T.6}$$

Non-Hermitian extension (Script 8, exceptional point):

$$g_{\text{EP}} = \frac{|\gamma_1 - \gamma_2|}{4}. \tag{T.7}$$

For $g < g_{\text{EP}}$: only widths split (undercritical); for $g > g_{\text{EP}}$: frequencies split (Rabi, overcritical). **Band-filter k_{crit} and cavity-QED "strong coupling" are special cases of this EP condition. [B]**

Angle	θ [°]	$\sin^2\theta$	$\kappa_{\text{mix}} = \tan 2\theta$
θ_W (Doc. 323: $\sin^2\theta_W = 0.2308$)	28.71	0.2308	1.57
Cabibbo θ_C ($\lvert V_{us}\rvert = 0.22452$, Doc. 041)	12.98	0.050	0.49
CKM θ_{13} ($\lvert V_{ub}\rvert = 0.00365$)	0.21	≈ 0	0.007
PMNS θ_{12} (Doc. 041: 33.44°)	33.44	0.304	2.34
PMNS θ_{23} (Doc. 041: 49.2°)	49.2	0.573	-6.77
PMNS θ_{13} (Doc. 041: 8.57°)	8.57	0.022	0.31

ξ-Level: θ_W sector closed [K]

$\sin^2\theta_W = 0.2308$ derived from ξ in Doc. 323 **[K]**. Hence $\kappa_{\text{mix}}(\theta_W) = 1.5652$ is ξ-derived by composition. Screening against 11 candidate constants: no hit (closest: $e/\sqrt{3}$, 2.7×10^{-3}).

GST texture and H_{12} [B]

Doc.-041 formulas of type $\theta = \arcsin\sqrt{m_1/m_2}$ match the Gatto–Sartori–Tonin texture at leading order ($m_1 \ll m_2$); exactly, the texture gives $\tan\theta = \sqrt{m_1/m_2}$, i.e. $\sin^2\theta = m_1/(m_1+m_2)$ (Script 7). The two forms differ by $y^{3/2}/2$ with $y = m_1/m_2$; for $m_1/m_2 = 1/20$, $\arcsin\sqrt{1/20} = 12.92°$ against $\arctan\sqrt{1/20} = 12.60°$ (2.5 %).

$$M = \begin{pmatrix} 0 & \sqrt{m_1 m_2} \\ \sqrt{m_1 m_2} & m_2 - m_1 \end{pmatrix}, \quad \text{eigenvalues } \{-m_1, m_2\}, \quad \tan\theta = \sqrt{m_1/m_2}. \tag{T.8}$$

$H_{12} = \sqrt{m_1 m_2}$ (geometric mean of masses). **[B]**

Four check-points from Doc. 041

Check-points for Doc. 190 — author decision required

(i) $\delta_{\text{CKM}} = \arcsin\left(2\sqrt{2}\sqrt{\xi}/3\right)$ gives 0.011 rad vs. 1.20 rad; candidate: $\arcsin\left(2\sqrt{2}/3\right) = 1.231$ rad.
(ii) $\theta_{13} = \arcsin\left(\xi^{1/3}\right)$ gives $2.93°$ vs. $8.57°$; candidate: $\arcsin\left((25\xi)^{1/3}\right) = 8.59°$.
(iii) $\lvert V_{us}\rvert$ with Doc. 006 masses: $5.4\,\%$ discrepancy.
(iv) Doc. 007 ↔ Doc. 041 tension already declared as `\begin{warning}` and `\begin{speculation}` in the source.

T.8 Delimitation

Delimitation from resonance-as-vocabulary

Schepis (IPI Letters 3(2) 2025; ICIP-2026) uses resonance as connecting vocabulary without computable quantities. This document takes the opposite direction: every bridge is verified by a script and carries a definite status marker. Observers appear only as read-out sub-circuits; no consciousness vocabulary.

T.9 Closability Balance

Bridge	Status
R-new-b	**[B]** Hermitian + non-Hermitian (EP), complete
R-new-c	**[K]** for θ_W (composition); **[B]** GST texture; four check-points open
R-new-d	**[B]** Duality $D^2 + V^2 = 1$, all pure pointers
R-new-a (qual.)	**[B]** K^q hierarchy, φ quantitatively outside tongues
R-new-a (quant.)	**[K]** $K_{\text{eff}} = 2\pi\xi$ from Doc. 007/008; orbifold derivation outstanding
[AXIOM]	Reducibility measurement → pointer; in principle unprovable

T.10 Verification Scripts

Script	Content	Result
pruef_328_1	Dictionary, sympy	**[B]**
pruef_328_2	Band-filter regimes, numpy	**[E]** confirmed
pruef_328_3	Adler, Arnold tongues	**[E]** confirmed
pruef_328_4	κ_{mix} table	Finding
pruef_328_5	Measurement κ, $D^2 + V^2$	**[B]**
pruef_328_6	ξ-composition, screening	**[K]**
pruef_328_7	GST texture, corpus check	**[B]**, check-points
pruef_328_8	EP, non-Hermitian	**[B]**
pruef_328_9	Pure pointers, K^q	**[B]**
pruef_328_10	$K_{\text{eff}} = 2\pi\xi$	**[K]**

Chapter U

Doc. 329 — Is the Stability Threshold n_{thresh} Universal?

Abstract

Douglas Matzke's hyperbit framework (IPI 2026) derives a stability threshold

$$n_{\text{thresh}} = \frac{2\pi}{\sqrt{2}\,\ln 2} = 6.4097 \text{ bits} \tag{U.1}$$

and reads it as a universal number: Cl(6) carries 6.000 bits and thus sits 0.41 bits below the collapse threshold – matter is stable because the smallest fermion algebra just barely fails to collapse. Doc. 326 raised the suspicion that this number hangs on the choice of a particular temperature. This document tests that numerically and settles it.

Main result [B]

With the system-dependent bit energy of FFGFT $E_{\text{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly

$$n_{\text{thresh}}(L) = \frac{L}{\sqrt{2}\,\ell_{\text{P}}} \tag{U.2}$$

– **linear in the scale**, not an invariant number. Matzke's 6.41 corresponds exactly to the scale

$$L_{\text{Matzke}} = \frac{2\pi\,\ell_{\text{P}}}{\ln 2} = 9.0647\,\ell_{\text{P}}. \tag{U.3}$$

The "universal" bit value *is* the FFGFT bit energy, evaluated at about nine Planck lengths. The universality is a concealed choice of scale.

What remains *untouched* is the robust core of Matzke's argument: the collapse condition Compton = Schwarzschild yields $M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ purely geometrically – without any reference to bits, Landauer, or a temperature. That part stands independently **[B]**.

Verification script: `326_nthresh_skalenabhaengigkeit.py` – 12 assertions, all passed.

U.1 The question

Doc. 326 works out the central divergence between Matzke's framework and FFGFT: universal vs. system-dependent bit values. The suspicion formulated there was that Matzke's stability threshold "hangs on the postulate $T = T_P$" and would itself be system-dependent under a system-dependent bit energy – classified as **[S]**.

This question is numerically decidable, and completely so: one recomputes Matzke's chain, extracts the scale-independent parts, replaces the bit value by the FFGFT bit energy, and sees whether a constant remains.

U.2 Matzke's chain

The Landauer bridge

The temperature used is not the Planck temperature itself but the **KMS temperature** of the Planck scale:

$$T_{\mathrm{KMS}} = \frac{T_P}{2\pi} = 2.2549 \times 10^{31}\ \mathrm{K}. \tag{U.4}$$

The 2π comes from the KMS periodicity – the same one that generates the Hawking and Unruh temperatures in Doc. 325. Landauer at this temperature:

$$E_{\mathrm{bit}} = k_B T_{\mathrm{KMS}} \ln 2 \quad \Longrightarrow \quad m_{\mathrm{bit}} = \frac{m_P \ln 2}{2\pi} = 0.1103\, m_P. \tag{U.5}$$

Numerically reproduced (script [1], ratio 1.000000000). **[B]**

The collapse condition

Compton wavelength equal to Schwarzschild radius:

$$\frac{\hbar}{Mc} = \frac{2GM}{c^2} \quad \Longrightarrow \quad M_{\mathrm{coll}} = \sqrt{\frac{\hbar c}{2G}} = \frac{m_P}{\sqrt{2}} = 1.5390 \times 10^{-8}\ \mathrm{kg}. \tag{U.6}$$

The scale-independent core [B]

$M_{\mathrm{coll}} = m_P/\sqrt{2}$ contains **no bit value**. The collapse condition is purely geometric and identical under any bit interpretation. Numerical check: $n_{\mathrm{thresh}} \cdot m_{\mathrm{bit}} = M_{\mathrm{coll}}$ to nine digits (script [2]).

The threshold

$$n_{\mathrm{thresh}} = \frac{M_{\mathrm{coll}}}{m_{\mathrm{bit}}} = \frac{m_P/\sqrt{2}}{m_P \ln 2/(2\pi)} = \frac{2\pi}{\sqrt{2}\ln 2} = 6.4097. \tag{U.7}$$

The number is thus a *quotient* of a geometric numerator and a posited denominator. Everything about it that is not geometric sits in the bit value.

U.3 The same computation with system-dependent bit energy

Instead of a universal bit value, FFGFT posits (Doc. 257, A271–A273):

$$E_{\text{bit}}(L) = \frac{\hbar c}{L} \quad \Longrightarrow \quad m_{\text{bit}}(L) = \frac{\hbar}{cL}. \tag{U.8}$$

Inserted into $n_{\text{thresh}} = M_{\text{coll}}/m_{\text{bit}}$:

$$n_{\text{thresh}}(L) = \frac{m_{\text{P}}}{\sqrt{2}} \cdot \frac{cL}{\hbar} = \frac{L}{\sqrt{2}\,\ell_{\text{P}}}, \tag{U.9}$$

since $\ell_{\text{P}} = \hbar/(c m_{\text{P}})$. Numerically confirmed for $L \in \{1, 5, 10\}\,\ell_{\text{P}}$ to machine precision (script [3]). **[B]**

What follows

n_{thresh} is a **linear function of the scale**, not a constant. It has no distinguished value as long as no scale is distinguished. The question "how many bits until collapse?" is unanswerable without specifying the scale – it is underdetermined.

Matzke's implicit scale

Solving $m_{\text{bit}} = \hbar/(cL)$ for L and inserting Matzke's bit value:

$$L_{\text{Matzke}} = \frac{\hbar}{c\,m_{\text{bit}}} = \frac{2\pi\,\ell_{\text{P}}}{\ln 2} = 9.0647\,\ell_{\text{P}} = 1.4651 \times 10^{-34}\ \text{m}. \tag{U.10}$$

And indeed Eq. (U.2) returns $n_{\text{thresh}} = 6.4097$ at this scale – exactly Matzke's value (script [4]). **[B]**

Core finding

Matzke's universal bit value is the FFGFT bit energy $E_{\text{bit}} = \hbar c/L$, evaluated at $L = 2\pi\ell_{\text{P}}/\ln 2$. Both frameworks compute with the same quantity; the difference is that FFGFT carries the scale along as a free parameter, whereas Matzke fixes one particular scale and calls the result universal.

U.4 Numerical range

n_{thresh} at the scales of the corpus (script [5]):

Scale	L [m]	n_{thresh}
$L_0 = \xi\ell_{\text{P}}$ (FFGFT minimal scale, Doc. 180)	2.155×10^{-39}	9.43×10^{-5}
ℓ_{P} (Planck length)	1.616×10^{-35}	0.7071
$2\pi\ell_{\text{P}}/\ln 2$ (Matzke's implicit scale)	1.465×10^{-34}	**6.4097**
1 fm (nuclear physics)	1.000×10^{-15}	4.38×10^{19}
$L_e = \hbar c/E_0$ (Doc. 257)	2.685×10^{-14}	1.17×10^{21}
1 nm (atomic physics)	1.000×10^{-9}	4.38×10^{25}

Range across these scales: a factor of 4.64×10^{29}. A quantity varying over 29 orders of magnitude is not a constant. **[B]**

Sensitivity to the temperature choice

Equivalently, the dependence can be expressed via the posited temperature (script [6]); n_{thresh} scales exactly inversely with it:

Posited temperature	$m_{\text{bit}}/m_{\text{P}}$	n_{thresh}
$0.5\,T_{\text{KMS}}$	0.0552	12.82
$0.9\,T_{\text{KMS}}$	0.0993	7.12
$1.0\,T_{\text{KMS}}$	0.1103	**6.41**
$1.1\,T_{\text{KMS}}$	0.1213	5.83
$2.0\,T_{\text{KMS}}$	0.2206	3.20

A deviation of just $10\,\%$ in the posited temperature shifts n_{thresh} by $1 - 1/1.1 = 9.1\,\%$ – and thereby moves $\mathrm{CI}(6)$ from 0.41 bits below the threshold to 0.17 bits above it (at $1.1\,T_{\text{KMS}}$: $n_{\text{thresh}} = 5.83 < 6.000$). The stability claim flips.

U.5 “It from bit” – with resolution depth

The finding of this document reads as criticism. It is, however, above all a *refinement*, and read that way it confirms Matzke’s underlying intuition rather than refuting it.

The analogy to digital technology

In computing, a bit has no absolute meaning. “16 bits” says nothing until it is clear: 16 bits *of what*? A 16-bit colour value, a 16-bit audio sample and a 16-bit address space carry the same bit count and denote entirely different physical quantities. What a bit *means* is fixed only by the quantisation – the spacing between two distinguishable states. For audio that is the amplitude resolution, for images the colour depth, for instruments the size of the least significant bit.

$E_{\text{bit}} = \hbar c / L$ is a resolution depth

The FFGFT bit energy does not say “a bit weighs this much” but “at this scale a distinction costs this much”. $\hbar c$ is the exchange rate between distinction and energy; L states at which level the distinguishing happens. A bit at nuclear scale is a different physical object from a bit at atomic scale – although both carry exactly one yes/no decision.

Wheeler’s *it from bit* is thereby not refuted but sharpened: yes, physics can be built from distinguishability – but information is nothing absolute, it is always relative to a resolution. The missing parameter in the classical formulation is the resolution depth.

In this language Matzke’s choice of a fixed bit value is not wrong but a fixing of the *word width*: one fixes the quantisation and computes – a familiar and legitimate

procedure from engineering practice. The only fallacy is to mistake the chosen word width for a constant of nature. What Matzke actually demonstrated – that the whole of black-hole physics can be formulated as pure bit bookkeeping – stands untouched and is the real result.

The floor of resolution

Unlike in digital technology, the resolution depth in FFGFT is bounded from below:

$$L \geq L_0 = \xi \cdot \ell_{\mathsf{P}} = 2.155 \times 10^{-39}\ \mathrm{m} \qquad \text{(Doc. 180 \textbf{[K]})}. \tag{U.11}$$

L_0 is the counterpart of the smallest representable quantisation step. Below it there are no distinctions left – not because one could not measure precisely enough, but because there is nothing left to distinguish. The maximal resolution depth is thus finite, and precisely from this it follows in Doc. 327 that the spectral operator $\hat{F}$ is bounded; it is self-adjoint on the $\mathbb{Z}_3$-invariant sector. There are no infinities in FFGFT. On why ℓ_{P} as a reference scale is not arbitrary – and where the justification of the factor ξ nevertheless remains open – see the following subsection.

Why ℓ_{P} is not an arbitrary scale

Here an obvious objection arises: if all scales are on an equal footing and only the resolution depth changes, why should there be a floor at all – and why precisely at ξ times the Planck length? Would that not be the very fallacy charged against Matzke: distinguishing a scale without justifying it?

The answer is in Section 3 of this document. ℓ_{P} is not one scale among many but precisely the one at which Compton wavelength and Schwarzschild radius coincide – that is, the collapse condition, which was shown there to be *purely geometric and free of any bit value* ($M_{\mathrm{coll}} = m_{\mathsf{P}}/\sqrt{2}$, with no reference to bits, Landauer, or temperature). The situation is therefore asymmetric:

Matzke's $L = 2\pi\ell_{\mathsf{P}}/\ln 2$	**FFGFT's** ℓ_{P}
comes from a Landauer evaluation at a posited temperature	is the scale at which geometry itself tips over: below it no distinguishable object exists, because it collapses
has no independent justification – *chosen*	follows from ξ via $G = \xi^2/(4m_e)$ – *derived*

What matters is the direction of the derivation chain (Doc. 180):

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) \implies G = \frac{\xi^2}{4m_e} \implies \ell_{\mathsf{P}} = \sqrt{\frac{\hbar G}{c^3}} \implies L_0 = \xi \cdot \ell_{\mathsf{P}}. \tag{U.12}$$

ξ is a spectral value, determined independently of any length (Docs. 322, 327). In FFGFT ℓ_{P} is a *derived* quantity, not an input. The chain runs from the algebra to the length, not the reverse.

The floor follows from the Lagrangian [B] (R89/R90)

The last step of the chain is better founded than Doc. 180 presents it. There $L_0 = \xi^1 \cdot \ell_P$ is justified by the coupling term $g_T^\ell = \xi m_\ell$ carrying one power of ξ – an argument by analogy. The exponent, however, follows *necessarily* from the mass term: the time field Δm is a massive mediator with

$$m_T = \frac{M_{\text{base}}}{\xi} \quad \Longrightarrow \quad \text{range} = \frac{\hbar}{m_T c} = \xi \cdot \lambda_C(M_{\text{base}}), \tag{U.13}$$

hence range $\sim \xi^1$, because $m_T \sim \xi^{-1}$ and $\lambda_C \sim 1/m$. In the fundamental sector ($M_{\text{base}} = m_P$) the range is exactly $\xi \ell_P = L_0$ **[B]**.
This also answers *why* a floor exists at all: because the mediator is massive. Below its Compton range the time field can no longer mediate – there "distance" loses its meaning. No postulate, but propagator physics.
The same rule accounts for the m_T values in the corpus only in part: Doc. 032 evaluates in the lepton sector ($M_{\text{base}} = m_e$); there the universal form gives $m_T = m_e/\xi \approx 3.83$ GeV. The 5.22 GeV of Doc. 032 (which itself calls it an *effective* torsion mass) follow only with additional factors. They result, moreover, only with $E_0 = 7500$ GeV from Doc. 032 ($E_0 \xi = 1$ GeV); with the corpus value $E_0 = 7.398$ MeV the same formula would give about 5 MeV. Universal form, sector-dependent value – structurally identical with $E_{\text{bit}} = \hbar c/L$.

What remains open [S]

The choice $M_{\text{base}} = m_P$ for the fundamental sector introduces no new parameter ($\lambda = 1$ in Planck units) but is not independently justified (R90).
The Schwarzschild check mentioned in Doc. 190 (Clarification 4) offers no support: an object with $M_0 = \xi m_P/2$ has Schwarzschild radius $\xi \ell_P$, but this is an *identity* – from $r_s = 2GM/c^2$ and $\ell_P = G m_P/c^2$ it follows for any value of ξ and any exponent (R92). It tests nothing.

For the arguments of this document and for Doc. 327 this has no consequences: both require only that *some* finite floor exists, not its precise value. The number of rungs of the scale ladder is finite for any $L_0 = \xi^n \ell_P$ with finite n, so $\sum_n r_n < \infty$ and hence the boundedness of $\hat{F}$ hold independently of the exponent. What remains open is the *location* of the floor, not its existence.

The difference from Matzke's choice of scale thus persists and is sharper than the preceding analysis suggested: with Matzke the scale is chosen, and the choice carries the result. With FFGFT the reference scale ℓ_P is derived from ξ and the factor below it follows from the mediator mass; what remains open is only why the base mass of the fundamental sector is m_P.

The rules are scale-invariant, the values are not

The decisive point: that the *values* are scale-dependent does not mean the *rules* are. In FFGFT the same structure holds at every scale:

Rule	Form	scale-invariant
Time-mass duality	$\tilde{T} \cdot m = 1$	yes
Bit energy	$E_{\text{bit}} = \hbar c / L$	yes (form), no (value)
KMS read-off rule	$T = \hbar / (k_B \tau)$	yes
Information unit	$\Delta w = 1$ (winding quantum)	yes
$\mathbb{Z}_3$ triality selection	$T_R = 0$	yes
Fractal dimension	$D_f = 3 - \xi$	yes

The KMS rule, for instance, tempers Unruh, Gibbons–Hawking and black holes with *one* formula across 22 orders of magnitude in τ (Docs. 313, 325). The triality selection $T_R = 0$ acts in hadron confinement just as it does at the horizon of a black hole (Docs. 321, 325). The winding quantum $\Delta w = 1$ is the same topological quantity at every scale – only its energy is scale-specific.

The actual finding

Scale-invariant *rules* with scale-dependent *values* are no contradiction but the normal case in any field theory – consider running couplings: $\alpha_s(\mu)$ changes its value with the scale, while the Callan–Symanzik equation describing that running holds scale-independently.

$E_{\text{bit}}(L) = \hbar c / L$ behaves in exactly this way: the *formula* is universal, the *value* is not. Whoever takes the value to be universal mistakes the rule for one of its evaluations.

With this, “it from bit” can be stated precisely in FFGFT: the world is built from distinctions, and the rules by which distinctions compose into physics are the same at all scales. What changes with the scale is only the price of a distinction – the resolution depth. Matzke’s 6.41 is then no error but the right number for the word width $L = 2\pi \ell_{\text{P}} / \ln 2$.

U.6 Placement

What falls

The reading of $n_{\text{thresh}} = 6.41$ as a *fundamental number* of the universe. It is the quotient of a geometric quantity and a posited bit value; the postulate is a choice of scale, and across scales the result varies by 29 orders of magnitude. The statement “Cl(6) lies 0.41 bits below the threshold” holds only at $L = 9.06\,\ell_{\text{P}}$ and does not survive beyond it. **[X]**

What stands

Considerably more than the criticism might suggest:

- $M_{\text{coll}} = m_{\text{P}} / \sqrt{2}$ from Compton = Schwarzschild – purely geometric, independent of any bit interpretation **[B]**.

- The Schwarzschild radius without field equations: Matzke's Novel Result #1 is untouched by this analysis **[B]**.
- Cl(6) as a fermion generation (Furey), the three falling out, the algebraic structure as a whole **[B]**.
- The Hawking mechanism as an algebraic picture – the structural correspondences to FFGFT are exact (Doc. 326).

Consequence for FFGFT

The analysis is not merely a refutation; it supports the FFGFT position of Doc. 257 and A271–A273 from the outside. If a framework that explicitly posits a universal bit value can be reconstructed as the evaluation of the scale-dependent formula $E_{\text{bit}} = \hbar c/L$ at one particular scale, then scale dependence is not the special case but the general form.

This fits with Doc. 325: there the Hawking temperature is derivable without a universal bit value ($k_B T_H = E_{\text{bit}}(4\pi r_s)$ exactly), and the numerical check in Doc. 326 gave $T_H(\text{Matzke})/T_H(\text{KMS}) = 1.000000$. Both findings say the same thing: the universal bit value is a reparametrization, not an independent physical input.

Worth noting is the origin of the 2π in Matzke's bit value: it comes from the KMS periodicity, i.e. from exactly the relation that generates the Hawking temperature in Doc. 325 *without* a bit value.

U.7 Status summary

Statement	Status
$m_{\text{bit}} = m_{\text{P}} \ln 2/(2\pi)$ from Landauer at $T_{\text{P}}/(2\pi)$ reproduced	**[B]**
$M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ purely geometric, bit-value free	**[B]**
$n_{\text{thresh}}(L) = L/(\sqrt{2}\ell_{\text{P}})$ – linear in the scale	**[B]**
Matzke's scale: $L = 2\pi\ell_{\text{P}}/\ln 2 = 9.0647\,\ell_{\text{P}}$	**[B]**
Range across corpus scales: factor 4.6×10^{29}	**[B]**
$n_{\text{thresh}} \propto 1/T$ – a $10\,\%$ temperature error flips the claim	**[B]**
$n_{\text{thresh}} = 6.41$ as a universal number	**[X]**
Compton = Schwarzschild, Cl(6) structure, r_s without GR	**[B]** (untouched)

Relation to the corpus

Building block	Doc.
Bit energy $E_{\text{bit}} = \hbar c/L$, information unit $\Delta w = 1$	257
Landauer as carrier operation, not abstract information	A271–A273
Elementary cells, bit capacity, partition invariance	302
ℓ_{P}, m_{P}, $L_0 = \xi \ell_{\text{P}}$	180
KMS relation, $k_B T_H = E_{\text{bit}}(4\pi r_s)$	325
Matzke comparison, bit-value divergence	326

References

1. D. Matzke, *Black Holes from Hyperbits*, IPI Annual Conference, August 2026.
2. J. Pascher, Doc. 326 *Black Holes from Hyperbits and from FFGFT*, FFGFT corpus, 2026.
3. J. Pascher, Doc. 325 *The FFGFT Hawking Mechanism*, FFGFT corpus, 2026.
4. J. Pascher, Docs. 257, 302, A271–A273 (bit energy, Landauer), FFGFT corpus.
5. J. Pascher, Doc. 180 *Derivation of* L_0, FFGFT corpus, 2026.
6. R. Landauer, IBM J. Res. Dev. **5**, 183, 1961.

© 2026 Johann Pascher · CC BY 4.0

Chapter V

Doc. 330 — Three Operations from T^4 to Observable Space

Abstract

This document consolidates the demarcation of FFGFT against discrete emergence models (here: HLV by Krüger, August 2026) and clarifies which operations mediate between T^4 and the observable space $\mathbb{R}^3 \times \mathbb{R}_t$.

Central claim: The failure of discrete lattice models (class-X4 overlaps, missing operator specificity, open continuum limit) is not a surprise, but a structural consequence of treating geometric auxiliary constructs as ontological building blocks. FFGFT avoids these failures not by choosing a different lattice, but because the fundamental level T^4 is compact and geometrically complete — and the three operations leading from it to the observable space have strictly different logical status.

The three operations (Doc. 270 **[K]**):

1. **Type I** — Duality decompactification: $S^1_m \to \mathbb{R}_t$ via $\tilde{T} \cdot m = 1$. Information-preserving (embedding, not projection).
2. **Type II** — Geometric projection: $D3 \to D2 \to D1$. Lossy, irreversible. The step $D3 \to D2$ is identified as holographic projection.
3. **Type III** — Representational translation: $T^4 \leftrightarrow \mathscr{H}$ (Hilbert space/matrix). Bijective, lossless — upward movement, no geometric dimension reduction.

V.1 Starting point: the HLV audit as case study

Marcel Krüger subjected his own Helix-Light-Vortex model (HLV) to four prospectively frozen mathematical tests in *Local-to-Global Compatibility on Aperiodic Cut-and-Project Carriers* (v1.4, August 2026). All key claims fail or remain open.

1. **Local-to-global compatibility:** 5,455 genuine volume overlaps (class X4) among 3,384 projected tetrahedra. Six witnesses certified with 120-digit arithmetic. *Failure.* **[E]**
2. **Operator specificity:** An ordinary periodic cubic lattice (CUBIC) outperforms HLV in spectral multiplet score (GAE-001A). *Rejected.* **[E]**

3. **Geometry-to-constitutive law:** Pure carrier geometry does not determine physical densities, masses, or couplings (Proposition 2.2). *Trivially negative.* **[E]**
4. **Finite-to-continuum:** A bounded graph has bounded spectrum, no UV limit. Mosco/resolvent convergence unproven. *Open.* **[E]**

Corollary (Krüger 4.5) [E]

Local validity of an operator does not imply global embeddability of the underlying carrier.

The structural cause

HLV attempts to use geometric bodies (3D tetrahedra from $\mathbb{R}^6$ projection) as ontological building blocks of physical space. It thereby inherits all four problems. The remedy is not a *better* lattice — it is a fundamental carrier that is geometrically complete and closed from the outset.

V.2 The FFGFT fundamental level: T^4 with D4 lattice

Structure of T^4

The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_m$$

with three spatial circles $S^1_{x,y,z}$ (radius R_H, quasi-continuous) and one mass circle S^1_m (radius $R_m \sim \xi \cdot \ell_P$, discrete). The fourth direction S^1_m is not a mass dimension alongside a separate time, but the substitute-time direction itself. Mass and time are two readings of *the same* dimension; $\tilde{T} \cdot m = 1$ is the translation dictionary (Doc. 270 **[K]**, Doc. 314 **[K]**). **[POSTULATE]**

The lattice on T^4 is the D4 lattice (Doc. 314 **[K]**): kissing number 24, $|\mathrm{Aut}(D_4)| = 1152$, triality as built-in $\mathbb{Z}_3$ symmetry (16 order-3 elements generate the $T^4/\mathbb{Z}_3$ orbifold with 9 fixed points). Negative result with theorem character: $5 \nmid 1152$, so $\mathrm{Aut}(D_4)$ contains no order-5 element **[K]**.

Why T^4 is flat — intrinsic vs. extrinsic curvature [K]

A ring torus embedded in $\mathbb{R}^3$ ($T^2 \hookrightarrow \mathbb{R}^3$) is curved: the outer side is convex (positive Gaussian curvature), the inner side concave (negative Gaussian curvature). By the Theorema Egregium this Gaussian curvature is *intrinsic*, an isometric invariant; on the ring torus it is pointwise non-zero, $K = \cos v/\big(r(R + r\cos v)\big)$, and only its integral vanishes ($\int K dA = 2\pi\chi = 0$, Gauss–Bonnet). Intrinsically flat ($K \equiv 0$) is the flat torus $\mathbb{R}^n/\Lambda$, which cannot be embedded isometrically in $\mathbb{R}^3$; $T^4 = \mathbb{R}^4/\Lambda$ is of this kind.

Intrinsically flat torus [K]

T^4 is intrinsically flat: an interior observer measures no curvature. Angle sums are π, parallel geodesics remain parallel. T^4 is locally isometric to $\mathbb{R}^4$ — with global boundary identification, but without local curvature corrections.

Two direct consequences for FFGFT:

(a) D4 lattice through topology, not curvature. What the D4 lattice carries into the Hilbert space is topological data: winding numbers, mode content, triality. The distinction $\mathbb{Z}^4/D_4$ is spectral, not geometric (Doc. 314 **[K]**: "The torus is flat anyway — curvature was never the difference").

(b) Fourier analysis without curvature corrections. On the flat torus, Fourier modes $e^{ik\cdot x}$ form the natural basis; Laplacian eigenvalues are $|k|^2$ without curvature correction. This is the basis for the bijective, lossless Type III translation $T^4 \leftrightarrow L^2(T^4)\otimes\mathbb{C}^3$: Fourier decomposition on a flat torus is a complete isometric map.

This structurally distinguishes FFGFT from approaches like HLV, which embed geometric bodies into $\mathbb{R}^3$ and thereby generate overlap problems: the embedding question does not arise in FFGFT, because T^4 as an intrinsically flat, self-contained carrier requires no embedding into an outer space.

How information is encoded in the D4 lattice [K]

The geometry does not disappear when moving to the Hilbert space — it migrates into spectrum and fibre (Doc. 314 **[K]**):

Spectrum = theta series. The degeneracies of the Laplacian on flat T^4 are exactly the theta series of the D4 lattice: $\theta_{D_4}(q) = 1 + 24q^2 + 24q^4 + \ldots$ The kissing number 24 becomes the degeneracy of the first excitation $|k|^2 = 2$. Two flat tori ($\mathbb{Z}^4$ vs. D_4) are distinguishable by spectrum alone, without curvature **[K]**.

Fibre = triality. Triality is built in, not appended **[K]**: $|\mathrm{Aut}(D_4)| = 1152$, quotient to $W(D_4)$ is $6 = |S_3|$. The 16 order-3 elements generate the $\mathbb{Z}_3$ orbifold and act on every mode orbit as a $\mathbb{C}^3$ circulant. The fibre $\mathbb{C}^3$ follows from the lattice automorphism structure, not an additional postulate.

Winding numbers = topology. The reciprocity relation $\lambda_4 \cdot m = 2\pi$ holds mode-by-mode exactly **[K]**: winding number = mass number. Winding numbers are topological — invariant under deformation; fractal corrections accumulate only when unrolled, not when rolled up.

In one sentence (Doc. 314): "In the Hilbert space the lattice becomes flat, but not without consequence — what geometry carried, spectrum and fibre carry, and the division of labour between them is exactly determinable."

Where $\tilde{T}\cdot m = 1$ appears in the Hilbert space [K]

The time-mass duality appears in $\mathscr{H} = L^2(T^4)\otimes\mathbb{C}^3$ not as time evolution, but in three complementary forms:

(a) As the fourth winding number n_4. The multi-index $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$ counts windings in all four torus directions **[B]** (Doc. 322). The mass dimension S^1_m carries the winding number n_4. The reciprocity relation $\lambda_4 \cdot m = 2\pi$ holds mode-by-mode exactly

[K] (Doc. 314): winding number = mass number. $\tilde{T} \cdot m = 1$ is thus the Kaluza-Klein relation of the fourth torus direction, readable as n_4 quantum number in the spectrum.

(b) As the mass operator eigenvalue equation.

$$\hat{M}\,|n_\theta, n_\phi, p\rangle = m_i\,|n_\theta, n_\phi, p\rangle \quad \textbf{[K]} \tag{V.1}$$

The eigenvalues are the particle masses (Doc. 322, Doc. 320). The duality appears as the spectrum of the operator.

(c) As radius-class separation in the spectrum. The splitting at $R_4 \neq R_{123}$ makes $\tilde{T} \cdot m = 1$ directly readable **[K]** (Doc. 314): the mass dimension generates a large mode spacing $\sim 1/R_m$, the spatial directions a quasi-continuous spacing $\sim 1/R_H$.

What does *not* appear in the Hilbert space

Time $\mathbb{R}_t$ itself. It is the result of the Type-I decompactification $S^1_m \to \mathbb{R}_t$ — a geometric downward operation *before* the Type-III translation. In the Hilbert space $L^2(T^4)$, only the torus lives; time appears only when leaving the Hilbert space and embedding the spectrum into $\mathbb{R}_t$. $\tilde{T} \cdot m = 1$ resides in the Hilbert space as n_4 winding number and mass operator eigenvalue structure, not as time evolution.

Where the fractal property resides in the Hilbert space [K] / [B]

The fractal property of FFGFT appears in the Hilbert space at three precisely separable locations:

(a) In the operator $\hat{F}$ itself — as self-similarity structure [B].

$$\hat{F} = \bigoplus_{n=1}^{N} S_n \circ L_n, \quad S_n = r_n U_n, \quad r_n \in (0,1) \tag{V.2}$$

Doc. 322 fixes only $r_n \in (0,1)$; the concrete choice is open (cf. Question 3). The scale operators S_n implement discrete scale transformations; each level n is a contraction of the previous by r_n — this is self-similarity as an operator. $\hat{F}$ encodes the sub-Planck geometry as a hierarchical convolution of scale levels down to the minimum length scale $L_0 = \xi \cdot \ell_P$ (Doc. 180, 327 **[K]**).

(b) In the fractal dimension $D_f = 3 - \xi$ — as spectral property [K].

$$D_f = \inf\left\{ s \,:\, \sum_i \lambda_i^s < \infty \right\} = 3 - \xi \approx 2.9999 \quad \textbf{[K]} \tag{V.3}$$

D_f is defined via the Hausdorff measure of the spectrum of $\hat{F}$ (Doc. 322): the spectrum on $L^2(T^4)$ has a fractal fine structure. The value is spatial ($3 - \xi$, three dimensions), not four-dimensional. Local action: $D_f^{\text{space}} = 3 - \xi$ with 6.7×10^{-5} relative (Doc. 314 **[K]**).

(c) In the interface factor K_{frac} — as scale bridge [K].

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9867 \tag{V.4}$$

K_{frac} is cumulative (Doc. 133): it accumulates over the RG running across 17 orders of magnitude and acts at the transition from geometric (bare) to measured (SI) mass: $m_i^{\text{SI}} = K_{\text{frac}} \cdot C_{\text{conv}} \cdot m_i^{\text{bare}}$.

Decisive separation [K]

In the *rolled-up* T^4 (Hilbert space) there are **no** fractal corrections — winding numbers are purely topological (Doc. 314 **[K]**). The fractal property appears only when unrolling (Type I: $S^1_m \to \mathbb{R}_t$) as an accumulated defect. In the Hilbert space itself, $\hat{F}$ is a bounded operator, self-adjoint with real spectrum on the $\mathbb{Z}_3$-invariant sector **[B]** (Doc. 327, Theorem 3) — the fractality resides in the scale hierarchy of the operator structure ($r_n \in (0,1)$), not in the carrier $L^2(T^4)$.

V.3 No embedding problem in T^4

T^4 is compact, complete, and geometrically self-contained. There is no embedding problem because T^4 does not need to be *embedded* in an outer space: it *is* the fundamental level. The class-X4 errors of HLV arise because projection elements must be embedded into an external $\mathbb{R}^3$. In FFGFT this question does not arise.

Type III: representational translation upward

The translation $T^4 \leftrightarrow \mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230, 314, 322) is a bijective, lossless upward movement (Type III, Doc. 270 **[K]**). It adds no geometric extra dimensions, but expresses the same T^4 content in a higher-dimensional language. The mass operator $\hat{M}$ (Doc. 322, section on masses) has the particle masses as eigenvalues; $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is the smallest eigenvalue of the D_4 sub-operator of $\hat{F}$ **[B]**, provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector, on which $\hat{F}$ is self-adjoint (Doc. 327, Theorem 3 and Corollary 7).

V.4 The three operations from T^4 to the observable space

Type I: duality decompactification

$$S^1_m \xrightarrow{\tilde{T}\cdot m=1} \mathbb{R}_t$$

The unrolling of the compact mass dimension into the non-compact time direction is an *embedding*, not a projection. No mode content is lost; time arises as an additional non-compact direction **[K]**.

Type II: geometric projection (lossy)

The further steps $D3 \to D2 \to D1$ are genuine geometric projections of spatial directions: lossy, irreversible. The step $D3 \to D2$ is identified as holographic projection (Doc. 270 **[K]**). It is not fundamental.

Classification of the holographic step

The step $D3 \to D2$ is *not* the fundamental level of FFGFT. It is a lossy observer projection — physically relevant for surface phenomena (black hole membrane, Doc. 325), but not for the derivation of masses and couplings anchored on T^4. The two-dimensional projection surface is not a fundamental object.

Three-way comparison

Type	Operation	Direction	Status
I	Duality decompactification $S^1_m \to \mathbb{R}_t$	geometrically down	information-preserving **[K]**
II	Geometric projection $D3 \to D2 \to D1$	geometrically down	lossy **[K]**
III	Representational translation $T^4 \leftrightarrow \mathscr{H}$	representationally up	bijective, lossless **[K]**

V.5 Open questions in the flat Hilbert space — status

Question 1: Haar measure and D4 lattice [B]

The canonical Haar measure on T^4 is compatible with the D4 spectral structure (scripts 330-1, all 7/7): Fourier modes are exactly orthogonal; D4 contains only even norm shells; $\mathrm{Aut}(D_4) \subset \mathrm{Iso}(T^4)$ leaves the Haar measure invariant. The $\mathbb{Z}_3$ orbifold breaks translation invariance and splits the 24-fold degeneracy to $9 \oplus 8 \oplus 4 \oplus 2 \oplus 1$ (Doc. 314 **[K]**) — but only as fine structure $\sim \xi \approx 10^{-4}$, below any measurement threshold. **[B]**

Question 2: Orbifold fixed points [K] / [S]

Nine fixed points of $T^4/\mathbb{Z}_3$ follow from the Lefschetz fixed-point formula (script 330-2, B1): $|\det(1-A)| = |(1-\omega)(1-\omega^2)|^2 = 9$. Neutrino localisation: $\nu_1 \leftrightarrow F_2$, $\nu_2 \leftrightarrow F_5$, $\nu_3 \leftrightarrow F_7$ **[K]** (Doc. 322). What remains **[S]**: back-reaction of Hawking emission on the fixed-point structure during $M \to M_{\mathrm{coll}} = m_{\mathrm{P}}/\sqrt{2}$ (explicitly open, Doc. 325).

Question 3: Fractal weighting r_n and $L^2(T^4)$ [B]

$r_n \in (0,1)$ (Doc. 322; concrete choice open) is position-independent: $\hat{F}$ does *not* break the translation invariance of $L^2(T^4)$ — only the $\mathbb{Z}_3$ orbifold does (and only as fine structure $\sim \xi$). $\hat{F}$ is bounded ($\|\hat{F}\| \le \sum r_n < \infty$); it is normal and self-adjoint on the $\mathbb{Z}_3$-invariant sector of $\mathscr{H}$ **[B]**, on the whole space open **[S]** (Doc. 327, Theorem 3). With $r_n = \xi^n$ one would have $\|\hat{F}\| \le \xi/(1-\xi) = 1.00013\,\xi$, so ξ would also be an upper spectral bound, and the $\hat{F}$ spectrum would leave no room for a mass hierarchy.

Status of the three questions

Question 1 (Haar/D4): **[B]** — fully resolved.
Question 2 (fixed points): **[K]** for 9 fixed points and ν-localisation; **[S]** for back-reaction during evaporation (explicitly open, Doc. 325).
Question 3 ($r_n/\hat{F}$): **[B]** — $\hat{F}$ bounded, self-adjoint on the $\mathbb{Z}_3$-invariant sector (whole space **[S]**); r_n does not break translation invariance.

V.6 Why Krüger's four problems do not arise in FFGFT

1. **Class-X4 overlaps:** arise because HLV embeds projection elements into $\mathbb{R}^3$. In FFGFT, T^4 is the fundamental level; no embedding into an outer space occurs.
2. **Missing operator specificity:** In FFGFT the spectrum is determined by the D4 lattice with built-in triality. The negative result $5 \nmid |\mathrm{Aut}(D_4)|$ is itself a specific theorem about the spectrum **[K]**.
3. **Geometry without physics:** T^4 alone also does not fix coupling constants. The physics comes from the combination T^4-spectrum $+ \xi = \lambda_{\min}(\hat{F}_{D_4})$.
4. **Continuum limit:** No limit problem, because Type I is an embedding (no limit passage needed) and Type III is bijective (no information loss). Only Type II is lossy — but this is known and documented (Doc. 270).

V.7 What remains from the comparative discussion

Structurally secured: The four HLV problems do not arise in FFGFT — due to T^4 as intrinsically flat, self-contained carrier, bijective Type-III translation, and Type-I embedding $S^1_m \hookrightarrow \mathbb{R}_t$.

Correct from the comparative discussion: The holographic projection $D3 \to D2$ (Type II) is lossy and not fundamental; mass and time are not encoded in the 2D surface but arise through Type I ($\tilde{T} \cdot m = 1$) and the recursion ξ^n.

Correction vs. uncritical adoption: There is no "two-dimensional projection surface as fundamental object" in FFGFT. The corpus has a clear ontological hierarchy: T^4 as primary reality (Doc. 152 **[POSTULATE]**), not a 2D surface. $\tilde{T} \cdot m = 1$ is the Kaluza-Klein relation of the mass dimension, not a "reading principle on a screen".

Verification scripts: `pruef_330_1_haar_d4.py` (7/7), `pruef_330_2_fixpunkte_rn.py` (9/9).

Status overview

© 2026 Johann Pascher · CC BY 4.0

Element	Status
$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_m$ as fundamental level	**[POSTULATE]**
D4 lattice on T^4; kissing number 24; $\lvert\mathrm{Aut}(D_4)\rvert = 1152$	**[K]**
T^4 intrinsically flat; spectral structure from topology	**[K]**
Information in spectrum (theta series) and fibre (triality)	**[K]**
$\tilde{T} \cdot m = 1$ as n_4 winding number and mass operator eigenvalue	**[K]**
Fractal property in $\hat{F}$ structure ($r_n \in (0,1)$), not in $L^2(T^4)$	**[B]**
$D_f = 3 - \xi$; no fractal corrections in rolled-up T^4	**[K]**
Three operation types (Doc. 270): Type I/II/III	**[K]**
Type III bijective ($T^4 \leftrightarrow \mathscr{H}$, Doc. 322/314)	**[K]**
HLV class-X4 errors structurally excluded (no embedding problem)	**[B]**
Continuum limit structurally resolved by Type I (embedding)	**[K]**

Chapter W

Doc. 332 — Two Roads from the Discrete to the Continuum

Source

This treatise references and comments on statements from: Krüger, M. (2026). *Dimension-Graded Geometry and Typed Emergence on Aperiodic Carriers: From Cellular Degree, Holonomy, and Regge Curvature Support to a Falsifiable Spacetime-Limit Programme in an HLV Testbed.* Zenodo, DOI: 10.5281/zenodo.22105698. All direct quotations in Part 1 are drawn from this preprint or the author's accompanying summary. The refinement in Section 2.1 ("two different meanings of continuum") follows a piece of feedback from M. Krüger (IPI, 26 August 2026) on an earlier version of this document; see Doc. 190, register entries R96, R97, and R98.

Introduction

In theoretical physics, one methodological task keeps recurring: starting from a discrete or compact structure — a lattice, a simplicial complex, a compact manifold — one wants to obtain a continuum that admits a physical interpretation (time, space, a gauge field, curvature). The passage from the discrete to the continuous is by no means automatically licensed. It requires an explicitly declared, mathematically controlled map that shows precisely *which* structure is preserved and *which* is lost. If this map is not supplied, a category error results: objects from different mathematical categories (for instance, chains of different degree in a chain complex) are tacitly identified without the identification itself being justified.

This treatise contrasts two different strategies for justifying such a passage: first, a careful, multi-stage construction within the discrete Regge-calculus and homological-lattice-vacuum tradition, which blocks the category error through explicit lemmas and *postulates* (rather than proves) the continuum limit via a refinement sequence with Mosco convergence; and second, a construction in which the continuum does not arise as the limit of a sequence but as a canonical pullback of functions along a fixed covering projection.

W.1 The Discrete Construction — Chain Complexes, Regge Curvature, and the Conditional Hyperbolic Limit

The Methodological Foundation: Chain Complexes Without Physical Presupposition

The starting point is a finite oriented chain complex K with boundary maps ∂_p satisfying $\partial^2 = 0$. The dual cochain spaces $C^p = \mathrm{Hom}(C_p, \mathbb{R})$ carry coboundary operators d_p with $d^2 = 0$. Crucially, the "degree" p here is "nothing more than this chain degree — no physical meaning is built in."

Category-Error Block

$C^p \cong C^q$ for $p \neq q$ is "not well-typed unless an explicitly declared map $\Phi : C^p \to C^q$ exists that demonstrably preserves the relevant operations." The lemma "reflexively blocks 'edge = time', 'face = gauge field', '3-cell = gravity' before anyone can assert this without supplying the bridge."

Why a Single Dimension Alone Carries No Curvature

A standard result states: "in 1D the Riemann tensor is a 2-form valued in the endomorphism bundle — every 2-form on a 1-manifold vanishes trivially." Importantly, "extrinsic curvature (a line bent in $\mathbb{R}^2$ or $\mathbb{R}^3$) is something different and does not contradict the theorem" — this blocks the fallacy that a visibly curved line automatically yields a one-dimensional gravitational degree of freedom.

Edge Transport, Face Holonomy, and the Limit of Pure Gauge Transformations

A $U(1)$ phase 1-cochain θ_{ij} on edges generates, via $d\theta$, a loop phase Φ_f on faces, invariant under vertex rephasing $\theta_{ij} \mapsto \theta_{ij} + \alpha_j - \alpha_i$. If $\theta = d\alpha$ is exact, then $d\theta = d^2\alpha = 0$ — "pure vertex phase differences therefore generate no nontrivial flux on a contractible region." This marks the first point at which "a face can for the first time carry algebraically different information than an edge" — whether this ever becomes a physical gauge field remains explicitly open.

Regge Curvature Sits on Codimension Two — and the Trap of Local Tetrahedralization

In Regge calculus, curvature in a d-dimensional piecewise-flat simplicial manifold concentrates on $(d-2)$-dimensional "hinges" — in 3D on edges. The deficit angle $\varepsilon_h = 2\pi - \sum_\sigma \theta_{\sigma h}$ is "defined only if a valid global piecewise-flat complex with compatible simplex metrics exists."

The Decisive Precondition

"Local tetrahedralization of individual projected cells is not sufficient to define a global Regge curvature field if different tetrahedra overlap with positive volume or fail to meet face-to-face — in that case the star of a putative hinge is not the star of a valid manifold at all, and the deficit angle no longer has a unique geometric interpretation."

If tetrahedra overlap with positive volume (rather than satisfying the so-called HFF condition — "face-to-face": intersections of distinct closed simplices are either empty or a shared face), the geometric meaning of the deficit angle collapses, even though the individual local computations remain error-free.

This restriction concerns the classical, piecewise-flat, face-to-face Regge calculus. Generalized Regge approaches exist in the literature — for instance, constructions using simplices of constant curvature rather than flat ones [4], or allowing a variable triangulation that changes over time [5] — but these solve a different problem: they replace the flatness assumption of individual simplices, or permit a changing discretization, without altering the requirement of a globally face-to-face-consistent triangulation, and they do not address the case discussed here of overlapping simplices that fail to meet face-to-face.

Separating Algebraic Chain Structure from Geometric Realizability

A previously observed empirical failure — referred to in the preprint itself as the "007F result" — is elevated to a construction principle: a numerical investigation reported there found, according to the author, 5455 certified cases in which projected cells overlap with positive volume. An explicit distinction is drawn between the abstract provenance chain complex K_{abs} (algebraically valid solely through orientation and $\partial^2 = 0$, independent of any embedding) and the embedded realization $\iota : K_{\text{abs}} \to E_{\|}$, which must satisfy the strict HFF condition.

Clean Isolation of the Failure

"$\partial^2 = 0$ in K_{abs} implies $d^2 = 0$ and all finite algebraic Hodge relations — regardless of whether a concrete realization ι satisfies the strict HFF condition. What fails is only the promotion of these algebraic cells to a globally embedded manifold."

Hodge Positivity

Standard DEC statement: $\langle \omega, \Delta_p \omega \rangle = \|d_p \omega\|^2 + \|\delta_p \omega\|^2 \geq 0$, so Δ_p is positive semi-definite and self-adjoint in the declared finite inner product.

Spectral Dimension: Triviality at Both Ends

For a finite positive semi-definite operator A on a space of dimension $N = \dim \mathscr{H} < \infty$ with eigenvalues $\lambda_1, \ldots, \lambda_N$, $P(\sigma) = N^{-1} \mathrm{Tr}\, e^{-\sigma A}$ is defined, from which $d_s(\sigma) = -2\sigma P'/P$ follows.

Unavoidable Consequence

$\lim_{\sigma\downarrow 0} d_s(\sigma) = 0$ (Taylor $P \approx 1-\sigma\bar{\lambda}$) and $\lim_{\sigma\to\infty} d_s(\sigma) = 0$ if A has a nontrivial null space. "A positive-dimensional plateau on a finite support can only be an intermediate-scale phenomenon — both endpoints are trivial."

The Continuum Commitment via Mosco Convergence — a Postulate, Not a Proof

A fixed support with bounded degree/edge weight has a bounded graph Laplacian — no intrinsic UV sequence $\lambda \to \infty$. With a shrinking scale $a_n \downarrow 0$, $A_n = a_n^{-2} L_n$, it is *postulated, not proved*: the Dirichlet forms $\mathcal{E}_n$ converge in the Mosco sense to $\mathcal{E}[f] = \int \nabla f^T C(x) \nabla f d^3x$ with $C(x) \approx c^2 \mathbb{1}$.

The Conditional Hyperbolic Limit

Under five unproven assumptions (Mosco convergence, uniform energy stability, convergent initial data, vanishing consistency errors, compactness), the weak limit yields $\partial_t^2 u - c^2 \Delta u = 0$ — "none of the five points is actually demonstrated in this preprint."

W.2 The Alternative Construction — Continuum as Canonical Pullback Rather Than as the Limit of a Sequence

Rather than considering a sequence of discrete approximations with shrinking lattice spacing, the compact structure is treated from the outset as fundamental. The passage to the noncompact description proceeds via a **canonical function pullback** along a fixed covering projection $p : \mathbb{R} \to S^1$.

The Pullback as an Exact Map

$\Phi = p^* : C(S^1) \to C(\mathbb{R})$ is an exact, infinite-dimensional, continuous, algebra-preserving map — not a family of approximations. Every periodic function on S^1 immediately yields a periodic function on $\mathbb{R}$; the lossy direction lies at $\mathbb{R} \to S^1$, not at the compactification itself.

Why This Is Methodologically Distinct from the Category-Error Block

The category-error block is directed against identifying two chain spaces of *different degree* within a discrete chain complex. The canonical function pullback does not structurally fall into this category: it does not identify two different chain degrees, but rather two *representations of the same function algebra*. The general methodological concern remains justified for genuine chain-degree changes; it simply does not describe the case of an explicit, declared, operation-preserving pullback between function spaces.

Refinement: two different meanings of "continuum"

The pullback $\Phi = p^* : L^2(S^1_m) \to \mathrm{Per}_{R_m}(\mathbb{R}_t)$, $(\Phi f)(t) = f(t \bmod R_m)$, delivers a *continuum of the coordinate representation*: a sufficiently regular representative of f can be evaluated, compared against non-periodic test functions, and (almost everywhere) differentiated at every point $t \in \mathbb{R}$ rather than only modulo R_m. Since these are throughout functions of a *single* real variable t (the case $n = 1$), the Sobolev embedding theorem already gives, for $f \in H^1$, the embedding $H^1(I) \hookrightarrow C^{0,1/2}(I)$ (Hölder-$1/2$-continuous) on any bounded interval I or on S^1_m itself: such a representative is continuous, indeed absolutely continuous, and therefore, by the fundamental theorem of calculus for Lebesgue integrals, classically differentiable *almost everywhere*, with derivative equal to the weak derivative a.e. The same example also shows the remaining gap: $f(t) = |t|$ lies in H^1, is continuous everywhere, but is *not* classically differentiable at every point ($t = 0$ excepted) and is not C^1. The gap from C^1 is twofold: (a) H^1 guarantees differentiability only almost everywhere, not at every point; (b) C^1 additionally requires continuity of the derivative. For higher dimensions $n \geq 2$ even the H^1 gain disappears (H^1 functions can be unbounded there); that case does not arise here. This property belongs to the representative, not to the bare L^2 element itself: $L^2(S^1_m)$ (likewise $\mathrm{Per}_{R_m}(\mathbb{R}_t)$) consists of equivalence classes of almost-everywhere-equal functions, for which a value at a fixed point t is simply not well defined without additional regularity — regardless of whether one stays on S^1_m or passes to the pullback on $\mathbb{R}_t$. It does *not* deliver a continuum of the state space: since Φf is R_m-periodic, $\int_{\mathbb{R}} |\Phi f|^2 \, dt = \infty$ for any $f \neq 0$ — Φf does not lie in ordinary $L^2(\mathbb{R}_t)$ with Lebesgue measure over the whole line. The correct target space $\mathrm{Per}_{R_m}(\mathbb{R}_t)$ (periodic L^2_{loc}, norm over one period) carries the same countable Fourier basis $\{e_n\}_{n \in \mathbb{Z}}$ and the same Parseval identity as $L^2(S^1_m)$ itself — the isometry is a change of representation within *isomorphic* Hilbert spaces, not an extension to a larger state space. Both statements — continuum of the coordinates, no continuum of the state space — are simultaneously true and do not contradict each other; they simply answer different questions. This distinction sharpens, rather than refutes, the property of the pullback as a pure change of representation described in the preceding paragraph.

The Difference from the Mosco-Convergence Strategy

The essential difference lies precisely in the "Mosco convergence" and "conditional hyperbolic limit" building blocks: there, a scaled refinement sequence with a *postulated, unproven* Mosco convergence is necessary. In the pullback construction this refinement machinery is dispensed with entirely — no sequence $a_n \downarrow 0$, no rescaled operator A_n, no Mosco commitment. The price: the compact structure itself is not derived from something more fundamental but is set as the starting point.

Note for the FFGFT context: The pullback introduced in this section is a *different* operation from the "unrolling" used so far in the FFGFT corpus (type-I decompactification, Doc. 285): there the winding number is discarded — a lossy measurement act. Here the pullback is a lossless change of representation that unrolls m and t together.

The role of K_{frak} in both cases, and the binding of the recursion to the $\tilde{T} \cdot m$ duality, are treated fully in Doc. 333.

Boundary Cases of the Pullback Construction: Massless Waves and Black Holes

The pullback isometry $L^2(S^1_m) \cong \text{Per}_{R_m}(\mathbb{R}_t)$ holds for every fixed mass $m > 0$. At the two limits of the $\tilde{T} \cdot m = 1$ curve it behaves structurally differently.

Limit $m \to 0$ (massless waves): Photons and gravitational waves exist physically. In FFGFT $m = 0$ is not an attainable state — $\tilde{T} \cdot m = 1$ has no solution at $m = 0$ (Doc. 306): FFGFT knows no empty state, only a thinnest one. Formally: $\tilde{T} \to \infty$, hence $R_m = \hbar/(mc^2) \to \infty$. The mass circle S^1_m dissolves — it becomes $\mathbb{R}$ itself. The pullback becomes trivial: rolled up and unrolled are identical, because no winding remains. Massless fields have no mass circle and hence no period R_m — K_{frak} does not arise, because the mass formula does not produce $m = 0$.

Limit $T \to T_0$ (black hole as extremum): At the other extreme — maximum mass $m \to E_P/\xi$, minimum time $T \to T_0 = \xi \cdot t_P$ — the mass circle is maximally compact. The corpus establishes (Doc. 306): the minimum time T_0 cannot reach zero; the horizon is the extremum, not the singularity — everything finite, exact closure $T_0 \cdot E_{\text{max}} = 1$. For the pullback: R_m has a minimum at $R_{m_{\text{max}}} = T_0$. The isometry holds exactly there too — but what is unrolled is the shortest possible $\mathbb{R}_t$ section per winding.

Symmetry of both limits: Both boundary cases are protected by $\tilde{T} \cdot m = 1$ — neither $m = 0$ nor $T = 0$ is attainable. The pullback remains exact in both cases; what changes is the size of R_m and hence the physical scale of the unrolled $\mathbb{R}_t$.

What the Triviality of the Spectral Dimension Means for an Operator with an Infinite Discrete Spectrum

The triviality theorem concerns explicitly a *finite* operator on a *fixed finite support*. An operator with, from the outset, infinitely many discrete eigenvalues on a compact manifold — the Laplace operator on S^1, with spectrum $\{k^2 : k \in \mathbb{Z}\}$, is a case in point — does not satisfy this precondition: $N = \dim \mathscr{H} = \infty$, and the spectrum diverges without bound rather than terminating at a finite maximum. Already the Taylor expansion $P(\sigma) \approx 1 - \sigma\bar{\lambda}$ breaks down, since a mean over infinitely many, unboundedly growing eigenvalues is not well defined. The triviality theorem therefore simply does not apply to such an operator — neither confirmed nor refuted, its precondition is absent.

W.3 Summary

	Discrete chain-complex strategy	**Canonical pullback strategy**
Starting point	finite cell complex	compact structure set as fundamental
Category error	blocked by explicit lemma	does not arise (representation change, not degree change)
Geometric failure	precisely localized (algebra remains valid)	does not arise (no embedding question)
Continuum passage	refinement sequence, Mosco convergence *postulated*	single exact pullback, no sequence needed
Open preconditions	five unproven assumptions for hyperbolic limit	compactness of the base structure itself unexplained (in the FFGFT context, however, compactness is enforced by $\tilde{T}\cdot m = 1$, not freely chosen)

Both strategies carry a price: the first pays with unproven convergence assumptions for the continuum limit; the second pays with the fact that the compactness of the fundamental structure itself stands as a starting assumption. Which price is the more acceptable one depends on the respective ambition of the construction. *Note:* In the FFGFT framework ($\tilde{T} \cdot m = 1$), this price is in fact not one — there, the compactness of the time dimension is not freely chosen but follows necessarily from mass and time forming a single geometric dimension.

W.4 References

[1] Krüger, M. (2026). *Dimension-Graded Geometry and Typed Emergence on Aperiodic Carriers: From Cellular Degree, Holonomy, and Regge Curvature Support to a Falsifiable Spacetime-Limit Programme in an HLV Testbed.* Zenodo. DOI: 10.5281/zenodo.22105698.

[2] Regge, T. (1961). General relativity without coordinates. *Il Nuovo Cimento*, 19(3), 558–571.

[3] Wilson, K. G. (1974). Confinement of quarks. *Physical Review D*, 10, 2445–2459. — The original lattice-gauge-theory paper; foundation for the edge-transport/face-holonomy building block (Section 1.3).

[4] Bahr, B., & Dittrich, B. (2010). Regge calculus from a new angle. *New Journal of Physics*, 12, 033010. arXiv:0907.4325. — Generalized Regge approach using constant-curvature rather than flat simplices; see the discussion in Section 1.4.

[5] Dittrich, B., & Höhn, P. A. (2010). From covariant to canonical formulations of discrete gravity. *Classical and Quantum Gravity*, 27, 155001. arXiv:0912.1817. — Generalized Regge approach with variable triangulation over time; see Section 1.4.

[6] Höhn, P. A. (2015). Canonical linearized Regge calculus: counting lattice gravitons with Pachner moves. *Physical Review D*, 91, 124034.

[7] Desbrun, M., Hirani, A. N., Leok, M., & Marsden, J. E. (2005). Discrete exterior calculus. arXiv:math/0508341. — Foundational paper for discrete exterior calculus and the Hodge positivity discussed in Section 1.6.

[8] Friedman, J. L., & Sorkin, R. D. (1980). Spin $1/2$ from gravity. *Physical Review Letters*, 44, 1100.

[9] Ambjørn, J., Jurkiewicz, J., & Loll, R. (2005). Spectral dimension of the universe. *Physical Review Letters*, 95, 171301. arXiv:hep-th/0505113. — Empirical illustration of the triviality theorem from Section 1.7: measurements on finite causal dynamical triangulations show exactly the intermediate-scale plateau described there, together with the finiteness-driven decay to zero at small and large diffusion times.

[10] Ambjørn, J., Görlich, A., Jurkiewicz, J., & Loll, R. (2012). Nonperturbative quantum gravity. *Physics Reports*, 519(4–5), 127–210. arXiv:1203.3591. — Review article on causal dynamical triangulations with extensive discussion of finite-size effects on the spectral dimension.

[11] Mosco, U. (1994). Composite media and asymptotic Dirichlet forms. *Journal of Functional Analysis*, 123(2), 368–421. — Original paper introducing the convergence notion used in Section 1.8.

[12] A treatment of Mosco convergence of discrete graph Dirichlet forms on the flat torus $\mathbb{T}^n$ toward a continuous target functional (arXiv:2511.16486, 2025). — Addresses exactly the construction type described in Section 1.8 (shrinking lattice spacing $\varepsilon \to 0$ on a toroidal lattice) and shows under which additional technical conditions such Mosco convergence is actually provable — a useful counterpoint to the unproven commitment referenced here.

[13] Klein, O. (1926). Quantentheorie und fünfdimensionale Relativitätstheorie. *Zeitschrift für Physik*, 37(12), 895–906. — Classic example of a pullback-type construction in physics: the Fourier decomposition of a field along a compact circular fiber, structurally related to the canonical function pullback discussed in Section 2.

Chapter X

Doc. 333 — K_{frak}, Unrolling and the $\tilde{T} \cdot m$ Duality

Preliminary note

In the FFGFT corpus, "unrolling" has hitherto always meant **type-I decompactification**: the measurement act selects the mass circle as time, $S^1_m \to \mathbb{R}_t$, discarding the winding number — a lossy step (Doc. 285, Doc. 248, Doc. 270). Doc. 332 introduces alongside this a different operation: the **canonical function pullback** $\Phi = p^* : L^2(S^1_m) \to \mathrm{Per}_{R_m}(\mathbb{R}_t)$, which is lossless and exact — a pure change of coordinates with no loss of information.

This document places both operations side by side, clarifies the role of K_{frak} in each case, and shows that the recursion follows necessarily from the $\tilde{T} \cdot m$ duality.

X.1 Two Meanings of "Unrolling" in the Corpus

Type I: the measurement act (lossy)

The meaning used so far in the corpus (Doc. 285 § "How FFGFT unrolls"): the embedded observer selects the mass circle S^1_m as the time axis. The map

$$S^1_m \longrightarrow \mathbb{R}_t, \qquad \tilde{T} \mapsto t,$$

is **lossy**: the winding number is discarded, the discrete spectrum of the compact circle does not appear in $\mathbb{R}_t$. This is not a mathematical operation of the theory but the bridge to observation — the torus remains the fundamental structure, $\mathbb{R}_t$ is the image (Doc. 285). At the same time the recursion ξ^n unrolls into a continuous dilation flow $t \mapsto \xi t$ with log-period $|\ln \xi| = \ln 7500 \approx 8.92$ (Doc. 285 § "The recursion unrolls along with it").

Type III: the coordinate change (lossless)

The meaning introduced in Doc. 332: the canonical function pullback

$$\Phi = p^* \; : \; L^2(S^1_m) \longrightarrow \mathrm{Per}_{R_m}(\mathbb{R}_t), \qquad (\Phi f)(t) = f(t \bmod R_m),$$

is an **exact isometry** — no loss of information, no winding number discarded, Fourier basis and Parseval identity preserved. It is a change of representation within isomorphic Hilbert spaces (R96, Doc. 332). The lossy direction lies at $\mathbb{R}_t \to S^1_m$, not at the pullback itself.

	Type I (measurement act)	**Type III (pullback)**
Operation	$S^1_m \to \mathbb{R}_t$	$L^2(S^1_m) \cong \mathrm{Per}_{R_m}(\mathbb{R}_t)$
Loss	winding number discarded	none
Character	physical act	mathematical change of representation
Winding number	not preserved	preserved
Information loss	yes	no
Corpus references	Doc. 285, 248, 270	Doc. 332 (R96)

X.2 The Role of K_{frak} in Both Cases

At the type-III pullback: invisible

A Laplace operator on the compact T^4 is local — it cannot carry a cumulative quantity (Doc. 314). K_{frak} therefore does not enter through the operator but through the **bridge constants** E_0, v at the scale conversion (Doc. 314, R72).

In the rolled-up state on S^1_m the period $R_m = \hbar/(mc^2)$ is exact in natural units — no K_{frak}. Only when m is substituted from the mass formula does R_m carry the correction:

$$m_i = r_i \cdot \xi_0^{p_i} \cdot v \cdot K_{\mathrm{frak}} \quad \Rightarrow \quad R_m^{\mathrm{corr.}} = \frac{\hbar}{m_i^{\mathrm{bare}} \cdot K_{\mathrm{frak}} \cdot c^2} = \frac{R_m^{\mathrm{bare}}}{K_{\mathrm{frak}}}.$$

K_{frak} sits in the *choice of* R_m, not in the pullback mechanism. The pullback itself is exact in both cases (bare or corrected).

At the type-I measurement act: visible as a scale correction

At the lossy unrolling, K_{frak} appears as a correction factor of the bridge constants: the ratio of the two E_0 values (bare and α-anchored) is $1/\sqrt{K_{\mathrm{frak}}}$ (R72, Doc. 314). The correction belongs to the *scale structure* — the transition between the compact and the observable representation — not to the time geometry itself.

	Type I	**Type III**
K_{frak}	visible in bridge constants (E_0 ratio, mass formula)	invisible at the isometry step; sits in R_m via m
Where exactly	at the SI transition	in the choice of period

X.3 Mass and Time Unroll Together

In the FFGFT compactification, $\tilde{T}$ and m are a **single geometric dimension**:

$$\tilde{T} \cdot m = 1 \qquad \text{(fundamental relation, Doc. 285)}.$$

They are not two independent quantities. At the type-III pullback, the pullback therefore unrolls m and t *together*: the entire mode $f \in L^2(S^1_m)$ is unrolled and the ratio $\tilde{T}/m$ remains constant. The physics does not change, only the coordinate representation.

Unrolling t alone — dropping R_m — would violate $\tilde{T} \cdot m = 1$ and is structurally inadmissible. K_{frak} is *not* the price of unrolling, but the price of the **artificial separation** of m and t at the transition to absolute SI units.

X.4 The Recursion Is Necessarily Bound to the Duality

Stable particle: Case B

For a stable, isolated particle ξ is frozen (Case B, Doc. 315, Doc. 313): the winding number is rational $74/75$, closure occurs after exactly 75 revolutions. $K_{\text{frak}} = 74/75$ is the closed bookkeeping of this rolled-up geometry — exact in fractional arithmetic, because ξ is set rationally (Doc. 315).

Dynamics: Case C and the recursion chain

As soon as dynamics enter — interaction, RG flow, mass change — a deeper consequence of $\tilde{T} \cdot m = 1$ comes into play: if m changes, $\tilde{T} = 1/m$ changes necessarily with it. The period $R_m = \hbar/(mc^2)$ changes, the time scale of the physics changes, and this feeds back onto the effective mass. The recursion (Doc. 295, Doc. 146) reads:

$$\xi_{n+1} = \xi_n\,(1 - 100\,\xi_n).$$

The closure deficit per revolution no longer remains constant but runs cumulatively. The rational 75-closure becomes a logarithmic spiral (Case C, Doc. 295):

$$\rho = 75\,e^{\beta/2\pi} \qquad \text{(equiangular spiral, self-similarity ratio } e\text{)}.$$

In Case C, K_{frak} is no longer a constant but an effectively scale-dependent quantity. The multiplicative form $(1-\xi)^{100}$ is the natural bookkeeping of this running case; its distance from $74/75$ — the binomial term $4950\,\xi^2$ (Doc. 315) — is precisely the difference between frozen and running recursion.

Why Case A is excluded

Case A ($K_{\text{frak}} = 1$, closure after one revolution) would mean: $\tilde{T} \cdot m = 1$ generates no recursion, because m never changes — a static, interaction-free universe with $D_f = 3$ exactly. This contradicts the observed particle structure and the basic framework.

Summary of the three cases

Case	ξ	K_{frak}	Binding to duality
A	constant, $d = 0$	$= 1$	no feedback, excluded
B	frozen	$= 74/75$ exact	stable mode, duality at rest
C	running (recursion)	scale-dependent	duality generates running feedback $m \leftrightarrow \tilde{T}$

$K_{\text{frak}} = 74/75$ is the **projection of Case C onto Case B**: exact in the rolled-up, static description; first approximation in the dynamic one.

X.5 Mode Specificity, Non-Closure and Superposition

Each mode carries its own period

The period $R_m = \hbar/(mc^2)$ is mode-specific: every particle i with torus mode $\mathcal{T}_i = (n_i, l_i, j_i)$ carries its own period. Unrolling (Type III or Type I) is therefore always particle-specific — it unrolls one particular mode, not a mixture. Simultaneously unrolling different periods R_{m_e} and R_{m_μ} would be structurally inadmissible, because $\tilde{T} \cdot m = 1$ would be violated for both at once.

The non-closure condition (Doc. 193)

The same mode specificity underlies the non-closure condition (Doc. 193): the prefactors r_i are not numbers in a field but labels of discrete geometric spaces — fully determined by $\mathcal{T}_i$. **Multiplicative** combination of different modes,

$$(r_i \cdot r_j,\ p_i + p_j) \notin \mathcal{P} \qquad \text{for } i \neq j,$$

yields no FFGFT particle — the product lies outside the set of admissible pairs **[B]**. K_{frak} sits in the mode-specific choice of R_m; a cross-mode K_{frak} would be the same inadmissible cross-product.

Why superposition is nevertheless permitted — and where the difference lies

In quantum mechanics, superpositions of different states are not only permitted but fundamental. An electron in the sp^3 hybrid orbital of a carbon atom is a linear combination of s and p states; an excited atom is in a superposition of ground state and excited state. The operation is **additive**:

$$|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle, \quad |c_1|^2 + |c_2|^2 = 1.$$

This produces no new (r, p)-pair, no new mass, no new torus mode — and therefore does not contradict the non-closure condition.

The crucial difference between QM and FFGFT: In standard QM, time t is not a quantum-mechanical operator and not part of the Hilbert space — it is simply *absent* as a physical quantity of the system. The Schrödinger equation treats t as a classical external parameter that evolves the state from outside, without itself belonging to the system. A superposition $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ therefore carries no mode-bound time structure — t does not belong to the superposition but stands outside it.

In FFGFT, time is geometrically bound to mass: $\tilde{T}_i = 1/m_i$. Two modes with different masses $m_1 \neq m_2$ carry different time scales $\tilde{T}_1 \neq \tilde{T}_2$. A superposition of such modes would not merely mix amplitudes but overlay incompatible time scales — structurally more problematic than in QM, where this conflict does not arise.

In the FFGFT corpus, superposition is therefore described as **intermediate torsion within a mode** (Doc. 174): a metastable intermediate state between stable torsion extremes of the *same* mode, not a cross-mode superposition. Decoherence drives this intermediate state back into one of the stable configurations.

	Standard QM	**FFGFT**
Time	absent: no operator, no Hilbert space element	$\tilde{T} = 1/m$, mode-bound, geometric
Superposition	additive, t stands outside	intermediate torsion within one mode
Cross-mode	algebraically permitted	incompatible $\tilde{T}$, structurally problematic
Non-closure	not relevant	$r_i \cdot r_j \notin \mathscr{P}$, multiplicatively forbidden **[B]**

The consistent boundary

The linearity of quantum mechanics — the superposition principle — and the non-closure condition protect the same boundary from two sides: atoms and bound states generate no new fundamental particles, no new torus modes, no new (r, p)-pair. The connection between FFGFT and established atomic physics runs through the derived quantities α and m_e (Doc. 011, 044, 006) — quantum mechanics takes these as input and derives binding energies and spectra from them, without touching the mode structure.

> *"Unrolling" in the FFGFT corpus has hitherto always meant the lossy measurement act (Type I, Doc. 285): winding number discarded, $S^1_m \to \mathbb{R}_t$. The function pullback from Doc. 332 is a different, lossless operation (Type III): an exact change of representation that unrolls m and t together and at which K_{frak} plays no role. K_{frak} sits in the choice of the period R_m via the mass formula and becomes visible only at the transition to absolute SI values. The recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ is necessarily bound to $\tilde{T} \cdot m = 1$: every mass change entails a time change, which feeds back recursively onto the mass. Case B (stable particle, $K_{frak} = 74/75$) is the approximation*

for frozen duality; Case C (running recursion, log-spiral) is the complete dynamic description.”

References: Doc. 133 (K_{frak} derivation), Doc. 146 (recursion), Doc. 248 (observer), Doc. 270 (type-I theorem), Doc. 285 (decompactification), Doc. 295 (log-spiral), Doc. 313 (three cases), Doc. 314 (local operator, bridge constants), Doc. 315 (form question K_{frak}), Doc. 332 (pullback isometry, R96).

Chapter Y

Doc. 334 — Superposition Without Time

Preliminary note

This document builds on Doc. 307 (time in the state space or beside it) and Doc. 333 (superposition in FFGFT), without repeating their content. The specific focus here: what does it mean concretely for the interpretation of *superposition, entanglement and collapse* that standard QM has no time operator? And what changes structurally when $\tilde{T} \cdot m = 1$ gives every mode its own time scale?

Y.1 The Classical Background — Instantaneity as an Inheritance

Classical mechanics: time runs, calculations are instantaneous

Instantaneity is not a QM-specific problem. It is an inheritance from classical physics that QM took over unchanged.

In Newtonian mechanics, gravity acts instantaneously over arbitrary distances. The Coulomb potential is instantaneous. Lagrangian and Hamiltonian mechanics describe the state $(q(t), p(t))$ at a fixed t — the calculation is a timeless algebraic evaluation over the instantaneous state. Nobody asks how long this calculation "takes", because calculations in this language have no duration. The time t runs in the background as an external parameter — the physics resides in the timeless structure of the equations.

This is the same structure as in QM. Pauli only made explicit what was tacit in the classical framework.

The hidden inconsistency of QM

QM inherits this instantaneity — and adds a new tension:

The state $\psi(t)$ is instantaneous. It is a timeless mathematical quantity at a fixed t — no duration, no process.

Evaluating $|\psi(t)|^2$ requires time. $|\psi(t)|^2$ is a probability density — to verify it one needs many measurements, hence real physical time. The statistics that confirm the state are distributed over a temporal process. The formalism, however, describes only the state at one moment.

This is the tacit inconsistency: *the formalism is instantaneous, but its verification is temporally extended.* The time required for probabilistic evaluation stands outside the formalism — it is neither operator nor variable, but what the experimenter expends in order to know the state at all.

Consequence for superposition

Superposition $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ is therefore doubly timeless: first as a state (no time operator, no intrinsic "when"), second as a statement (probabilistic, requires time for verification, which lies outside the formalism). What "exists" in superposition can only be confirmed through repeated measurements over real time — and each individual measurement collapses the state, thus destroying the superposition.

In FFGFT: the period $\tilde{T}_i = 1/m_i$ gives the mode an intrinsic time scale. Superposition is not timeless — it has a geometrically defined scale on which it is stable or unstable.

Y.2 The Absence of Time in QM — an Underestimated Consequence

The Pauli theorem and its consequences

Wolfgang Pauli showed in 1933 that for a Hamiltonian with a spectrum bounded from below there can be no self-adjoint time operator canonically conjugate to $\hat{H}$ (Doc. 307, Doc. 306). The consequence was tacit but far-reaching: time was kept outside the Hilbert space and treated as a classical external parameter.

This is not merely a technical decision. It means: **time is not an observable in QM.** One cannot ask "when is the system in state ψ?" and obtain an answer from the formalism — t must be supplied from outside.

What is rarely stated in textbooks

Most textbooks introduce time as a parameter without emphasising what the absence of a time operator means. It means concretely:

- Superposition $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ is a state *without time structure* — there is no intrinsic "clock" saying how long the system remains in this superposition.
- Entanglement between two particles is *timeless* — the joint state has no built-in notion of simultaneity.
- Collapse is *without duration* — it "happens" in the external parameter t, but t itself is not part of the collapse process.

This is not a failure of QM — it is a structural property that is rarely made explicit.

Y.3 Three Consequences in Detail

Superposition is timeless

An electron in the state $|\psi\rangle = c_\uparrow |\uparrow\rangle + c_\downarrow |\downarrow\rangle$ has no intrinsic time scale. The question "how long is it in this superposition?" is not answerable within the QM formalism — one would need a time operator, which does not exist. What does exist is the coherence time T_2 — but that is a phenomenological quantity describing interaction with the environment, not an intrinsic property of the state.

In FFGFT this is different: every mode $\mathcal{T}_i = (n_i, l_i, j_i)$ carries $\tilde{T}_i = 1/m_i$ as a geometric quantity. A state *is* its time scale — not as the duration of a superposition, but as a structural property of the mode.

Entanglement without simultaneity

Two entangled particles share a joint state $|\Psi\rangle \in \mathscr{H}_1 \otimes \mathscr{H}_2$. This joint state has no built-in time structure. The question of the simultaneity of a measurement on both particles is therefore not a QM question but a classical question about the external parameter t — and hence a relativistic question (relativity of simultaneity).

This is the deepest reason why QM is compatible with Lorentz invariance: because "simultaneity" stands outside the formalism and can therefore be transformed by the Lorentz group without touching the QM formalism.

In FFGFT: if two different modes with $m_1 \neq m_2$ were entangled, they would carry different $\tilde{T}_1 \neq \tilde{T}_2$. A cross-mode entanglement therefore generates incompatible time scales — structurally more problematic than in QM, where this conflict does not arise because t stands outside.

Collapse without duration

The measurement process in QM describes a transition $|\psi\rangle \to |\psi_k\rangle$ (projection onto an eigenstate). This transition has *no duration* in the QM formalism — it is instantaneous in the parameter t. How long the collapse takes is a question that lies outside standard QM (it is discussed in theories of the measurement process, decoherence, or quantum gravity).

This is not an inconsistency but a direct consequence of the absence of the time operator: without $\hat{t}$ there is no observable "collapse duration".

In FFGFT the type-I step (measurement act, Doc. 285/333) is not instantaneous in the abstract sense but a geometric transition — the mass circle S^1_m becomes the time axis $\mathbb{R}_t$, the winding number is discarded. This is a loss of information that has a natural "direction".

Y.4 What $\tilde{T} \cdot m = 1$ Changes Structurally

Time as a property of the mode, not a parameter

The fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 285, 306, 307) sets time not as an external parameter but as a geometric quantity of every mode: the period $R_m = \hbar/(mc^2)$ is intrinsic — it belongs to the particle as its mass does.

This has three direct consequences for the points above:

Superposition: A superposition within one mode — intermediate torsion (Doc. 174) — has a natural time scale $\tilde{T}_i$. Superposition *across* different modes would overlay incompatible time scales $\tilde{T}_i \neq \tilde{T}_j$ — this is the structural reason why cross-mode superpositions are problematic in FFGFT (Doc. 333).

Entanglement: Within one mode, simultaneity is defined by $\tilde{T}$. Across modes there is no shared time concept — similar to QM, but for a different reason: not because t stands outside, but because different modes carry different geometric time scales.

Collapse (Type I): The measurement act (Doc. 285) has a natural direction — lossy from S^1_m to $\mathbb{R}_t$ — and a natural asymmetry: the reverse direction $\mathbb{R}_t \to S^1_m$ requires the discarded winding number. The arrow of time is thereby not an external additional assumption but a consequence of the geometry of measurement.

Pauli theorem and compact time

The Pauli theorem applies to an *open* time parameter conjugate to a Hamiltonian bounded from below. For a *compact* time dimension the argument does not apply (Doc. 307): compact time behaves like an angle variable, its conjugate has a discrete spectrum (winding numbers), and the Pauli condition is not met.

$\tilde{T} \cdot m = 1$ realises exactly this situation: time is compact, the conjugate spectrum is the discrete mass ladder. The theorem does not exclude this case — it virtually forces it as the only consistent way to integrate time into the Hilbert space.

Y.5 Summary and Demarcation

	Standard QM	**FFGFT ($\tilde{T} \cdot m = 1$)**
Time	absent: no operator, no Hilbert-space element; external parameter	geometric: $\tilde{T}_i = 1/m_i$, compact, mode-bound
Superposition	timeless, no intrinsic clock	intermediate torsion within one mode; $\tilde{T}_i$ as scale
Cross-mode	algebraically permitted, no time conflict (t external)	incompatible $\tilde{T}_i \neq \tilde{T}_j$, structurally problematic
Entanglement	timeless, simultaneity external (Lorentz)	mode-specific $\tilde{T}$; cross-mode: no shared time concept
Collapse	instantaneous in t, no duration in formalism	Type-I: directed information loss, arrow of time geometric
Pauli theorem	blocks time operator for open t	does not apply for compact $\tilde{T}$ (angle variable)

This document does not claim to replace standard QM. It shows that its structural property — time as absent — is rarely made explicit, and that $\tilde{T} \cdot m = 1$ offers a geometric alternative that describes the same phenomena (superposition, entanglement, collapse) from a different basic structure.

References: Doc. 174 (superposition as torsion), Doc. 285 (type-I measurement act), Doc. 306 (native time–energy reciprocity), Doc. 307 (time in the state space or beside it), Doc. 333 (mode specificity and superposition).

Chapter Z

Doc. 335 — Mass-Bound Time Scales and Interferometry

Z.1 Mass-Bound Time Scales and Interferometry

Abstract

*From the FFGFT fundamental relation $\tilde{T} \cdot m = 1$ follow mass-bound Compton frequencies $f_m = mc^2/h$ as the only physically grounded eigenfrequencies of a discrete substrate **[K]**. For all stable particles these lie at least 12 orders of magnitude above the measurement frequencies of current interferometers (MHz range). This yields a precise, derivable statement about interferometry experiments for the detection of space quantisation: not a design-induced blind spot but the mass-bound nature of the time scale $\tilde{T} = 1/m$ explains why Holometer-type experiments at their current sensitivity cannot find a discreteness signal **[K]**. A K_{frak} correction to the Compton frequency of $\Delta f_e/f_e = 100\xi_0 \approx 1.3\%$ acts only on the SI conversion; it is not a measurable spectral line, neither spectroscopically nor interferometrically **[K]**.*

Background: Two Approaches to Space Quantisation

Experiments such as the Fermilab E-990 Holometer ($L = 40\,\text{m}$, $f_c = c/(2L) \approx 3.75\,\text{MHz}$) and the planned QUEST experiment ($L = 1.84\,\text{m}$, $f_c \approx 81.5\,\text{MHz}$) search for a quantisation signal of spacetime. The null results of these experiments have been interpreted from various quarters as evidence for or against specific discrete substrate models.

One concrete claim is that the interferometers are by design blind at exactly the frequency at which a discrete D4 lattice with lattice constant $a = \ell_P$ would resonate. This claim presupposes that the relevant substrate frequency is $f \sim c/(2L)$.

The FFGFT framework provides a more precise answer to the question of the relevant frequency scales — from first principles, without parameter fitting.

FFGFT Eigenfrequencies from $\tilde{T} \cdot m = 1$

The Mass-Bound Time Scale

[ASSUMPTION] In FFGFT, time is not an independent external quantity but is bound to mass by the fundamental relation $\tilde{T} \cdot m = 1$ (in natural units; in SI: $\tilde{T} \cdot m = \hbar/c^2$).

[K] Every massive particle therefore carries an intrinsic time scale:

$$\tilde{T}_m = \frac{\hbar}{mc^2}, \qquad f_m = \frac{mc^2}{h}. \tag{Z.1}$$

This is the Compton frequency — not an additional postulate but a geometric consequence of the T^4 structure and the duality relation.

External Inputs

[ASSUMPTION] This document uses the following standard values as external inputs; they are declared here and not derived:

Quantity	Value	Source
m_e	9.1094×10^{-31} kg	PDG
m_μ	1.8836×10^{-28} kg	PDG
m_p	1.6726×10^{-27} kg	PDG
m_n	1.6749×10^{-27} kg	PDG
h	6.6261×10^{-34} J s	SI (derived from ξ)
c	2.9979×10^{8} m/s	SI (derived from ξ)

Numerical Values

[K]

Compton frequencies of stable particles (SI)

Particle	$f_m = mc^2/h$	$R_m = \hbar/(mc^2)$
Electron	1.2356×10^{20} Hz	1.288×10^{-21} s
Muon	2.5548×10^{22} Hz	6.230×10^{-24} s
Proton	2.2687×10^{23} Hz	7.015×10^{-25} s
Neutron	2.2719×10^{23} Hz	7.005×10^{-25} s

Comparison with Interferometer Frequencies

[K]

$$\frac{f_e}{f_c^{\text{Holometer}}} = \frac{1.2356 \times 10^{20}\,\text{Hz}}{3.747 \times 10^{6}\,\text{Hz}} \approx 3.3 \times 10^{13}, \tag{Z.2}$$

$$\frac{f_e}{f_c^{\text{QUEST}}} \approx 1.5 \times 10^{12}. \tag{Z.3}$$

The smallest mass-bound frequency (electron) thus lies 12 to 13 orders of magnitude above the measurement frequencies of current interferometers.

Lattice Dispersion Correction at the FFGFT Scale

[K] For a discrete substrate with lattice constant a, the lattice dispersion correction to linear dispersion is of order $(k \cdot a)^2$. At $k_1 = \pi/L$ this gives:

$$\left.\frac{\delta\omega}{\omega}\right|_{\text{lattice}} \sim \left(\frac{\pi a}{L}\right)^2. \tag{Z.4}$$

[K]

Lattice scale a	Holometer ($L = 40\,\text{m}$)	QUEST ($L = 1.84\,\text{m}$)
$\ell_P = 1.62 \times 10^{-35}\,\text{m}$	$\sim 10^{-72}$	$\sim 10^{-70}$
$\hbar/(m_e c) = 3.86 \times 10^{-13}\,\text{m}$	$\sim 10^{-27}$	$\sim 4 \times 10^{-25}$
$\hbar/(m_p c) = 2.10 \times 10^{-16}\,\text{m}$	$\sim 3 \times 10^{-34}$	$\sim 10^{-31}$

Even at the Compton scale of the electron — which in the FFGFT framework is the *physical* fundamental scale, not ℓ_P — the lattice dispersion correction is $\sim 10^{-27}$, far beyond any measurable sensitivity.

The K_{frak} Correction and Its Measurability

[K] The fractal correction $K_{\text{frak}} = 1 - 100\xi_0$ with $\xi_0 = 4/30000$ acts exclusively on absolute SI values, not on the pullback isometry itself (Doc. 190, R96, R98). Applied to the Compton frequency of the electron:

$$\Delta f_e = f_e \cdot 100\xi_0 = f_e \cdot \frac{400}{30000} \approx 1.65 \times 10^{18}\,\text{Hz}. \tag{Z.5}$$

This corresponds to a relative deviation of $100\xi_0 \approx 1.33\%$ from the Compton frequency. The value is arithmetically correct; but since K_{frak} acts only on absolute SI values, i.e. on the SI anchoring, the shift is not a directly measurable spectral line – it is accessible neither to spectroscopy (precision spectroscopy at $\sim 10^{20}$ Hz) nor to a Michelson interferometer in the MHz range.

The Central Statement on Interferometry Experiments

Conclusion: Why Holometer-type experiments find no signal

[K] In FFGFT the relevant frequency scale for space quantisation is not $f \sim c/(2\ell_P)$ (universal Planck frequency) but $f_m = mc^2/h$ (mass-bound Compton frequency). This scale differs for every particle and lies for all stable particles at least 12 orders of magnitude above the measurement frequencies of current interferometers.
It follows directly: Holometer-type experiments find nothing because the time scale of discreteness is not universal but mass-bound — not because a design-induced blind spot of the instrument lies exactly at the substrate resonance frequency.
The statement is falsifiable: an interferometer signal at frequency f would require a massless or ultralight field with $m = hf/c^2$. For $f = 3.75\,\text{MHz}$ this gives $m \approx 2.8 \times 10^{-44}\,\text{kg} \approx 3 \times 10^{-14}\,m_e$ — no known particle.

Relation to the D4 Lattice Argument

[K] The D4 lattice with lattice constant $a = \ell_P$ has 24 nearest neighbours (8 of type $(\pm 1, 0, 0, 0)$ and 16 of type $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$). The dispersion relation of a scalar field on this lattice is:

$$\omega^2(\mathbf{k}) = \frac{c^2}{\ell_P^2} \sum_{\boldsymbol{\delta}} \big(2 - 2\cos(\mathbf{k} \cdot \boldsymbol{\delta}\,\ell_P)\big). \tag{Z.6}$$

For $k_1 = \pi/L$ with $L = 40\,\text{m}$ one has $k_1\ell_P \approx 1.3 \times 10^{-36}$, hence $k_1\ell_P \ll 1$: the dispersion relation is exactly linear in this regime, identical to the continuum up to order $(k_1\ell_P)^2 \sim 10^{-72}$. No special lattice resonance exists at $f_c = c/(2L)$ — the free spectral range $c/(2L)$ is an optical property of the interferometer, not a lattice property.

FFGFT replaces the D4 lattice argument with a stronger statement: not the lattice dispersion but the mass-bound time structure $\tilde{T} \cdot m = 1$ determines at which frequency a signal is to be expected.

Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

Verification script: `pruef_335_paul_frequenzen.py` (5/5 assertions).
Declared 27 August 2026

Part IV

Part IV — The Galois Bridge: GF(9), GF(27) and the Zeta Function of the Torus

Chapter aa

Doc. 336 — The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Abstract

Two independently developed frameworks — the Fundamental Fractal Geometric Field Theory (FFGFT) with its compact $T^4/\mathbb{Z}_3$ geometry and Matzke's GALG with ternary algebra $\mathbb{Z}_3\mathbb{C}$ — arrive at the same $\mathbb{Z}_3$ structure by entirely different routes. This document shows algebraically why *this is not a coincidence, and derives new physical relations from it.*

The key is the Frobenius automorphism $x \mapsto x^3$ *in GF(9): it simultaneously realises the FFGFT sector pairing $k \leftrightarrow -k$, Matzke's vacuum conjugation $V_s = V^\dagger$, and the field conjugation in $\mathbb{Z}_3\mathbb{C}$* ***[B]****.*

The main new results:

- *The Wolfenstein parameter of the CKM matrix satisfies $\lambda = \xi^{1/6}$ (0.5% deviation)* ***[K]****; the exponent $1/6 = 1/(2 \cdot 3)$ is a motivated stipulation: 6 is the degree of the compositum $\mathrm{GF}(729) = \mathrm{GF}(9) \cdot \mathrm{GF}(27)$ over $\mathrm{GF}(3)$, since $\mathrm{GF}(9) \not\subset \mathrm{GF}(27)$.*
- *The CKM element $|V_{ub}| = \sqrt{\xi}/3 = \sqrt{\xi} \cdot (C_2 - 1)$ (0.8% deviation), where $C_2 = 4/3$ is the Casimir value of SU(3)* ***[K]****.*
- *GF(9) and GF(27) classify the particle spectrum topologically — but exact mass values require $\xi \in \mathbb{R}$* ***[B]****.*

Notation: *Symmetric $\mathbb{Z}_3\mathbb{C}$ after Matzke: $a + bi$ with $a, b \in \{-1, 0, +1\}$, $+1 + 1 = -1$ (ternary inline carry).*

Guide: what this document shows

The basic problem. The three quark generations mix with strongly different strengths. The Wolfenstein parameter $\lambda \approx 0.225$ describes this hierarchy empirically — but why exactly this value? Why three generations? Why do the muon and electron have such different masses, even though both are leptons?

What GF(9) and GF(27) contribute. The finite field GF(9) has exactly 9 elements. Its Frobenius automorphism $x \mapsto x^3$ partitions them into three classes: fixed points $\{0, +1, -1\}$ (the values of the triality Δw), and three conjugate pairs $(k, -k)$ (the sector

pairing). GF(27) extends this: it has degree 3 over GF(3) (GF(9) is not a subfield of GF(27); degree 6 is reached only by the compositum GF(729)), and the 8 three-cycles correspond to 8 quark-colour states in 3 generations.

What ξ does that GF(9) cannot. GF(9) is a finite field — it has exactly 9 elements, and none of them equals $m_e/m_P \approx 4 \times 10^{-23}$. Exact particle masses cannot be represented in a finite field. The transition to physical scales occurs through $\xi = 4/30\,000$, whose denominator 3 forms a singularity in GF(3) — this is algebraically the transition from discrete topology to continuous physics.

The surprising part. Despite this, the combination of the field degrees of GF(9), GF(27) and ξ yields precise numerical predictions for CKM parameters that were previously known only empirically. This suggests that the Wolfenstein hierarchy is not a free parametrisation but follows from algebraic structure.

Notation and symbols

Symbol	Meaning	Source
Status markers		
[B]	algebraically proved, verification script available	Doc. 190
[K]	derivable from ξ (internally consistent)	Doc. 190
[S]	open bridge (sketch, no proof yet)	Doc. 190
Fields and algebras		
GF(p^n)	Finite field with p^n elements	Galois theory
$\mathbb{Z}_3\mathbb{C}$	Sym. GF(9): $a+bi$, $a,b \in \{-1,0,+1\}$	[1, 2]
$\phi : x \mapsto x^3$	Frobenius automorphism in GF(9)	this doc.
$\varphi : N \mapsto N \bmod 3$	Ring homomorphism $\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$	this doc.
FFGFT quantities		
ξ	$= 4/30\,000$, geometric base parameter	[3]
Δw	Winding quantum (triality), $\Delta w \in \{0,+1,-1\}$	[4]
$T^4/\mathbb{Z}_3$	Compact 4-torus with $\mathbb{Z}_3$ orbifold	[3]
$k \leftrightarrow -k$	Sector pairing on $T^4/\mathbb{Z}_3$	[4]
Matzke GALG		
$V_s = V^\dagger$	Vacuum conjugation (adjunction)	[11]
$T_k^3 = \mathbf{1}$	Trine relation (tripotent condition)	[11]
$N = \sum q_k^\dagger q_k$	Number operator, $\sigma(N) = \{0,1,2,3\}$	[4, 1]
CKM notation		
$\lambda \approx 0.225$	Wolfenstein parameter (Cabibbo angle)	[6]
$C_2 = 4/3$	Casimir value $SU(3)_{\text{fund}}$	Lie algebra
$\lvert V_{ub}\rvert$	CKM element: $u \to b$ transition	[6]

Symmetric $\mathbb{Z}_3\mathbb{C}$: Elements $a+bi$ with $a,b \in \{-1,0,+1\}$ use the symmetric representation of GF(3) around zero. $+1+1=-1$ (ternary inline carry) and $-1-1=+1$ (ternary inline borrow). The generator $(1+i)^n$ has order 8.

aa.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1+N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod 3$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is not an approximation; it does not, however, follow from characteristic 3: Doc. 324 computes over $\mathbb{C}$ with $T_k = \mathbf{1} + (\omega - 1)P$, $P^2 = P$, and $T_k^3 = \mathbf{1} + (\omega^3 - 1)P = \mathbf{1}$ holds exactly because $\omega^3 = 1$ **[B]**. Over $\mathbb{C}$ one has $(1+N)^3 = 1+3N \neq 1$, and $(\omega - 1)P$ is not nilpotent; the reason is $\omega^3 = 1$ together with $P^2 = P$.

aa.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δw

Previously open [[S]]; now partially closed.

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, +1, +1, +1, -1, -1, -1, 0\} \tag{aa.1}$$

with particle classification:

$$\begin{aligned} N \bmod 3 = 0 &: \text{ colour-neutral (neutrino, positron)} \\ N \bmod 3 = 1 &: \text{ anti-down quarks (3 colours)} \\ N \bmod 3 = 2 &: \text{ up quarks (3 colours)} \end{aligned}$$

This is *exactly* the $\mathbb{Z}_3$ class decomposition by the winding quantum $\Delta w \in \{0, +1, -1\}$ (triality, see below).

Ring homomorphism [B]:

$$\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta w \tag{aa.2}$$

is the canonical quotient; $\varphi(N+M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: [B] (homomorphism), [K] (physical assignment)

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta w$ (Furey's number operator ↔ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Triality and orbifold sectors. $\Delta w = N \bmod 3$ is an additive $\mathbb{Z}_3$ charge, the triality; the ring homomorphism φ is unaffected by this. It is, however, not the sector index k of the orbifold action τ of Doc. 321: the winding sum is invariant under τ, and every three-element orbit contains states of all three sectors $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$ (Doc. 321). The two $\mathbb{Z}_3$ are compatible, since the triality is invariant under τ, but they are different quantities. Which of the two is the physical colour charge remains open; Docs. 346 to 357 use the triality. The Frobenius fixed points $\{0, \pm 1\}$ in Section aa.3 are likewise values of the triality. Verification script: `pruef_341_zwei_z3.py`.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open [[S]]; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into GF(3) (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \tfrac{1}{3}, \tfrac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad \iota(Q) = (3Q \bmod 3,\ \lfloor Q \rfloor) \tag{aa.3}$$

with values:

Q	$\iota(Q)$	Particle
0	$(0, 0)$	Neutrino
$1/3$	$(1, 0)$	Anti-Down
$2/3$	$(2, 0)$	Up
1	$(0, 1)$	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as $\mathbb{Q}$ number)

$N \bmod 3$ fully classifies $Q = N/3$ **[B]**. Q itself (as rational number $1/3$, $2/3$) requires $\mathbb{Q}$ and is not a GF(3) object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was [S]; now [B].

The Frobenius automorphism in GF(9):

$$\phi : x \mapsto x^3, \quad \phi(a + bi) = a - bi \quad \text{[field conjugation]} \tag{aa.4}$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k \bmod 3$ ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)
- `pruef_326_gf9_orbit_masse_weinberg.py`: Fixed points **[B]**, norm hierarchy **[B]**, Weinberg topology **[K]**
- `pruef_328_gluonen_tripotente_gf27_noether.py`: gluons **[B]**, tripotents **[B]**, GF(27) **[K]**, Noether **[B]**
- `pruef_329_ckm_xi_gf27.py`: $\lambda = \xi^{1/6}$, $|V_{ub}| = \sqrt{\xi}/3$ [K]

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero GF(9) elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in GF(3)

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in GF(3) is singular. $\xi = 4/30000 = 1/7500$ has no residue class mod 3 (denominator $\equiv 0$ mod 3, not invertible); $\equiv 1$ holds only for the numerator 4 of the notation $4/30000$ (or $\xi = (4/3)/10^4$ with $10^4 \equiv 1$, only the factor $1/3$ is singular); since $3 \mid 7500$, $v_3(\xi) = -1$. The singularity marks the transition GF(3)$\to \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod 3$, so $\sqrt{2} = i \in \text{GF}(9)$ **[B]**. $n_{\text{thresh}} = 2\pi/(\sqrt{2}\ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

aa.3 Three further structural results

Frobenius fixed points = triality values (Question 3, extended)

The complete orbit structure under $\phi(x) = x^3$ in GF(9):

- **Fixed points** ($x^3 = x$): $\{0, +1, -1\} = \text{GF}(3)$ — exactly the values of the triality $\Delta w \in \{0, +1, -1\}$ **[B]**
- **Non-fixed points**: 6 elements in 3 conjugate 2-cycles **[B]**: $(i \leftrightarrow 2i)$, $(1{+}i \leftrightarrow 1{+}2i)$, $(2{+}i \leftrightarrow 2{+}2i)$ — exactly the FFGFT pairs $(k, -k)$ for $k \neq 0$.

Stable states under Frobenius are the triality values; unstable states are the conjugate pairs.

Norm hierarchy GF(3) $\to$ GF(9) $\to \mathbb{R}(\xi)$ for masses

In GF(3), $1^2 \equiv 2^2 \equiv 1 \pmod 3$ — muon and tau are *algebraically degenerate* as squares **[B]**. The GF(3) structure distinguishes only:

$$n^2 \in \{0,\ 1\} \pmod 3 : \quad 0 \leftrightarrow \text{massless}, \quad 1 \leftrightarrow \text{massive (generations 2+3 degenerate)} \tag{aa.5}$$

In GF(9), the norm $\|x\| = (a^2 + b^2) \bmod 3$ gives two classes **[B]**:

$$\|x\| = 1 : \{+i, -i, +1, -1\} \quad \text{— purely real/imaginary (light)}$$
$$\|x\| = -1 : \{+1+i, +1-i, -1+i, -1-i\} \quad \text{— mixed (heavy)}$$

The full mass hierarchy requires ξ **[K]**: ξ breaks the GF(3) degeneracy and gives $m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$ as continuous numbers.

Status: [B] (degeneracy), [K] (mass values)

Hierarchy: GF(3) gives whether massive/massless; GF(9) gives light/heavy; $\xi \in \mathbb{R}$ gives exact mass ratios. The singularity $C_2 = 4/3$ in GF(3) (Question 4) marks exactly the transition GF(3)$\to \mathbb{R}$.

Weinberg angle: GF(9) gives topology, ξ gives value

Gal(GF(9)/GF(3)) $= \mathbb{Z}_2$ explains *why* there are two gauge groups $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ — a degree-2 extension gives exactly two components **[K]**.

The obvious GF(9) quotient $|\mathrm{GF}(3)^*|/|\mathrm{GF}(9)^*| = 2/8 = 0.250$ is not the numerical value of $\sin^2\theta_W$; it lies 8 % above the $\overline{\mathrm{MS}}$ value 0.231. The on-shell Weinberg angle $1 - M_W^2/M_Z^2$ is matched within $0.4\,\%$ by the icosahedral matrix element $|\langle v_0|R_5|v_1\rangle|^2 = 2/9$ (Doc. 372 §3.1); for pole masses from $\tilde{T}\cdot m = 1$ the on-shell scheme is the natural one. The $\overline{\mathrm{MS}}$ value is described by the scale flow from ξ (Doc. 323). The GF(9) statement of this section — a degree-2 extension explains the two gauge groups — is unaffected by the choice of numerical value.

Status [K]

GF(9) provides the algebraic justification for two gauge groups (degree-2 Galois extension). The numerical value follows in the on-shell scheme from the A_5 matrix element $2/9$ (Doc. 372), in the $\overline{\mathrm{MS}}$ scheme from the ξ scale flow (Doc. 323).

aa.4 Status overview

Table aa.1: Algebraic bridge questions FFGFT ↔ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N \bmod 3$	canonical, proved	**[B]**
1b	$\varphi(N) = \Delta w$ physical	same threefold split	**[K]**
2a	$N \bmod 3$ classifies Q	injection ι proved	**[B]**
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	**[K]**
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	**[B]**
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = GF(3)$\to\mathbb{R}$	**[K]**
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	**[B]**
P3b	Fixed pts. = triality values, $(k,-k)$ pairs	orbit structure	**[B]**
P4b	$1^2 = 2^2 = 1$ in GF(3)	Gen. 2+3 degenerate	**[B]**
P4c	Norm $\in \{1,2\}$ in GF(9)	light/heavy	**[B]**
P5b	$\mathrm{Gal}(\mathrm{GF}(9)/\mathrm{GF}(3)) = \mathbb{Z}_2$	two gauge groups; value $2/9$ (Doc. 372)	**[K]**

aa.5 Five algebraic computations (script 328)

Gluon representations from bilaterals

With the normalised number operator $N = \sum_k q_k^\dagger q_k$ (Furey normalisation $q_k = a_{p,k}/2$), the spectrum is $\sigma(N) = \{0,1,2,3\}$ **[B]** — exactly Furey's result.

The *bilaterals* $q_i q_j^\dagger$ yield gluon representations:

- 6 off-diagonal $(q_i q_j^\dagger,\ i \neq j)$: traceless, rank 2 **[B]**

- 2 diagonal from $(N_i - N_j)/\sqrt{2}$: SU(3) Cartan generators **[B]**

The 8 matrices are linearly independent (rank 8) **[B]** — 8 gluon representations matching the 8 generators of SU(3).

Tripotent classification

All tripotents T with $T^3 = T$ in the 8-dimensional representation:

- $T_k = P_{p,k} - P_{n,k}$: eigenvalues $\{+1, -1\}$ in sym. $\mathbb{Z}_3\mathbb{C}$ **[B]**
- $P_{p,k}$, $P_{n,k}$ themselves: tripotents with eigenvalues $\{0, +1\}$ **[B]**
- 6 tripotent products $T_i \cdot T_j$ $(i \neq j)$ **[B]**

Generator comparison: FFGFT trine vs. Matzke's $(1+1j)^n$

FFGFT trine: $T_k^3 = \mathbf{1}$ to 6.5×10^{-16} **[B]** (algebraically exact over $\mathbb{C}$ because $\omega^3 = 1$, $P^2 = P$, Doc. 324; not from characteristic 3, see section "Motivation"). Matzke's $(1+1j)$ analogue in 8×8: $D^8 = 256 \cdot \mathbf{1}$ (scaling factor 256 from the matrix representation). In sym. $\mathbb{Z}_3\mathbb{C}$: $(1+i)^8 = 1$ (order 8) **[B]**.

GF(27) and three generations

$\mathrm{GF}(27) = \mathrm{GF}(3)[x]/(x^3 + 2x + 1)$ (irreducible, also $x^3 + 2x + 2$) **[B]**.

Orbit structure of Frobenius $x \mapsto x^3$ in GF(27): 3 fixed points $= \mathrm{GF}(3)$, $(27-3)/3 = 8$ three-cycles **[B]**.

Status [K]

8 three-cycles in GF(27) correspond to 8 quark-colour states in 3 generations. The Galois group $\mathrm{Gal}(\mathrm{GF}(27)/\mathrm{GF}(3)) = \mathbb{Z}_3$ may describe generation mixing (CKM matrix) **[S]**.

Noether conserved charge of the tripotent symmetry

The tripotent condition $X^3 = X$ is a $\mathbb{Z}_3$-affine symmetry. By Noether's theorem, every symmetry generates a conserved charge.

The corresponding Noether charge is **[B]**:

$$Q_{\mathbb{Z}_3} = N \bmod 3 = \Delta w \tag{aa.6}$$

This is identical to the ring homomorphism φ from §2.1, i.e. the triality. It is compatible with the $\mathbb{Z}_3$ orbifold action τ on $\mathrm{T}^4/\mathbb{Z}_3$ (invariant under τ), but it is not its sector index (see Question 1).

Status [B]

Noether charge of the tripotent symmetry ($X^3 = X$) = winding quantum $\Delta w = N \bmod 3$ (triality). It is compatible with the $\mathbb{Z}_3$ orbifold action but a different

quantity from its sector index; which of the two is the physical colour charge is open.

aa.6 CKM Wolfenstein hierarchy from ξ and GF(27) (script 329)

From the degrees of GF(9) and GF(27) over GF(3) and ξ, three precise relations emerge for the CKM Wolfenstein parameters.

$\lambda = \xi^{1/6}$

GF(9) and GF(27) have degrees $[\mathrm{GF}(9) : \mathrm{GF}(3)] = 2$ and $[\mathrm{GF}(27) : \mathrm{GF}(3)] = 3$ over GF(3). There is no tower $\mathrm{GF}(3) \subset \mathrm{GF}(9) \subset \mathrm{GF}(27)$: $\mathrm{GF}(3^d) \subset \mathrm{GF}(3^n)$ holds only for $d \mid n$, and $2 \nmid 3$. Degree 6 is reached only by $\mathrm{GF}(729)$: $2 \cdot 3 = \mathrm{lcm}(2,3) = 6$ is the degree of the compositum $\mathrm{GF}(9) \cdot \mathrm{GF}(27) = \mathrm{GF}(729)$ over $\mathrm{GF}(3)$. The exponent $1/6$ is therefore a motivated stipulation (a justification via $\mathrm{GF}(729)$ is outstanding); the numerical value $\xi^{1/6} = 0.22602$ ($0.46\,\%$ deviation) is not affected **[K]**:

$$\lambda = \xi^{1/6} = \xi^{1/(2\cdot 3)} = 0.2260 \qquad (\text{exp.: } \lambda = 0.2250,\ \text{dev. } 0.5\,\%) \tag{aa.7}$$

The Wolfenstein hierarchy is the ξ hierarchy in steps of $1/6$: $|V_{us}| \sim \xi^{1/6}$, $|V_{cb}| \sim \xi^{1/3}$, $|V_{ub}| \sim \xi^{1/2}$.

$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$

With $C_2 = 4/3$ (Casimir value $\mathrm{SU}(3)_{\mathrm{fund}}$), $C_2 - 1 = 1/3$ **[B]**, and:

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \frac{\sqrt{\xi}}{3} = 3.85 \times 10^{-3} \qquad (\text{PDG: } 3.82 \times 10^{-3},\ \text{dev. } 0.8\,\%) \quad \textbf{[K]} \tag{aa.8}$$

This relation connects the GF(9) extension ($\sqrt{\xi}$) with the SU(3) Casimir value.

Consistency: $A \cdot |\vec{\rho}| \approx 1/3$

From $\lambda = \xi^{1/6}$ and $|V_{ub}| = \sqrt{\xi}/3$ it follows necessarily **[K]**:

$$A\sqrt{\rho^2 + \eta^2} = \frac{|V_{ub}|}{\lambda^3} = \frac{\sqrt{\xi}/3}{\xi^{1/2}} = \frac{1}{3} = C_2 - 1 \tag{aa.9}$$

Numerically: $A\sqrt{\rho^2 + \eta^2} = 0.316$ vs. $1/3 = 0.333$ (5% dev.).

Status [K]

CKM Wolfenstein hierarchy from the field degrees of GF(9), GF(27) and ξ:

$$\lambda = \xi^{1/(\deg(\mathrm{GF}9)\cdot\deg(\mathrm{GF}27))} = \xi^{1/[\mathrm{GF}(729):\mathrm{GF}(3)]} = \xi^{1/6}$$

$$|V_{ub}| = \sqrt{\xi}\cdot(C_2 - 1) = \sqrt{\xi}/3$$

The exponent is a motivated stipulation (degree of the compositum, not of a tower up to GF(27)). Verification script: `pruef_329_ckm_xi_gf27.py` (16 assertions, all OK).

aa.7 ξ as a Galois Number

The mass-ratio identity from §aa.6 leads to a new result: $\xi = 4/30000$ is not a free constant but a Galois number **[K]**.

Core identity

The FFGFT formula (Doc. 006) and the Galois group orders from §aa.3 yield:

$$\frac{(r_\mu/r_e)^2}{\xi} = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \qquad \text{(aa.10)}$$

These are not two computations of the same value but a single algebraic equation. From it ξ follows:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{1}{7500} = \frac{4}{30000} \quad [\mathrm{K}] \qquad \text{(aa.11)}$$

Winding numbers from GF(3^n)

The quantum numbers $r_g = n_{\phi,g}^2/n_{\theta,g}$ (Doc. 317) follow from a recursion with a Galois initial condition:

Winding number	Value	Galois origin
$n_{\theta,1}$	3	$\mathrm{char}(\mathrm{GF}(3^n))$
$n_{\phi,1}$	2	$\lvert\mathrm{GF}(3)^*\rvert$
$n_{\theta,2}$	5	recursion $= 3 + 2$; largest or primitive prime factor of $\lvert\mathrm{GF}(81)^*\rvert$
$n_{\phi,2}$	4	$\varphi(5)$, Euler-φ of the prime factor
$n_{\theta,3}$	9	recursion $= 5 + 4$
$n_{\phi,3}$	5	largest or primitive prime factor of $\lvert\mathrm{GF}(81)^*\rvert = 80$

Since $80 = 2^4 \cdot 5$, 5 is the largest and the only primitive (Zsigmondy) prime factor of $3^4 - 1$; it is new compared to the earlier stages only in this sense, since $|\mathrm{GF}(27)^*| = 26$ already contributes 13.

All factors in the core identity are Galois orders: $8 = |\mathrm{GF}(9)^*|$, $27 = |\mathrm{GF}(27)|$, $5 =$ largest or primitive prime factor of $|\mathrm{GF}(81)^*|$. Verification scripts: `pruef_330` (mass ratio), `pruef_331` (ξ derivation).

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = \mathbf{1}$ (over $\mathbb{C}$ from $\omega^3 = 1$, $P^2 = P$) **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All algebraic results were independently verified by Python verification scripts; the author bears sole responsibility for the content.

The methodological principles governing the use of AI in FFGFT documents are described in the A-series (in particular Docs. A001–A010): AI-generated content is only incorporated into the corpus after independent review and verification by scripts. External criticism and AI outputs are not adopted without scrutiny (Doc. 190, methodology note).

References

[1] D. J. Matzke, *Quantum Computation Using Geometric Algebra*, Ph.D. Dissertation, University of Texas at Dallas, 2002. https://www.matzkefamily.net/doug/papers/PHD/phd_matzke_final.pdf

[2] D. J. Matzke, *Introducing Existons/Anti-Existons (conscious hyperbits) as discrete unit of physicality and discrete unit of consciousness*, in: *Combinatorial Foundations: Scientific Essays in Honor of Louis Kauffman on His 80th Birthday*, ANPA, 2026.

[3] D. J. Matzke, *GALG: Geometric Algebra over* $\mathbb{Z}_3\mathbb{C}$, IPI Talk, 26 July 2025. https://informationphysicsinstitute.blogspot.com/2025/07/ipi-talk-doug-matzke-wwwquantumdougcom.html

[4] C. Furey, *Charge quantization from a number operator*, Phys. Lett. B **742** (2015), 195–199. arXiv:1603.04078. DOI: 10.1016/j.physletb.2015.01.023

[5] C. Furey, *Generations: Three Prints, in Colour*, JHEP **10** (2014), 046. arXiv:1405.4601. DOI: 10.1007/JHEP10(2014)046

[6] C. Furey, *Standard model physics from an algebra?* Ph.D. Dissertation, University of Waterloo, 2016. arXiv:1611.09182

[7] L. Wolfenstein, *Parametrization of the Kobayashi-Maskawa Matrix*, Phys. Rev. Lett. **51** (1983), 1945. DOI: 10.1103/PhysRevLett.51.1945

[8] Particle Data Group, R. L. Workman et al., *Review of Particle Physics*, Prog. Theor. Exp. Phys. **2022**, 083C01 (2022). https://pdg.lbl.gov

[9] J. Pascher, *Doc. 257: $T^4/\mathbb{Z}_3$ Geometry and FFGFT*, FFGFT Corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality

[10] J. Pascher, *Doc. 321: SU(3) and $\mathbb{Z}_3$ Emergence*, FFGFT Corpus, ORCID 0009-0000-6518-4064.

[11] J. Pascher, *Doc. 324: G(6,$\mathbb{Z}_3\mathbb{C}$) and FFGFT Comparison*, FFGFT Corpus, ORCID 0009-0000-6518-4064.

Chapter ab

Doc. 337 — Time Emergence: FFGFT and GALG Compared

Abstract

This document offers two suggestions for Doug Matzke's GALG framework [1, 1], supported by direct quotations and algebraic verification.

Proposal 1 (time): *Matzke's co-exclusion primitive generates proto-time through operator change [1] (p. 37), in direct parallel to FFGFT decompactification (unrolling, Doc. 333). Both frameworks share the same three-layer emergence structure. This is not a weakness of GALG but an unspoken parallel that strengthens it.*

Proposal 2 (G): *The Schwarzschild derivation in [1] starts from the hyperbit $e_i^2 = +1$ — a purely algebraic, G-free initial condition. The SI interface is introduced only through T_{Planck}, from which, with $T = T_{Planck}/(2\pi)$, the value $M_{bit} = m_P \cdot \ln 2/(2\pi)$ follows. Since $m_P = \sqrt{\hbar c/G}$, this interface contains G — but this is structurally unavoidable: any theory that produces SI units must build a bridge to measured constants somewhere. T_{Planck} is that bridge, and it is legitimate. The suggestion: restrict the claim* "without free parameters" *to the algebraic layer $e_i^2 = +1 \to Cl(6)$, and name T_{Planck} explicitly as the SI interface. In FFGFT, ξ and m_e serve as the SI interface (Doc. 012) — a different route, but the same structural necessity.*

Both proposals are documented through direct quotations from Matzke's publications and verified by Python scripts.

Guide

The basic problem. Observable time is a macroscopic phenomenon. Quantum mechanics and algebraic field theories operate at a level where time is either absent or appears only as a discrete primitive. How does continuous, observable time arise from a timeless algebraic base structure?

What this document shows. Two suggestions, supported by direct quotations from his own publications: (1) Time emergence in GALG has the same structure as in FFGFT — this strengthens Doug's framework but has not yet been made explicit. (2) The G derivation in [1] is algebraically necessary from the SI interface T_{Planck}

onward — the claim *"without free parameters"* holds precisely for the algebraic layer, not for the SI embedding.

Notation

Symbol	Meaning	Source
Status markers		
[B]	algebraically proved, script available	Doc. 190
[K]	derivable from ξ	Doc. 190
[S]	open bridge, sketch	Doc. 190
FFGFT		
$\tilde{T} = 1/m$	intrinsic time scale of a mode	Doc. 321 [4]
$T^4/\mathbb{Z}_3$	compact 4-torus geometry	Doc. 257 [3]
Decompactification	unrolling of the compact dimension	Doc. 333 [4]
Matzke GALG		
Co-occurrence	addition $a + b$: spatial primitive	[1] p. 37
Co-exclusion	multiplication ab: temporal primitive	[1] p. 37
Proto-time	emergent time order from co-exclusion	[1] p. 14

ab.1 Time in FFGFT: compactified and emergent

The compactified time scale $\tilde{T} = 1/m$

In FFGFT, each particle of mass m carries an intrinsic time scale (Doc. 321 [4]):

$$\tilde{T} = \frac{1}{m} \qquad \text{(natural units, } \hbar = c = 1\text{)} \tag{ab.1}$$

This time scale lives on the compact geometry $\mathrm{T}^4/\mathbb{Z}_3$. It is mode-specific — each mode has its own time scale, and cross-mode superpositions generate incompatible time scales (Doc. 334 [6]) **[B]**.

Decompactification as the emergence mechanism

The compact dimension is not directly observable. Observable time arises only through unrolling: the transition from compact geometry to macroscopic spacetime (Doc. 333 [4]).

Doc. 333 distinguishes two unrolling types:

- **Type I** (measurement act): lossy, irreversible — wavefunction collapse as decompactification.
- **Type III** (pullback): lossless, reversible — geometric retraction to the compact manifold.

Observable time is a Type-I phenomenon: it arises through the irreversible unrolling act, not from the compact geometry itself.

Core statement FFGFT [B]: Before decompactification, only $\tilde{T} = 1/m$ exists as a mode-bound geometric quantity. Classical, measurable time t is the result of unrolling.

ab.2 Time in GALG: proto-temporal and emergent

Co-occurrence and co-exclusion

Matzke [1] defines two fundamental primitives for geometric algebra (Ch. 4.2, pp. 36–39):

Co-occurrence (spatial primitive): two states exist simultaneously, expressed as GA addition $a + b$. Direct quote [1], p. 37:

> "Co-occurrence is defined as two or more states 'occurring' simultaneously [...] commutative properties of addition reflects the lack of temporal ordering characteristic of simultaneity."

Co-occurrence is static — no temporal ordering.

Co-exclusion (temporal primitive): an operator changes the state through GA multiplication ab. Direct quote [1], p. 37:

> "Since co-exclusion infers an operator for change and implies temporal sequence, it is considered to be the temporal primitive. Conversely, co-occurrence is labeled the spatial primitive since it is static."

Proto-time emerges from operators

Classical time is not encoded in the GALG framework, but emerges from the interaction of disjoint operators. Direct quote [1], p. 14:

> "Interaction of disjoint finite sets of dimensions (operators and resulting co-exclusions) creates a synchronization-based proto-time from which classical time and energy metrics ultimately emerge."

On the need for a different temporal concept in the quantum domain (p. 12):

> "The quantum domain must have a different kind of primitive temporal framework, if for no other reason than it exists in a completely different spatial environment. Conceptually, this proto-temporal framework is identical to the synchronization tokens and mechanisms used by asynchronous logic."

Core statement GALG: Before emergence, only the algebraic primitive of co-exclusion (multiplication) exists. Classical time and energy arise from the interaction of disjoint operator sets.

ab.3 The structural parallel

Layer	FFGFT	GALG
Timeless base layer	$\tilde{T} = 1/m$ compactified	co-occurrence (addition)
Temporal primitive	torsion / winding on T^4	co-exclusion (multiplication ab)
Emergence mechanism	decompactification (unrolling)	interaction of disjoint operators
Result	observable time t, mode-bound	classical time + energy

Both frameworks share the same three-layer structure: (1) a timeless algebraic base layer, (2) a temporal primitive that implies sequence without being continuous itself, (3) an emergence mechanism that produces classical, measurable time.

ab.4 Open question

Is Matzke's co-exclusion the algebraic expression of the same structure that FFGFT describes as decompactification?

Concretely: in FFGFT, unrolling Type I (measurement act, collapse) is an irreversible transition — just as co-exclusion describes a state change through an operator without possibility of reversal (Matzke [1], p. 145: *"The process is irreversible because a multiplicative inverse does not exist for* $(1 \pm a_0)$ *and* $(1 \pm a_1)$*"*).

This parallel — irreversibility as a shared feature — could be the algebraic core of the connection **[S]**.

Verification would require:

1. Explicit construction of the unrolling operator in GALG notation.
2. Proof that co-exclusion + irreducibility reproduces the FFGFT mass formula $\tilde{T} = 1/m$.

ab.5 Mass and gravitational constant: derived or implicit?

Matzke's Schwarzschild derivation and the implicit G

In his IPI Letters paper 2026 [1], Matzke presents the derivation as follows:

> "This paper derives the complete black hole picture — Schwarzschild radius, Bekenstein entropy, Hawking radiation, collapse threshold, and information conservation — from a single algebraic axiom with no general relativity and no free parameters."

The derivation uses $M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$ as starting point (Landauer at $T_{\text{Planck}}/(2\pi)$, then $E = mc^2$). From the collapse condition Compton = Schwarzschild:

$$\frac{\hbar}{M_{\text{coll}}\, c} = \frac{2G\, M_{\text{coll}}}{c^2} \quad \Rightarrow \quad G = \frac{\hbar c}{2\, M_{\text{coll}}^2} \tag{ab.2}$$

with $M_{\text{coll}} = n_{\text{thresh}} \cdot M_{\text{bit}} = m_P/\sqrt{2}$ (numerically verified: 0.00% deviation).

The implicit G from this construction:

$$G_{\text{doug}} = \frac{\hbar c}{2\,M_{\text{coll}}^2} = \frac{\hbar c}{m_P^2} = 6.6743 \times 10^{-11}\ \text{m}^3/(\text{kg}\cdot\text{s}^2) \tag{ab.3}$$

Deviation from experimental value: $0.00\,\%$.

The SI interface and what it achieves

Matzke's initial condition is purely algebraic and G-free:

$$e_i^2 = +1 \quad \text{(hyperbit, } +1 \text{ signature)} \tag{ab.4}$$

From this follows the Witt decomposition into nilpotents $\{ap, an\}$ with $ap^2 = 0$, then Cl(6), then the Planck voxel — all without G.

The SI interface enters only through T_{Planck}:

$$M_{\text{bit}} = \frac{k_B\,(T_P/2\pi)\,\ln 2}{c^2} = \frac{m_P \ln 2}{2\pi} \tag{ab.5}$$

Only with this value does $n_{\text{thresh}} \cdot M_{\text{bit}} = m_P/\sqrt{2}$ hold, with $n_{\text{thresh}} = 2\pi/(\sqrt{2}\ln 2) \approx 6.41$ (not an integer); with $k_B T_P \ln 2/c^2 = m_P \ln 2$ one would obtain $2\pi/\sqrt{2} \cdot m_P = 4.44\,m_P$. The starting value thus requires the Landauer temperature $T = T_P/(2\pi)$. Since $m_P = \sqrt{\hbar c/G}$, this interface contains G — but this is structurally **unavoidable**: any theory producing dimensionful SI quantities must insert measured constants somewhere. T_{Planck} is the minimal such bridge.

Matzke himself names this clearly in his IPI mail of 27 Aug 2026:

> "the fact that M_{bit} is directly related to M_{Planck} was a happy coincident based on Planck units consolidation only."

The suggestion: restrict the claim *"without free parameters"* explicitly to the algebraic layer $e_i^2 = +1 \to \text{Cl}(6)$. The SI embedding via T_{Planck} is legitimate and necessary — but should be named as such, not presented as part of the parameter-free derivation.

FFGFT: G as genuine derivation

In FFGFT, G is derived from ξ and m_e (Doc. 012 [8]), *without* m_P as input:

$$G = \frac{\xi^2}{4\,m_e} \cdot C_{\text{conv}} \cdot K_{\text{frak}} \qquad \text{with } \xi = \frac{4}{30\,000},\ K_{\text{frak}} = \frac{74}{75} \tag{ab.6}$$

with $C_{\text{conv}} = 7.783 \times 10^{-3}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}\,\text{MeV}$ and $m_e = 0.511$ MeV; this gives $G = 6.679 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ ($+0.07\,\%$; with $K_{\text{frak}} = 0.986$: 6.6745×10^{-11}, $+0.003\,\%$).

Here ξ is fixed by the $\text{T}^4/\mathbb{Z}_3$ geometry and m_e by the mass formula, both independent of G. This is a genuine derivation: G is output, not input. This holds for the form of G and its order of magnitude, which follow from ξ and the single anchor; C_{conv} is the SI conversion of the single-anchor chain (Doc. 383), and the agreement with the CODATA value to the stated accuracy arises only with the small residual of this chain that is contained in C_{conv}. It is therefore not an independent proof of precision.

Comparison

Quantity	GALG (Matzke)	FFGFT
Input	m_P (contains G!)	ξ, m_e
Mass	$M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$	$\tilde{T} = 1/m$, then ξ
Gravitation	$G = \hbar c/m_P^2$ (tautology)	$G = \xi^2 C_{\text{conv}}/(4m_e) \cdot K$
Status	algebraically consistent **[S]**	genuine derivation **[K]**

Both frameworks recover the measured value of G (FFGFT: $+0.07\,\%$ with $K_{\text{frak}} = 74/75$, $+0.003\,\%$ with $K_{\text{frak}} = 0.986$, the last digit via the residual of the single-anchor chain in C_{conv}), and both require an SI interface. GALG chooses T_{Planck} — minimal and physically well motivated. FFGFT chooses ξ and m_e — derived geometrically from the torus. Both routes are legitimate; the difference is the choice of bridge, not whether a bridge is needed.

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All direct quotations were taken from Matzke's PhD [1] and verified with page references. The author bears sole responsibility for the content. Methodological principles: A-series, Doc. 190.

References

[1] D. J. Matzke, *Quantum Computation Using Geometric Algebra*, Ph.D. Dissertation, University of Texas at Dallas, 2002. https://www.matzkefamily.net/doug/papers/PHD/phd_matzke_final.pdf
Direct quotes in this doc.: p. 12 (proto-temporal framework), p. 14 (proto-time from co-exclusion), p. 37 (temporal primitive), p. 145 (irreversibility).

[2] D. J. Matzke, *GALG: Geometric Algebra over $\mathbb{Z}_3\mathbb{C}$*, IPI Talk, 26 July 2025. https://informationphysicsinstitute.blogspot.com/2025/07/ipi-talk-doug-matzke-wwwquantumdougcom.html

[3] J. Pascher, *Doc. 257: $T^4/\mathbb{Z}_3$ Geometry and FFGFT*, FFGFT Corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality

[4] J. Pascher, *Doc. 321: SU(3) and $\mathbb{Z}_3$ Emergence; intrinsic time scale $\tilde{T} = 1/m$*, FFGFT Corpus, ORCID 0009-0000-6518-4064.

[5] J. Pascher, *Doc. 333: K_{frak}, Unrolling and the $\tilde{T} \cdot m$ Duality*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.

[6] J. Pascher, *Doc. 334: Superposition without Time*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.

[7] D. J. Matzke, *Black Holes from Hyperbits: The Event Horizon as Nilpotent Algebra, Planck Voxels, and High-Dimensional Corner Geometry*, IPI Letters 2026, Vol. 4(3): N1–N22. https://ipipublishing.org/index.php/ipil/article/download/378/201/1385

[8] J. Pascher, *Doc. 012: Derivation of the gravitational constant G from ξ*, FFGFT Corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality

Chapter ac

Doc. 338 — Mass Calculations and Galois Comparison

Abstract

This document places the FFGFT mass calculations of the three lepton generations alongside the corresponding Galois structures in GF(3^n).

*The main result: $(m_\mu/m_e)^2 = 43200$ is exact and identical in both frameworks — FFGFT from quantum numbers (r_i, p_i), Galois from group orders $|GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)|$. The equality is not a numerical coincidence but an algebraic identity from which $\xi = 4/30000$ follows as a Galois number **[K]**.*

The remaining deviation of bare masses from experiment (~0.5%) is neither a measurement error nor K_{frac}; it is an empirical discrepancy. The candidate is the fractal-recursive correction of higher order from the torus winding numbers; the physical attribution is open. Verification scripts: `pruef_330`, `pruef_331`.

ac.1 FFGFT Mass Formula and Quantum Numbers

The mass of fermion i in FFGFT (Doc. 006):

$$m_i^{\text{bare}} = r_i \cdot \xi^{p_i} \cdot v, \qquad v = 246 \text{ GeV}, \quad \xi = \frac{4}{30000} \tag{ac.1}$$

The quantum numbers (r_i, p_i) follow from the winding numbers $(n_{\theta,g}, n_{\phi,g})$ of the T^4 torus (Doc. 317):

$$r_g = \frac{n_{\phi,g}^2}{n_{\theta,g}}, \qquad p_g = \frac{3}{2}\left(\frac{2}{3}\right)^{g-1} \quad (g = 1, 2, 3) \tag{ac.2}$$

The values $3/2, 1, 2/3$ form the geometric sequence with $p_1 = D/2$ and ratio $2/3$ (Docs. 158, 018).

Table ac.1: FFGFT quantum numbers of leptons (Doc. 006, 317)

Lepton	Gen. g	n_θ	n_ϕ	r_i	p_i
Electron	1	3	2	$4/3$	$3/2$
Muon	2	5	4	$16/5$	1
Tau	3	9	5	$25/9$	$2/3$

ac.2 Lepton Masses: Bare Prediction vs. Experiment

Table ac.2: Lepton masses: FFGFT bare vs. PDG experiment

Lepton	m_i^{bare} (MeV)	m_i^{exp} (MeV)	Dev. (%)	Correction
Electron	0.5050	0.5110	-1.18	fractal-recursive
Muon	104.960	105.658	-0.66	fractal-recursive
Tau	1783.4	1776.9	$+0.37$	fractal-recursive

The bare prediction lies systematically 0.5–1.2% from experiment. This deviation is *not*:

- Measurement error (PDG errors $< 10^{-4}$ %)
- K_{frac} correction (K_{frac} cancels exactly in ratios, Doc. 070)

It is an empirical discrepancy: for the square $(m_\mu/m_e)^2$ the residual is $43200 - 42753.1 = 446.9$, far beyond any measurement uncertainty (Doc. 352). The candidate is the **fractal-recursive correction of higher order** from the torus winding numbers; the physical attribution is open.

K_{frac} is fixed by the comparison with the fine-structure constant (Doc. 070); the agreement of α itself is a property of the anchor E_0:

$$\alpha = \xi \cdot E_0^2/K_{\text{frac}}, \quad E_0 = \sqrt{m_e m_\mu} \approx 7.348 \text{ MeV} \quad \Rightarrow \quad \alpha^{-1} = 137.06 \text{ (dev. 0.02 \%)} \qquad \text{(ac.3)}$$

ac.3 Mass Ratios (K_{frac}-free)

In ratios, K_{frac} cancels exactly (Doc. 070):

$$\frac{m_\mu}{m_e} = \frac{r_\mu}{r_e} \cdot \xi^{p_\mu - p_e} = \frac{12}{5} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \sqrt{7500} = 120\sqrt{3} \approx 207.846 \quad (\text{exp: } 206.768, \text{ dev. } 0.52\,\%) \qquad \text{(ac.4)}$$

Table ac.3: Mass ratios: FFGFT bare vs. experiment

Ratio	FFGFT bare	Experiment (PDG)	Dev. (%)
m_μ/m_e	$120\sqrt{3} = 207.846$	206.768	$+0.52$
m_τ/m_e	3531.6	3477.2	$+1.56$
m_τ/m_μ	16.99	16.82	$+1.04$

ac.4 Galois Comparison

Core identity: $(m_\mu/m_e)^2 = 43200$

Squaring the mass ratio removes the irrational $\sqrt{3}$ (undefined in GF(3^n) since $3 = 0$ there) and yields an exact integer identity **[K]**:

$$\underbrace{\frac{(r_\mu/r_e)^2}{\xi}}_{\text{FFGFT}} = \underbrace{|\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|}_{\text{Galois}} = 8^2 \cdot 25 \cdot 27 = 43200 \tag{ac.5}$$

Table ac.4: FFGFT vs. Galois for $(m_\mu/m_e)^2$

	FFGFT	Galois
Input	$r_e = 4/3,\ r_\mu = 16/5,\ \xi$	$\lvert\text{GF}(9)^*\rvert = 8,\ \lvert\text{GF}(27)\rvert = 27,\ 5$
Formula	$(r_\mu/r_e)^2/\xi$	$8^2 \cdot 5^2 \cdot 27$
Result	43200	43200
Status	**[K]**	**[K]**

The factors on the Galois side:

- $8 = |\text{GF}(9)^*| = 3^2 - 1$: multiplicative group of the 2nd generation
- $27 = |\text{GF}(27)| = 3^3$: field of the 3rd generation (all elements)
- $5 =$ new prime($|\text{GF}(81)^*| = 80$): only new prime in $|\text{GF}(3^4)^*|$

Derivation of ξ from the identity

Since both sides equal the same number, ξ follows algebraically **[K]**:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{144/25}{43200} = \frac{1}{7500} = \frac{4}{30000} \tag{ac.6}$$

ξ is not a free constant — it follows necessarily from the Galois group orders and the winding numbers of the T^4 torus.

Winding numbers from GF(3^n)

Table ac.5: Winding numbers: derivation from GF(3^n) **[K]**

Quantity	Value	Galois origin	Note
$n_{\theta,1}$	3	char(GF(3^n))	axiomatic
$n_{\phi,1}$	2	$\lvert\mathrm{GF}(3)^*\rvert$	mult. group of prime field
$n_{\theta,2}$	5	recursion $3+2$	also new prime factor of $\lvert\mathrm{GF}(81)^*\rvert = 80$
$n_{\phi,2}$	4	$\varphi(5)$	Euler-φ of prime factor
$n_{\theta,3}$	9	recursion $5+4$	also char^2
$n_{\phi,3}$	5	new prime factor of $\lvert\mathrm{GF}(81)^*\rvert$	

Recursion law: $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$. GF$(3^4)^* = 80 = 2^4 \cdot 5$ supplies the new prime factor 5, absent from GF$(3^1)^*$, GF$(3^2)^*$, GF$(3^3)^*$. 5 is new only relative to the prime factors of these levels: $\lvert\mathrm{GF}(27)^*\rvert = 26 = 2 \cdot 13$ already brings the new prime factor 13, and the smallest prime factor of 80 is 2.

Complete comparison at ratio level

Table ac.6: Complete comparison at the ratio level

Method	$(m_\mu/m_e)^2$	m_μ/m_e	Dev. exp.
Experiment (PDG)	42 753.1	206.768	—
FFGFT bare $(r_\mu/r_e)^2/\xi$ **[K]**	43 200	207.846	$+0.52$ %
Galois $8^2 \cdot 25 \cdot 27$ **[K]**	43 200	207.846	$+0.52$ %

FFGFT and Galois give exactly the same bare value, so the deviation from experiment is the same in both frameworks. The comparison is made in the integer form $(m_\mu/m_e)^2$; the $0.52\,\%$ residual in m_μ/m_e is exactly half the quadratic residual and not an independent finding. The physical attribution of the residual is open (Doc. 352).

ac.5 Fine-Structure Constant from Galois Numbers

Two key identities

Combining two Galois identities eliminates ξ from the α-formula entirely.

Identity 1 (pruef_331, Doc. 317):

$$\xi = \frac{(r_\mu/r_e)^2}{\lvert\mathrm{GF}(9)^*\rvert^2 \cdot 5^2 \cdot \lvert\mathrm{GF}(27)\rvert} = \frac{144/25}{43200} = \frac{1}{7500} \tag{ac.7}$$

Identity 2 (new observation):

$$m_e \cdot m_\mu = 53.991\ \mathrm{MeV}^2 \approx 54 = \lvert\mathrm{GF}(3)^*\rvert \cdot \lvert\mathrm{GF}(27)\rvert\ \mathrm{MeV}^2 \quad (\text{dev. } 0.016\,\%) \tag{ac.8}$$

This holds for the experimental (fractally corrected) masses; bare masses give $0.505 \times 104.96 = 53.00$ MeV^2 (dev. 1.85%).

Derivation — $|\mathrm{GF}(27)|$ cancels

Substituting both identities into $\alpha = \xi \cdot E_0^2 / K_{\mathrm{frac}}$:

$$\begin{aligned} \alpha &= \frac{1}{7500} \cdot 54\ \mathrm{MeV}^2 / K_{\mathrm{frac}} \\ &= \frac{|\mathrm{GF}(3)^*| \cdot |\mathrm{GF}(27)|}{|\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|} \cdot \frac{12^2}{5^2} / K_{\mathrm{frac}} \\ &= \frac{|\mathrm{GF}(3)^*| \cdot 12^2}{5^2 \cdot |\mathrm{GF}(9)^*|^2 \cdot 5^2} \cdot \frac{1}{K_{\mathrm{frac}}} = \frac{2 \cdot 144}{40000} \cdot \frac{75}{74} = \frac{27}{3700} \end{aligned} \tag{ac.9}$$

$|\mathrm{GF}(27)| = 27$ cancels between numerator and denominator.

Result

$$\boxed{\frac{1}{\alpha} = \frac{3700}{27} = 137.037037\ldots \quad (\text{exp: } 137.035999,\ \text{dev. } 0.00076\,\% \ =\ 7.6\ \text{ppm})} \tag{ac.10}$$

All inputs are Galois group orders:

- $2 = |\mathrm{GF}(3)^*|$, $8 = |\mathrm{GF}(9)^*|$, $27 = |\mathrm{GF}(27)|$, $5 = \text{new prime}(|\mathrm{GF}(81)^*|)$
- $12 = n_{\phi,\mu} \cdot n_{\theta,e} = 4 \cdot 3$ (winding numbers, Galois)
- $K_{\mathrm{frac}} = 74/75$, $74 = 2 \cdot 37$, $37 \,|\, |\mathrm{GF}(3^{18})^*|$ (Galois)

No ξ, no v, no m_e as explicit input. The only SI bridge is the MeV unit (implicit in $E_0^2 = 54$ MeV^2). The agreement with the measured α is therefore a property of the anchor $E_0^2 = 54$ MeV^2, not an independent prediction.

Table ac.7: Precision comparison: Galois results vs. experiment

Quantity	Galois result	Dev. from exp.
$(m_\mu/m_e)^2 = 43200$	43200 (exact)	$+1.04\%$ (bare)
$m_\mu/m_e = 120\sqrt{3}$	207.846	$+0.52\%$ (bare)
$1/\alpha = 3700/27$	137.037	0.00076% (7.6 ppm)

The α-formula is ~700× more precise than the bare mass ratio because the identity $m_e m_\mu = 54$ MeV^2 holds for the experimentally measured (fractally corrected) masses.

The identity $m_e \cdot m_\mu \approx 54$ MeV^2 is not an empirical coincidence. Doc. 011 derives $E_0 = \sqrt{m_e m_\mu}$ as the logarithmic average on the T0 torus geometry (there as the fundamental definition of E_0); fractally corrected it is $E_0 = \sqrt{m_e m_\mu / K_{\mathrm{frak}}} = 7.398$ MeV (R72). The identification with $|\mathrm{GF}(3)^*| \cdot |\mathrm{GF}(27)|$ is a new Galois-side expression of the same quantity **[K]**. Verification script: `pruef_332_alpha_galois.py`.

Verification scripts

- `pruef_330_galois_massenverhaeltnis.py`: $(m_\mu/m_e)^2 = 43200$ (FFGFT = Galois), K_{frac} cross-check
- `pruef_331_xi_aus_galois.py`: $\xi = 4/30000$ fully derived from GF(3^n)

Note on AI assistance

This document was prepared with support from Claude (Anthropic, A-series). All algebraic calculations were verified by Python scripts with exact rational arithmetic (sympy). The physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 006: Particle masses in FFGFT*, FFGFT corpus, ORCID 0009-0000-6518-4064. https://github.com/jpascher/T0-Time-Mass-Duality

[2] J. Pascher, *Doc. 070: Mathematical structure of T0 theory*, FFGFT corpus.

[3] J. Pascher, *Doc. 317: KSAU-FFGFT lepton synthesis*, FFGFT corpus, August 2026.

[4] J. Pascher, *pruef_330: Galois mass ratio* m_μ/m_e, FFGFT corpus, August 2026.

[5] J. Pascher, *pruef_331:* ξ *from Galois fields*, FFGFT corpus, August 2026.

[6] Particle Data Group, *Review of Particle Physics*, Phys. Rev. D **110** (2024).

Chapter ad

Doc. 339 — Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$

Abstract

*The multiplicative group $GF(27)^* \cong \mathbb{Z}_{26}$ decomposes under the Frobenius automorphism $\phi : x \mapsto x^3$ into exactly two types: two fixed points $\{+1, -1\}$ and eight three-element orbits. Via the decomposition $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (CRT), four $\mathbb{Z}_{13}$ three-orbits times two $\mathbb{Z}_2$ parities yield the eight gluons of the adjoint $SU(3)$ representation. The orbit decomposition itself is algebraically forced. For the fixed points $\{+1, -1\}$ there are two readings: as the massive $\mathbb{Z}_2$ sector (particle/antiparticle) and, as $GF(3)^*$ of the fixed field $GF(3)$, as the $U(1)$ sector (photon). Since it is the same set, this double assignment is not resolved; the assignment is open **[S]**. Whether the massless modes carry spin 1 does not follow from the Frobenius structure (Doc. 231, open). Status: **[K]**.*

ad.1 Starting Point

In Doc. 338

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$$

is shown for the bare mass ratio as an algebraically exact relation **[B]**. The fundamental relation $\tilde{T} \cdot m = 1$ describes a duality: massive sector ($m > 0$, compact torus S^1_m) and massless sector ($m \to 0$, $\tilde{T} \to \infty$). In Doc. 285 the $3'$ extension is identified as a kinetic (massless) window. This note asks: what algebraic structure separates both sectors within $GF(3) \subset GF(9)$, $GF(27) \subset GF(729)$? ($GF(9)$ is not a subfield of $GF(27)$, since $2 \nmid 3$; the only subfields of $GF(27)$ are $GF(3)$ and $GF(27)$, and the smallest common extension of $GF(9)$ and $GF(27)$ is $GF(729)$.)

ad.2 Decomposition of $\mathbb{Z}_{26}$ under the Frobenius

Basic structure [B]

$GF(27)^* \cong \mathbb{Z}_{26}$, cyclic of order $3^3 - 1 = 26 = 2 \cdot 13$. Since $\gcd(2, 13) = 1$:

$$\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}.$$

The Frobenius $\phi : x \mapsto x^3$ acts:

- On $\mathbb{Z}_2$: $3 \equiv 1 \pmod 2$ — ϕ is *trivial*.
- On $\mathbb{Z}_{13}$: $\mathrm{ord}(3, \mathbb{Z}_{13}) = 3$ (since $3^3 = 27 \equiv 1 \pmod{13}$) — ϕ has order 3.

Orbits of ϕ on $\mathbb{Z}_{13}$ [B]

Type	Elements
Fixed point	$\{0\}$
Three-orbit 1	$\{1, 3, 9\}$
Three-orbit 2	$\{2, 5, 6\}$
Three-orbit 3	$\{4, 10, 12\}$
Three-orbit 4	$\{7, 8, 11\}$

Complete orbit table in $\mathbb{Z}_{26}$ [B]

$\mathbb{Z}_2$ parity	$\mathbb{Z}_{13}$ orbit	$\mathbb{Z}_{26}$ orbit
0	$\{0\}$	$\{0\}$ (fixed point)
1	$\{0\}$	$\{13\}$ (fixed point)
0	$\{1, 3, 9\}$	$\{14, 16, 22\}$
1	$\{1, 3, 9\}$	$\{1, 3, 9\}$
0	$\{2, 5, 6\}$	$\{2, 6, 18\}$
1	$\{2, 5, 6\}$	$\{5, 15, 19\}$
0	$\{4, 10, 12\}$	$\{4, 10, 12\}$
1	$\{4, 10, 12\}$	$\{17, 23, 25\}$
0	$\{7, 8, 11\}$	$\{8, 20, 24\}$
1	$\{7, 8, 11\}$	$\{7, 11, 21\}$

Check: $2 + 8 \times 3 = 26 = |GF(27)^*|$. ✓

ad.3 Physical Identification [K]

Massive sector: fixed points $= \mathbb{Z}_2$

The two fixed points $\{+1, -1\}$ form the massive sector: every massive particle has an antiparticle ($\mathbb{Z}_2$ structure), consistent with the compact torus sector.

Massless $SU(3)$ sector: eight gluons

$$8 \text{ gluons} = 4\ \mathbb{Z}_{13}\text{-orbits} \times 2\ \mathbb{Z}_2\text{-parities} = |GF(9)^*|.$$

The three elements of each $\mathbb{Z}_{13}$ orbit correspond to the three colours $\{\mathrm{R}, \mathrm{G}, \mathrm{B}\}$; the $\mathbb{Z}_2$ parity distinguishes colour from anticolour. This is the adjoint representation of $SU(3)$, derived from Galois structure alone.

$U(1)$ sector: photon from the fixed field $GF(3)$

The fixed field of the Frobenius on $GF(27)$ is $GF(3)$ (Fermat: $x^3 = x$ for $x \in GF(3)$). The projection $GF(27) \to GF(3)$ gives:

Structure	Sector
$GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$	Photon: $U(1)$, own antiparticle
$27 - 3 = 24$ remaining elements	$SU(3)$ sector (quarks, gluons)

$GF(3)^* = \{1, 2\}$ is the same set as the fixed points $\{+1, -1\}$ of the previous section. The assignment of this set to the photon and to the massive sector is not resolved and remains open **[S]**.

ad.4 Algebraic Necessity [B]

$$\text{Non-fixed elements in } GF(27)^* = 26 - 2 = 24, \tag{ad.1}$$

$$24 = 8 \cdot 3 = |GF(9)^*| \cdot |\text{colours}| = (3^2 - 1) \cdot 3 = 3^3 - 3, \tag{ad.2}$$

$$8 \text{ gluons} = \tfrac{26-2}{3} = 8. \tag{ad.3}$$

All three relations follow solely from $|GF(27)| = 3^3$ and the order 3 of the Frobenius on $\mathbb{Z}_{13}$.

ad.5 Complete Picture

Galois structure	Physical sector	Symmetry
$GF(3)^* = \mathbb{Z}_2$	Photon	$U(1)$
Fixed points $\{+1, -1\} \subset GF(27)^*$	Massive particle/antiparticle	$\mathbb{Z}_2$
4×2 three-orbits in $GF(27)^*$	8 gluons	$SU(3)_{\text{adj}}$

The first two rows concern the same set $GF(3)^* = \{+1, -1\}$; which of the two assignments holds is open **[S]**.

The relation $8 \times 3 = 24 = \text{Kissing}(D_4)$ is algebraically forced ($3^3 - 3 = 24$). Whether Kissing(D_4) has a deeper geometric connection to this Galois structure is subject to further investigation.

Verification script: `pruef_339_frobenius.py` (6/6 assertions).

ad.6 Scales of the Duality and Effective Masslessness [K]

The two poles of $\tilde{T} \cdot m = 1$

The fundamental relation $\tilde{T} \cdot m = 1$ has two excluded edges (Doc. 306), both finite and quantified:

Pole	Length	Time	Energy per cell
Dense (derived)	$L_0 = \xi \ell_P$	$T_0 = \xi t_P$	$E_{\max} = E_P/\xi \approx 9.2 \times 10^{22}$ GeV
Thin (calibrated)	$R_H = \hbar/E_H$	$T_H = \hbar/E_H$	$E_H = E_0\, \xi^{41/4} \approx 1.4 \times 10^{-33}$ eV

At both ends the reciprocity closes exactly: $T_0 \cdot E_{\max} = t_P \cdot E_P = 1$ and $T_H \cdot E_H = 1$ (natural units). There is no infinity — neither above nor below. The dense pole is derived from ξ **[B]**; the thin pole is calibrated to the measured H_0 (Doc. 182, P35/P36) **[S]**.

What “massless” means in FFGFT

$\tilde{T} \cdot m = 1$ has no solution at $m = 0$ — FFGFT knows no empty state, only a thinnest one (Doc. 306). The minimal rest mass per cell is:

$$m_0 = \frac{E_H}{c^2} \approx 1.4 \times 10^{-33}\ \text{eV}/c^2.$$

Gluons and photons are modes in the limit $m \to m_0$, not at $m = 0$. The time circle S^1_m opens to the time axis $\mathbb{R}_t$ in this limit (type-III pullback, Doc. 332, R96).

The deviation of the dispersion relation from the massless formula $\omega = |\mathbf{k}|$ is:

$$\frac{\delta\omega}{\omega} = \frac{m_0^2}{2\,k^2} \approx \begin{cases} 2.5 \times 10^{-67} & k = 2\,\text{eV (visible light)} \\ 1.7 \times 10^{-59} & k = 2.4 \times 10^{-4}\,\text{eV (CMB)} \\ 9.8 \times 10^{-85} & k = 1\,\text{GeV (QCD)} \end{cases}$$

The experimental upper bound on the photon rest mass is $m_\gamma < 10^{-18}$ eV/c^2 (PDG). The FFGFT minimum mass m_0 lies 10^{15} times below this — beyond any conceivable measurement.

Conclusion

“Massless” in FFGFT is not an algebraic exactness ($m \neq 0$) but a structural statement: the rest mass lies below the Hubble energy scale E_H/c^2 and is indistinguishable from zero for all physical purposes. Both poles of the duality are finite and quantified — there is no infinity, but both limits lie far beyond any experimental reach.

ad.7 Frobenius Structure, Masslessness and Spin [K]

Derivation

The derivation proceeds in four steps.

Step 1 [K]: The Frobenius map $\phi : x \mapsto x^3$ on $GF(27)^*$ corresponds in the type-III pullback picture (Doc. 332, R96) to the conformal scaling of the wave vector:

$$k \;\mapsto\; 3k, \qquad \omega(3k) = 3\,\omega(k).$$

This follows directly from the massless dispersion relation $\omega = c|k|$. Massive modes do not satisfy this relation: for $m > 0$, $\omega^2 = c^2k^2 + (mc^2/\hbar)^2$, so $\omega(3k) \neq 3\,\omega(k)$. Fixed points transform trivially: $\phi(x) = x$ corresponds to $k \mapsto k$.

Step 2 [B, standard result]: Wigner (1939) classifies the representations of the Poincaré group by m^2 and little group; massless particles are distinguished by their helicity h. The linear dispersion $\omega(\lambda k) = \lambda\,\omega(k)$ characterises masslessness for every spin (0, $\frac{1}{2}$, 1, ...) and does not fix the spin. Massive fields do not satisfy it (in the non-relativistic limit $k \to 0$ one has $\omega \to mc^2/\hbar$).

Step 3 [K]: The 8 three-orbits of $GF(27)^*$ carry the conformal scaling $k \mapsto 3k$ and therefore correspond to massless modes; that they are spin-1 fields would have to be shown via the winding numbers on $L^2(T^3, S)$ (Doc. 231, open). The 2 fixed points $\{+1, -1\}$ transform trivially and correspond to the massive sector.

Step 4 [K]: The fixed field $GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$ is conformally invariant (fixed under the projection $GF(27) \to GF(3)$) and carries a massless singlet, identified with the photon; spin 1 is open (Doc. 231). Step 3 places the fixed points $\{+1, -1\}$ in the massive sector; Step 4 and the table in the section on the $U(1)$ sector assign the same set $GF(3)^* = \{1, 2\} = \{+1, -1\}$ to the photon. This double assignment is not resolved; the assignment is open **[S]**.

Compatibility with the FFGFT corpus

In FFGFT, spin arises from winding numbers (n_θ, n_ϕ) on the spin bundle $L^2(T^3, S)$ (Doc. 231), not from an external Poincaré representation. The Wigner classification in Step 2 is therefore a standard result [B] that is consistent with the FFGFT spin-bundle framework but does not replace it. A massless spin-1 field has the two helicities ± 1 (three states only in the massive case); an assignment to the three-orbits of $GF(27)^*$ under $\mathbb{Z}_3$ therefore does not follow directly and remains as an open bridge to Doc. 231.

Result

Frobenius type	Sector	Spin	Physical object
Fixed points $\{+1, -1\}$	Massive, $k \mapsto k$	–	Leptons, quarks
8 three-orbits	Massless, $k \mapsto 3k$	open	8 gluons
Fixed field $GF(3)^*$	Massless, $k \mapsto k$	open	Photon (singlet)

Note: The spin of the massive fixed points (leptons: $\frac{1}{2}$, quarks: $\frac{1}{2}$) does not follow from the Frobenius structure but from the winding numbers on $L^2(T^3, S)$ (Doc. 231). Likewise, spin 1 of the gluons and the photon has to be shown via Doc. 231. The first and third rows concern the same set $\{+1, -1\} = GF(3)^*$; their assignment is open.

ad.8 Context: Connection to HLV and the Hilbert Space Doubling [K]

The fork from Doc. 307

Doc. 307 poses the fundamental question: is time a parameter *beside* the state space (unrolled picture) or a compact dimension *inside* it (rolled-up picture)? Both descriptions use the same operator formalism; the difference lies in where time is located.

The rolled-up picture ($\tilde{T} \cdot m = 1$, $L^2(T^4) \otimes \mathbb{C}^3$) is the massive side: discrete modes, winding numbers (n_θ, n_ϕ), lepton masses from $\xi = 4/30000$. The unrolled picture ($L^2(\mathbb{R}_t)$) is the massless side — and exactly there, as the Frobenius decomposition of $GF(27)^*$ shows algebraically, the gluons live.

FFGFT, HLV and the doubling $3 \to 6$

In Doc. 285 (FFGFT-HLV dimensional bridge) the doubling $\mathbf{6} = \mathbf{3} \oplus \mathbf{3}'$ is specified from the Hilbert space perspective:

- **FFGFT sector (3, compact torus S^1_m):** $\mathbb{Z}_3$-crystallographic, 3-fold symmetry, lives natively in 3D/4D. Protected invariant: $\theta = 2/9$ (Koide), $r = \sqrt{2}$ ($Q = 2/3$). This is the massive sector.
- **HLV sector (3′, type-III pullback to $\mathbb{R}_t$):** A_5-icosahedral, 5-fold symmetry, requires 6D embedding. The unrolling referred to here is the lossless mathematical type (isometry $L^2(S^1_m) \cong \mathrm{Per}_{R_m}(\mathbb{R}_t)$, Doc. 332 R96, type III), *not* the measurement act (type I, lossy, Doc. 333). Protected invariant: φ (golden ratio), $r = 2/\sqrt{5}$. This is the massless candidate.

The 6D splitting is forced by the crystallographic restriction: 5-fold symmetry is forbidden in 3D ($2\cos 72^\circ = 1/\varphi \notin \mathbb{Z}$) but exists in 6D. The **3′** sector is geometrically present in FFGFT but physically empty without dynamics.

What the Frobenius decomposition contributes to the bridge [K]

The Frobenius decomposition of $GF(27)^*$ provides a new algebraic precision that Doc. 285 did not yet have:

Structure	Hilbert space picture	Galois picture
Massive sector (**3**)	$L^2(S^1_m)$, rolled-up	Fixed points $\{+1,-1\}$ of ϕ
Massless sector (**3′**)	$L^2(\mathbb{R}_t)$, type-III pullback (R96)	8 three-orbits of ϕ
Koide $\theta = 2/9$	$\mathbb{Z}_3$ circulant	$GF(9)^*$ structure (Doc. 338)
Golden ratio φ	A_5 icosahedron	not anchored in $GF(27)^*$

The golden ratio φ does not appear in the $\mathbb{Z}_{13}$ picture: $\mathbb{Z}_{13}$ carries no 5-fold symmetry (since $5 \nmid 12$). This is consistent with φ being the HLV invariant, not the FFGFT invariant.

The Frobenius structure of $GF(27)^*$ confirms algebraically what Doc. 285 shows geometrically: FFGFT is the $\mathbb{Z}_3$-crystallographic core (massive fixed points), HLV would be the A_5 sector (massless **3′** orbits).

ad.9 Extension: Two Pictures, One Structure [K]

Unrolled picture and conformal symmetry

The Frobenius map $\phi : x \mapsto x^3$ on $GF(27)^*$ corresponds in the unrolled picture ($\mathbb{R}_t$ instead of S^1_m) to a conformal scaling of the wave vector:

$$k \mapsto 3k, \qquad \omega \mapsto 3\omega.$$

This is consistent with the massless dispersion relation $\omega = c|k|$:

$$\omega(3k) = c \cdot |3k| = 3 \cdot c \cdot |k| = 3\,\omega(k). \quad \checkmark$$

For massive modes, $\omega^2 = c^2k^2 + (mc^2/\hbar)^2$ gives $\omega(3k) \neq 3\,\omega(k)$ — massive modes are *not* conformally invariant. The Frobenius structure thus explains the separation algebraically:

Frobenius type	Physical sector	Symmetry
Fixed points $\{+1,-1\}$	Massive, $\omega(3k) \neq 3\omega(k)$	no conformal inv.
Three-orbits (8 total)	Massless, $\omega(3k) = 3\omega(k)$	conformal inv.

Accessible calculations per picture

Content	Picture	Status
Lepton masses, winding numbers (n_θ, n_ϕ)	Compact	**[B]**
Galois identity $43200 = 64 \cdot 25 \cdot 27$	Compact	**[B]**
$K_{\text{frak}}^{-36} \approx 16/\pi^2$	Compact	**[B]**
Gluon dispersion $\omega = c\|k\|$	Unrolled	**[K]**
Frobenius = conformal scaling $k \mapsto 3k$	Unrolled	**[K]**
Hilbert space isometry $L^2(S^1_m) \cong \text{Per}_{R_m}(\mathbb{R}_t)$	Both	**[B]**
$\xi = 4/30000$, mass ratios	Both	**[B]**

The conformal dimension of gluons ($\Delta = 1$, spin-1 vector bosons) remains to be derived from the FFGFT field theory — noted as an open point.

Declared 29 August 2026

ad.10 Note on Authorship

This document was developed on 29 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits, $\mathbb{Z}_2 \times \mathbb{Z}_{13}$ decomposition, conformal symmetry test) were carried out by the assistant and verified by the author. The physical interpretation and all status markers (**[B]**, **[K]**) are the author's responsibility.

ad.11 Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

Chapter ae

Doc. 340 — Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

Abstract

*Starting from the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (Doc. 339), the measured neutrino mass differences Δm^2_{atm} and Δm^2_{sol} are derived from algebraic properties of the four Frobenius orbits in $\mathbb{Z}_{13}$. The atmospheric mass difference $\Delta m^2_{atm} = (11^2 - 1) m_\nu^2 = 120\, m_\nu^2$ (1.0 % deviation) and the solar $\Delta m^2_{sol} = (|\mathbb{Z}_{13}| - 2)/3 \cdot m_\nu^2 = (11/3) m_\nu^2$ (0.46 % deviation) follow from the number 11, the maximum element of the fourth Frobenius orbit $\{7, 8, 11\}$ in $\mathbb{Z}_{13}$. Orbit 3 $\{4, 10, 12\}$ is algebraically self-inverse and carries Majorana character. The coherent venting amplitude $p_{vent} = 5/24$ is proved algebraically from the branching rule $\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}$ and the index $|A_5 : A_4| = 5$. The SICC venting formula is equivalent to the standard oscillation formula. Status: **[B]** for the group theory; **[K]** for the mass differences.*

ae.1 Starting Point

In Doc. 339 the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ is established **[B]**. The four three-element orbits of $\mathbb{Z}_{13}$ under the Frobenius $\phi : x \mapsto 3x$ are:

Orbit	Elements	Sum	Product mod 13
1	$\{1, 3, 9\}$	13	1
2	$\{2, 5, 6\}$	13	8
3	$\{4, 10, 12\}$	26	12
4	$\{7, 8, 11\}$	26	5

The FFGFT neutrino base mass is speculatively set as $m_\nu = \xi^2/2 \cdot m_e \approx 4.54\,\text{meV}$ **[S]** (Doc. 047). The present calculations test whether the measured mass differences follow from the Galois structure.

ae.2 Algebraic Properties of the Orbits [B]

Orbit 4 as a multiple of Orbit 1

$$\{7, 8, 11\} = 7 \cdot \{1, 3, 9\} \pmod{13},$$

since $7 \cdot 1 = 7$, $7 \cdot 3 = 21 \equiv 8$, $7 \cdot 9 = 63 \equiv 11 \pmod{13}$. Since $2^{-1} \equiv 7 \pmod{13}$:

$$\boxed{\text{Orbit}_4 = 2^{-1} \cdot \text{Orbit}_1 \pmod{13}.}$$

Duality Orbit 2 ↔ Orbit 4

The minimal elements and their multiplicative inverses in $\mathbb{Z}_{13}$:

Orbit	min	$\text{min}^{-1} \pmod{13}$	lies in
1	1	1	Orbit 1
2	2	7	Orbit 4
3	4	10	Orbit 3
4	7	2	Orbit 2

Orbits 2 and 4 are algebraically inverse: $2^{-1} = 7$, $7^{-1} = 2$. Orbit 3 is self-inverse: $4^{-1} = 10$, $10^{-1} = 4$, $12^{-1} = 12 \pmod{13}$. The self-inverse nature of Orbit 3 corresponds algebraically to Majorana character. The sum: $4 + 10 + 12 = 26 = |GF(27)^*|$.

ae.3 Neutrino Masses from the Orbit Structure [K]

The factor 11

The maximum element of the fourth orbit, $\max(\{7, 8, 11\}) = 11$, follows from:

$$11 = |\mathbb{Z}_{13}| - |\text{Frobenius fixed points in } \mathbb{Z}_{13}| - 1 = 13 - 1 - 1 = 11.$$

This factor determines the mass hierarchy (normal ordering):

State	Galois origin	Mass	Value
ν_1	Orbit 1 (D4, 4D)	m_ν	4.54 meV
ν_2	Orbit 3 (self-inverse, E6, solar)	$\sqrt{14/3}\, m_\nu$	9.81 meV
ν_3	Orbit 4 (E8, atm., heaviest)	$11\, m_\nu$	49.96 meV

The sum $\sum m_i = (1 + \sqrt{14/3} + 11)\, m_\nu \approx 64.3$ meV is consistent with the Planck-2018 bound $\sum m_i < 120$ meV.

Atmospheric mass difference

$$\Delta m^2_{\text{atm}} = m^2_{\nu_3} - m^2_{\nu_1} = (11^2 - 1)\, m_\nu^2 = 120\, m_\nu^2 = 2.476 \times 10^{-3}\ \text{eV}^2.$$

Measured: $2.500 \times 10^{-3}\ \text{eV}^2$. **Deviation: 1.0 %.**

The factor $120 = |A_5 \times \mathbb{Z}_2| = 11^2 - 1$ is the order of the full icosahedral symmetry group – consistent with the A_5 symmetry of the HLV sector (Doc. 285).

Solar mass difference

$$\Delta m^2_{\text{sol}} = m^2_{\nu_2} - m^2_{\nu_1} = \left(\frac{14}{3} - 1\right) m_\nu^2 = \frac{11}{3}\, m_\nu^2 = 7.565 \times 10^{-5}\ \text{eV}^2.$$

Measured: $7.530 \times 10^{-5}\ \text{eV}^2$. **Deviation: 0.46 %.**

The algebraic derivation of $11/3$ **[B]**:

$$\frac{11}{3} = \frac{|\mathbb{Z}_{13}| - |\text{fixed points of } GF(27)^*|}{|\text{Frobenius order}|} = \frac{13 - 2}{3}.$$

Numerator: $13 - 2 = 11$ (the two fixed points $\{+1, -1\}$ of $GF(27)^*$ subtracted from $|\mathbb{Z}_{13}|$). Denominator: 3 (the order of $\phi : x \mapsto x^3$).

ae.4 Kappa and the SICC Connection [K]

Relation κ – Δm^2_{atm}

From Doc. 339 (and Paul Masic's SICC derivation):

$$\kappa = \frac{8}{r_e} \cdot m_e \xi^2 = 6\, \xi^2\, m_e = 12\, m_\nu.$$

It then follows immediately:

$$\Delta m^2_{\text{atm}} = (11^2 - 1)\, m_\nu^2 = 120\, m_\nu^2 = \frac{5}{6}\, \kappa^2.$$

The ratio:

$$\frac{\kappa^2}{\Delta m^2_{\text{atm}}} = \frac{144}{120} = \frac{6}{5}.$$

The coherent venting amplitude $p_{\text{vent}} = 5/24$ [B]

If W-venting events accumulate as coherent quantum phases rather than energy loss, the oscillation phase reads:

$$\phi_{\text{venting}} = p_{\text{vent}} \cdot \frac{\kappa^2\, L}{\hbar c\, E_\nu}.$$

For $\phi_{\text{venting}} = \Delta m^2_{\text{atm}} \cdot L/(4\,\hbar c\, E_\nu)$ (standard phase, as in the equivalence section below), one requires:

$$p_{\text{vent}} = \frac{\Delta m^2_{\text{atm}}}{4\,\kappa^2} = \frac{120\, m_\nu^2}{4 \cdot 144\, m_\nu^2} = \frac{120}{576} = \frac{5}{24}.$$

The factor 4 of the standard phase is essential here: without it one would obtain $p = 120/144 = 5/6$.

Algebraic derivation of $p = 5/24$ from the group structure [B]

The branching rule for restricting the 4D representation ρ_4 of A_5 to the subgroup A_4 (tetrahedral group, order 12) reads:

$$\boxed{\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}.}$$

This follows from the A_4 character table: the characters of ρ_4 on the A_4 classes $(e, (123), (132), (12)(34))$ are $(4, 1, 1, 0)$, and the scalar products with A_4 characters yield multiplicity 1 for both $\rho_1^{A_4}$ and $\rho_3^{A_4}$.

Consequence: A_4 leaves a one-dimensional sub-representation (the W-direction) fixed in ρ_4. That is, A_4 is the *stabiliser* of the W-direction in A_5.

In A_5 there are exactly

$$|A_5 \,:\, A_4| = \frac{60}{12} = 5$$

conjugates of A_4 – corresponding to the 5 tetrahedra embedded in the icosahedron. Each conjugate stabilises a different W-direction. The W-venting selects exactly one of these 5 directions (the physical 4th spatial direction), hence:

$$\boxed{p_{\text{vent}} = \frac{|A_5 \,:\, A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.}$$

Equivalence to the standard oscillation formula

The complete SICC venting formula reads:

$$\phi_{\text{venting}} = \frac{5}{24} \cdot \frac{\kappa^2 L}{\hbar c\, E_\nu} = \frac{5}{24} \cdot \frac{144\, m_\nu^2 L}{\hbar c\, E_\nu} = \frac{120\, m_\nu^2 \cdot L}{4\, \hbar c\, E_\nu} = \frac{\Delta m^2_{\text{atm}} \cdot L}{4\, \hbar c\, E_\nu}.$$

This is *identical* to the standard oscillation formula. All parameters – κ, p_{vent}, Δm^2_{atm} – derive from the Galois structure $GF(27)^*$ and the group structure $A_5 \supset A_4$. No free parameter, no DUNE calibration.

The complete derivation chain:

$$GF(27)^* \xrightarrow{[B]} m_\nu \xrightarrow{[K]} \kappa = 12\, m_\nu \xrightarrow{[K]} \Delta m^2_{\text{atm}} = 120\, m_\nu^2 \xrightarrow{[B]} p_{\text{vent}} = \frac{|A_5 \,:\, A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

Non-closure of the angular structure: $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ [B]

The SICC simulation quantises directions onto the 24 kissing vectors of D_4; the Frobenius structure supplies orbit phases $2\pi k/13$. The two angular lattices are coprime:

$$\gcd(13, 24) = 1, \qquad \text{lcm}(13, 24) = 312.$$

The combined angular structure therefore closes only after 312 steps, not 24 – a discrete spiral. This is the non-closure structure of Doc. 193 in angular form: there at the level of mass products $r_i \cdot r_j$, here at the level of $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$.

Continuously, the spiral also appears across the beam spectrum: $\phi(E) \propto 1/E$ is not periodic in E. Over $[0.5,\ 5]$ GeV (DUNE) the phase winds through 1.155 turns with remainder 55.7° mod 2π – the curve does not close. This is invisible in a 1D phase histogram; in a polar plot (radius $=$ spectral weight, angle $=\ \phi$ mod 2π) the spiral becomes visible.

Fractal correction: not applicable to ratios

The fractal correction $K_{\text{frak}} = 1 - 100\,\xi$ (Doc. 011/133) applies to absolute masses, not to dimensionless ratios (Doc. 306). $\Delta m^2_{\text{atm}}/m^2_\nu = 120$ is a ratio; K_{frak} must not be applied here. As a check: applied, the deviation of Δm^2_{atm} would rise from $1.0\,\%$ to $2.3\,\%$; the effect on DUNE disappearance would be 0.2 percentage points – below detector resolution.

ae.5 Mixing Angles [K] / [S]

Two mixing angles can be read off from Orbit-2 elements with $< 6\,\%$ deviation; these are order-of-magnitude statements (see below) **[S]**:

$$\sin^2\theta_{12} \approx \cos^2\!\left(\frac{2\pi\cdot 2}{13}\right) = 0.323 \quad (\text{measured: } 0.307,\ \text{dev. } 5.1\,\%)$$

$$\sin^2\theta_{23} \approx \cos^2\!\left(\frac{2\pi\cdot 5}{13}\right) = 0.560 \quad (\text{measured: } 0.545,\ \text{dev. } 2.8\,\%)$$

The reactor angle θ_{13} has two independent Galois derivations **[K]**:
(a) Galois formula from Doc. 328 (stronger):

$$\theta_{13} = \arcsin\!\left((25\,\xi)^{1/3}\right), \quad 25\,\xi = \frac{1}{300} = \frac{1}{5\cdot|A_5|},$$

$$\sin^2\theta_{13} = \left(\frac{1}{300}\right)^{2/3} \approx 0.02230 \quad (\text{measured: } 0.0220,\ \text{dev. } 1.4\,\%)$$

The factor $25 = 5^2$ comes from the Galois identity $43200 = |GF(9)^*|^2\cdot 5^2\cdot|GF(27)| = 64\cdot 25\cdot 27$ (Doc. 338), where 5^2 is the square of the maximal A_5 representation dimension.
(b) Sum rule from R_ν (independent confirmation):

$$\sin^2\theta_{13} = \frac{2}{3}\,R_\nu = \frac{2}{3}\cdot\frac{11}{360} = \frac{11}{540} \approx 0.02037 \quad (\text{dev. } 7.4\,\%) \quad [1]$$

The factor $2/3$ comes from the sum rule of the source; there is no Galois reading of this factor.

The same source independently gives:

$$\sin^2\theta_{12} = \frac{1}{3-2R_\nu} \approx 0.340 \quad (\text{dev. from Galois value } 0.323:\ 5.4\,\%)$$

The CP-violation phase $\delta_{CP} \approx -\pi/2$ (experimental hint) corresponds to Orbit-3 element $k = 10$: $\arg(e^{2\pi i\cdot 10/13}) = -83.1^\circ$, deviation 6.9° **[S]**.

The $\cos^2(2\pi k/13)$ readings for θ_{12} and θ_{23} in this section are order-of-magnitude statements: the twelve values $\{\cos^2(2\pi k/13),\ 1-\cos^2(2\pi k/13)\}$ are so dense that at $6\,\%$ tolerance almost any parameter between 0 and 1 finds a neighbour. For θ_{23}, Doc. 372 gives the sharper statement $\sin^2\theta_{23} \in \{4/9,\ 5/9\}$ from the A_5 matrix-element spectrum (prediction $|\sin^2\theta_{23} - 1/2| = 1/18$); $\cos^2(10\pi/13) = 0.560$ and $5/9 = 0.556$ are algebraically distinct (degree 6 versus rational) and only numerically close **[B]**. The derivation $\sin^2\theta_{13} = (25\xi)^{2/3}$ above stands; it uses $|A_5|$ and ξ, not the $\cos^2$ reading. For θ_{12} only the order of magnitude remains **[S]**.

Literature connection: T_{13} as a flavour symmetry group

The Frobenius group $T_{13} = \mathbb{Z}_{13} \rtimes \mathbb{Z}_3$ is known in the literature as a candidate for the flavour symmetry of neutrinos [2]. It is isomorphic to the $\mathbb{Z}_{13}$ part of $GF(27)^*$ together with the Frobenius action, i.e. to the Frobenius structure on the orbits of $\mathbb{Z}_{13}$ used in this document. The literature treated T_{13} as an abstract symmetry (without physical derivation); FFGFT provides the algebraic origin: $T_{13} \cong (\mathbb{Z}_{13}$ part of $GF(27)^*) \rtimes \langle\phi\rangle$ with the Frobenius automorphism $\phi : x \mapsto x^3$; the full group is $GF(27)^* \rtimes \langle\phi\rangle \cong \mathbb{Z}_2 \times T_{13}$. ($|T_{13}| = 39$, the full group has order 78; the Frobenius fixes the $\mathbb{Z}_2$ part.)

ae.6 Orbit 3 as the Majorana Sector [B]

Orbit 3 $\{4, 10, 12\}$ is fully self-inverse in $\mathbb{Z}_{13}$:

$$4^{-1} \equiv 10, \quad 10^{-1} \equiv 4, \quad 12^{-1} \equiv 12 \pmod{13}.$$

The self-inverse nature means the associated particles are their own antiparticles – Majorana character. Additionally $4 + 10 + 12 = 26 = |GF(27)^*|$ **[B]**.

Doc. 343 (Theorem D‴) shows the quadratic character is the general criterion: Orbit 1 is also self-inverse (Majorana, ν_1); Orbit 4 is not (Dirac, ν_3). A separate prediction of sterile neutrinos from the Galois algebra does not follow **[B]**.

ae.7 Neutrinoless Double Beta Decay [K]

With measured PMNS mixing angles and Galois masses:

$$m_{ee} = \left|c_{12}^2 c_{13}^2\, m_1 + s_{12}^2 c_{13}^2\, m_2\right| \approx 6.0\,\text{meV}$$

(Majorana phases zero; ν_3 Dirac ⇒ no m_3 contribution, Doc. 343 **[B]**). The KamLAND-Zen bound $m_{ee} < 28$–$122\,$meV (PRL 135 (2025) 262501; NME-dependent) is satisfied with large margin. Upper bound without ν_3: $m_{ee} \le m_{\nu_1} + m_{\nu_2} \approx 14.4\,$meV. Minimum (destructive interference): $m_{ee,\min} \approx 0.13\,$meV. With $s_{12}^2 = 0.307$, $s_{13}^2 = 0.022$, $m_1 = 4.54$, $m_2 = 9.81\,$meV the two terms are 3.08 and $2.95\,$meV; their sum gives 6.0, their difference $0.13\,$meV. A term $s_{13}^2 m_3$ does not occur, since ν_3 is a Dirac neutrino.

ae.8 Conjugation Structure of the Orbits [B]

The four Frobenius orbits in $\mathbb{Z}_{13}$ form two conjugate $3/\bar{3}$ pairs under $x \mapsto -x \pmod{13}$, matching the conjugacy classes $C_{3_1}, C_{\bar{3}_1}, C_{3_2}, C_{\bar{3}_2}$ of T_{13} [2]:

$$-\text{Orbit}_1 = \text{Orbit}_3, \quad -\text{Orbit}_2 = \text{Orbit}_4.$$

Together with the inversion duality ($x \mapsto x^{-1}$, Doc. 339), the orbits carry a complete $\mathbb{Z}_2 \times \mathbb{Z}_3$ symmetry in $\mathbb{Z}_{13}$. Orbit 1 is also self-inverse under inversion ($3^{-1} = 9$, $9^{-1} = 3$, both in Orbit 1).

ae.9 Model Discrepancies for Context [S]

The A_4/TM_2 model [1] that provides the θ_{13} sum rule makes predictions that deviate from both the Galois picture and experiment:

- $\sin^2\theta_{23} = 0.433$–$0.441$ (lower octant) – measured: 0.545–0.560 (upper octant); Galois: 0.560 (upper octant).
- $\delta_{CP} \approx \pm 51^\circ$ – T2K hint: $\approx -90^\circ$; Galois candidate ($k = 10$ from Orbit 3, justification open **[S]**): $\approx -83^\circ$.
- $m_{ee} = 5.1$–5.3 meV (with Majorana phases) – Galois (zero phases): 6.0 meV.

For θ_{23} and δ_{CP}, the FFGFT Galois values agree better with experiment than the A_4 model. The θ_{13} sum rule is adopted; the remaining A_4 predictions are not.

ae.10 Falsifiable Predictions

1. **Neutrino masses (normal ordering):** $m_1 = 4.54$, $m_2 = 9.81$, $m_3 = 49.96$ meV, $\sum = 64.3$ meV. Testable: CMB-S4, Euclid (cosmology at 1 meV level). Independent convergence: Razzaghi et al. [1] extract from A_4 texture and experiment $m_2 = (9.21$–$9.92)$, $m_3 = (49.91$–$51.42)$, $\sum = (63.97$–$64.55)$ meV; our Galois values m_2, m_3 and $\sum$ lie within these ranges.
2. **Mass-difference ratio [K]**:
$$\frac{\Delta m^2_{\text{atm}}}{\Delta m^2_{\text{sol}}} = \frac{3(11^2-1)}{11} = 32.73 \quad \text{(measured: } 33.20\text{)}.$$
Once m_ν is measured to 1 meV precision, the ratio $\Delta m^2/m_\nu^2$ is directly testable.
3. ν_3 **is a Dirac neutrino [B]**: No Majorana phase for ν_3; $m_{ee} \leq m_{\nu_1} + m_{\nu_2} \approx 14.4$ meV. Inverted hierarchy more strongly excluded. Testable: nEXO ($m_{\beta\beta} \approx 5$–20 meV, NME-dependent); KamLAND-Zen-800: < 28–122 meV.
4. **Venting rate and DUNE observable [K]**: The coherent SICC venting amplitude $p_{\text{vent}} = 5/24 \approx 20.8\,\%$ per coherence step is a model-internal prediction. The DUNE observable depends on the beam spectrum: monochromatic 2.5 GeV gives $99.8\,\%$ ν_μ disappearance (first oscillation maximum, $\phi = 1.61$ rad); a Gaussian spectrum 2.5 ± 1.0 GeV clamped to $[0.5, 5]$ GeV gives an average of $76\,\%$ (SICC V3: $76.16\,\%$). For a direct prediction the real LBNF flux table must be used as weight.

ae.11 Open Points

1. Connection of the discrete orbit phases $2\pi k/13$ to the full PMNS U-matrix **[S]**.
2. Algebraic justification for why $k = 10$ from Orbit 3 is distinguished for δ_{CP}; candidate $\arg(e^{2\pi i \cdot 10/13}) = -83.1^\circ$ (T2K hint $\approx -90^\circ$, dev. 6.9°) **[S]**.

ae.12 Note on Authorship

This document was developed on 30 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits in $\mathbb{Z}_{13}$, branching rule $\rho_4 \downarrow_{A_4}$, character theory of A_5 and A_4, numerical verification of the mass differences) were carried out by the assistant and verified by the author. The physical interpretation, the connection to the SICC framework, and all status markers (**[B]**, **[K]**, **[S]**) are the author's responsibility.

ae.13 Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

ae.14 Verification Scripts

`pruef_340_neutrino_galois.py` (13/13 assertions passed);
`pruef_340b_spirale_breitband.py` (6/6 assertions passed: non-closure $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$, spiral, broadband DUNE, K_{frak}).

Declared 30 August 2026, supplemented 31 August 2026 (θ_{13}, T_{13} literature, non-closure, broadband)

[1] N. Razzaghi, S. M. M. Rasouli, P. Parada, P. V. Moniz, *Two-zero textures based on A_4 symmetry and unimodular mixing matrix*, arXiv:2203.05753 [hep-ph] (2022).

[2] C. Hartmann, A. Zee, *Neutrino mixing and the Frobenius group T_{13}*, arXiv:1106.0333 [hep-ph] (2011).

Chapter af

Doc. 341 — GF(27) in GALG and FFGFT: Why G(3) Contains No GF(27) but G(6) Does

Abstract

Doug Matzke reports in his IPI mail of 1 September 2026 that a systematic search in GALG G(3) (6561 attempts) finds no element of order 26. This document explains algebraically why, shows where such elements exist in G(6), and draws the consequences for the FFGFT mass layer.

Three theorems: ***(A)*** *$G(3) \cong M_2(GF(9))^2$: order 26 is algebraically impossible in G(3) — not a search error, but a theorem.* ***(B)*** *$G(6) \cong M_8(GF(9))$: order-26 elements exist; a 5-blade example can be verified directly in GALG.* ***(C)*** *GF(27) is present in G(6) as an order structure, not as a subfield (since $3 \nmid 8$) — exactly the FFGFT statement from Doc. 338.*

Consequence: $1/\alpha = 3700/27$ lives on the G(3)/GF(9) level; the lepton mass layer $(m_\mu/m_e)^2 = 43200$ needs G(6) with order-26 elements. Verification script: `pruef_341_gf27_in_galg.py` (all assertions ***[B]****).*

af.1 Background and Notation

This section provides the necessary background for readers not immediately familiar with all the algebraic structures involved. It can be skipped by specialists.

Status markers

This document uses the standard FFGFT corpus status markers:

- **[K]** — *Confirmed*: algebraically or numerically fully proved; all assertions in the verification script passed.
- **[B]** — *Established*: numerically secured by sampling or explicit construction; analytic proof present or straightforward.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Galois fields GF(p^n)

A *Galois field* GF(p^n) is the unique finite field with p^n elements, where p is prime (the *characteristic*) and $n \geq 1$. Its multiplicative group GF(p^n)$^* =$ GF(p^n) $\setminus \{0\}$ is cyclic of order $p^n - 1$.

- GF(3): the integers $\{0, 1, 2\}$ modulo 3. Multiplicative group order 2.
- GF(9) = GF(3^2): elements $a + bi$ with $a, b \in \{0, 1, 2\}$ and $i^2 = -1 \equiv 2$. Multiplicative group order 8. This is Doug's symmetric Z3C.
- GF(27) = GF(3^3): 27 elements, multiplicative group order $26 = 2 \cdot 13$.
- GF(81) = GF(3^4): 81 elements, multiplicative group order $80 = 2^4 \cdot 5$.

The *order* of an element x in a group is the smallest positive k with $x^k = 1$. If the group has order m, then every element order divides m (Lagrange's theorem). This is the key fact used throughout.

Clifford algebra Cl(n) and GALG G(n)

The Clifford algebra Cl(n) is generated by n basis vectors $e_1, \ldots, e_n$ with the rule $e_i^2 = +1$ (hyperbits in Matzke's language) and $e_i e_j = -e_j e_i$ for $i \neq j$. A *blade* of grade r is a product $e_{i_1} e_{i_2} \cdots e_{i_r}$ with $i_1 < i_2 < \cdots < i_r$; together with the scalar 1 (grade 0) they form a basis of 2^n elements. *GALG G(n)* is Cl(n) over the coefficient field GF(9) = Z3C (i.e. each blade carries a Z3C = GF(9) coefficient). As a matrix algebra: Cl(3) $\cong M_2$(GF(9))^2 and Cl(6) $\cong M_8$(GF(9)), where $M_k(F)$ denotes $k \times k$ matrices over the field F.

Why GF(27) and order 26?

The lepton mass ratio $(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$ involves $27 = |\text{GF}(27)|$ (Doc. 338, pruef_330/331). The multiplicative group GF(27)* has order $26 = 2 \cdot 13$. An element of order 26 in any algebra generates a copy of GF(27)*. Doug's search for order-26 elements in G(3) is therefore a direct test of whether GF(27) structure is present in G(3).

Companion matrix

Given a monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, its *companion matrix* is the $n \times n$ matrix with 1's on the subdiagonal and the coefficients $-a_0, \ldots, -a_{n-1}$ in the last column. Its characteristic polynomial is $f(x)$, so its eigenvalues are the roots of f and its order in GL$_n$ equals the multiplicative order of the roots.

af.2 Occasion and Starting Point

Doug Matzke (IPI, 1 Sept. 2026) reports finding in GALG G(3): GF(9) in the generator $(a + ja)^8 = 1$; GF(81) in G(3) with $(1 + a + b + c + (a \wedge b))^{80} = 1$; and element orders $\{2, 3, 4, 8, 16, 32, 40, 80\}$ — of these, $2, 4, 8, 16, 40, 80$ divide $80 = |\text{GF}(81)^*|$, consistent with the

$\mathrm{Cl}(3) \cong M_2(\mathrm{GF}(9))^2$ structure where eigenvalues live in $\mathrm{GF}(81)$; 3 is unipotent (char. 3); 32 is incompatible with $M_2(\mathrm{GF}(9))^2$ (presumably the number $\varphi(80) = 32$ of generators).

An explicit search for elements of order 26 (which would signal $\mathrm{GF}(27)^*$ structure):

```
gasolve([a,b,c], is_root26): Attempted 6561 with 0 found.
```

$6561 = 3^8$ is the number of elements in a 4-trit search space — exhaustive at that depth.

This is not a search error or a limitation of the tool. It is an algebraic theorem: order 26 requires the prime factor 13, and 13 does not appear in $80 = |\mathrm{GF}(81)^*|$. The following sections prove this precisely and show where $\mathrm{GF}(27)$ structure does appear — in $\mathrm{G}(6)$, not $\mathrm{G}(3)$.

af.3 Theorem A: G(3) Contains No GF(27)

Structure of G(3)

GALG $\mathrm{G}(n) = \mathrm{Cl}(n)$ over the symmetric Z3C $= \mathrm{GF}(9) = \mathrm{GF}(3)[i]$, where $i^2 = -1$, characteristic 3. For odd $n = 3$:

$$G(3) = \mathrm{Cl}(3) \cong M_2(\mathrm{GF}(9)) \oplus M_2(\mathrm{GF}(9)) \tag{af.1}$$

Every irreducible representation is 2-dimensional over $\mathrm{GF}(9)$.

Eigenvalue Structure

The eigenvalues of a 2×2 matrix over $\mathrm{GF}(9)$ lie in the algebraic closure — in the smallest extension field $\mathrm{GF}(9^2) = \mathrm{GF}(81)$.

$$|\mathrm{GF}(81)^*| = 80 = 2^4 \cdot 5 \tag{af.2}$$

The number 13 does not divide 80. Therefore:

- No element of $M_2(\mathrm{GF}(9))$ has an order divisible by 13.
- Order 26 requires $13 \mid \mathrm{ord}(X)$ — impossible in $\mathrm{G}(3)$.
- Doug's search space $\{a, b, c\}$ spans $\mathrm{G}(3)$ — the result 0/6561 is algebraically forced **[B]**.

Consistency with Doug's Findings

Table af.1: Orders in GALG G(3) and their Galois origin

Order	Origin	Divides
2	$\lvert\mathrm{GF}(3)^*\rvert$	80
4	$\varphi(5) = 4$	80
5	odd prime factor $\lvert\mathrm{GF}(81)^*\rvert$	80
8	$\lvert\mathrm{GF}(9)^*\rvert$	80
16, 20, 40, 80	subgroups of $\mathrm{GF}(81)^*$	80
3	unipotent (char. 3)	—
26	$\lvert\mathrm{GF}(27)^*\rvert$, needs factor 13	**not** 80

A numerical sample of 3000 random Cl(3) elements confirms: no order is divisible by 13 **[B]**.

af.4 Theorem B: G(6) Contains Order-26 Elements

Structure of G(6)

$$G(6) = \mathrm{Cl}(6) \cong M_8(\mathrm{GF}(9)) \tag{af.3}$$

Irreducible representations are 8-dimensional over GF(9).

Construction via Irreducible Polynomial

The polynomial $f(x) = x^3 + 2x + 1$ over GF(3) (equivalent to $x^3 - x + 1$) has no root in GF(3) and is therefore irreducible. Since $\gcd(3, 2) = 1$, f remains irreducible over GF(9); its roots lie in $\mathrm{GF}(27) \setminus \mathrm{GF}(3)$. The companion matrix

$$C_f = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathrm{GF}(3)) \tag{af.4}$$

has order 26 in $\mathrm{GL}_3(\mathrm{GF}(3))$ (f is primitive) **[B]**.

Embedding in G(6) as a GALG Element

The 8×8 element $X = C_f \oplus C_f \oplus (-I_2)$ satisfies:

- $X^{26} = 1$, $X^{13} \neq 1$, $X^2 \neq 1$ — order exactly 26 **[B]**
- Blade decomposition over 38 Z3C blades from $\{e_1, \ldots, e_6\}$

Minimal Example: 5 Blades, Directly Verifiable in GALG

A systematic search shows: order-26 elements need at least 5 blades; with 4 blades there are none. The following 5-blade element can be verified directly in Doug's GALG tool:

$$X = -(1+i)(e_1e_2) + (1+i)(e_1e_2e_3e_6) + (1-i)(e_1e_2e_5e_6) + (1+i)(e_1e_4e_5e_6) - (e_1e_2e_3e_4e_6) \quad \text{(af.5)}$$

GALG syntax:

```
X = -(1+1j)*(e1^e2) + (1+1j)*(e1^e2^e3^e6)
+ (1-1j)*(e1^e2^e5^e6) + (1+1j)*(e1^e4^e5^e6)
- (e1^e2^e3^e4^e6)
X**26 == 1 # True
X**13 == 1 # False
```

[B] (pruef_341, all assertions passed)

Frequency in G(6)

Table af.2: Fraction of elements with $13 \mid \mathrm{ord}(X)$ in G(6) by blade support

Support (number of blades)	Hits / Invertible	Fraction
≤ 4	0 / ∼450	0 %
5	rare	< 1 %
6	85 / 478	17.8 %
10	190 / 511	37.2 %

Order-26 elements are common in G(6) — Doug's search in G(3) (span$\{a, b, c\}$) could not find them for two reasons: algebraically (Theorem A) and because G$(3) \subsetneq G(6)$.

af.5 Theorem C: GF(27) as Order Structure, not Subfield

Subfield Criterion

A unital embedding GF$(27) \hookrightarrow M_n(\mathrm{GF}(9))$ makes GF$(9)^n$ a GF$(27)\otimes$GF$(9) =$ GF(729)-vector space. Since $[\mathrm{GF}(729) : \mathrm{GF}(9)] = 3$, we need $3 \mid n$.

Table af.3: GF(27) as subfield in M_n(GF(9))

n	$3 \mid n$	Subfield possible?
2	no	no (G(3))
8	no	no (G(6))
3	yes	yes
6	yes	yes
9	yes	yes

For G(6) with $n = 8$: no GF(27) subfield. But the minimal polynomial of X factors:

$$\chi_X(x) = (x^3 + 2x + 1)^2 \cdot (x + 1)^2 \quad \in \mathrm{GF}(3)[x] \tag{af.6}$$

Two cubic factors — eigenvalues in GF(27) — and one linear factor from GF(3). GF(27) appears as splitting field, not as subalgebra.

Connection to Docs. 330/338

This is exactly the FFGFT statement from Docs. 330 and 338:

$$\mathrm{GF}(9) \not\subset \mathrm{GF}(27) \quad (\text{since } \gcd(2,3) = 1) \tag{af.7}$$

Both fields sit in parallel in $\mathrm{GF}(3^6) = \mathrm{GF}(729)$, not in a tower. The mass-ratio identity $(m_\mu/m_e)^2 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$ uses both fields *in parallel* — consistent with Theorem C.

af.6 Vacuum Structure and Z3 Separation (pruef_341_vakuum_witt_z3)

Doug's IPI mail of 1 September 2026 (22:05) not only verifies the order-26 element but discloses the full structure of six vacuum states $\mathrm{Vss}[0 \ldots 5]$ in G(6) and their grade-2 (bivector) content. The following results are verified in `pruef_341_vakuum_witt_z3.py`.

Witt Pairs and the Number Operator N

The three grade-2 bivectors

$$B_1 = -i(e_1 e_3), \quad B_2 = -i(e_2 e_6), \quad B_3 = -i(e_4 e_5) \tag{af.8}$$

satisfy $B_k^2 = +1$, commute pairwise, and commute with jE_7. They form the three *Witt pairs* $\{13\}, \{26\}, \{45\}$ — the decomposition of Cl(6) into three complex planes (Fock space of 3 modes). The number operator $N = B_1 + B_2 + B_3$ is simultaneously the grade-2 part of $\mathrm{Vss}[0]$ and its constructor **[B]**:

$$(1 + N)(1 \pm jE_7) = \mathrm{Vss}[0] \quad (\text{sign depends on basis orientation of } E_7) \tag{af.9}$$

Triality Built into Z3C

Over $\mathbb{C}$, the Fock number operator $N = B_1 + B_2 + B_3$ would have eigenvalues $\{-3, -1, +1, +3\}$. Over GF(9) in characteristic 3: $-3 \equiv 0 \equiv +3$ **[B]**.

$$N \bmod 3 \ \in \ \{0,\ +1,\ -1\} \quad \text{with multiplicities} \quad \{2,\ 3,\ 3\} \tag{af.10}$$

The vacuum ($n = 0$) and the fully occupied state ($n = 3$) *coincide* in characteristic 3. The $\mathbb{Z}_3$ cycle is closed without any extra input. This is the additive $\mathbb{Z}_3$ charge of FFGFT, the triality (Doc. 336, $Q_{\mathbb{Z}_3} = N \bmod 3$). The $+1$-eigenspace is 3-dimensional

— Doug's fermion-generation candidate. The triality is not the sector index of the cyclic permutation: under σ (next section) $N = B_1 + B_2 + B_3$ remains unchanged, as does the winding sum under τ in Doc. 321, and all three eigenphases of σ occur in the eigenspaces $N = \pm 1$. The fixed-point/orbit separation by σ is the other $\mathbb{Z}_3$ (verification script `pruef_341_zwei_z3.py`).

Z3 Separation of Even and Odd Vacuums

The cyclic Witt-pair rotation

$$\sigma : \quad e_1 \to e_2 \to e_4 \to e_1, \quad e_3 \to e_6 \to e_5 \to e_3 \tag{af.11}$$

cycles $B_1 \to B_2 \to B_3 \to B_1$ while leaving E_7 invariant (even permutation). Applied to Doug's six vacuums **[B]**:

$$\begin{aligned} &\sigma(\mathrm{Vss}[0]) = \mathrm{Vss}[0], \quad \sigma(\mathrm{Vss}[2]) = \mathrm{Vss}[2], \quad \sigma(\mathrm{Vss}[4]) = \mathrm{Vss}[4] \\ &\sigma(\mathrm{Vss}[1]) = \mathrm{Vss}[3], \quad \sigma(\mathrm{Vss}[3]) = \mathrm{Vss}[5], \quad \sigma(\mathrm{Vss}[5]) = \mathrm{Vss}[1] \end{aligned} \tag{af.12}$$

Even vacuums $[0, 2, 4]$: individual $\mathbb{Z}_3$-fixed points.
Odd vacuums $[1, 3, 5]$: a single $\mathbb{Z}_3$-orbit of three.

This algebraic separation is the *same* as the Frobenius fixed-point/orbit split from Doc. 339: fixed points = massive sector (fermion generations), orbit = other sector (Doug's dark energy/mass candidate). Doug finds this split via grade parity; FFGFT finds it via Frobenius eigenstructure. **They agree [B].**

Bilaterals as Sector-Changers (Gluon Role)

The Witt raisers $ap_k = (e_a + i\,e_b)/2$ for each pair satisfy **[B]**:

$$[N,\ ap_i \cdot ap_j] = +1 \cdot (ap_i \cdot ap_j) \tag{af.13}$$

The bilaterals $ap_i \cdot ap_j$ shift the N-sector by ± 1 mod 3, i.e. they are triality changers. This is the gluon role in Doc. 339 (three-fold orbit transmutation between sectors). Doug's question "how do gluons transmute quarks using bilaterals?" is answered: they shift the triality N mod 3 by ± 1.

Projector Properties

$$\mathrm{Vss}[0]^2 = -\mathrm{Vss}[0], \qquad N \cdot \mathrm{Vss}[0] = 0 \tag{af.14}$$

$-\mathrm{Vss}[0] = 2 \cdot \mathrm{Vss}[0]$ is idempotent over GF(9); $\mathrm{Vss}[0]$ projects onto the $N = 0$ sector (vacuum + fully occupied state).

af.7 Higher Roots of Unity: root7, p27 Projectors, Coefficient Purity

Doug's follow-up questions (IPI, 4–7 Sept. 2026) concern 7th roots of unity, projectors with $X^{27} = X$, and a structural objection to the order-26 element. Verified in `pruef_341_primordnung.py`, `pruef_341_root7_gf9.py`, `pruef_341_r26_rein_p27.py`.

Entry Order of Primes

The smallest k with $p \mid 3^k - 1$ (i.e. $k = \mathrm{ord}_p(3)$) gives the first field GF(3^k) in which p-th roots of unity exist **[K]**:

p	k	$\lvert\mathrm{GF}(3^k)^*\rvert$
13	3	$26 = 2 \cdot 13$
5	4	$80 = 2^4 \cdot 5$
11	5	$242 = 2 \cdot 11^2$
7	6	$728 = 2^3 \cdot 7 \cdot 13$

Within G(3) (eigenvalues in GF(81)) root5 is indeed the first odd-prime root; globally 13 enters before 5. In no GF$(3^k)^*$ is there an element of order 3, since $3^k - 1 \equiv 2 \pmod 3$ for all k: tripotents $t^3 = t$ with $t \neq 0, \pm 1$ are a matrix phenomenon, never a field phenomenon **[K]**.

root7 Exists Only in GF(9)-Arithmetic

$7 \nmid \lvert\mathrm{GF}(9)^*\rvert = 8$, and $7 \mid 3^k - 1$ first at $k = 6$. A genuine complex 7th root of unity has coefficients in $\mathbb{Q}(e^{2\pi i/7})$ (degree 6 over $\mathbb{Q}$); $\cos(2\pi/7)$ has minimal polynomial $8x^3 + 4x^2 - 4x - 1$ and lies neither in $\mathbb{Q}(i)$ nor in GF(9). Constructions of the form $\cos\theta + B\sin\theta$ (Euler form over $\mathbb{R}$) are therefore not GALG elements **[K]**.

What does exist is an order-7 element in G(6) over GF(9): $\Phi_7(x) = x^6 + \cdots + 1$ is irreducible over GF(3); its companion matrix C has order 7 in $\mathrm{GL}_6(\mathrm{GF}(3))$, and $X_7 = C \oplus I_2 \in M_8(\mathrm{GF}(9))$ has order 7 with 47 blades over all grades, all coefficients in $\{\pm 1, \pm i\}$ **[K]**. A compact alternative: take any Y with $7 \mid \mathrm{ord}(Y)$ and set $X = Y^{\mathrm{ord}(Y)/7}$. Example **[B]**:

```
Y = (1j)*(e1^e2^e3) - (e1^e3^e6) + (e2^e3^e4) - (e2^e4^e6)
- (1j)*(e4^e5^e6) - (e1^e2^e4^e5)
ord(Y) = 182 = 2*7*13;  X = Y**26;  X**7 == 1, X != 1
```

X has 10 blades, all coefficients pure (± 1 or $\pm i$).

p27 Projectors as a Field-Tower Hierarchy

Since $x^{27} - x = \prod_{a \in \mathrm{GF}(27)} (x - a)$, the condition $X^{27} = X$ holds exactly when X is semisimple with spectrum in GF(27). This generalises tripotents ($X^3 = X$, spectrum in GF(3)) along the field tower **[K]**:

	$X^3=X$	$X^9=X$	$X^{27}=X$	$X^{81}=X$	$X^{729}=X$
N (tripotent)	✓	✓	✓	✓	✓
r_{26} (order 26)	–	–	✓	–	✓
X_7 (order 7)	–	–	–	–	✓

Doug's "power-of-27 projector" is thus precisely: spectrum in GF(27). X_7 needs GF(729) because $7 \nmid 26$.

Coefficient Purity and the Order-26 Element

Doug objects that r_{26} contains $-(1+j)(e_1e_2)$, mixing e_1e_2 (square -1) with je_1e_2 (square $+1$), which he reads as two signature choices. Algebraically, $1+j$ is simply the generator of GF$(9)^*$ (order 8, norm -1) — Doug's own $(a+ja)^8 = 1$ — so $-(1+j)B$ is scalar times blade. The objection is a *physical* purity requirement: each blade coefficient in $\{\pm 1\}$ or $\{\pm j\}$, never $a+bj$ with $a,b \neq 0$. That requirement is satisfiable **[B]**: about 2.5 % of random 5-blade elements with pure coefficients have order exactly 26, e.g.

```
X = (1j)*e3 + (1j)*(e2^e5) + (1j)*(e3^e6) + (e4^e6) + (e3^e5^e6)
X**26 == 1,  X**13 != 1
```

X_7 above is already pure (no mixed coefficient).

Unipotent Orders and the Jordan Decomposition

Roots of unity of order 3 cannot be semisimple: $3 \nmid 3^k - 1$ for every k. They arise instead from nilpotents. Doug's construction $N = (a+b+c)(abc) = e_1e_2 - e_1e_3 + e_2e_3$ has $N^2 = 0$, hence in characteristic 3

$$(1+N)^3 = 1, \qquad (-1+N)^3 = -1, \qquad (-1+N)^6 = 1 \tag{af.15}$$

(all binomial coefficients $\binom{3}{1}, \binom{3}{2}$ vanish mod 3) **[K]**. These are *unipotent* elements (eigenvalue 1 with a Jordan block). N commutes with every element built from disjoint generators (e_4, e_5, e_6) but not with e_1 and not even with e_1e_2 **[K]**— this is why a product with an order-91 element built from other generators needs disjoint index sets.

The general statement is the Jordan decomposition: every $X \in \mathrm{GL}_8(\mathrm{GF}(9))$ is $X = SU$ with S semisimple (spectrum in some GF(3^k), order dividing $3^k - 1$), U unipotent (order 3^j), $[S,U] = 0$, and

$$\mathrm{ord}(X) = \mathrm{ord}(S) \cdot \mathrm{ord}(U). \tag{af.16}$$

In G(6) this is realised by a 6-dimensional block carrying S (order $\mid 728$) and a 2-dimensional Jordan block J_2 carrying U (order 3). With C the companion matrix of the primitive polynomial $x^6 + x^5 + 2$ over GF(3) (order 728) **[K]**:

S	$\mathrm{ord}(S \oplus J_2)$	blades
C^8 (order 91)	$273 = 3 \cdot 91$	35
C^4 (order 182)	$546 = 3 \cdot 182$	38
C (order 728)	$2184 = 3 \cdot 728$	35

So r_{273} and r_{546} exist inside G(6); nine generators are not required. The same works starting from Doug's own $r_{182} = Y$: on its 2-dimensional complement Y has the double eigenvalue $+1$, so its centraliser (dimension 10 over GF(9)) contains $M_2(\mathrm{GF}(9))$ and hence a nilpotent M with $[M, Y] = 0$, $M^2 = 0$ (40 pure blades). With $U = 1 + M$, $U^3 = 1$: $r_{273} = UY^2$ and $r_{546} = UY$, both of order as stated, in G(6) **[K]**. Conversely r_{729} does *not* exist in G(6): $729 = 3^6$ is a pure 3-power, so the element would have to be unipotent, and $(1+N)^{3^j} = 1 + N^{3^j}$ with $N^8 = 0$ in M_8 forces $(1+N)^9 = 1$. The maximal unipotent order in $M_n(\mathrm{GF}(9))$ is the smallest power of 3 that is $\geq n$: 9 for M_8 (G(6)), 27 for M_{16}, 81 for M_{32}, 729 only from $M_{256} = \mathrm{Cl}(16)$ **[K]**. The number 729 is therefore unrelated to GF(729), whose multiplicative group has order 728: the 728-structure is semisimple, the 3^j-structure unipotent, and the Jordan decomposition keeps the two apart.

Finally, Doug's counts — 32 of the 80 elements of $\langle r_{80} \rangle$ and 72 of the 182 elements of $\langle r_{182} \rangle$ are again generators — are Euler's totient: $\varphi(80) = 32$, $\varphi(182) = 72$; X^m generates $\langle X \rangle$ iff $\gcd(m, n) = 1$ **[K]**. Verified in `pruef_341_unipotent_jordan.py`.

A Convention Note: the Sign of E_7

GALG defines $E_7 = -(e_1 \wedge \cdots \wedge e_6)$. An element transmitted with a term "$(+i)\,E_7$" meaning $+i\,e_1 \cdots e_6$ is read in GALG as $-i\,e_1 \cdots e_6$. For X_7 this sign flip yields a non-invertible element with $X^{27} = X$ and no $X^n = 1$ — exactly the behaviour Doug observed **[K]**. With Doug's E_7 the last term of X_7 must be entered as `(-1j)*E7`.

af.8 Consequences for FFGFT and GALG

Table af.4: GF(27) structure and vacuum separation in GALG and FFGFT

Level	GALG	FFGFT	Status
α	G(3)/GF(9)	$1/\alpha = 3700/27$, cancels	**[B]**
Lepton masses	G(6), ord. 26 needed	$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$	**[K]**
GF(27) subfield	absent in G(3), G(6)	parallel in GF(729)	**[B]**
Vacuum split	Vss$[0,2,4]$ vs. Vss$[1,3,5]$	massive vs. other sector	**[B]**
$\mathbb{Z}_3$ structure	fixed pts vs. orbit	Frobenius split (Doc. 339)	**[B]**
N eigenvalues	$\{0,+1,-1\}$, mult. $\{2,3,3\}$	triality $\in \{0,+1,-1\}$	**[B]**
Bilaterals	$[N, ap_i a p_j] = +1$	triality changer, gluon role	**[B]**

The fine-structure constant $1/\alpha = 3700/27$ lives on the GF(9)/G(3) level: $|\mathrm{GF}(27)| = 27$ cancels in the derivation. The lepton mass layer and the vacuum structure both need G(6). Doug's two-family split of vacuums and FFGFT's Frobenius separation find the same $\mathbb{Z}_3$ orbit structure from different starting points **[B]**.

af.9 Theorem D: Vacuum Structure and $\mathbb{Z}_3$ Orbit Decomposition

The six vacuum states Vss$[0,\ldots,5]$ split under the $\mathbb{Z}_3$ rotation σ (Doc. 339) into two classes **[B]**:

- **Even vacua** Vss$[0,2,4]$: σ-fixed points (massive sector); $N \bmod 3 = 0$ (multiplicity 2).
- **Odd vacua** Vss$[1,3,5]$: $\mathbb{Z}_3$ three-cycle (other sector); $N \bmod 3 \in \{+1,-1\}$ (multiplicity 3 each).

The bilateral products $ap_i \cdot ap_j$ shift the N-sector by $\pm 1 \bmod 3$ — they are triality changers and play the gluon role of Doc. 339 **[B]**. Doc. 344 (Theorem F) shows: this decomposition agrees exactly with the tripotent property of the even vacua and the GALG grade construction — two independent derivations of the same structure.

Appendix: For the Reader Without an Algebra Background

This appendix explains the key ideas in plain language, assuming no prior knowledge.

What is a finite field?

A field is a set of numbers in which you can add, subtract, multiply and divide (except by zero). Familiar examples are the real numbers $\mathbb{R}$ and the rational numbers $\mathbb{Q}$. A *finite* field has only finitely many elements. You compute "modulo" a prime p: discard all multiples of p. Example GF(3): the numbers $\{0,1,2\}$ with $1+2=0$, $2+2=1$, etc.

What does GF(p^n) and GF$(p^n)^*$ mean?

GF(p^n) is the unique finite field with exactly p^n elements. The *star* GF$(p^n)^*$ denotes all elements *except zero* — analogous to $\mathbb{Z}^* = \{\ldots,-1,+1,\ldots\}$ vs. $\mathbb{Z}$:

$$|\mathrm{GF}(27)| = 27 \qquad |\mathrm{GF}(27)^*| = 26 = 27 - 1$$

Group order and Lagrange's theorem

GF$(p^n)^*$ is always a cyclic group of order $p^n - 1$. The *order* of an element x is the smallest k with $x^k = 1$. By Lagrange's theorem, every element order *divides* $p^n - 1$. Think of a clock hand that returns every 4 hours: it cannot fit on a 6-hour clock because 4 does not divide 6.

Concretely:

- GF$(81)^*$: order $80 = 2^4 \cdot 5$. Prime factors: 2 and 5 only. Order $26 = 2 \cdot$ **13** is impossible — 13 does not divide 80.
- GF$(27)^*$: order $26 = 2 \cdot 13$. Elements of order 13 and 26 exist here.

Why does GF(27) appear in the lepton masses?

$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 25 \cdot 27$. The 27 is |GF(27)| (total field size including zero); the 8 is |GF(9)*| (group order of GF(9)). These arise because the FFGFT winding numbers on the T^4 torus are algebraically linked to these fields.

Why does Doug find no GF(27) in G(3)?

G(3) uses 2×2 matrices over GF(9); eigenvalues live in GF(81). GF(81)* has order 80, and $13 \nmid 80$, so order 26 is impossible. In G(6) with 8×8 matrices, order 26 is achievable. A 5-blade example is given in §af.4 and directly verifiable in GALG.

Verification Script

`pruef_341_gf27_in_galg.py` — Theorems A, B, C verified numerically (Cl(6) over GF(9) as 8×8 matrices; exact order determination; blade decomposition with Z3C coefficients; samples of 3000 and 600 elements). All assertions passed **[B]**.

`pruef_341_primordnung.py` — entry order of primes $\mathrm{ord}_p(3)$; $3^k - 1 \equiv 2 \pmod 3$ (no field tripotents); trine-product counterexample in $M_2(\mathrm{GF}(3))$.

`pruef_341_root7_gf9.py` — non-existence of root7 with Z3C coefficients; order-7 element $X_7 = C_{\Phi_7} \oplus I_2$ over GF(9); contrast with $\cos\theta + B\sin\theta$.

`pruef_341_r26_rein_p27.py` — $1+j$ as GF(9)* generator; pure-coefficient root26; $X^q = X \Leftrightarrow$ spectrum in GF(q); E_7 sign discrepancy; compact root7 Y^{26}.

`pruef_341_unipotent_jordan.py` — unipotent trine/anti-trine $N^2 = 0$; commutation of N; Jordan decomposition $r_{273}, r_{546}, r_{2184}$ in G(6); impossibility of r_{729} below M_{256}; Euler-φ counts.

`pruef_341_vakuum_witt_z3.py` — Vacuum structure verified: constructor identity, Witt pairs, N eigenvalues, $\mathbb{Z}_3$ separation of Vss$[0 \ldots 5]$, bilaterals as sector-changers, projector properties. All assertions passed **[B]**.

`pruef_341_zwei_z3.py` — triality N mod 3 and eigenphase of the cyclic permutation σ as two different $\mathbb{Z}_3$; N invariant under σ; all three σ eigenphases in the eigenspaces $N = \pm 1$.

Note on Computational Tools

This document is the work of Johann Pascher and Doug Matzke. Johann Pascher brings years of theoretical development in FFGFT (documented in over 350 corpus documents), poses the research questions, reads and corrects all results, and draws the physical conclusions. Doug Matzke brings the GALG framework, its implementation, and the real-time verification of every algebraic claim in his own tool. The correspondence between both authors drives the direction of inquiry.

Claude (Anthropic) was used as a tool for several tasks that are impractical to do by hand:

- **Computation:** Writing and executing the Python verification scripts (`pruef_341_*`); performing randomised blade searches (thousands of candidates per run); computing order determinations that require iterated 8×8 matrix multiplication mod 3 over exponent spaces up to $\sim 10^{30}$; computing blade decompositions.
- **Writing and typesetting:** Drafting the LaTeX source of this document from the authors' results, notes, and corrections; maintaining the corpus file structure, changelogs, and register entries; preparing correspondence drafts that Johann Pascher reviews and approves.
- **Iteration:** Incorporating corrections, checking consistency across the corpus, and updating cross-references — tasks that scale with the size of a 350-document corpus.

The tool executes; the authors decide what to compute and write, judge every result, correct errors, and determine what the results mean physically.

A central methodological concern is the prevention of fabricated content. The workflow enforces this at two levels: (1) every algebraic claim is backed by a Python verification script with explicit assertions — if an assertion fails, the claim is not used; (2) targeted instructions direct the tool to read the existing corpus before writing, never to introduce facts not present in the sources, and to mark all results explicitly as confirmed (**[K]**), established (**[B]**), or speculative (**[S]**). Invented content is the primary risk when using language models for scientific work; the scripts and instructions are designed to make it detectable and avoidable.

[1] D. Matzke, *IPI mail: GF(27) in GALG*, 1 September 2026 (personal correspondence).

[2] J. Pascher, *Doc. 330: GF(9)-FFGFT bridge*, FFGFT corpus.

[3] J. Pascher, *Doc. 338: Mass calculations and Galois comparison*, FFGFT corpus, 28 August 2026.

[4] J. Pascher, *pruef_341_gf27_in_galg.py*, FFGFT corpus, 1 September 2026.

[5] D. Matzke, *G(6,$\mathbb{Z}_3\mathbb{C}$) and GALG framework*, IPI Letters 2022ff.

Chapter ag

Doc. 342 — Factorisation Classes in $G(6)$ and the Harmonics of the Primes

Abstract

Doc. 341 shows that $GF(27)$ enters $G(6)$ as an order structure through the characteristic polynomial $\chi_X = (x^3+2x+1)^2(x+1)^2$. This document asks which further factorisations χ admits in $G(6) \cong M_8(GF(9))$, and shows that the same factorisation rule organises the harmonic patterns of the primes (Doc. 057, Doc. 159).

*Four results: **(A)** Over $GF(3)$ there are eight irreducible cubic polynomials, four of them primitive (order 26) and four of order 13; they generate five factorisation classes of χ in $G(6)$ **[B]**. **(B)** A prime p enters $|GF(3^k)^*|$ exactly at $k = \mathrm{ord}_p(3)$; the harmonic primes enter in the order $13\,(k{=}3)$, $5\,(k{=}4)$, $11\,(k{=}5)$, $7\,(k{=}6)$; from $k = 7$ on, no new prime ≤ 13 enters (later ones: $17, 19, 23, 29, 31, 37, \ldots$) **[B]**. **(C)** Element orders in $G(6)$ are partitions of 8 with an lcm rule — the algebraic form of a chord of overtones **[B]**. **(D)** From $k \geq 4$ on, the Galois grid is finer than the fractal-recursive correction ($1.05\,\%$, Doc. 338); Galois products then become statistically indistinguishable from higher torus modes. This is not a contradiction but the algebraic form of the mode convergence of Doc. 159 **[B]**.*

*Negative result: the tau sector is not a Galois product at $0.05\,\%$ **[S]**. Check scripts: `pruef_342_faktorisierung_primzahlen.py`, `pruef_342b_hoehere_harmonien.py`, `pruef_342c_dichte_hoehere_harmonien.py`.*

ag.1 Starting point: algebraic equals geometric

The following status markers are used throughout:

[K] *Confirmed* — algebraically or numerically fully proved; all check-script assertions passed; derived directly from ξ or a corpus-established derivation.
[B] *Established* — numerically secured by sampling or explicit construction; analytical proof present or trivial.
[S] *Speculative* — indication or conjecture, not yet fully proved.

Docs. 338 and 341 showed that mass ratios from torus winding numbers (r_i, p_i) and from Galois group orders yield the same number: $(m_\mu/m_e)^2 = 43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$. The question whether the geometric formulation (vectors on T^4) is "more precise" than the algebraic one is ill-posed: winding numbers on S^1 and elements of $\mathrm{GF}(3^k)$ are both integer, exact objects. The bridge is the Frobenius automorphism $g \mapsto g^3$ on $\mathrm{GF}(3^k)$, which represents the $\mathbb{Z}_3$ rotation of the orbifold $T^4/\mathbb{Z}_3$ (Doc. 314, Doc. 336).

The algebraic route has one methodological advantage: polynomial factorisation can be enumerated systematically, geometric search cannot. This document exploits that.

ag.2 Theorem A: Factorisation classes in G(6)

Irreducible cubic polynomials over GF(3)

A monic cubic over $\mathrm{GF}(3)$ is irreducible iff it has no root in $\{0, 1, 2\}$. There are eight such polynomials; their companion matrices have order 26 or 13 in $\mathrm{GL}_3(\mathrm{GF}(3))$ **[B]**:

Table ag.1: Irreducible cubics over $\mathrm{GF}(3)$ and their order

Label	Polynomial	Order
f_1	$x^3 + 2x + 1$	26
f_2	$x^3 + 2x^2 + 1$	26
f_3	$x^3 + 2x^2 + x + 1$	26
f_4	$x^3 + x^2 + 2x + 1$	26
g_1	$x^3 + 2x + 2$	13
g_2	$x^3 + x^2 + 2$	13
g_3	$x^3 + x^2 + x + 2$	13
g_4	$x^3 + 2x^2 + 2x + 2$	13

The number four is forced: $\mathrm{GF}(27)^*$ has $\varphi(26) = 12$ primitive elements in Frobenius orbits of length 3, hence $12/3 = 4$ minimal polynomials. Doc. 341 fixes only order 26 for the leptons, not a class; the assignment of charged leptons to the Dirac layer $\{f_3, f_4\}$ and neutrinos to the Majorana layer $\{f_1, f_2\}$ follows in Doc. 343, Theorem D‴ **[S]**.

The five classes

In $\mathrm{G}(6) \cong M_8(\mathrm{GF}(9))$, χ splits over $\mathrm{GF}(3)$ into factors whose degrees sum to 8. With cubic and linear factors this gives five classes **[B]**:

Table ag.2: χ factorisation classes with $\mathrm{GF}(27)$ content in $\mathrm{G}(6)$

Type	$\chi(x)$	Count	Order
Doc.-341 type	$f_i^{\,2}\,(x{+}1)^2$	4	26
Mixed primitive	$f_i\,f_j\,(x{+}1)^2$, $i \neq j$	6	26
Order-13 type	$g_i^{\,2}\,(x{+}1)^2$	4	26 (sign partners of the f_i, Doc. 343)
Primitive×13	$f_i\,g_j\,(x{+}1)^2$	16	26
Thin	$f_i\,(x{+}1)^5$	4	26

The name of the order-13 type refers to g_i, whose roots have order 13; the factor $(x+1)^2$ contributes the eigenvalue -1 (order 2), so in the semisimple case the order is $\mathrm{lcm}(13, 2) = 26$, with a Jordan block 78. Order 13 would arise only with $(x-1)^2 = (x+2)^2$.

The four primitive polynomials are the four Frobenius conjugacy classes $f_1 = \{g, g^3, g^9\}$, $f_2 = \{g^{17}, g^{23}, g^{25}\}$, $f_3 = \{g^5, g^{15}, g^{19}\}$, $f_4 = \{g^7, g^{11}, g^{21}\}$ of a generator g of $\mathrm{GF}(27)^*$ (root of f_1). Doc. 341 fixes only order 26 for the leptons, not a class; the assignment of charged leptons to the Dirac layer $\{f_3, f_4\}$ and neutrinos to the Majorana layer $\{f_1, f_2\}$ follows in Doc. 343, Theorem D‴ **[S]**.

ag.3 Theorem B: Entry rule of the harmonic primes

Doc. 057 describes primes as elementary relational steps: 2 octave, 3 fifth, 5 major third, 7 natural seventh, 11 eleventh, 13 thirteenth. A p-limit system admits primes up to p as interval building blocks.

Table ag.3: Order of $\mathrm{GF}(3^k)^*$ and entry of new primes

k	$\lvert\mathrm{GF}(3^k)^*\rvert$	prime factors	new	harmonics
1	2	2	2	octave
2	8	2^3	—	$\mathrm{GF}(9)$, Matzke's $\mathbb{Z}_3\mathbb{C}$
3	26	$2 \cdot 13$	13	thirteenth; leptons (Doc. 338)
4	80	$2^4 \cdot 5$	5	major third; Doug's $\mathrm{G}(3)$ orders
5	242	$2 \cdot 11^2$	11	eleventh; neutrinos (Doc. 340)
6	728	$2^3 \cdot 7 \cdot 13$	7	seventh; with 13 on one carrier
7	2186	$2 \cdot 1093$	1093	no small harmonics left
8	6560	$2^5 \cdot 5 \cdot 41$	41	—
12	531440	$2^4 \cdot 5 \cdot 7 \cdot 13 \cdot 73$	73	all four united

Entry rule [B]: A prime $p \neq 3$ divides $3^k - 1$ iff $\mathrm{ord}_p(3) \mid k$; it therefore first enters at $k = \mathrm{ord}_p(3)$. This yields the order $13, 5, 11, 7$ for $k = 3, 4, 5, 6$. All four harmonic primes

above 3 enter between $k = 3$ and $k = 6$; from $k = 7$ on, no new prime ≤ 13 enters (the smallest later ones are $17, 19, 23, 29, 31, 37, \ldots$; $k = 7$ brings 1093, a Wieferich prime, $k = 8$ brings 41). By the entry rule itself, e.g. $\mathrm{ord}_{23}(3) = 11$ ($3^{11} - 1 = 2 \cdot 23 \cdot 3851$), $\mathrm{ord}_{17}(3) = 16$, $\mathrm{ord}_{19}(3) = \mathrm{ord}_{37}(3) = 18$, $\mathrm{ord}_{29}(3) = 28$, $\mathrm{ord}_{31}(3) = 30$.

This is the algebraic reason why practically relevant harmonics end at the 13-limit, and why the FFGFT corpus has so far needed only the primes $\{2, 3, 5, 7, 11, 13\}$. The neutrino eleven of Doc. 340 ($\Delta m^2_{\mathrm{atm}} = (11^2 - 1)\, m_\nu^2$) sits at $k = 5$; this had not been located in the corpus before.

$\mathbb{Z}_3$ compatibility: for $p \equiv 1 \pmod 3$, $\mathbb{Z}_3 \subset \mathrm{GF}(p)^*$. This holds for 7 and 13, not for $2, 5, 11$. The two $\mathbb{Z}_3$-compatible primes are exactly those that appear as structural numbers in Doc. 289 (top quark) and Docs. 338/341 (leptons) **[B]**.

ag.4 Theorem C: Chords as partitions of 8

For semisimple $X \in M_8(\mathrm{GF}(9))$ with χ split over $\mathrm{GF}(3)$ into factors of degrees $d_1 + \cdots + d_r = 8$, the eigenvalues lie in $\mathrm{GF}(3^{d_i})$ and $\mathrm{ord}(X)$ divides $\mathrm{lcm}(3^{d_i} - 1)$ **[B]**. A partition of 8 is thus the algebraic form of a chord of overtones $k \in \{d_i\}$; the order is the common period.

Table ag.4: Partitions of 8 and maximal semisimple order

Partition	$\mathrm{lcm}(3^{d_i}-1)$	reading
$(3, 3, 1, 1)$	26	pure $\mathrm{GF}(27)$ chord (Doc. 341)
$(3, 3, 2)$	104	$= 2^3 \cdot 13$: $\mathrm{GF}(9)$ with $\mathrm{GF}(27)$
$(4, 3, 1)$	1040	$= 2^4 \cdot 5 \cdot 13$: third with thirteenth
$(6, 2)$, $(6, 1, 1)$	728	$= 2^3 \cdot 7 \cdot 13$: seventh with thirteenth
$(5, 3)$	3146	$= 2 \cdot 11^2 \cdot 13$: eleventh with thirteenth
(8)	6560	$= 2^5 \cdot 5 \cdot 41$: maximal

The number of irreducible resp. primitive polynomials of degree k over $\mathrm{GF}(3)$ grows quickly (Möbius formula): degree 3: $8/4$, degree 4: $18/8$, degree 6: $116/48$, degree 8: $810/320$. The space of possible chords is therefore large — which leads directly to the question of discriminating power.

ag.5 Theorem D: Discriminating power and the boundary of the Galois layer

Density of Galois products

Forming all products of up to three building blocks $\{3^k, 3^k - 1, \text{prime divisors}\}$ for $k \leq k_{\max}$ with exponents $\{-1, 1, 2\}$, the mean logarithmic spacing of neighbouring values in the window $10 < x < 10^4$ drops as follows **[B]**:

Table ag.5: Mean log spacing of Galois products

$k_{\max}$	values in window	mean spacing	expected chance hits at $0.5\,\%$
2	54	$12.8\,\%$	0.08
3	218	$3.17\,\%$	0.32
4	650	$1.06\,\%$	0.94
5	1423	$0.49\,\%$	2.06
6	2503	$0.28\,\%$	3.62
8	5311	$0.13\,\%$	7.69

From $k_{\max} \approx 5$ on, the product set is denser than $0.5\,\%$: every number is "hit". A control test confirms this: at $k_{\max} = 6$, ten physical mass ratios have 2.9 hits on average, two hundred random numbers 3.7. The hit rate is at chance level **[B]**.

The threshold: Galois grid versus fractal correction

Doc. 338 quantifies the fractal-recursive higher-winding correction between bare value and measurement at $1.05\,\%$ for $(m_\mu/m_e)^2$. The Galois grid is coarser than twice this correction up to $k = 3$ ($3.17\,\% > 2.10\,\%$) and hence discriminating; from $k = 4$ on it lies below ($1.06\,\% < 2.10\,\%$) **[B]**.

Consequence. The exact Galois layer ends structurally where the fractal correction begins. Up to GF(27) the algebra is exact and the $1\,\%$ is a higher-mode correction; above that, the algebra itself is finer than the correction and dissolves into the higher winding modes. Galois products with $k \geq 4$ are statistically indistinguishable from higher torus modes.

This is not a contradiction. It is the same mechanism as in the overtone series: the interval $(n{+}1) : n$ shrinks from 1200 cents ($n = 1$) via 105 cents ($n = 16$) to below 7 cents at $n \approx 250$ and below 5 cents from $n \approx 350$; the series becomes quasi-continuous. Doc. 159 describes exactly this mode convergence on the fractal torus; here it appears in algebraic form.

Methodological rule

- $k \leq 3$ (GF(3), GF(9), GF(27)): identity layer. A hit with few building blocks is significant and may carry **[K]/[B]** if grounded geometrically in ξ (as 43200 in Doc. 338).
- $k = 4$ (GF(81), 5-limit): transition region; only with additional justification.
- $k \geq 5$: no status from product search. A relation holds here only if derived geometrically from ξ.

ag.6 Negative result: tau sector

At a tolerance of $0.05\,\%$, none of the ratios m_τ/m_μ, m_τ/m_e or their squares is a product of up to three Galois building blocks with $k \leq 6$ **[S]**. Hits at $0.5\,\%$ exist (e.g. $7 \cdot 11^2/3$ for $(m_\tau/m_\mu)^2$ at $0.17\,\%$) but lie at chance level by Theorem D. The tau ratio remains

grounded in the corpus through winding numbers (Doc. 258, Doc. 310), not through a Galois identity.

ag.7 Summary

Table ag.6: Results and status

Result	Content	Status
A	8 irreducible cubics, 4 primitive; 5 χ classes	**[B]**
B	entry rule $k = \mathrm{ord}_p(3)$; order $13, 5, 11, 7$	**[B]**
B′	$7, 13$ are $\mathbb{Z}_3$-compatible ($p \equiv 1 \bmod 3$)	**[B]**
C	chords $=$ partitions of 8, order $=$ lcm	**[B]**
D	Galois grid $< 2\times$ fractal correction from $k = 4$	**[B]**
D′	hit rate physical $=$ random for $k \geq 5$	**[B]**
—	tau sector no Galois product at $0.05\,\%$	**[S]**
—	physics of the classes f_2, f_3, f_4	**[S]**

Check scripts

Directory `2/python/Dok342_Skripte/`, wrapper `run_all_342.sh`:

- `pruef_342_faktorisierung_primzahlen.py` — Theorem A, p-limit assignment
- `pruef_342b_hoehere_harmonien.py` — Theorems B, C; product search with look-elsewhere note; Koide
- `pruef_342c_dichte_hoehere_harmonien.py` — Theorem D; density, random control, overtone analogue, threshold

All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All counts and order determinations are verified by Python check scripts with exact arithmetic. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 057: Relational number system*, FFGFT corpus.

[2] J. Pascher, *Doc. 159: Harmonic structure of the torus*, FFGFT corpus.

[3] J. Pascher, *Doc. 338: Mass calculations and Galois comparison*, FFGFT corpus, 28 August 2026.

[4] J. Pascher, *Doc. 340: Neutrino mass hierarchy from GF*$(27)^*$, FFGFT corpus, 30 August 2026.

[5] J. Pascher, *Doc. 341: GF*(27) *in GALG and FFGFT*, FFGFT corpus, 1 September 2026.

Chapter ah

Doc. 343 — The Zeta Function of the T^4/Z_3 Torus: Orbifold Projection and Dirichlet L-Functions

Abstract

The Galois factorisation classes of Doc. 342 and the harmonic series on the T^4 torus of Doc. 159 together yield precise statements about the Dirichlet L-function $L(s,\chi_0)$ and its relation to the spectral zeta function of the T^4/Z_3 orbifold.

Four results: ***(A)*** *In the one-dimensional analogue (decomposition by $n \bmod 3$) the Z_3 projection transforms the full Riemann zeta into a Dirichlet L-function: $L(s,\chi_0) = (1-3^{-s})\zeta(s)$* ***[B]****. It is not the spectral zeta function of the orbifold itself (fixed-point-free Z_3 on D_4, Docs. 314/330): that is one third of the four-dimensional Epstein zeta of D_4, with Weyl pole at $s=2$ and without the factor $(1-3^{-s})$* ***[B]****.* ***(B)*** *The Euler product of this L-function is hierarchically ordered by the entry depth $k = \mathrm{ord}_p(3)$; the physically visible primes $\{2,3,5,7,11,13\}$ are exactly those with $k \le 6$* ***[B]****.* ***(C)*** *The factor $(1-3^{-s})$ generates additional zeros of $L(s,\chi_0)$ at $s = 2\pi i k/\ln 3$ ($k \in \mathbb{Z}\setminus\{0\}$) on the imaginary axis $\mathrm{Re}(s) = 0$* ***[B]****. (B) and (C) are statements about $L(s,\chi_0)$ and arithmetic modulo 3, not about the spectrum of the orbifold: its Euler factors have the same form for every odd prime, including $p=3$; no GF(3^k) ordering occurs there.* ***(D)*** *The four Frobenius conjugacy classes of primitive elements in GF$(27)^*$ (Doc. 342, Theorem A) are paired into* two *classes by the FFGFT sector pairing $k \leftrightarrow -k$ (Doc. 336; on GF$(27)^*$: $x \mapsto x^{-1}$), $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$; negation $x \mapsto -x = x^{13}$ couples each primitive class to exactly one order-13 class* ***[B]****. The Galois character of* Gal(GF(27)/GF(3)) $\cong \mathbb{Z}_3$ *does not distinguish the classes; the distinction lies in $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$. Sector pairing and the Majorana criterion $k \mapsto k^{-1}$ (Doc. 340) partition the two layers identically: $\{f_3, f_4\}$ Dirac (charged leptons), $\{f_1, f_2\}$ Majorana (ν_1, ν_2 active; ν_3 in Dirac orbit O_4)* ***[B]****; prediction: ν_3 has no Majorana phase* ***[B]****.*

Theorem D‴: quantum numbers of the second layer

What quantum numbers would the two layers carry? The algebra answers this without physical assumption (`pruef_343f_zweite_schicht.py`). Each primitive class f_i lies in exactly one $\mathbb{Z}_{13}$ orbit of the Frobenius (Doc. 339); with the exponents of Table 3: $f_1 = \{1,3,9\} \mapsto O_1$, $f_2 = \{17,23,25\} \mapsto O_3 = \{4,10,12\}$, $f_3 = \{5,15,19\} \mapsto O_2 = \{2,5,6\}$, $f_4 = \{7,11,21\} \mapsto O_4 = \{7,8,11\}$. Doc. 340 takes the self-inverseness of an orbit under $k \mapsto k^{-1} \pmod{13}$ as the Majorana criterion. Both involutions together give ***[B]****:*

Table ah.1: Quantum numbers of the four primitive classes from the algebra

Class	$\mathbb{Z}_{13}$ orbit	$k \mapsto -k$	$k \mapsto k^{-1}$	profile	layer
f_1	O_1	$\to f_2$	self-inverse	Majorana	$\{f_1, f_2\}$
f_2	O_3	$\to f_1$	self-inverse	Majorana	$\{f_1, f_2\}$
f_3	O_2	$\to f_4$	$\to O_4$	Dirac	$\{f_3, f_4\}$
f_4	O_4	$\to f_3$	$\to O_2$	Dirac	$\{f_3, f_4\}$

The sector pairing $k \mapsto -k$ and the Majorana criterion $k \mapsto k^{-1}$ partition the four classes identically: $\{f_1, f_2\}$ ***is the Majorana layer,*** *$\{f_3, f_4\}$* ***the Dirac layer [B]****. Clarification: The Majorana criterion is an algebraic property of the Galois orbit (self-inverseness under $k \mapsto k^{-1}$ mod 13). Whether this implies the physical Majorana condition ($\nu = \bar{\nu}$) is an additional assumption* ***[S]****; see Doc. 344 for the open anti-neutrino question. With the negation $f_i \leftrightarrow g_i$ the numbering of Doc. 342 is thus the natural one: sector pairing pairs adjacent indices, negation equal indices. Both layers carry $\mathbb{Z}_2$ parity 1 (massive sector, Doc. 339), no colour, and three states per class (three-cycle).*

Interpretation [B]/[S]. *Doc. 341 fixes order 26 in $G(6)$ for the leptons; the class follows here: charged leptons are Dirac particles and belong to $\{f_3, f_4\}$* ***[B]****. The algebraic Majorana layer $\{f_1, f_2\}$ is fully occupied by active neutrinos: Doc. 340 places ν_1 in $O_1 = f_1$ (orbit-algebraically Majorana) and ν_2 in $O_3 = f_2$ (orbit-algebraically Majorana). Doc. 340 also places ν_3 in O_4 — a Dirac orbit (not self-inverse). This is not a contradiction: the quadratic character mod 13 determines the property of the field. ν_3 in O_4 is therefore a* ***Dirac neutrino [B]****. There is no empty slot and no prediction of sterile neutrinos from this algebra.*

Clarification on anti-neutrinos ***[S]****. The Majorana criterion is an algebraic property of the Galois orbit (self-inverseness under $k \mapsto k^{-1}$ mod 13)* ***[B]****. Whether this implies the physical Majorana condition ($\nu = \bar{\nu}$) is an additional assumption. In the Furey/GALG construction (Doc. 344) the anti-neutrino Sd[0] = Vss[1] ≠ Vss[0] = Su[0], so neutrinos there are Dirac particles* ***[S]****.*

Testable prediction [B]: *ν_3 has no Majorana phase; m_{ee} depends only on ν_1 and ν_2:*

$$m_{ee} = |m_1 U_{e1}^2 + m_2 U_{e2}^2| \leq m_{\nu_1} + m_{\nu_2} \approx 14.4\,\text{meV}.$$

With the values of Doc. 340 ($m_1 = 4.54$, $m_2 = 9.81$ meV, $s_{12}^2 = 0.307$, $s_{13}^2 = 0.022$) this gives $m_{ee} \approx 6.0$ meV (zero phases, without ν_3; with the Galois angles $0.323/0.0223$: 6.10 meV). The inverted hierarchy is more strongly excluded because the ν_3 contribution to m_{ee} vanishes. Testable at nEXO ($m_{\beta\beta} \approx 5$–20 meV, NME-dependent); KamLAND-Zen-800: < 28–122 meV (2025).

Distinction from Doc. 006. *Transferring the lepton ladder to neutrinos via an additional double-ξ suppression does not hold, because the neutrino values given with it (9.1, 1.9, 18.0 meV) follow from three incompatible prescriptions; Doc. 006 does not compute neutrino masses. Authoritative for neutrinos is Doc. 340 ($m_{\nu_1} = 4.54$, $m_{\nu_2} = 9.81$, $m_{\nu_3} = 49.9$ meV); its Δm^2 values agree with experiment to 0.5–1 %.*

Two further results: ***(E)*** *The harmonic primes $\{5, 7, 11, 13\}$ are uniquely forced by the prime factorisations of $3^k - 1$ for $k = 3, 4, 5, 6$; $p = 13$ is the worldwide unique prime with $\text{ord}_p(3) = 3$* ***[B]****.* ***(F)*** *The Euler factor threshold at $p^* \approx 9.76$ ($s = 2$) marks the 7-limit exactly; from $p_{\max} \approx 30$ the Galois product density lies at chance level; the Weyl obstruction (Doc. 316) shows that the limit lies in the object; Shor's circuit model is digital and unaffected, but the physical apparatus is a resonant system subject to the resonance limits as long as the threshold theorem is not demonstrated at scale* ***[S]****; empirically no quantum advantage has been demonstrated.*

Check scripts: `pruef_343_zeta_galois.py`, `pruef_343b_klassen_symmetrie.py`, `pruef_343g_epstein_satzA.py`, `pruef_343h_gf3k_vierdimensional.py`, `pruef_344_primzahlmuster.py`.

ah.1 Status markers and starting point

The following status markers are used throughout:

- **[K]** *Confirmed* — fully proved from ξ or a corpus-established derivation; all check-script assertions passed.
- **[B]** *Established* — numerically secured; analytical proof present or trivial.
- **[S]** *Speculative* — indication or conjecture, not yet fully proved.

Doc. 159 shows that the spectral zeta function of the Laplace operator on a torus T^d is the Epstein zeta function, whose one-dimensional special case is the Riemann zeta $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Doc. 342 shows that primes appear in the GF(3^k) hierarchy of the T^4/Z_3 orbifold ordered by entry depth $k = \mathrm{ord}_p(3)$. This document combines both: what does the GF(3^k) hierarchy imply for the zeta function itself?

ah.2 Theorem A: Z_3 decomposition in the one-dimensional analogue and Dirichlet L-function

Decomposition of the zeta by Z_3 classes

On T^4/Z_3 the allowed modes $n \in \mathbb{Z}^4$ are structured by Z_3 invariance. In the one-dimensional analogue $\zeta(s)$ splits into three partial sums by n mod 3 **[B]**:

$$\zeta(s) = \underbrace{\sum_{\substack{n=1 \\ 3 \mid n}}^{\infty} \frac{1}{n^s}}_{=3^{-s}\zeta(s)} + \underbrace{\sum_{\substack{n=1 \\ n \equiv 1 (3)}}^{\infty} \frac{1}{n^s}}_{L_1(s)} + \underbrace{\sum_{\substack{n=1 \\ n \equiv 2 (3)}}^{\infty} \frac{1}{n^s}}_{L_2(s)} \tag{ah.1}$$

The fixed-point sector ($n \equiv 0$) contributes exactly $3^{-s}\zeta(s)$; the twist sectors ($n \equiv \pm 1$) together $(1 - 3^{-s})\zeta(s)$.

Principal character χ_0 mod 3

The Dirichlet character χ_0 mod 3 with $\chi_0(n) = 0$ if $3 \mid n$, otherwise $\chi_0(n) = 1$, gives **[B]**:

$$L(s, \chi_0) = \sum_{\gcd(n,3)=1} \frac{1}{n^s} = (1 - 3^{-s})\,\zeta(s) = \prod_{p \neq 3} \frac{1}{1 - p^{-s}} \tag{ah.2}$$

This is the zeta function of the one-dimensional analogue: the characteristic prime $p = 3$ does not appear as an Euler factor.

The spectral zeta function of the orbifold is a different one. For the corpus orbifold (fixed-point-free $\mathbb{Z}_3$ on D_4, Docs. 314/330) the spectral zeta function is one third of the four-dimensional Epstein zeta of D_4, with Weyl pole at $s = 2$ and without the factor $(1 - 3^{-s})$ **[B]** (`pruef_343g_epstein_satzA.py`). Theorem A therefore holds as a statement about the one-dimensional analogue, i.e. about $L(s, \chi_0)$ **[B]**; $L(s, \chi_0)$ is not to be identified with the spectral zeta function of the orbifold.

Non-trivial character χ_1 mod 3

The character χ_1 mod 3 with $\chi_1(1) = +1$, $\chi_1(2) = -1$, $\chi_1(0) = 0$ measures the *asymmetry* of the twist sectors **[B]**:

$$L(s, \chi_1) = \sum_{\substack{n=1 \\ n \equiv 1 (3)}}^{\infty} \frac{1}{n^s} - \sum_{\substack{n=1 \\ n \equiv 2 (3)}}^{\infty} \frac{1}{n^s} \tag{ah.3}$$

with the exact value (Dirichlet 1837): $L(1, \chi_1) = \pi/(3\sqrt{3})$.

Physical meaning **[S]**: $L(s, \chi_1)$ measures the difference between winding numbers $w = +1$ and $w = -1$ on the torus, i.e. the $\mathbb{Z}_3$ sector asymmetry. Whether $L(s, \chi_1)$ has a direct physical observable in FFGFT (particle–antiparticle asymmetry in the twisted sector?) remains open **[S]**.

ah.3 Theorem B: Hierarchical Euler product by $k = \mathrm{ord}_p(3)$

The primes in $L(s, \chi_0) = \prod_{p \neq 3} (1 - p^{-s})^{-1}$ are not equivalent. By the entry-depth rule $k = \mathrm{ord}_p(3)$ from Doc. 342, Theorem B, the Euler product is hierarchically ordered **[B]**:

$$L(s, \chi_0) = \prod_{k=1}^{\infty} \prod_{\substack{p \neq 3 \\ \mathrm{ord}_p(3) = k}} \frac{1}{1 - p^{-s}} \tag{ah.4}$$

Table ah.2: Hierarchy of Euler factors by $k = \mathrm{ord}_p(3)$

k	new prime(s)	Euler factor at $s = 2$	harmonics	FFGFT role
1	2	$4/3 = 1.333$	octave	$\mathbb{Z}_2$ fixed point, massive sector
3	13	≈ 1.006	thirteenth	$\mathrm{GF}(27)^*$, leptons
4	5	≈ 1.042	major third	factor in $(m_\mu/m_e)^2$
5	11	≈ 1.008	eleventh	neutrinos (Doc. 340)
6	7	≈ 1.021	seventh	top quark (Doc. 289)
≥ 7	$17, 19, 23, 29, 31, 37, 41, \ldots, 1093, \ldots$	$\approx 1 + p^{-2}$	no small harmonics	suppressed

Theorem B is a statement about $L(s, \chi_0)$ and arithmetic modulo 3, not about the spectrum of the orbifold: the Euler factors of the orbifold's spectral zeta function have the same form for every odd prime, including $p = 3$; an ordering by $\mathrm{ord}_p(3)$ does not occur there (`pruef_343h_gf3k_vierdimensional.py`).

Fractal suppression of higher Euler factors

On the fractal T^4 torus (Doc. 159) high-mode contributions are suppressed by the damping factor $\tau^{(1+\xi)k}$. Transferring this to the Euler factors of $L(s,\chi_0)$, the *weighted* Euler product reads

$$L^{\text{frac}}(s,\chi_0) \;\cong\; \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \text{ord}_p(3)=k}} \left(\frac{1}{1-p^{-s}}\right)^{\tau^{k(1+\xi)}} \tag{ah.5}$$

Since $\tau^{k(1+\xi)} \to 0$ as $k \to \infty$ (with $\tau < 1$), each Euler factor is pushed toward 1 for large k: primes with $\text{ord}_p(3) \geq 7$ make no physical contribution **[B]**. This is Theorem D of Doc. 342 in the language of the zeta function.

ah.4 Theorem C: Zeros from the factor $(1-3^{-s})$

Zeros of $(1-3^{-s})$

The factor $(1-3^{-s})$ in $L(s,\chi_0) = (1-3^{-s})\zeta(s)$ has zeros at **[B]**:

$$3^{-s} = 1 \quad\Longleftrightarrow\quad s = \frac{2\pi i\, k}{\ln 3}, \qquad k \in \mathbb{Z}\setminus\{0\} \tag{ah.6}$$

These lie on $\text{Re}(s) = 0$ (imaginary axis), *not* on the critical line $\text{Re}(s) = 1/2$ of the Riemann hypothesis. They are absent from the Riemann zeta spectrum; they arise solely from removing the multiples of 3. They are a statement about $L(s,\chi_0)$ and arithmetic modulo 3, not about the spectrum of the orbifold, whose spectral zeta function does not contain the factor $(1-3^{-s})$ (Theorem A).

The zero spacing $2\pi/\ln 3 \approx 5.72$ encodes the periodicity of the factor 3^{-s}. A resonance test of the first eight known Riemann zeros γ_n against this unit shows no significant coincidence (deviations 0.15–0.47) **[B]**: the full Riemann zeta encodes the distribution of *all* primes, not only the Z_3-compatible ones.

Complete zero structure of $L(s,\chi_0)$

$L(s,\chi_0)$ has three types of zeros:

- **Trivial zeros of** $\zeta(s)$: at $s = -2, -4, -6, \ldots$
- **Non-trivial Riemann zeros**: on $\text{Re}(s) = 1/2$ (assumed), unchanged
- **Zeros of** $(1-3^{-s})$: at $s = 2\pi i k/\ln 3$, $k \neq 0$ — new, generated by the decomposition modulo 3 **[B]**

ah.5 Theorem D: The four classes under sector pairing

The Galois character does not distinguish the classes

The Galois group $\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \langle F\rangle \cong \mathbb{Z}_3$ with $F\colon x \mapsto x^3$ has three characters $\chi_j(F) = \omega^j$, $j = 0,1,2$. All four primitive classes $f_1, \ldots, f_4$ of Doc. 342 are *regular* F-orbits

of length 3: F acts on each as a 3-cycle, and the associated character is the same for all four. The Galois character therefore does not separate the classes **[B]**.

The distinction lies in the discrete logarithm: with a generator x of GF$(27)^*$, each class is an orbit $\{x^k, x^{3k}, x^{9k}\}$ with $\gcd(k, 26) = 1$, i.e. a coset of $\langle 3\rangle = \{1, 3, 9\}$ in $(\mathbb{Z}/26)^* \cong \mathbb{Z}_{12}$. There are $12/3 = 4$ cosets **[B]**:

Table ah.3: The four primitive classes as cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$ (basis $x^3 + 2x + 1 = 0$)

Class	Minimal polynomial	Exponents k	Role
f_1	$x^3 + 2x + 1$	$\{1, 3, 9\}$	Majorana layer (Theorem D‴)
f_2	$x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	$= f_1^{-1}$, Majorana layer
f_3	$x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	Dirac layer
f_4	$x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	$= f_3^{-1}$, Dirac layer

The quotient $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$ is cyclic; inversion $k \mapsto -k$ is its element of order 2.

Sector pairing reduces four to two

Doc. 336 identifies the FFGFT sector pairing $k \leftrightarrow -k$ with the Frobenius automorphism; at the level of elements of GF$(27)^*$ it is inversion $x \mapsto x^{-1} = x^{25}$. It permutes the classes as **[B]**

$$f_1 \longleftrightarrow f_2, \qquad f_3 \longleftrightarrow f_4. \tag{ah.7}$$

If sector pairing is a symmetry of the mass layer — as it is in the corpus — there are not four but **two** physically distinguishable GF$(27)^*$ layers: $\{f_1, f_2\}$ and $\{f_3, f_4\}$ (assignment in Theorem D‴).

Negation couples order 26 to order 13

Since $-1 = x^{13}$ in GF(27), the map $x \mapsto -x$ sends every element of order 26 to one of order 13 and vice versa. The four order-13 classes $g_1, \dots, g_4$ of Doc. 342 are therefore not an independent structure but the sign partners of the primitive classes **[B]**:

$$f_i \leftrightarrow g_i \qquad (i = 1, \dots, 4). \tag{ah.8}$$

The candidate “sub-lepton structure” from the order-13 polynomials, left open in Doc. 342, is thereby eliminated.

Open question

The reduction $4 \to 2$ sharpens the question of Doc. 342: not “do f_2, f_3, f_4 carry further layers”, but “there are exactly two GF$(27)^*$ layers $\{f_1, f_2\}$ and $\{f_3, f_4\}$ — which particles carry which?” Theorem D‴ answers this from the algebra.

ah.6 Theorem E: Prime patterns in the GF(3^k) hierarchy

The harmonic primes are uniquely forced

A prime p satisfies $\mathrm{ord}_p(3) = k$ if and only if p divides $3^k - 1$ and $p \nmid 3^j - 1$ for all $j < k$. The prime factorisations of the first relevant values give **[B]**:

Table ah.4: Prime factors of $3^k - 1$ and the forced physical primes

k	$3^k - 1$	prime factors	phys. prime	note
1	2	$\{2\}$	2	octave
3	26	$\{2, 13\}$	**13**	unique prime with $\mathrm{ord}_p(3) = 3$ worldwide
4	80	$\{2, 5\}$	**5**	only non-2 factor
5	242	$\{2, 11\}$	**11**	only non-2 factor
6	728	$\{2, 7, 13\}$	**7**	(13 has $\mathrm{ord}_{13}(3) = 3$, not 6)

The strongest result concerns $p = 13$ **[B]**: since $3^3 - 1 = 26 = 2 \cdot 13$ and $\mathrm{ord}_2(3) = 1$, $p = 13$ is the *unique* prime worldwide with $\mathrm{ord}_p(3) = 3$. There is no other, and there can be none. The physical primes $\{5, 7, 11, 13\}$ are not selected by harmonic considerations, but are completely and uniquely forced by the prime-factor structure of $3^k - 1$ for $k = 3, 4, 5, 6$.

Necessary congruence condition

From $\mathrm{ord}_p(3) = k$ with $3 \mid k$ it follows that $3 \mid (p - 1)$, i.e. $p \equiv 1 \pmod 3$ **[B]**. In particular $p = 13$ (at $k = 3$) and $p = 7$ (at $k = 6$) are both $\equiv 1 \pmod 3$ — exactly the Z_3-compatible primes of the corpus (Doc. 342, Theorem B).

ah.7 Theorem F: Limits of distinguishability and computational cost

Exact threshold: Euler factor vs. fractal correction

The Euler factor $1/(1 - p^{-s})$ exceeds the fractal correction $1.05\,\%$ (Doc. 338, bare to measured) exactly for $p < p^*$ with $p^* = (0.0105)^{-1/s}$ **[B]**:

Table ah.5: Exact threshold p^* by value of s

s	p^*	primes $\le p^*$	harmonic limit	Euler factor $\ge 1.05\,\%$
1	95.2	$\{2, 3, 5, \ldots, 89\}$	—	all small primes
2	9.76	$\{2, 3, 5, 7\}$	7-limit	octave, fifth, third, seventh
3	4.57	$\{2, 3\}$	3-limit	octave and fifth only

For $s = 2$ (reference value of the spectral zeta) the threshold is $p^* \approx 9.76$: primes $\{2, 5, 7\}$ have Euler factors exceeding the fractal correction; $p = 11$ and all larger primes fall below it. This is exactly the harmonic 7-limit. $p = 11$ appears in the corpus (neutrinos, Doc. 340) via the structural formula $\Delta m^2 = (11^2 - 1)\, m_\nu^2$, not as a Galois product hit.

Galois products vs. random products: hit rate

Forming all products of up to three Galois building blocks with exponents $\{-1, 1, 2\}$ for $p \le p_{\max}$, the product set grows rapidly and the hit rate for arbitrary targets reaches chance level **[B]**:

Table ah.6: Hit rate for physical vs. random targets at $\pm 0.5\,\%$

$p_{\max}$	values	43200	m_μ/m_e	random avg
13	1334	2	0	0.92
30	5802	3	2	3.60
50	8946	3	2	5.33
100	13398	4	3	7.66
300	24992	8	12	14.15

From $p_{\max} \approx 30$ the hit rate lies at chance level ($\pm 0.5\,\%$); at the strict threshold $\pm 0.05\,\%$ already from $p_{\max} \approx 13$. The identity $43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$ (Doc. 338) uses only building blocks with $p \le 5$ and is derived *geometrically* from ξ, not found by product search.

Structural computational limit: Weyl obstruction and quantum computers

The distinguishability threshold at $p_{\max} \approx 13$ is a *physical* limit (fractal correction). Independently of this, the corpus contains a sharper *mathematical* limit that applies to every technology **[B]**:

Weyl obstruction, Doc. 316

There is no compact Riemannian manifold — of any dimension, curvature, topology, or orbifold structure — whose Laplace spectrum consists of the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

From Doc. 147, an n-qubit register is a compact state space with eigenvalue density $N(\lambda) \sim \lambda^{d/2}$; arithmetic counting densities grow as $T \ln T$. A finite state space cannot fully carry such a spectrum, *independently of the register size* **[B]**.

Doc. 075 names the consequence for Shor's algorithm:

Method	Limit	Type
Trial division	$O(\sqrt{n})$	computational
Pollard ρ	$O(n^{1/4})$	computational
Shor's QFT	resource limit	Weyl obstruction

The central statement of Doc. 316 holds technology-independently:

> *The limit lies not in the tool but in the object.* Transformation (Fourier, QFT, optics) projects and generates no information. Relaxation finds what is already in the potential. Both presuppose that the sought structure is already present.

Physical limit ($p^* \approx 9.76$, Theorem F.1) and mathematical limit (Weyl obstruction, Theorem F.3) thus appear in the same picture: beyond a certain point the prime structure is hidden in the noise, and no analogue tool retrieves it; Shor's algorithm is digital as a model, a resonator as an apparatus — and thus, until proven otherwise, bound as well (Theorem G′).

ah.8 In plain terms: what the limit means

Question. So arbitrarily dense harmonics can be computed — but beyond a certain depth they can no longer be distinguished?

Answer. Yes, exactly that.

Computation can go arbitrarily far. GF(3^k) for $k = 7, 8, 12, 100$ — the algebra does not stop. Every k yields new primes, new polynomial classes, new chords. At $k = 12$ all four harmonic primes $5, 7, 11, 13$ sit in one field simultaneously; at $k = 7$ the Wieferich prime 1093 enters. These structures are exactly computable and mathematically all present.

But beyond a certain depth they are physically meaningless, for two reasons that coincide.

First, the grid becomes too fine. For $k \leq 3$ the reachable values lie about $3\,\%$ apart — if 43200 hits a mass value to $1\,\%$, that is a result. For $k \geq 5$ they lie less than $0.5\,\%$ apart. Then *every* number hits something, and a hit means nothing. The control test in Theorem F shows it: two hundred random numbers hit as often as the actual mass ratios.

Second — and this is the physical reason — the fractal correction of about $1\,\%$ (Doc. 338, bare to measured) is itself already a statement about higher winding modes. Once the Galois grid is finer than this correction, nature itself no longer distinguishes neighbouring Galois values. The structure is not gone; it has dissolved into the noise of the higher modes.

The analogy with harmonics is exact: everyone hears the overtones $2:1$ and $3:2$. $17:16$ is a semitone, $33:32$ a quarter-tone. From roughly the 250th overtone the spacing falls below 5 cents — mathematically exactly present, acoustically no longer distinguishable. The series becomes a continuum.

The consequence for FFGFT. The exact Galois layer reaches to GF(27), i.e. $k = 3$: the lepton layer, 43200, the four classes, $p = 13$ as the unique prime with $\mathrm{ord}_p(3) = 3$.

Above that ($k = 4$, GF(81), $p = 5$) is already the transition. From $k = 5$ on, anything found by search is indistinguishable from chance — and must receive no status unless it comes directly from ξ.

This is not a deficiency of the theory but its consistency: Doc. 159 says mode convergence, Doc. 316 says Weyl obstruction, Doc. 342 and this document say the same thing algebraically. There is a limit, it lies at $k \approx 3$–4, and it comes from the subject itself, not from our computational capacity.

ah.9 In plain terms II: synthesis, analysis, factorisation — and coupling

Question. What does the limit mean for synthesis and analysis — and for factorisation? And how does coupling enter: Doc. 328 has three coupling regimes; do pure numbers show the same picture as physical processes?

Synthesis is unlimited, analysis is limited

Synthesis — adding overtones, multiplying primes, forming GF(3^k) chords from partitions — goes arbitrarily far and arbitrarily dense. Nothing prevents superposing the 250th overtone with the 251st or multiplying two 300-digit primes. Synthesis does not know the limit of Theorem F.

Analysis — which overtones are in the sound, which primes in the number, which Galois class carries the mass — hits exactly this limit. It is not a matter of computational cost but of resolution: to separate overtone n from $n{+}1$ one needs a time window of about n periods; to find the period r of a number, a frequency resolution $\Delta f \approx f_0/n^2$ (Doc. 075, precision barrier). Beyond a certain density the spacing lies below the noise, and the decomposition is no longer unique — not because it does not exist, but because many decompositions fit equally well. That is the chance-level finding of Theorem F in other words.

Factorisation is analysis

Factorising $N = p \cdot q$ means reading the two fundamentals out of a beat. For small p, q — the 7-limit range, $k \leq 3$ — this works; those are exactly the primes visible in the FFGFT spectrum at all. RSA primes with 300 digits sit at $k = \mathrm{ord}_p(3)$ of the order of p itself, i.e. at a depth undercutting the grid spacing of Theorem F by hundreds of orders of magnitude. Doc. 316 puts it thus: Fourier, QFT and optics project, they generate no information; the period must already be in the prepared states, and this preparation (modular exponentiation, 95 % of the gate cost, Doc. 147) is exactly the problem one wanted to solve.

Two things follow. First: FFGFT does not break RSA — and that is not a deficiency but a consistent result. The asymmetry synthesis-easy / analysis-hard on which RSA lives is the same asymmetry that ends the Galois layer at $k \approx 3$–4. Second, for instrument building: an instrument can *produce* overtones arbitrarily densely; ear and analysis *hear* them sharply only up to about the 7-limit. What lies above acts as

timbre, not as pitch — as the higher winding modes in FFGFT act as a $1\,\%$ correction, not as a separate mass layer.

Caveat [S]. Transferring Theorem F to $\mathbb{Z}/N\mathbb{Z}$ is an analogy. Doc. 075 records that ξ has no function in $\mathbb{Z}/N\mathbb{Z}$ and the period is fixed purely arithmetically by $\mathrm{lcm}(p-1, q-1)$. The analogy holds; the fractal correction as the *mechanism* of the factorisation limit does not.

Coupling: the same threefold division, not the same law

Doc. 328 classifies couplings relative to loss rates: subcritical ($\kappa < 1$, one maximum), critical ($\kappa = 1$, maximally flat), supercritical ($\kappa > 1$, splitting). Locking onto a frequency ratio p/q occurs in Arnold tongues of width $\Delta\Omega(p/q) \propto K^q$.

For pure numbers the tolerance ε takes the role of the coupling strength: a real number "locks" onto p/q if $|x - p/q| < \varepsilon$. Fractions with denominator $\leq Q$ cover the interval with measure $\approx 2\varepsilon \cdot 3Q^2/\pi^2$; coverage 1 gives the critical denominator height **[B]**

$$Q_c = \frac{\pi}{\sqrt{6\,\varepsilon}}. \tag{ah.9}$$

For $\varepsilon = 1.05\,\%$ (fractal correction), $Q_c \approx 12.5$ — the same order as $p_{\max} \approx 13$ in Theorem F. The Galois version confirms it numerically (`pruef_343c`): at $\varepsilon = 1.05\,\%$ the Galois grid with $p_{\max} = 7$ is subcritical (coverage 0.52), with $p_{\max} = 13$ critical (0.92), from $p_{\max} \approx 30$ supercritical (1.00). **The limit $k \approx 3$–4 of Theorem F is the critical point $\kappa = 1$ in the sense of Doc. 328 [B]**.

So the threefold division is the same in arithmetic and physics. *The law behind it is not* **[B]**: Arnold tongues fall exponentially in q ($\Delta\Omega \propto K^q$), Farey spacings only polynomially ($1/q^2$). At $K = 0.3$ and $q = 5$ the tongue is already 10^{-5} narrower than the arithmetic capture interval; at $q = 13$ by 10^{-15}. Physical locking is exponentially selective, arithmetic distinguishability polynomial. Physics "hears" fewer harmonics than arithmetic permits — the edge is the same, but physically sharper.

FFGFT is dynamically deeply subcritical

With $K_{\mathrm{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}$ (Doc. 328) the widest non-trivial Arnold tongue is $\Delta\Omega(1/2) \approx 5.6 \times 10^{-8}$, the total measure of all tongues $\approx 8 \times 10^{-4}$. Both lie five to two orders of magnitude below the fractal correction **[B]**. No mass ratio is pulled onto a p/q by resonant locking.

This yields a clarification that was so far only implicit in the corpus: the exact rationals — 43200, $8^2 \cdot 5^2 \cdot 27$, the winding numbers (r_i, p_i) — are *topological* (integer windings on T^4), not *dynamical* (locking). Discreteness and coupling are two separate mechanisms in FFGFT. This is consistent with Doc. 328, where φ lies outside all tongues: the coupling is too weak to hold ratios; what is held, topology holds.

ah.10 Theorem G′: ξ as resolution floor — the limit depends on the level

Question. Is it excluded that ξ plays a role in resonance finding or analysis?

Answer. Not absolutely. Three roles must be distinguished. As a *selection criterion* in a search function of the form $\exp(-(\omega-\omega_0)^2/4\xi)$, ξ is excluded: the centre ω_0 does the work, ξ only sets the width **[B]**. As a *dynamical coupling*, ξ is excluded: $K_{\text{eff}} = 2\pi\xi$ gives Arnold tongues of 5.6×10^{-8} (Theorem G″) **[B]**. As a *resolution floor*, however, ξ is the natural scale: Doc. 328 sets $g/\omega = \xi$ as the linewidth of the torus modes, Doc. 162 the fractal damping per operation as $K_{\text{frak}} = 1 - 100\xi$.

The 1.05 % is one fractal level

The correction bare → measured of Doc. 338 ($1.05\,\%$) and the damping level $100\xi = 1.33\,\%$ of Doc. 162 stand in the ratio 0.79 **[B]**. The thresholds in Theorems F and G were computed with $\varepsilon = 1.05\,\%$ — they are thus the thresholds of *one* fractal level, not of the theory. If this level is subtracted analytically (it is computable per Docs. 338/162), the resolving power moves to ξ:

Table ah.7: Thresholds of Theorems F/G for the corpus resolution levels (`pruef_343d_xi_aufloesung.py`)

Resolution ε	value	Q_c	p^*	primes $\le p^*$	Galois resolved to
$1.05\,\%$ (Doc. 338)	1.05×10^{-2}	12.5	9.8	4	$k=3$
100ξ (Doc. 162)	1.33×10^{-2}	11.1	8.7	4	$k=3$
ξ (Doc. 328)	1.33×10^{-4}	111	87	23	$k\ge8$
$1/Q_{\text{TFLN}}$ (Doc. 186)	10^{-6}	1283	1000	168	$k\ge8$
ξ^2	1.78×10^{-8}	9619	7500	950	$k\ge8$

At $k_{\max} = 8$ the Galois grid still has a mean spacing of $0.128\,\%$ — ten times coarser than $\xi = 0.0133\,\%$. At ξ resolution the layer remains resolved at least to $k = 8$, and primes up to 83 would have distinguishable Euler factors **[B]**.

Photonics as access to the ξ floor

Doc. 186 gives $Q > 10^6$ for TFLN resonators. Resolving ξ requires only $Q > 1/\xi = 7500$; resolving ξ^2 would require $Q > 5.6\times10^7$ **[B]**. Photonic hardware thus reaches down to the ξ floor, not below — whereas NISQ hardware, per Doc. 034, loses the ξ corrections in noise. This is the only place where ξ "plays a role in analysis": not as the tool that finds the resonance, but as the scale down to which anything can be distinguished at all. ξ has the same role everywhere in the corpus: $43200 = (r_\mu/r_e)^2/\xi$ says *where* the resonance is, not how to find it.

Caveats [S]. Whether the 100ξ level is actually subtractable is asserted in the corpus, not carried out. Doc. 186 gives $Q > 10^6$ for the optical phase, not for mass ratios; the transfer $1/Q \to \varepsilon$ is an analogy. The check of Doc. 186 itself

(`pruef_343e_dok186_check.py`) confirms the TFLN quality but refutes its "locking" onto φ^{-k} (contradiction to Doc. 328 [K]), the Fibonacci condition for winding numbers ($r_\mu = 16/5$, $16 \notin F$) and the B-meson interpretation ($\alpha = \xi$ would need 9×10^8 events for 4σ).

Consequence for factorisation: $N_{\max} \approx 1/\varepsilon$

A resonance-based factoriser must separate the period r from $r{+}1$ (resolution $1/r$), or, by the precision barrier of Doc. 075 ($\Delta f \approx f_0/n^2$), isolate the factor n (resolution $1/n^2$). The resolution floor ε therefore caps it at $N_{\max} \approx 1/\varepsilon$ **[B]**:

Table ah.8: Resolution floor and ceiling of a resonance-based factoriser

ε	factors up to $n_{\max}$	numbers up to $N_{\max}$	bits
$1.05\,\%$ (one fractal level)	10	95	7
ξ	87	7.5×10^3	13
$1/Q_{\text{TFLN}}$	10^3	10^6	20
ξ^2	7.5×10^3	5.6×10^7	26
RSA-2048	—	would need $\varepsilon \approx 2^{-1024}$	—

The $p^* \approx 87$ of the threshold table *is* the factorisation statement: a resonance method at the ξ floor separates factors up to about 87. Even with ξ^2 that is 26 bits — trial division does it in milliseconds. What moves with the ξ floor is the Galois layer of physics ($k \approx 3 \to k \geq 8$), not factorisation: physics gains resolving power for mass ratios; number theory gains nothing, because its numbers carry no ξ scale (Doc. 075).

Why classical methods reach further: state versus process

Classical methods resolve nothing — they count. $a^r \bmod N = 1$ is true or false; there is no ε. Every bit is a $\mathbb{Z}_2$ state, in the corpus a winding number on T^4 (Doc. 247): a *state* that exists without energy and is re-thresholded to 0 or 1 after every step. That is the topological discreteness of Theorem G″. A resonance method, by contrast, carries its information as a *process* — phase, frequency — and has no state to threshold back to; the floor ε stays with it. The price of exactness is enumeration:

Table ah.9: Classical limits are counting limits (operations), not resolution limits

bits	trial division $\sqrt{N}$	Pollard-ρ $N^{1/4}$	NFS	
13	90	10	9×10^2	resonance ceiling at ξ
26	8×10^3	90	3×10^4	resonance ceiling at ξ^2
512	10^{77}	3×10^{38}	2×10^{19}	hours
2048	—	1.3×10^{154}	1.5×10^{35}	5×10^{12} years at 10^{15} op/s

On energy: Landauer's bound $k_B T \ln 2$ holds in the corpus unchanged and without ξ modification (Doc. 184), but *only for the erasure* of a bit (Doc. 247); Doc. 255 derives

it as the frozen energy of a minimal winding difference $\Delta w = 1$ from T^4 geometry. It caps dissipation per irreversible step, not the number of steps; reversible steps are exempt. The classical wall is therefore time, not energy **[B]**.

The two method types are dual: **exactness costs enumeration, resolution costs precision.** Resonance has constant cost per candidate and a floor at $1/\varepsilon$; counting has no floor and a cost growing subexponentially in N. Both are limits; the counting limit merely lies hundreds of orders of magnitude further out. Doc. 316 says it generally: the arithmetic structure cannot be projected into a continuum without loss of information.

Quantum computers: the empirical yardstick

Two levels must be separated. The *circuit model* of Shor's algorithm is digital and untouched by the Weyl obstruction (Theorem F): the QFT runs over $\mathbb{Z}_Q$, $Q = 2^{2n}$, with equally spaced eigenfrequencies and linear counting density $N(\lambda) \propto \lambda$; the number theory sits in the period r of a^x mod N, not in the spectrum **[B]**. That holds on paper.

The *physical apparatus* is something else. A qubit is a two-level resonator, every gate a resonant drive (Rabi rotation), a photon a mode with continuous phase. Quantum computers and photonics do not compute digitally and do not enumerate; they are resonant systems and are subject to the resonance limits: the resolution floor ε (Theorem G′) and, wherever the arithmetic is to be carried in a spectrum, the Weyl obstruction. That resonators become "digital" is the claim of the threshold theorem (re-thresholding by projective measurement at gate error $< 10^{-3}$, approximate QFT with phase resolution $1/\mathrm{poly}(n)$, connectivity, magic-state distillation, modular exponentiation at $95\,\%$ of the gates, Doc. 147); each condition has been demonstrated individually, the product at scale has not. Until it is, the apparatus obeys what every resonator obeys. The substrate is immaterial — photonics, ions, superconductors differ only in the error channel (for TFLN, photon loss: < 1 dB per facet per Doc. 186 is up to $20\,\%$ per transition, above the loss thresholds of fusion-based schemes). The photonic operations of Doc. 186 (analogue matrix convolution, resonator locking) are analogue in any case and sit, with $\varepsilon = 1/Q \approx 10^{-6}$, at $N_{\max} \approx 10^6$ (20 bit).

There are thus several limits — resolution, Weyl, resources — and the corpus has no mechanism that frees a physical resonator from them (Doc. 034: ξ effects below noise; Doc. 162: 100ξ damping per gate, character open). Doc. 075 denies the obstruction for Shor; it is an older document, and its denial holds for the model, not for the apparatus. Whether the threshold theorem holds physically at scale is **[S]** and is decided not in theory but on the factoring curve.

Table ah.10: Factoring with Shor's algorithm versus classical records and building blocks (as of September 2026)

Year	Shor on hardware	classical (NFS)	building blocks
2001	$N = 15$	—	—
2012	$N = 21$ (with foreknowledge)	—	—
2019	attempt at $N = 35$ failed	—	—
2020	—	RSA-250 (829 bit)	—
2024	—	—	error correction below threshold (Google Willow)
2025	—	—	48 logical qubits (Quantinuum Helios)
2026	—	—	94 logical qubits $< 10^{-4}$; logical QFT (12 log. qubits)

The larger "quantum records" ($56\,153$, $4\,088\,459$, 2.6×10^{14}, 1.1×10^{12}) concern numbers of special structure (factors differing in a few bits), solved by annealing or QAOA after classical pre-processing; Gutmann and Neuhaus (IACR ePrint 2025/1237) replicated all of them with a 1981 VIC-20, an abacus and a dog. The largest *simulated* Shor run (39 bit) ran classically on 2048 GPUs.

The Shor curve has stood at $N = 21$ since 2012; the classical curve at 829 bit. The yardstick is therefore empirical: **an advantage must become visible on the Shor factoring curve, not on component benchmarks.** That is the corpus position of Doc. 075 — no general exponential quantum advantage demonstrated — and it remains valid in September 2026. The theoretical possibility (Shor polynomial; Weyl does not apply; whether another physical obstruction exists is open) remains **[S]**; its burden of proof lies with experiment. Resolution floor (G′), counting limit (state/process) and resource limit (Shor) are three different ceilings; for factoring only the third acts today, and it has not been broken.

ah.11 Summary

Table ah.11: Results and status

Theorem	Content	Status
A	1D analogue: $(1-3^{-s})\zeta(s) = L(s,\chi_0)$; orbifold zeta = $\frac{1}{3}$ Epstein zeta of D_4, without $(1-3^{-s})$	**[B]**
A′	$L(s,\chi_1)$ measures twist-sector asymmetry	**[S]**
B	Euler product of $L(s,\chi_0)$ hierarchical by $k = \mathrm{ord}_p(3)$; no such ordering in the orbifold spectrum	**[B]**
B′	Fractal damping suppresses Euler factors $k \geq 7$	**[B]**
C	Zeros of $(1-3^{-s})$ at $s = 2\pi i k/\ln 3$ on $\mathrm{Re}(s) = 0$	**[B]**
C′	No resonance of Riemann zeros with Z_3 unit	**[B]**
D	Galois character does not separate; classes = $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$	**[B]**
D′	Sector pairing $k \leftrightarrow -k$: $f_1 \leftrightarrow f_2$, $f_3 \leftrightarrow f_4$ — 4 classes → 2	**[B]**
D″	Negation couples order 26 to 13; no sub-lepton layer	**[B]**
D‴	$\{f_1, f_2\}$ Majorana layer, $\{f_3, f_4\}$ Dirac layer (sector pairing = Majorana criterion)	**[B]**
D⁻	ν_3 Dirac (O4); m_{ee} from ν_1, ν_2 only (≲14.4 meV); IH more strongly excluded	**[B]**
E	$p = 13$ unique prime with $\mathrm{ord}_p(3) = 3$ worldwide; primes forced by $3^k - 1$	**[B]**
E′	$\mathrm{ord}_p(3) = k$, $3 \mid k \Rightarrow p \equiv 1 \pmod 3$ (Z_3-compatible)	**[B]**
F	Euler factor threshold $p^* \approx 9.76$ ($s = 2$) = 7-limit boundary	**[B]**
F′	Product density: chance level from $p_{\max} \approx 30$ (±0.5 %)	**[B]**
F″	Weyl obstruction: limit lies in the object, not the tool (Doc. 316)	**[B]**
G	Coupling: $Q_c = \pi/\sqrt{6\epsilon} \approx 12.5$ at 1.05 %; limit $k \approx 3$–4 = critical point $\kappa = 1$	**[B]**
G′	Same threefold division, different law: Arnold K^q vs. Farey $1/q^2$	**[B]**
G″	FFGFT dynamically deeply subcritical ($K_{\mathrm{eff}} = 2\pi\xi$); discreteness topological, not dynamical	**[B]**
G‴	Theorem F on $\mathbb{Z}/N\mathbb{Z}$ (RSA) only as analogy	**[S]**
H	1.05 % (Doc. 338) ≈ 100ξ (Doc. 162): limit $k \approx 3$ is that of one fractal level	**[B]**
H′	At $\epsilon = \xi$: $Q_c \approx 111$, $p^* \approx 87$, Galois resolved to $k \geq 8$	**[B]**
H″	TFLN $Q > 10^6$ resolves ξ, not ξ^2; ξ = resolution floor, not selector	**[B]**
H‴	Subtractability of the 100ξ level; Doc. 186: locking/Fibonacci/B-meson refuted	**[S]**
I	Resonance factoriser capped at $N_{\max} \approx 1/\epsilon$ (13–26 bit); classical: counting limit, Landauer erasure only	**[B]**
I′	Circuit model digital (Weyl inapplicable); physical apparatus = resonator, subject to ϵ/Weyl until threshold theorem demonstrated; curve at $N = 21$ since 2012	**[S]**

The central statement: in the one-dimensional analogue the decomposition modulo 3 filters the full Riemann zeta to $L(s,\chi_0) = (1-3^{-s})\,\zeta(s)$ — a version in which the 3-Euler factor is separated out and the primes are ordered by the Galois hierarchy $k = \mathrm{ord}_p(3)$. This is the zeta-function form of the entry rule of Doc. 342, i.e. a statement about arithmetic modulo 3. The spectral zeta function of the T^4/Z_3 orbifold is one third of the four-dimensional Epstein zeta of D_4; its Euler factors have the same form for every odd prime, and no GF(3^k) ordering occurs there.

Check scripts

Directory `2/python/Dok343_Skripte/`, wrapper `run_all_343.sh`:

- `pruef_343_zeta_galois.py` — Theorems A–D; partial zeta decomposition, L-functions, zeros, resonance test, weighted Euler product.
- `pruef_343b_klassen_symmetrie.py` — Theorem D; explicit Frobenius classes in $\mathrm{GF}(27)^*$, action of inversion and negation, cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$, exact values of $L(s,\chi_1)$.
- `pruef_344_primzahlmuster.py` — Theorems E–F; patterns in $\mathrm{ord}_p(3)$, exact threshold p^*, Galois vs. random hit rate, Euler factors.
- `pruef_343c_kopplung_zahlen.py` — Plain-terms II section; Farey coverage and $Q_c(\varepsilon)$, law comparison Arnold K^q vs. Farey $1/q^2$, regime of $K_{\text{eff}} = 2\pi\xi$, Galois coverage by $p_{\max}$.
- `pruef_343f_zweite_schicht.py` — Theorem D‴; classes, orbits, involutions $k \mapsto -k$ and $k \mapsto k^{-1}$, Dirac/Majorana partition.
- `pruef_343d_xi_aufloesung.py` — Theorem G′; threshold table for $\varepsilon \in \{1.05\,\%,\ 100\xi,\ \xi,\ 1/Q_{\text{TFLN}},\ \xi^2\}$.
- `pruef_343e_dok186_check.py` — check of Doc. 186: TFLN quality, Arnold tongues at Fibonacci approximants, 50-GHz ladder, B-meson statistics, winding numbers vs. Fibonacci.
- `pruef_343g_epstein_satzA.py` — Theorem A against the four-dimensional spectral zeta function (Epstein zeta of D_4, Weyl pole at $s = 2$).
- `pruef_343h_gf3k_vierdimensional.py` — Euler factors of the orbifold zeta; no ordering by $\mathrm{ord}_p(3)$, no special role of $p = 3$.

  All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All numerical computations are verified by Python check scripts. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 057: Relational number system*, FFGFT corpus.

[2] J. Pascher, *Doc. 159: Harmonic structure of the torus*, FFGFT corpus.

[3] J. Pascher, *Doc. 034: T0 formalism and QM optimisation*, FFGFT corpus.

[4] J. Pascher, *Doc. 162: T0 optimisation strategies compared*, FFGFT corpus.

[5] J. Pascher, *Doc. 186: FFGFT photonics analysis*, FFGFT corpus.

[6] J. Pascher, *Doc. 184: The p-bit in FFGFT*, FFGFT corpus.

[7] J. Pascher, *Doc. 247: Information as state and as process*, FFGFT corpus.

[8] J. Pascher, *Doc. 255: Information kinetics — Landauer and Vopson from T^4 geometry*, FFGFT corpus.

[9] J. Pascher, *Doc. 328: Coupling regimes, resonance and synchronisation*, FFGFT corpus, August 2026.

[10] J. Pascher, *Doc. 330: Three operations from T^4 to the observation space*, FFGFT corpus, August 2026.

[11] J. Pascher, *Doc. 336: The algebraic bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$*, FFGFT corpus, August 2026.

[12] J. Pascher, *Doc. 075: RSA and period-finding*, FFGFT corpus, August 2026.

[13] J. Pascher, *Doc. 147: Quantum computing and T0 time-mass duality*, FFGFT corpus, August 2026.

[14] J. Pascher, *Doc. 316: FFGFT and the Riemann zeta function*, FFGFT corpus, August 2026.

[15] J. Pascher, *Doc. 342: Factorisation classes in $G(6)$ and the harmonics of the primes*, FFGFT corpus, 2 September 2026.

[16] P. Gutmann, S. Neuhaus, *Replication of Quantum Factorisation Records with an 8-bit Home Computer, an Abacus, and a Dog*, IACR ePrint 2025/1237.

[17] C. Gidney, M. Ekerå, *How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits*, arXiv:1905.09749; C. Gidney (2025), resource estimate $< 10^6$ qubits.

[18] P. G. L. Dirichlet, *Beweis des Satzes, dass jede unbegrenzte arithmetische Progression, deren erstes Glied und Differenz ganze Zahlen ohne gemeinschaftlichen Factor sind, unendlich viele Primzahlen enthält*, Abh. Kgl. Preuß. Akad. Wiss. (1837) 45–81.